

Extension of the discrete structure of the continuum: vertex operator algebras, Moonshine, T-duality, and M-theory

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Abstract

An extension of the discrete structure of the continuum is constructed that relates its numerical coordinates π, φ, e, α to exceptional incidence, vertex operator algebras and the duality of a compact sector. Triadic Modular Holography (HMT) begins with *All Possible Paths* (APP): the support $(\mathbb{Z}/9\mathbb{Z})^2$, its additive Cayley graph with cardinal-direction generators $\pm(1, 0), \pm(0, 1)$, and the sum and product evaluations, with residues and quotients retained. The signature $\text{TRIT} = \{-1, 0, +1\}$ distinguishes orientation and regime; the *Topological Phase Kernel* (TPK) composes evaluation selection, transport and memory update to extend compatible states. The same tree of histories supports cylinder measure, sheet transport, oriented balance, incidence and memory complexes with their refinement maps.

The generated dodecaphasic register K links two realisations. Its incidence determines a Witt flag which, with the specified gluing, metric and chamber, leads to a Leech neighbour. The vertex-algebra construction theorems applicable to this lattice identify the extensions of orders two and three with $V^\natural$, whose automorphism group is the Monster; chosen isomorphisms transport products, grading and traces. In the twelve-position representation of A_5 , the projection of K onto the three-dimensional sector is nonzero and determines the orthogonal reduction $12 \rightarrow 11 \rightarrow 10$. The cyclic orbit of the register identifies operators with stability bounds; bilateral balance provides state and memory recovery under its isometric hypotheses.

Comparison of the stars of B_K, B_φ and B_{APP} distinguishes the polarised signatures $(3, 1, 0), (1, 0, 3)$ and $(1, 1, 2)$. The ordered frame of the four supports $H_{\bar{\alpha}}, H_e, H_\varphi, H_{\text{APP}}$ has trivial stabiliser in the Witt action on the twelve positions. Alongside the regional residues, the numerical reading retains the integer quotients of sum and difference: their composition satisfies a cocycle identity, and their reductions supply additional algebraic components beyond the affine evaluation of the inputs. These results specify the complementary roles of incidence, orientation and carry memory within the common extension.

The Weyl realisation of phase generates the matrix algebras at each depth; full-fibre inclusions yield a uniformly hyperfinite algebra and a tracial factor of type II_1 . The action section and angular pair fix an oriented ellipse and a unique normalised radius. Integer-coordinate exchange and radius inversion preserve its quadratic form and unitarily intertwine the self-adjoint Hamiltonians, including their domains. Transport of the residual relation ensures covariance under a change of section. A screen morphism coordinates this duality with the K -directed reduction and the isotropic lattice quotient $26 \rightarrow 24$.

Composition of trajectories with boundary data retained defines an additive action cocycle. Its extension towards M-theory specifies the codomain of eleven-dimensional fields and the realisation maps that must preserve action, grading and duality. Under the stated intertwining and domain conditions, the supercharge identity is transported to the isometric image. The conditions for a complete physical identification are distinguished from these algebraic and operator-theoretic results.

The arithmetic preceding these realizations includes the APP defect laws under changes of modulus and dimension, the invariant index-two sublattice, and nonadic encoding with carry. The emitter determines an alphabet of 42 blocks and explicit regional junctions; the ordered register retains correlations distinct from its histogram. Quadratic correlation proves the boundary-chart identity, and selection separates saturation, dual neutrality, and orientation. The level-three rational identity also establishes optimal uniform divisibility by 3^9 in the formal power-series ring.

Keywords: Triadic Modular Holography; discrete structure of the continuum; dodecaphasic register; vertex operator algebras; Moonshine; Monster; T-duality; Weyl representation; dimensional screens; M-theory.

Contents

1	Introduction	5
2	Formal kernel of Triadic Modular Holography	8
2.1	APP: positional support and arithmetic evaluations	8
2.2	TRIT: orientation and quadratic regimes	13
2.3	TPK: Topological Phase Kernel	20
2.4	Accumulated memory and transport resolvents	26
3	Topological Phase Kernel and generation of the seed catalogue	30
3.1	Phase selection, transport and enriched state	30
3.2	Initial conditions and reversible enumeration	31
3.3	Sheet evaluation and arithmetic record	32
3.4	Six windows and decimal block emission	33
3.5	Complete calculation of an emission beginning with 501	36
3.6	Factorisation of the finite census	37
3.7	Oriented ternary reading and preservation of fibres	39
3.8	Dihedral action and the forty-three orbits	40
3.9	Self-contained enumeration and verification algorithm	41
3.10	Preservation of the record and the function of the seeds	41
3.11	Cyclic correlation and preservation of path order	41
4	Development of the formal kernel: initial conditions, geometry, and refinement	43
4.1	Initial states and the emission recurrence	43
4.2	Calendar closure and memory advancement	44
4.3	Observable compression and orthogonal conservation	46
4.4	Local development: commutation, Lorentzian signature, and discrete curvature	46
4.5	Cubic completion and two notions of boundary	48
4.6	Cubic incidence and Lorentzian realization	51
4.7	Compatibility of modular and positional reductions	52
4.8	Number as a compatible section and value as a reading	53
4.9	Composition of histories and memory multiplier	53
4.10	Double variance of refinement and conservation of unity	54
4.11	Phase return, Weyl representation, and two readings	55
5	Correlative construction of the continuum: histories, transport, incidence and terminal	56
5.1	Admissible histories and memory by composition	56
5.2	Tree refinement and measure of histories	57
5.3	Sheetwise transport and exact accounting of outputs	57
5.4	Oriented balance and cancellation under refinement	59
5.5	Ternary sheet incidence and rectangular complex	59
5.6	Memory complex and compatibility of incidences	60
5.7	Assembly with terminal paths	61
5.8	Determinants, limit and arithmetic readout	62
5.9	Provenance, verification and scope	63
6	Coinductive generation of π, φ, e to arbitrary depth	65
6.1	Orbits, regions, and characters of closure, propagation, and self-scaling	65
6.2	Incidences between halves of regional emissions	66
6.3	Finite transport and the prolongation boundary	69
6.4	Correlated construction and ordering of the readers	70
6.5	Deep memory and compatibility of the cylinders	70
6.6	Five closure regions and angular compensation	72
6.7	Propagation reader and normalization of unity	72
6.8	Modal incidence and generation of self-scaling	73
6.9	Iteration of the ratio reader and phase recovery	76
6.10	Exponential realizations of the generated coordinates	77
6.11	Compatible cylinders and arbitrary depth	79
6.12	Specular involution and synchronization of the regions	80
6.13	Sturmian helical suspension	80
6.14	Nonadic connection, holonomy, and monodromic representation	83
6.15	Block clock and capacity clock	84
6.16	Symbolic suspension and aperiodic time crystal	85
6.17	Specular symmetry of the regions and double realization	86
6.18	Temporal transport and tritic exponential composition	89

6.19	Multiplicative irreducibility and return periods	89
7	Regional generation and refinement	92
7.1	Positional emission through rational separation of cells	92
7.2	Regional blocks of six trits at arbitrary depth	93
7.3	Transition registers and reconstruction of linear lifts	94
7.4	Simultaneous refinement in ternary and decimal bases	95
7.5	Selection of the next block by the regional reader	96
7.6	Ternary signature, carry, Witt incidence, and reading clock	97
7.7	The carry cocycle and the integer lift of the regional signature	98
7.8	Integer clock of decimal reading	101
7.9	Composition theorem for the block, cylinder, and signature	101
7.10	Calendar census and simultaneous compatibility of the regional biographies	102
8	Regional determination of the dodecaphasic descriptor and selection of the coupling register	103
8.1	Regional bands, encoding, and integer signatures	103
8.2	Conversion clock and first zero advance	104
8.3	Oriented construction of the twenty-four-trit descriptor	105
8.4	Functional panel and terminal repertoire	106
8.5	Comparative readers, affine reader, and temporal incidence	110
8.6	Enumeration of cylinders and decimal candidates	111
8.7	Exceptional incidence and heterotypy condition	113
8.8	Regional composition and continuity of the realizations	115
8.9	Executable certification and traceability of the operations	116
9	Dodecaphase record and the rational phase constant	117
9.1	Cyclic extension and memory grading	118
9.2	The oriented record and the integral $1 + 5 + 6$ chart	118
9.3	Incidence evaluation preceding the dodecaphasic record	119
9.4	Conjugation of counts and boundary terms	120
9.5	Complete cylinder census and terminal-boundary selection	121
9.6	Visible reading and signed record of the terminal state	125
9.7	From the channel record to the signed vector	126
9.8	Orbitwise Hadamard transformation and integer reconstruction	126
9.9	Cyclic differences and the uniform component	128
9.10	Integral compatibility and determination of the transverse record	128
9.11	Two rational readouts: κ_{36} and κ_{per}	131
9.12	Reversibility of the rational constant and phase change	132
9.13	Periodicity and the cokernel of the transport operator	133
10	Derivation of the fine-structure constant	134
10.1	First route: positional determination by integer closure	134
10.2	Second route: analytic determination through vacancies	141
10.3	Compatibility of the dodecaphase and analytic representations of alpha	143
11	Exceptional prolongation: from dodecaphasic incidence to Moonshine	151
11.1	Compatibility of the Archimedean and incidence realizations	151
11.2	The finite realization and the ternary code	152
11.3	Dodecaphasic flag associated with K and α	156
11.4	Isometric intertwining of the incidence representations	158
11.5	The direction selected by the dodecaphase record	159
11.6	The Witt star and the generation of M_{12}	161
11.7	Three regional stars and the polarization of their completions	162
11.8	Polarized classification of the 495 four-subsets	164
11.9	Rigidity of the ordered four-support frame	165
11.10	Finite incidence topology and preservation of its domain	165
11.11	The binary residue and the five regions of π	166
11.12	From the ternary code to the Niemeier lattice	168
11.13	Incidence selection of the rootless unimodular neighbor	169
11.14	The vertex operator algebra and the Monster	170
11.15	Ternary orientation of the Leech neighbor	171
11.16	Ternary trace, twisted conformal weight, and order-three extension	172
11.17	Equivalence of the order-two and order-three presentations	173
11.18	Eta functions, the Fricke involution, and vacuum normalization	174
11.19	The Fricke correspondence and differential recovery	176
11.20	Compatibility of the level-two and level-five modular readers	177
11.21	Exceptional incidence and coordination of the screens	179

11.22	Graded Moonshine traces and representation of the Monster	179
12	Hilbert-space realization of the nonadic phase and preservation of the unit	181
12.1	Phase fiber, labeled basis, and inner product	181
12.2	Translation and phase: finite Weyl representation	182
12.3	Two lifts and two different normalizations	184
12.4	Inductive closure and tracial realization	184
12.5	Integer ranks and continuous tracial dimension	185
12.6	Polar transport of incidence and preservation of readers	186
13	Arithmetic–geometric coupling and bilateral recovery	188
13.1	Exact encoding and cyclic reference	188
13.2	Conservation and reconstruction of two channels	189
13.3	Transport of observables and incoming memory	189
14	Determinantal structure and oriented action sections	191
14.1	Local quotient and determinantal norm	191
14.2	Action character and regional continuation	192
14.3	Positive sections and action per turn	193
15	Angular representation and channel orientation	194
15.1	Circular charts and covariance of the action	194
15.2	Positive channels and reversal of the odd coordinate	195
16	Dodecaphasic germ and recovery of unimodal branches	196
16.1	Normalization and analytic profile	196
16.2	Branchwise inversion and itinerary memory	197
16.3	Scope of the rescaled return	197
17	Action ellipse and determination of the normalized radius	198
17.1	Domain and labelled semiaxes	198
17.2	Parametric phase and oriented area	198
17.3	Shape reciprocity and symplectic orientation	199
17.4	Normalized radius and lattice-readout matrix	199
17.5	Oriented depth moment	200
17.6	Pell dilation, Catalan moment and oriented return	201
18	T-duality: integer coordinates, change of section, and operator equivalence	203
18.1	Stable domain of the interchange	203
18.2	Compact Hamiltonian and self-adjoint domain	203
18.3	Specialization to the radius of the action ellipse	204
18.4	Change of representative and quotient memory	205
18.5	Radius normalization in the string realization	205
18.6	Relationship with the modular transformation	206
19	Dimensional screens and coordinated state reduction	207
19.1	Channels, relations, and effective dimension	207
19.2	Ternary puncturing and syndrome screen	207
19.3	The dodecaphase direction and rank reduction	208
19.4	Lorentzian lattice screen	208
19.5	Joint screen morphism	209
19.6	Coordination with the exceptional realization	209
20	Action on trajectories and eleven-dimensional dynamical realization	210
20.1	Descent of an evolution and sufficiency of the observed datum	210
20.2	Cocycle of directional and arithmetic changes	210
20.3	Extended persistence and worldsheet	211
20.4	Eleven-dimensional realization and graded operators	211
20.5	Compatible compactification and conclusion of the development	212
21	Conclusions	213
A	Finite enumerations of emissions and Witt stars	215
A.1	Initial-condition indices and signature recurrence	215
A.2	Explicit generators and closure of the stars	217
	References	218

1 Introduction

How can a single discrete structure determine coordinates of the continuum, support an exceptional realisation and preserve a duality on passing to dimensional representations? The question requires more than collecting values, dimensions and groups: it requires the construction of intermediate objects and the maps that transmit their information. This article develops that connection in Triadic Modular Holography (HMT). Its organising result is a twofold extension of the genealogical state: incidence leads to a vertex operator algebra isomorphic to $V^{\natural}$, while phase, action and channel representations determine a compact duality and a screen morphism. Coordination of these extensions then specifies the algebraic domain and transport conditions for an M-theory realisation.

The construction begins before real-valued evaluation. *All Possible Paths* (APP) denotes the paths on the support $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. Its Cayley graph uses the additive law and the cardinal-direction generators $\pm(1, 0), \pm(0, 1)$. Sum and product are evaluated on the same positions; residual reduction retains the quotient of each evaluation. The additive and multiplicative sheets are therefore two evaluations on a common support. The signature $\text{TRIT} = \{-1, 0, +1\}$ distinguishes transport orientation and the hyperbolic, parabolic and elliptic regimes of its quadratic realisation. The *Topological Phase Kernel* (TPK) is the effective composition of evaluation selection, cardinal-direction transport and register update that extends compatible states. These states retain positions, sheets, orientation, residues, quotients, carries, prefixes, boundary data and memory. Selection does not replace displacement, and phase return does not erase the history traversed.

The operations connecting arithmetic support with incidence admit explicit intermediate proofs. Changing the modulus, repeating the support at fixed modulus, and varying the dimension have their own sum-product defect laws. Compression into three classes produces an index-two sublattice and a quadratic unit of norm one. Within the emitter, the decimal alphabet, half-emission junctions, and the contrast between histogram and ordered memory preserve the distinctions between value, incidence, and trajectory. The multiplicative observable is examined before regional selection.

The emitter's labelled multigraph and its phase quotient place these realizations on explicit incidences. The five minimal eight-edge closures cover the APP, propagation, autoscale, and individual closure-region anchors. Minimum boundary occupancy and the parity of memory readers further specify representative selection and the transformation of records under route reversal.

The first result is the joint organisation of the discrete structure of the continuum. In Section 5, a single tree of executions determines cylinder measure and sheet transport, oriented balance, ternary incidence, the memory complex and its terminal assembly with determinant line. Maps are defined on generators, and their compatibility with history restrictions and modular reductions is proved. The measure uses compatible positive weights on a finitely branching tree without terminal leaves; the Gram identity retains the terminal survival remainder; the determinant construction retains its degrees and bases, together with the additional hypotheses required by a scalar evaluation. The five constructions thus retain a common origin and the particular conditions of each operation. They are not five domains selected after the coordinates have been produced.

Sections of compatible cylinders determine the numerical coordinates of the correlated generation of π, φ, e, α under the nesting, contraction and boundary conditions of their Archimedean realisation. The exceptional projection instead retains the incidence data realised as codes, lattices and graded algebras. The areal and volumetric sheets and their measures, developed in Section 6.8.1, and the vacancies associated with refinement, in Section 6.13.1, enter the construction of α and the exceptional extension together with the regional registers. Electron mass uses a different composition of these common operations; its evaluation is not the subject of this article.

The chart grouping two trits into a nonadic digit has an inverse and a transported addition with carry; the projection of the ambient cube of 729 words onto its first two coordinates has uniform nine-element fibres. The quadratic-correlation proof then determines the identity of the six-position chart. At the boundary, saturation leaves eight possibilities, dual neutrality leaves two, and orientation selects the fixed representative. Closure controls retain the endpoints of the telescoping identity.

The exceptional connection has a concrete antecedent in the dodecaphasic register K . Its twelve components arise from the signed register and the TPK operations. Rational encoding preserves the

blocks when base, length, phase origin and orientation are fixed; Hadamard inversion recovers the corresponding signed coordinate. The supports of the register enter a marked flag of the Witt design. The ternary code constructs the gluing of the Niemeier lattice with root system A_2^{12} ; the flag and origin, together with the specified metric and positive chamber, select the rootless neighbour identified with the Leech lattice. Each operation transmits a datum needed by the next object.

Section 11.7 specifies the passage from regional registers to incidence through a comparison of three stars. The intersections of P_π with H_e , H_φ and H_{APP} determine the four-subsets B_K , B_φ and B_{APP} ; the sign partition of the pre-carry cochain determines the polarisation of their completions. The census of all 495 four-subsets locates these configurations within the complete design, and the stabiliser of the ordered four-support frame characterises its rigidity within G_W . This rigidity concerns the action on positions; the numerical determination of K retains its integer construction, phase origin and previously fixed orientation.

Section 7.7 develops the integer component accompanying the residue reading. Euclidean division produces the residues q, a and quotients c, h , whose composition retains the memory of multiples of three. The cocycle, polynomial representations over $\mathbb{F}_3$ and transport identities specify which data pass through each change of register. Their connection to compatible cylinders retains the boundary bounds and the fibres of the regional reading. The star comparison and carry law thus have specified incidence and numerical roles in the connection that this article extends to lattices and algebras.

The lattice algebra is constructed on this lattice with oscillators, central extension and vertex product. The order-two extension of Frenkel–Lepowsky–Meurman and the order-three extension associated with the ternary isometry realise $V^\natural$, once the hypotheses of the recognition theorems used have been verified [11, 17, 18]. The chosen isomorphisms between the presentations preserve the vacuum, conformal vector, grading and products, and transport automorphisms and traces. The Monster is the automorphism group of this algebra; its action has domain $V^\natural$, not the digits of α . Borcherds’s Moonshine theorem determines the corresponding graded traces [12]. Lattice construction, algebraic extension and recognition of these traces are distinct stages of one chain of maps.

Upon reaching the level-three traces, formal factorization of the rational identity determines divisibility at every degree, with its optimal exponent.

The register K also has a directional realisation. In the specified representation of A_5 on the twelve positions, its projection onto the three-dimensional sector determines a vector $u_K \neq 0$ in $V_{11} = \mathbf{1}^\perp$. The nonzero vector determines the line ℓ_K , and its exact squared norm normalizes the rank-one term of the projector onto $V_{10} = V_{11} \cap u_K^\perp$, with specified kernel and covariance. The reduction $12 \rightarrow 11$ removes the uniform mode; the reduction $11 \rightarrow 10$ removes the direction selected by the complete register. Its cyclic orbit is also an invertible reference for identifying operators through their responses. Section 13 establishes the stability bounds and bilateral balance: two channels constructed from isometries allow the state to be recovered; the unitary extension includes incoming memory, and preservation of an observable requires the commutation condition in its statement.

The Hilbert-space realisation supplies the operator-theoretic support for duality. At each depth, phases label an orthonormal basis and the shift and character operators generate its full matrix algebra. Their Weyl relation represents the central extension of translations with the sign of the oriented area form fixed. Inclusion through the identity of the entire new fibre preserves the unit and normalised trace: the limit is the nonadic uniformly hyperfinite algebra, and its tracial closure is a factor of type II_1 . Fixing one continuation instead gives a corner inclusion and a different closure. The refinement map therefore determines the type of realisation, not merely its dimension.

The action section and angular coordinates fix the radius that specialises the compact duality. The determinantal norm, the coefficients of the action character and normalised regional renewal determine the positive sections before and after return. The angular charts retain orientation and winding number. Their composition defines a labelled ellipse whose semiaxis product preserves action and whose ratio determines a unique dimensionless radius. Positivity conditions place the generated coordinates in the elliptic chart; semiaxis exchange produces the reciprocal radius while preserving the common action scale.

On $\mathbb{Z}^2 \times \mathbb{R}_{>0}$, integer-coordinate exchange and radius inversion form an involution preserving the compact quadratic form. The diagonal Hamiltonian is self-adjoint, nonnegative and has compact resol-

vent; coordinate permutation unitarily conjugates it to the reciprocal-radius Hamiltonian, including the operator and quadratic-form domains. A change of residual section also transports the relation L_9 that preserves the integer. The minimum energy of each class is well defined and is evaluated using two integer candidates. Normalisation by the string scale expresses the same duality in dimensional radii. On the imaginary axis of the upper half-plane, modular inversion is related to it by an explicit semiconjugacy of parameters. This map does not identify the exchange operator with the automorphisms of orbifold sectors.

The dimensional screens coordinate the K -direction, incidence and compact sector. Puncturing the ternary code produces a length-eleven code with distance five and a perfect partition. The orthogonal reduction $12 \rightarrow 11 \rightarrow 10$ and the isotropic lattice reduction $26 \rightarrow 24$ retain their own presentations, relations and kernels. The morphism $\mathcal{R}_M^{\text{alg}}$ identifies their quotient presentations and intertwines duality; the graph of the projection retains the coordinate in V_{11} together with its image in V_{10} . The same operations restrict to the collection of the two radii determined by the ellipse.

The dynamical extension begins with trajectory composition, not with the choice of fields. Retaining the preceding datum at the boundary makes changes of direction and arithmetic type an additive action cocycle. The worldsheet and Polyakov action then specify a variational realisation. The map to the M-theory realisation receives the screens in an eleven-dimensional codomain equipped with a metric, three-form, gravitino, grading, action and constraints. Under the intertwining and domain-transport hypotheses, the supercharge theorem preserves $Q^2 = H$ on the image of the isometry. Complete physical identification additionally requires tensions, oscillators and amplitudes, anomaly control and the limits stated in Section 20.5.

This hierarchy determines the development: kernel and joint continuum; coinductive generation, register K and the two constructions of α ; incidence and exceptional algebra; Hilbert-space realisation, action and radius; duality, screens and dynamical transport. Section 9 determines the register used in 11.5; the realisation in 12 supports the operators in 18; and 19 composes the reductions in the morphism received by the eleven-dimensional realisation. The argument thus connects structure, value, representation and equation through their effective maps.

2 Formal kernel of Triadic Modular Holography

Triadic Modular Holography, abbreviated HMT, begins with a finite structure of positions and operations. The order of construction matters: positions receive arithmetic evaluations, transitions acquire a ternary signature, and the dynamics preserves the information needed to prolong them. Numerical realizations are formed only afterwards. A constant recognized at the end of this process does not replace the rule that produced its region or the history that determines its evaluation.

The name *All Possible Paths*, APP, expresses the role of paths on the support. It is adopted as the name of the construction, with a conceptual reference to the path formulation of quantum mechanics; it does not imply that its arithmetic law is Feynman's path integral. The *Topological Phase Kernel*, TPK,¹ denotes the composition of selection, transport, memory update, and prolongation. The acronyms are retained only after the objects they represent have been defined.

2.1 APP: positional support and arithmetic evaluations

Consider the nine interior marks on a line graduated from zero to ten,

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Their horizontal and vertical arrangement produces the eighty-one positions of D_9^2 . The interior marks must be distinguished from the ten unit intervals of the reference line. This arrangement fixes an alphabet and two directions; it does not yet use a real expansion to determine a trajectory.

Cyclic identification of the positions yields the support

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The chart of positive representatives retains the symbol nine for the zero class. For every positive integer, define

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n). \quad (2.1)$$

The positive digital root is therefore a choice of representative of the residue class. By itself, it does not preserve the integer before reduction; the coordinate q_9 supplies precisely the missing information.

At each position (i, j) , accumulating i , repeated j times, produces ij . The two integer evaluations and their projections are

$$S(i, j) = i + j, \quad P(i, j) = ij, \quad \Sigma(i, j) = \rho_9(i + j), \quad \Pi(i, j) = \rho_9(ij).$$

The symmetry $(i, j) \mapsto (j, i)$ interchanges the directions of accumulation and preserves both evaluations. This is a property of a single support, not a subsequent identification between two independent matrices.

Definition 2.1 (Lifted arithmetic evaluation). The APP evaluation that preserves both sheets is

$$\mathcal{A}(i, j) = ((R_{ij}^+, q_{ij}^+), (R_{ij}^\times, q_{ij}^\times)) = ((\rho_9(i + j), q_9(i + j)), (\rho_9(ij), q_9(ij))). \quad (2.2)$$

Here the term *sheet* denotes an evaluation on the same support: additive or multiplicative. The two quotient coordinates belong to the arithmetic state and are not discarded when the residues are reduced.

Proposition 2.2 (Reconstruction of the evaluations). *The two pairs in (2.2) determine $i + j$ and ij exactly. The residues alone do not determine these integer evaluations or their quotients.*

Proof. The first statement is (2.1) applied to each sheet. For the second, $10 = 1 + 9$ and $19 = 1 + 2 \cdot 9$ have the same residue and different quotients. On the two-dimensional support, positions $(5, 5)$ and $(8, 2)$ give the same sum and the same product residue, but products $25 = 7 + 18$ and $16 = 7 + 9$. The residue projection identifies data that the lifted evaluation distinguishes. $\square$

¹The authors also dedicate the initials TPK to Tan Pengkiang. This dedication is independent of the mathematical definition.

	1	2	3	4	5	6	7	8	9	One support: 81 positions.
1	1	2	3	4	5	6	7	8	9	Two evaluations: $S(i, j) = i + j$, $P(i, j) = ij$.
2	2	4	6	8	1	3	5	7	9	Each row is generated by accumulation: $P(i, j + 1) - P(i, j) = i$.
3	3	6	9	3	6	9	3	6	9	
4	4	8	3	7	2	6	1	5	9	The chart represents $0 \pmod{9}$ by 9.
5	5	1	6	2	7	3	8	4	9	The highlighted block is $B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$, with eigenvalues 9 and 5.
6	6	3	9	6	3	9	6	3	9	
7	7	5	3	1	8	6	4	2	9	
8	8	7	6	5	4	3	2	1	9	
9	9	9	9	9	9	9	9	9	9	

Figure 1: APP multiplication table in positive representatives. Non-unit rows and columns are distinguished by gray shading; the frame identifies the adjacent diagonal block with equal diagonals. The displayed table is an observable of the arithmetic torus; its representation edges do not constitute an intrinsic topological boundary of the torus.

2.1.1 Cayley graph, toroidal topology, and winding number

The cardinal translations are given, in row and column coordinates, by

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \quad T_d(x) = x + \nu_d.$$

The first coordinate increases southwards and the second eastwards. The graph

$$\mathcal{G}_9 = \text{Cay}(X_9, \{\nu_N, \nu_E, \nu_S, \nu_O\})$$

is the two-dimensional cyclic lattice: it has eighty-one vertices and four outgoing oriented edges per vertex. It is a Cayley graph for the *additive* structure of the support. Multiplication of residues has zero divisors and does not turn all APP rows into a multiplicative group.

A cardinal word $w = d_1 \cdots d_r$ and an origin x_0 determine

$$x_k = x_0 + \sum_{j=1}^k \nu_{d_j} \pmod{9}.$$

There are $81 \cdot 4^r$ oriented paths of length r , including returns and backtracking. When the path closes, its integer displacement before reduction defines

$$\text{wind}(w) = \frac{1}{9}(\#S(w) - \#N(w), \#E(w) - \#O(w)) \in \mathbb{Z}^2. \quad (2.3)$$

Path reversal changes the sign of this vector. Returning to the same residue position therefore does not require the winding to vanish. This is the first example of a distinction that will reappear in phase return: one coordinate may close while another records a nontrivial evolution.

2.1.2 Transport quotient and cocycle identity

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \rightarrow D_9$ be the section of positive representatives. The transport quotient associated with additive composition, also called the *carry* in positional arithmetic, is

$$c_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9}.$$

The compositional character of this memory is expressed by an exact identity.

Lemma 2.3 (Cocycle identity). *For $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a, b) + c_9(a + b, c) = c_9(b, c) + c_9(a, b + c).$$

Proof. Both sums equal

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$

Associativity of the sum of lifted representatives requires precisely this additional term when composing in the quotient. $\square$

With positive representatives, this cocycle is not normalized at the identity: $c_9(0, a) = 1$. If s_0 takes representatives in $\{0, \dots, 8\}$, its normalized carry c_0 satisfies

$$c_9(a, b) = c_0(a, b) + \mathbf{1}_{a=0} + \mathbf{1}_{b=0} - \mathbf{1}_{a+b=0}.$$

The difference is the change of section; it does not alter the cocycle identity. The subsequent decomposition of the calendar into phase and window will use representatives from zero to eight.

The totals of both evaluations distinguish the visible contribution from that retained in the quotient:

$$\begin{aligned} \sum S &= 810, & \sum P &= 2025, & \sum \Sigma &= 405, & \sum \Pi &= 459, \\ \sum q^+ &= 45, & \sum q^\times &= 174. \end{aligned}$$

It follows that

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \quad (2.4)$$

The sum-product difference is not exhausted by the visible defect 54. The identity also preserves the quotient defect 129. Removing it would change the arithmetic being transported, even if the residue table remained identical.

2.1.3 Modulus refinement and preservation of the two readers

Identity (2.4) preserves the integer defect 1215 through residue 54 and quotient 129. Its generalization requires separate specifications of the reading modulus, the support size and the number of coordinates. These three operations produce distinct counting laws from the same APP sheets.

For $N \geq 2$, fix the positive representative

$$\rho_N(s) = 1 + ((s - 1) \bmod N) \in \{1, \dots, N\}, \quad s = \rho_N(s) + NQ_N(s), \quad Q_N(s) \in \mathbb{Z}.$$

On $\{1, \dots, N\}^2$, the full-modulus readings are $\rho_N(i + j)$ and $\rho_N(ij)$. Their sums are denoted by $S_+^{\text{full}}(N)$ and $S_\times^{\text{full}}(N)$.

Proposition 2.4 (Defect law for an integer modulus). *For every $N \geq 2$,*

$$S_+^{\text{full}}(N) = \frac{N^2(N + 1)}{2}, \quad (2.5)$$

$$S_\times^{\text{full}}(N) = \frac{N}{2} \left(N^2 + \sum_{i=1}^N \gcd(i, N) \right), \quad (2.6)$$

$$\Delta_N^{\text{full}} = \frac{N}{2} \left(\sum_{i=1}^N \gcd(i, N) - N \right). \quad (2.7)$$

Proof. In an additive row, translation by i permutes the N residue classes and their positive representatives. Each row sums to $N(N + 1)/2$, proving (2.5). For a multiplicative row, let $g = \gcd(i, N)$. The image of $j \mapsto ij \pmod{N}$ contains the N/g residue classes divisible by g , and each has g preimages. Their positive representatives are $g, 2g, \dots, N$, whose sum is

$$g \frac{(N/g)(N/g + 1)}{2} = \frac{N(N + g)}{2g}.$$

Multiplicity g gives $N(N + g)/2$ per row. Summation over i yields (2.6); subtracting (2.5) gives (2.7). $\square$

Corollary 2.5 (Prime-power moduli). *If $N = p^m$, where p is prime and $m \geq 1$, then*

$$\sum_{i=1}^{p^m} \gcd(i, p^m) = p^m + m(p-1)p^{m-1}, \quad \Delta_{p^m}^{\text{full}} = \frac{m(p-1)}{2} p^{2m-1}. \quad (2.8)$$

Proof. For each $0 \leq k < m$, there are $p^{m-k} - p^{m-k-1}$ indices with exact p -adic valuation k . Their greatest common divisor with p^m is p^k , so this stratum contributes $(p-1)p^{m-1}$. The m strata contribute $m(p-1)p^{m-1}$, whereas the index $i = p^m$ contributes p^m . Substitution into (2.7) gives the second formula. $\square$

The triadic specialization is $\Delta_{3^m}^{\text{full}} = m3^{2m-1}$. The first moduli display both sums together with their difference:

N	S_+^{full}	$S_\times^{\text{full}}$	Δ_N^{full}
3	18	21	3
9	405	459	54
27	10206	10935	729
81	269001	277749	8748

Periodic refinement instead retains ρ_9 and enlarges the support to $\{1, \dots, 9r\}^2$, with integer $r \geq 1$. Each pair of residues modulo nine has r^2 representatives in this support, for both addition and multiplication. Consequently,

$$S_+^{\text{per}}(9r; 9) = r^2 405, \quad S_\times^{\text{per}}(9r; 9) = r^2 459, \quad \Delta^{\text{per}}(9r; 9) = r^2 54. \quad (2.9)$$

For $r = 3$, the sums are 3645 and 4131, and the defect is 486. The full-modulus reading at 27 produces 729 because it updates both the modulus and its representatives; the periodic reading produces 486 because it retains modulus nine and multiplies its incidences by nine. In both readings, the divisions $i + j = R^+ + NQ^+$, $ij = R^\times + NQ^\times$, with the modulus actually employed, preserve the integer identity at each cell and after summation. This specifies the datum accompanying each defect.

2.1.4 Dimensional variation at fixed modulus nine

A second extension retains the modulus and increases the number of factors. For $d \geq 1$, let

$$\mathcal{A}_d = \{1, \dots, 9\}^d, \quad \Sigma_d(x) = \rho_9\left(\sum_{j=1}^d x_j\right), \quad \Pi_d(x) = \rho_9\left(\prod_{j=1}^d x_j\right).$$

Proposition 2.6 (Dimensional count for accumulation and multiplication). *The sums of the two readers over $\mathcal{A}_d$ are*

$$S_{+,d} = 45 \cdot 9^{d-1}, \quad S_{\times,d} = 9^{d+1} - 9(d+3)6^{d-1}. \quad (2.10)$$

Therefore,

$$\frac{S_{\times,d} - S_{+,d}}{9^d} = 4 - (d+3)\left(\frac{2}{3}\right)^{d-1} \longrightarrow 4. \quad (2.11)$$

Proof. For each fixed choice of the first $d-1$ coordinates, the final coordinate traverses every additive residue once. Each positive representative occurs 9^{d-1} times, and their sum is 45; this proves the additive formula.

For multiplication, distinguish the units $U = \{1, 2, 4, 5, 7, 8\}$, the multiples of three of valuation one $T = \{3, 6\}$, and the zero representative 9. The 6^d words whose factors all belong to U distribute their products uniformly over the six units: fixing $d-1$ factors leaves a permutation of the group under multiplication by the last factor. Their contribution is $27 \cdot 6^{d-1}$.

The words with exactly one factor in T and all other factors in U number $2d \cdot 6^{d-1}$. For each position of the singular factor and each choice of the remaining units, its two possible values 3, 6 produce each

of the residues 3, 6 once. Their contribution is $9d6^{d-1}$. Every remaining word contains a factor 9, or at least two factors in T , so its product has representative 9. Their number is $9^d - 6^d - 2d6^{d-1}$. Thus

$$S_{\times,d} = 27 \cdot 6^{d-1} + 9d6^{d-1} + 9(9^d - 6^d - 2d6^{d-1}),$$

which simplifies to (2.10). Dividing by 9^d and subtracting the additive mean 5 gives (2.11); the term $(d+3)(2/3)^{d-1}$ tends to zero. $\square$

Dimension two recovers $S_{+,2} = 405$, $S_{\times,2} = 459$ and total defect 54. The limit 4 belongs to the mean defect under increasing dimension. Total defect, preserved quotient and dimensional mean are specified readers of APP arithmetic; TRIT and TPK subsequently transport these data with their orientation, modulus and mathematical residence.

2.1.5 Ternary reduction and multiplicative radical

The ring $R = \mathbb{Z}/9\mathbb{Z}$ has unit group $\{1, 2, 4, 5, 7, 8\}$ and maximal ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\}, \quad \mathfrak{m}^2 = 0.$$

Every additive row is a permutation of the nine representatives. A multiplicative row is such a permutation exactly when its index is a unit. Rows three and six contain three repeated values; row nine contains only the representative of the zero class. Thus multiplicative folding already distinguishes the units from the nilpotent sector before any continuous geometry is introduced.

The map $x \mapsto 3x$ has kernel and image both equal to $\mathfrak{m}$. It induces an isomorphism of vector spaces over $\mathbb{F}_3$,

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \quad [x] \longmapsto 3x,$$

but not a ring isomorphism. In particular, the visible sequence 3, 6, 9 admits two different readings: its direct reduction modulo three is 0, 0, 0, whereas extracting the factor three and then reducing its internal coordinate gives 1, 2, 0. This second operation preserves a coordinate of the ideal; it is not the same projection as directly reducing the original number.

2.1.6 Latin realization of the additive evaluation

The difference between the two evaluations can also be expressed through operators. Let $(e_r)_{r \in \mathbb{Z}/9\mathbb{Z}}$ be the orthonormal basis of $\mathbb{C}^9$. The assignment $v_{ab} = e_{a+b}$ produces an orthonormal basis in every row and column. The projectors $P_{ab} = v_{ab}v_{ab}^*$ satisfy

$$\sum_b P_{ab} = I_9 = \sum_a P_{ab}, \quad P_{ab}^2 = P_{ab}.$$

Indeed, each translation $b \mapsto a+b$ permutes the nine residues. This yields a realization of a Latin square by rank-one projectors, in the terminology of quantum Latin squares [4, 6]. This construction starts from the additive sheet already defined and makes explicit the map to its operator representation.

The multiplicative evaluation preserves different content: for $a = 3$, the vectors e_{3b} repeat the residues 0, 3, 6, and

$$\sum_{b \in \mathbb{Z}/9\mathbb{Z}} |e_{3b}\rangle\langle e_{3b}| = 3(|e_0\rangle\langle e_0| + |e_3\rangle\langle e_3| + |e_6\rangle\langle e_6|).$$

The multiplicity is recorded in the operator and characterizes the multiplicative radical. Thus the toroidal support, the Latin property, and the operator resolution of the identity are properties requiring different definitions. The quantum magic squares studied by De las Cuevas, Drescher, and Netzer require positive entries and row and column sums equal to the identity; their theory also distinguishes dilations with commuting entries [5]. This reference provides a precise language for comparing realizations of APP.

2.2 TRIT: orientation and quadratic regimes

Definition 2.7 (Ternary signature and local lift). The TRIT signature uses the balanced identification

$$\mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1.$$

For $n \in \mathbb{Z}$, its lift preserves $n = r + 3c$, where $r \in \{-1, 0, +1\}$ and $c \in \mathbb{Z}$. In a transition of bounded local amplitude, the three states $(0, -1)$, $(0, 0)$, $(0, +1)$ may exhibit the same zero residue and different carries.

The signature does not replace the trajectory. A zero residue does not mean an absence of orientation, phase, or quotient information. Nor does it introduce a physical quantity by itself. Its function is to distinguish algebraically the transport classes that will be realized next.

2.2.1 Three quadratic regimes and a common exponential

Once the ternary signature has been obtained, its quadratic realization is defined by

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \quad \tau \in \{-1, 0, +1\}. \quad (2.12)$$

If J_τ is the class of X , then $J_\tau^2 = -\tau I$. The structure distinguishes the elliptic type $\tau = +1$, the parabolic type $\tau = 0$, and the hyperbolic type $\tau = -1$. The real realization does not generate the signature: it expresses the types already produced by oriented reduction.

Proposition 2.8 (Exponentiation of the three types). *In each algebra (2.12),*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad (2.13)$$

where

$$C_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n}}{(2n)!}, \quad S_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n+1}}{(2n+1)!}.$$

The three realizations are

τ	J_τ^2	$\exp(tJ_\tau)$
+1	$-I$	$\cos t I + \sin t J_\tau$
0	0	$I + tJ_\tau$
-1	I	$\cosh t I + \sinh t J_\tau$

Proof. Separating the even and odd terms of the exponential series and substituting $J_\tau^{2n} = (-\tau)^n I$ and $J_\tau^{2n+1} = (-\tau)^n J_\tau$ gives the formulas. For $\tau = 0$, all terms of degree at least two vanish. In the other two cases, the trigonometric or hyperbolic series are recovered. $\square$

The exponential is common to all three regimes: it is not assigned exclusively to the parabolic branch. The invariants

$$C_\tau(t)^2 + \tau S_\tau(t)^2 = 1$$

follow from $\exp(tJ_\tau) \exp(-tJ_\tau) = I$. The same algebraic principle preserves unity through realizations that differ in their quadratic type.

In the hyperbolic regime, $P_\pm = (I \pm J_{-1})/2$ are complementary projectors and

$$\exp(tJ_{-1}) = e^t P_+ + e^{-t} P_-.$$

Growth and contraction appear jointly in two eigendirections. In the elliptic regime, the parameter describes a rotation; in the parabolic regime, a nilpotent translation. The temporal interpretation adopted by HMT associates +1 with return with memory, 0 with the threshold, and -1 with bidirectional opening. This interpretation requires an ordering of the parametrization; it does not add a fourth algebraic type.

2.2.2 Balanced division and composition of lifts

Ternary reduction expresses an integer evaluation of APP through a visible coordinate and a prolongation coordinate. Its function is not merely to abbreviate the notation for an integer: it provides a composition rule determining how the quotient changes when evaluations are added or multiplied. The balanced choice also makes the compatibility of this rule with sign reversal explicit.

Definition 2.9 (Balanced ternary coordinates). For $n \in \mathbb{Z}$, let

$$q_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \quad r_3(n) = n - 3q_3(n).$$

The map

$$\mathcal{B} : \mathbb{Z} \longrightarrow \{-1, 0, +1\} \times \mathbb{Z}, \quad n \longmapsto (r_3(n), q_3(n))$$

is called the balanced ternary decomposition. Its inverse is $\mathcal{L}(r, c) = r + 3c$.

Indeed, $r_3(n) \in \{-1, 0, +1\}$. If two pairs represent the same integer, $r + 3c = s + 3d$, then $r - s$ is a multiple of 3 and belongs to the interval $[-2, 2]$; necessarily $r = s$ and $c = d$. The representation is therefore unique. Division can be repeated on $q_3(n)$, so that each level preserves the data needed to reconstruct the preceding one. The quotient is not a perturbation of the residue: it is its complementary coordinate.

The signature obtained from an evaluation F , with $F = S$ or $F = P$ in APP, is $r_3 \circ F$. When reduction acts on the phase cursor, the same balanced section produces the classes $\{1, 4, 7\}$, $\{2, 5, 8\}$, and $\{3, 6, 9\}$, with values $+1, -1, 0$. The TPK selector defined below uses these classes to determine the operating mode. Reading an evaluation and selecting a mode thus share the same ternary reduction, but have different domains and functions.

Proposition 2.10 (Arithmetic of balanced pairs). Let $n = r + 3c$ and $m = s + 3d$, with $r, s \in \{-1, 0, +1\}$. Define

$$\chi(r, s) = \frac{r + s - r_3(r + s)}{3}.$$

Integer operations are transported to the pairs by

$$(r, c) \boxplus (s, d) = (r_3(r + s), c + d + \chi(r, s)), \quad (2.14)$$

$$(r, c) \boxtimes (s, d) = (rs, rd + sc + 3cd). \quad (2.15)$$

The map $\mathcal{L}$ preserves both operations.

Proof. The equality $r + s = r_3(r + s) + 3\chi(r, s)$ gives

$$n + m = r_3(r + s) + 3(c + d + \chi(r, s)).$$

On the other hand,

$$nm = rs + 3(rd + sc + 3cd).$$

Since the product of two balanced representatives again belongs to $\{-1, 0, +1\}$, it requires no additional residue reduction. Uniqueness of the decomposition proves both formulas. $\square$

For example, $2 = (-1) + 3$ and $4 = 1 + 3$. Their sum and product are reconstructed as

$$(-1, 1) \boxplus (1, 1) = (0, 2), \quad (-1, 1) \boxtimes (1, 1) = (-1, 3).$$

The resulting evaluations are 6 and 8. Adding the quotients suffices in this additive example; the product involves the cross terms rd , sc , and $3cd$. The discrepancy between the two operations also resides in the prolongation coordinate, not only in the residue.

The term χ is a cocycle of the balanced section. If $r \oplus s = r_3(r + s)$, it satisfies

$$\chi(r, s) + \chi(r \oplus s, t) = \chi(s, t) + \chi(r, s \oplus t).$$

Both sides represent the quotient of the same sum of three integers. This identity ensures that regrouping the operations does not change the reconstructed evaluation. At the same time, projection onto the first component suppresses this accounting: $1 + 1 = -1 + 3$, whereas the residue reading retains only -1 . The information removed is determined, in this case, by $\chi(1, 1) = 1$.

The relation with modulus 9 preserves a second discrete scale. The map

$$\beta : \{-1, 0, +1\} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad (r, \bar{c}) \longmapsto r + 3c \pmod{9}$$

is well defined and bijective. Changing the representative of $\bar{c}$ adds a multiple of 9; conversely, equality of images fixes r first, by reduction modulo 3, and then $\bar{c}$. Addition transported by this bijection contains the term $\chi(r, s)$ from (2.14), now reduced modulo 3. Thus the two ternary levels do not compose by independent coordinatewise addition: normalization of the first level modifies the second.

In particular, the classes 3, 6, and 0 of $\mathbb{Z}/9\mathbb{Z}$ have first component 0, but second components 1, 2, and 0, respectively. The distinction already observed between direct reduction and extraction of the ideal coordinate appears as a property of β . Preserving this coordinate prevents the zero residue sector from being artificially collapsed into a single position. Modulus 1000, used later to organize decimal blocks, serves a different positional-representation function; its quotient is preserved through the corresponding division. A change of base preserves the integer when it retains residue and quotient, whereas coincidence of residues in different bases does not by itself identify their transport states.

2.2.3 Inversion, residue class, and non-visible coordinates

Additive inversion lifts unambiguously:

$$\mathcal{B}(-n) = (-r_3(n), -q_3(n)).$$

In particular, 3, 0, and -3 have coordinates $(0, 1)$, $(0, 0)$, and $(0, -1)$. They share a zero signature but represent different evaluations. In a transition restricted to a quotient of maximum amplitude 1, these are the three possibilities of the visible fiber over zero; in a general integer lift, that fiber contains all pairs $(0, c)$, with $c \in \mathbb{Z}$.

The quotient coordinate recovers the evaluated integer, but does not by itself recover the trajectory that produced it. Two APP paths may share evaluation, residue, and quotient while differing in orientation, sequence of modes, or winding. Thus decomposition $\mathcal{B}$ is integrated into the transport state rather than replacing it. The arithmetic memory of a division and the memory of a trajectory are related coordinates, not synonyms.

For a signature sequence of length n , oriented reversal is

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1).$$

Applying it twice restores the sequence. The operation reverses the order of positions and the sign of each signature; it preserves the length and the zero positions in reversed order. In the adopted parametrization, it interchanges return with memory, associated with $+1$, and bidirectional opening, associated with -1 , while retaining the threshold 0. Transport of the remaining trajectory data is specified by the corresponding TPK involution.

This signature reversal must be distinguished from conjugation within a quadratic algebra. The former may change the type τ ; the latter, constructed below, acts within a type already fixed. This distinction prevents attributing an algebraic equivalence between different regimes to a sign reversal.

2.2.4 Quadratic product, conjugation, and algebraic norm

An element of $\mathbb{A}_\tau$ has a unique expression $aI + bJ_\tau$. The relation $J_\tau^2 = -\tau I$ determines its product:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \quad (2.16)$$

This is a real realization of the type already selected. The coordinates a, b are not APP residues, and the product (2.16) does not replace (2.15); it expresses another operation in the declared quadratic codomain.

Definition 2.11 (Conjugation and algebraic norm). Conjugation of type τ and its algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \quad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 2.12 (Multiplicativity and invertibility). For $z, w \in \mathbb{A}_\tau$,

$$z\bar{z} = N_\tau(z)I, \quad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

Moreover, z is invertible if and only if $N_\tau(z) \neq 0$, in which case

$$z^{-1} = \frac{\bar{z}}{N_\tau(z)}.$$

Proof. The product of $aI + bJ_\tau$ and $aI - bJ_\tau$ is $(a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so that $(zw)\bar{z}\bar{w} = (z\bar{z})(w\bar{w})$. If the norm does not vanish, the stated formula supplies the inverse. Conversely, if $zw = I$, multiplicativity implies $N_\tau(z)N_\tau(w) = 1$. $\square$

The quadratic form N_τ is positive definite only in the elliptic type. For $\tau = 0$, it equals a^2 , and the element J_0 is nonzero although its square and norm are zero. For $\tau = -1$, it equals $a^2 - b^2$; the nonzero elements $I + J_{-1}$ and $I - J_{-1}$ have zero norm and their product vanishes. Here the algebraic norm is a multiplicative quadratic form: it does not presuppose a positive vector-space norm.

The three types admit faithful realizations by

$$aI + bJ_\tau \mapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}.$$

The determinant of this matrix is $N_\tau(aI + bJ_\tau)$. The complex structure, the dual-number structure, and the split real structure are thus represented by a single matrix expression. Each preserves its own law and zero divisors. The form of signature $(1, 1)$ in the last case admits a Lorentzian interpretation in dimension $1 + 1$; identification of a physical metric additionally requires a specific space, field, and transport rule.

2.2.5 Multiplicative conservation and composition of evolutions

The exponential already obtained belongs to the set of norm-one elements:

$$N_\tau(\exp(tJ_\tau)) = 1.$$

This conservation persists under composition of parameters. Indeed, all factors are functions of the same generator, and

$$\exp(tJ_\tau)\exp(sJ_\tau) = \exp((t+s)J_\tau).$$

Comparing the two components gives the addition laws

$$C_\tau(t+s) = C_\tau(t)C_\tau(s) - \tau S_\tau(t)S_\tau(s), \quad (2.17)$$

$$S_\tau(t+s) = S_\tau(t)C_\tau(s) + C_\tau(t)S_\tau(s). \quad (2.18)$$

Conjugation changes t to $-t$ and coincides with inversion within this subgroup. It does not change τ : it reverses an evolution of the chosen regime.

In the elliptic type, the preserved form is $a^2 + b^2$, and the exponential trajectory traverses the unit-norm circle. In the hyperbolic type, the eigencoordinates are $u = a + b$, $v = a - b$, with $uv = N_{-1}(z)$. The exponential gives

$$(u, v) = (e^t, e^{-t}), \quad uv = 1.$$

Growth of one coordinate is accompanied by contraction of the other. Conservation expresses a relation between the two coordinates; it does not assert that each remains constant.

In the parabolic type,

$$(I + tJ_0)(I + sJ_0) = I + (t + s)J_0, \quad (I + tJ_0)^{-1} = I - tJ_0.$$

The evolution does not stop because $J_0^2 = 0$. Nilpotency removes terms of order at least two, but preserves the linear term that transports the parameter. The visible symbol 0, the zero element of the algebra, and the nonzero generator J_0 are distinct. This distinction is necessary to describe the threshold without identifying it with an absence of state or dynamics.

Proposition 2.13 (Composition of affine coordinates). *In the chart where the scalar component does not vanish, represent a direction by $I + uJ_\tau$. If $1 - \tau uv \neq 0$, the direction of its product with $I + vJ_\tau$ has coordinate*

$$u \star_\tau v = \frac{u + v}{1 - \tau uv}.$$

Proof. We have

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u + v)J_\tau.$$

Dividing by the scalar component gives the stated expression. $\square$

The affine chart preserves the ratio of components, not the scalar factor divided out. When $1 - \tau uv = 0$, the product must be examined before changing charts: it may have a well-defined direction with zero scalar component, or be the zero element in the presence of zero divisors. The composition law still belongs to the full algebra; a coordinate singularity does not automatically imply a singularity of the object. This formula will reappear in the elliptic regional reading. Its use requires retaining the denominator and, where applicable, the normalization factor.

2.2.6 Transport of type and subsequent realizations

The algebras $\mathbb{A}_{+1}$, $\mathbb{A}_0$, and $\mathbb{A}_{-1}$ are not isomorphic as unital real algebras. The first is a field; the second contains a nonzero nilpotent; the third has nontrivial idempotents and is isomorphic to $\mathbb{R} \times \mathbb{R}$. Thus a change of signature organizing a route cannot be interpreted as an undifferentiated isomorphism between the three realizations. The TPK preserves the regime index, the domains, and the effective operator acting at each transition.

2.2.7 Commutator norm of the arithmetic projections

The electronic prefactor is obtained without using α , torsion, or a dimensional unit. The domain is the third realization of APP, $\mathcal{A}_3 = \{1, \dots, 9\}^3$, equipped with the uniform measure. The function space is $\ell^2(\mathcal{A}_3; \mathbb{R})$, with inner product

$$\langle f, g \rangle = \frac{1}{729} \sum_{x \in \mathcal{A}_3} f(x)g(x).$$

In this block, $[P, Q] = PQ - QP$ denotes the additive operator commutator. We define

$$\Sigma(x) = \rho_9(x_1 + x_2 + x_3), \quad \Pi(x) = \rho_9(x_1 x_2 x_3).$$

The conditional expectations R_Σ and R_Π are the orthogonal projections onto functions constant on each additive and multiplicative fiber, respectively. Thus, for a function f ,

$$(R_\Sigma f)(x) = \frac{1}{|\Sigma^{-1}(\Sigma(x))|} \sum_{y: \Sigma(y)=\Sigma(x)} f(y),$$

and analogously for R_Π . The two projections act on the same space; they do not represent two independent arithmetic supports.

Lemma 2.14 (Two projections and principal angles). *If $s \in [0, 1]$ is a singular value of the pairing between orthonormal bases of the ranges of two projections P, Q , its nontrivial block contributes $s\sqrt{1-s^2}$ to $\|[P, Q]\|$.*

Proof. On the corresponding principal plane, one can write

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} s^2 & s\sqrt{1-s^2} \\ s\sqrt{1-s^2} & 1-s^2 \end{pmatrix}.$$

Their commutator is

$$[P, Q] = s\sqrt{1-s^2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The final matrix is orthogonal, so its norm is one. Common or orthogonal blocks have zero commutator. The orthogonal decomposition into principal planes gives the claim and reduces the total norm to the maximum of these contributions. $\square$

Proposition 2.15 (Exact incompatibility norm). *On $\ell^2(\mathcal{A}_3)$,*

$$\|[R_\Sigma, R_\Pi]\| = \frac{\sqrt{3}}{4}. \quad (2.19)$$

Proof. Let $N_{ab} = |\Sigma^{-1}(a) \cap \Pi^{-1}(b)|$. Integer enumeration of the three symbols yields

$$N = 9 \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 0 & 2 & 1 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 1 & 2 & 0 & 0 & 2 & 1 & 0 & 3 \end{pmatrix}.$$

Each row sums to 81; the column sums are

$$(m_1, \dots, m_9) = (36, 36, 108, 36, 36, 108, 36, 36, 297).$$

The normalized indicator functions of the fibers have pairing matrix $C_{ab} = N_{ab}/\sqrt{81m_b}$. Consequently,

$$M = CC^\top, \quad M_{ac} = \sum_{b=1}^9 \frac{N_{ab}N_{cb}}{81m_b}$$

is a rational matrix whose spectrum consists of the squares of the required singular values.

Exact multiplication shows that $(1, \dots, 1)$, $(1, -1, 0)^3$, and $(1, 1, -2)^3$ are eigenvectors with eigenvalues 1, $1/4$, and $7/132$, respectively. The subspace of vectors supported on $\{3, 6, 9\}$ whose sum is zero has dimension two and eigenvalue $1/18$. The zero-sum vectors supported on $\{1, 4, 7\}$ or $\{2, 5, 8\}$ form a four-dimensional kernel. These subspaces are independent and their dimensions sum to nine. The complete spectrum is therefore established:

$$\text{spec}(M) = \left\{1, \frac{1}{4}, \frac{7}{132}, \frac{1}{18}, \frac{1}{18}, 0, 0, 0, 0\right\}.$$

The maximum of $\lambda(1-\lambda)$ on this set is $3/16$, attained at $\lambda = 1/4$. The preceding lemma gives $\sqrt{3/16} = \sqrt{3}/4$. $\square$

The mode $(1, -1, 0)^3$ is precisely the distribution of values of the tritic phase selector in the nonadic chart. The scalar norm retains a measure of incompatibility, but does not by itself preserve the orientation of that mode or reconstruct the two projectors. That memory remains in the route state accompanying the scalar reading.

Proposition 2.16 (Algebra of the local electronic block). *On the principal plane attaining (2.19), let*

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}, \quad S = 2P - I, \quad J = \frac{4}{\sqrt{3}}[P, Q].$$

Then $S^2 = I$, $J^2 = -I$, $SJ = -JS$, and the real algebra generated by S, J is $M_2(\mathbb{R})$, a realization of $\text{Cl}_{1,1}$. Moreover,

$$n_{\pm} = \frac{S \pm J}{2}, \quad n_{\pm}^2 = 0, \quad n_+ n_- + n_- n_+ = I.$$

Proof. We have $S = \text{diag}(1, -1)$ and $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Multiplication verifies the three relations. The matrices I, S, J, SJ are linearly independent and form a basis of $M_2(\mathbb{R})$, proving the claim about the algebra. Finally, $(S \pm J)^2 = S^2 + J^2 \pm (SJ + JS) = 0$, and the sum of the cross products is $(S^2 - J^2)/2 = I$. $\square$

In this block, an elliptic direction, a hyperbolic direction, and two parabolic null directions coexist. The result describes a local matrix representation of the arithmetic of the two sheets. It does not identify all of APP with $M_2(\mathbb{R})$, or with a quantum Latin square; such claims would require other domains and other relations. Nor does the exponential belong exclusively to the parabolic regime: it can be applied to any of the generators of this block.

Proposition 2.16 establishes that the pair of projections of the APP sheets produces a normalized commutator whose square is $-I$. The same block contains an involution of square I and two nilpotent directions. The joint realization is proved for that concrete pair of projections; the preceding classification explains why the three quadratic relations can appear within one matrix algebra without being identified with one another.

In the exceptional prolongation, the Paley matrix reduced modulo 3 satisfies $A_W^2 = -I_6$. This relation endows the ternary space with a structure of dimension 3 over $\mathbb{F}_9$. Its connection to a real operator root is expressed through the integral presentation

$$\mathbb{Z}[u]/(u^2 + 1).$$

Base changes to $\mathbb{F}_3$ and $\mathbb{R}$ produce representations of the same relation. Each retains its characteristic, dimension, and incidence information. Developing the respective operators establishes the connection, not equality of their cardinalities.

Norm-one conservation consequently describes the quadratic realizations of evolutions. Conservation of quotients, orientation, and trajectory belongs to the transported arithmetic state. Both properties participate in the construction, with different functions: one determines the identities of the representations; the other makes it possible to compose, reconstruct, and prolong the operations that give rise to them.

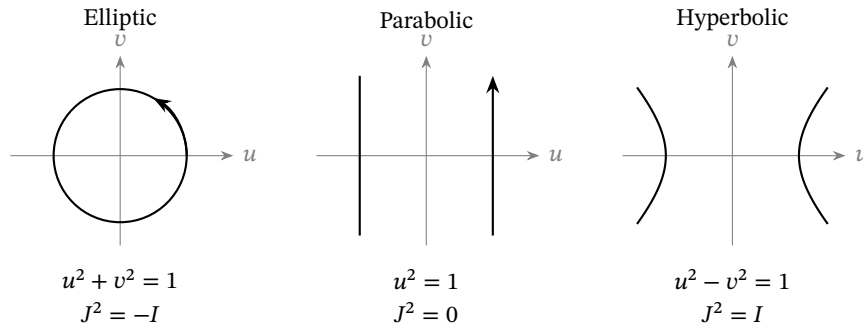


Figure 2: Quadratic realizations of TRIT. For $J_\tau = \begin{pmatrix} 0 & -\tau \\ 1 & 0 \end{pmatrix}$, the exponential preserves $u^2 + \tau v^2$. Parabolic degeneration preserves u and transports v linearly. The three diagrams represent orbits of quadratic forms; their geometric type corresponds to the algebraic relation of the generator.

2.3 TPK: Topological Phase Kernel

The dynamics uses the phase selector

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Its value determines which evaluation operates. Cardinal transport moves the corresponding cursor; the record preserves the previous value, the new quotient, the orientation, and the transition. Selection and transport are therefore different functions: a ternary sign is not a displacement on the torus.

We shall denote by $\hat{X}$ the space of arithmetic states with trajectory memory. An element contains the positions and directions of both cursors, the active mode, the TRIT signature, the phase, the residues and quotients of both evaluations, the carries, the constructed prefix, and the boundary data. When a refinement cylinder is introduced, it forms part of the state. This notation avoids replacing the object by a reduced version that retains only the visible symbol.

Definition 2.17 (TPK update). Let Sel_t be mode selection through M_{ph} , Tra_t the lifted cardinal transport, and Mem_t the record update. The transition is

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (2.20)$$

Each application preserves the fields it does not modify and updates the remaining fields by Euclidean division and trajectory concatenation.

The notation (2.20) expresses the order of composition; it does not replace the effective rules. The following section specifies its finite two-cursor realization completely. Prolongation will then be obtained not by erasing its memory and repeating the first emission, but by transporting the prefixes and their compatible boundaries. Thus APP supplies the evaluations, TRIT distinguishes their regimes, and TPK composes their evolution.

2.3.1 Dynamic state, observables, and operational dependencies

The Topological Phase Kernel organizes the evolution of the two APP sheets with the orientation determined by TRIT. Its definition comprises the state on which it acts, the transition rules, and the relations that make prolongations compatible. Equation (2.20) acquires operational content when the coordinates received and modified by each factor are specified. Position belongs to the toroidal support; phase belongs to the calendar; orientation determines transport; quotients record the arithmetic evaluation before reduction. These coordinates are related through the transition and retain different mathematical functions.

The observable projection of a state is

$$q = (x_+, d_+, x_-, d_-, \theta) \in \mathcal{Q}_{\text{obs}} := (X_9 \times \mathcal{D})^2 \times \Theta_{108}, \quad \mathcal{D} = \{N, E, S, O\}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z}. \quad (2.21)$$

The subscripts distinguish the additive and multiplicative cursors. The fibers of complete states over q preserve the ordered trajectory, the integer evaluations of both sheets, their residues and quotients, and the orientation and boundary records. A precision prolongation adds the prefix and the compatible cylinder. Thus two states with the same projection q may retain different records.

Operational dependencies have three levels. At the local level, the selector activates a sheet, transport changes its position, and the update computes the new arithmetic coordinates. At the trajectory level, these transitions compose and retain the records of preceding steps. At the refinement level, extension relations determine which continuation preserves the predecessor's prefix, signature, and boundary. The finite calendar describes the phase of these operations; holonomy describes their composition on the history.

2.3.2 Tritic selection and lifted cardinal transport

Let $\bar{\theta} \in \{0, \dots, 107\}$ be the calendar representative. The operating phase and the dodecaphase sector are

$$r(\theta) = 1 + (\bar{\theta} \bmod 9), \quad \beta(\theta) = \left[\left[\frac{\bar{\theta}}{9} \right] \right]_{12}. \quad (2.22)$$

The previously defined selector may be written $\varepsilon_9 = M_{\text{ph}}$. Its value vector in the ordered chart of D_9 is

$$m_{\text{ph}} = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The function $M_{\text{ph}} : D_9 \rightarrow \{-1, 0, +1\}$ and this vector are two presentations of the same selector, with their respective domains.

Define the activation indicators

$$a_+(\theta) = \mathbf{1}_{\{1,4,7\}}(r(\theta)), \quad a_{\times}(\theta) = \mathbf{1}_{\{2,5,8\}}(r(\theta)).$$

Cursor transport is then

$$x'_+ = x_+ + a_+(\theta)v_{d_+}, \quad x'_\times = x_\times + a_{\times}(\theta)v_{d_\times} \quad \text{in } X_9. \quad (2.23)$$

In phases 3, 6, 9, both positions remain unchanged; the neutral event forms part of the chronology and its record. Cardinal directions determine the vectors v_d , whereas the phase signature determines which vector is applied.

The integer lift preserves boundary crossing. If $s : X_9 \rightarrow D_9^2$ chooses positive representatives and $x' = x + av_d$, with $a \in \{0, 1\}$, define

$$b_d(x, a) = \frac{s(x) + av_d - s(x')}{9} \in \mathbb{Z}^2. \quad (2.24)$$

Divisibility by 9 follows from equality in X_9 . If $z \in \mathbb{Z}^2$ records previous crossings, the update is $z' = z + b_d(x, a)$. Thus

$$s(x') + 9z' = s(x) + 9z + av_d.$$

The integer transport data remain reconstructible after the position has been reduced on the torus.

Proposition 2.18 (Conservation of lifted displacement). *For a cardinal trajectory of n steps, with activation indicators a_t ,*

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t v_{d_t}.$$

The same equality is preserved under concatenation of two compatible trajectories.

Proof. The one-step identity is (2.24). Summing it over the n steps cancels the intermediate coordinates. Splitting the summation interval into two segments produces exactly the concatenation law. For a closed trajectory, the difference $z_n - z_0$ is the previously defined winding number. $\square$

The same principle governs APP evaluations. If a sheet takes the integer value N_t , its record simultaneously preserves $N_t = R_t + 9q_t$. The difference between two steps satisfies

$$N_{t+1} - N_t = (R_{t+1} - R_t) + 9(q_{t+1} - q_t).$$

Spatial-crossing memory z_t and arithmetic quotient q_t have different definitions; preserving them jointly maintains the geometric and arithmetic provenance of every transition.

2.3.3 Two-cursor chronology and composite returns

The cardinal involution j interchanges N with S and E with O . After applying (2.23), both directions are reversed at the end of a block of 27 steps. Writing

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete rule on the observable projection is

$$F_{\text{obs}}(q) = (x'_+, j^{h(\theta)} d_+, x'_\times, j^{h(\theta)} d_\times, \theta + 1). \quad (2.25)$$

The reading that forms the emission uses the positions before transport, according to the explicit recurrence in Section 4. The observable update and emission therefore compose in a fixed order.

Proposition 2.19 (Reversibility and period of the observable calendar). *The map F_{obs} is a permutation of $\mathcal{Q}_{\text{obs}}$ and satisfies $F_{\text{obs}}^{108} = I$. Within a block of 54 steps, positions and directions are restored; the coordinate in Θ_{108} advances by 54.*

Proof. Given the final state, the preceding phase is first recovered by subtracting one in Θ_{108} . That phase determines whether directions were reversed and which cursor moved. Applying the corresponding involution and subtracting the cardinal vector uniquely recovers the initial state.

Each interval of 27 steps contains nine activations of each cursor. The following block preserves the activations and reverses the directions, so the integer displacements cancel over 54 steps. The rule has period 54 in its position and orientation increments; the same cancellation holds for any calendar origin. After 108 steps, θ is also restored. $\square$

We reserve

$$\mathcal{K}_{27} = F_{\text{obs}}^{27} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}$$

for the iterate of 27 transitions. Its fourth iterate is the identity on the observable state defined in (2.21). Projecting history onto the calendar makes this finite check possible. The trajectory record, its evaluations, and the new prefixes remain in the dynamic state on which $\mathcal{U}_t$ acts; observable equality does not reinitialize them.

2.3.4 Affine memory transport and composition of states

The observable projection admits fibers of arithmetic and geometric data. Over a configuration q , we write a state in the form

$$x = (q, o, h, c, \sigma, C, b, \lambda).$$

Here o preserves orientation and sheet; h , residues and quotients; c , additive transport memory; σ , the boundary signature; C , the precision cylinder; b , its boundary label; and λ , the recorded trajectory. The signature is determined by a map

$$\text{Sig}_q : H_q \times C_q \times \text{Hist}_q \longrightarrow S_q, \quad \sigma = \text{Sig}_q(h, c, \lambda).$$

The symbols H_q , C_q , Hist_q , and S_q denote, respectively, the sets of residue–quotient data, the abelian transport-memory group, the recorded trajectories, and the boundary signatures. Realizations of this signature are specified component by component in the regional and terminal constructions. This dependence prevents treating the signature as a free parameter once the data producing it are fixed.

The elementary residue–quotient chart is explicit. For $n \in \mathbb{Z}$, let

$$r = 1 + ((n - 1) \bmod 9), \quad u = \frac{n - r}{9}.$$

Then $n = r + 9u$, with $r \in D_9$ and $u \in \mathbb{Z}$. Uniqueness follows because two representatives in D_9 congruent modulo 9 are equal. This chart supplies local integer reconstruction. The precision towers additionally incorporate the truncation maps and cylinders developed in 4 and 6.

Let $\gamma : q \rightarrow q'$ be an observable trajectory. Fiber transport is expressed through maps

$$H(\gamma) : H_q \rightarrow H_{q'}, \quad C(\gamma) : C_q \rightarrow C_{q'}, \quad \text{Hist}(\gamma) : \text{Hist}_q \rightarrow \text{Hist}_{q'}.$$

The maps respect identities and composition, and $C(\gamma)$ is an abelian-group homomorphism. The increment $k_{\text{tr}}(\gamma) \in C_{q'}$ satisfies

$$\begin{aligned} k_{\text{tr}}(\text{id}_q) &= 0, \\ k_{\text{tr}}(\gamma_2 \circ \gamma_1) &= k_{\text{tr}}(\gamma_2) + C(\gamma_2)k_{\text{tr}}(\gamma_1). \end{aligned} \tag{2.26}$$

Here k_{tr} denotes the transport cocycle; the rational coordinate κ of record K retains its notation in the dodecaphase-closure chapter.

Transportable coordinates are updated by

$$\begin{aligned} h' &= H(\gamma)h, & c' &= C(\gamma)c + k_{\text{tr}}(\gamma), \\ \lambda' &= \gamma \circ \lambda, & \sigma' &= \text{Sig}_{q'}(h', c', \lambda'). \end{aligned} \tag{2.27}$$

The second equation is an affine action: the linear term transports previous memory and the cocycle adds the increment produced by the trajectory. In the spatial-crossing realization for each cursor separately, $C(\gamma) = I$ and $k_{\text{tr}}(\gamma)$ is the sum of the vectors in (2.24). This concrete instance connects the abstract law with the cardinal operations already defined.

Proposition 2.20 (Compatibility of the update with concatenation). *Updating by (2.27) along γ_1 , followed by updating along γ_2 , coincides with updating along $\gamma_2 \circ \gamma_1$.*

Proof. For the affine coordinate, two updates give

$$c'' = C(\gamma_2)C(\gamma_1)c + C(\gamma_2)k_{\text{tr}}(\gamma_1) + k_{\text{tr}}(\gamma_2).$$

Functoriality of C and (2.26) turn this expression into $C(\gamma_2 \circ \gamma_1)c + k_{\text{tr}}(\gamma_2 \circ \gamma_1)$. The statement for h follows from composition of H , and that for λ from associativity of trajectories. The final signature agrees because it is evaluated on the same three reconstructed coordinates. $\square$

The cylinder and its boundary complete this law through the relations Ext_γ , defined in 2.3.7. Their elements determine which of the preceding updates preserve an admissible prolongation. States, together with the triples (γ, x, x') admitted by these relations, form a category over observable trajectories:

$$(\gamma_2, x', x'') \circ (\gamma_1, x, x') = (\gamma_2 \circ \gamma_1, x, x'').$$

The identity of x is (id_q, x, x) . Composition preserves arithmetic transport and boundary prolongability simultaneously. This is the structure in which returns with memory and successive refinements take place.

2.3.5 Dodecaphase record, phase, and accumulated memory

The 108 transitions are distributed over the ordered windows

$$E_m = \{9m - 8, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The record preserves the phase origin, orientation, and data produced during each window. Its map is

$$\mathcal{R}_{12}(x) = \left((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m) \right)_{m=1}^{12},$$

where τ_m is the window's subroute, σ_m its orientation, κ_m the integer boundary increment, and Λ_m the local incidence record. We adopt the chart developed in 9.2; its window indices correspond to $m = \beta + 1$. The domain comprises compatible TPK histories, and the codomain their oriented temporal records. Composition of segments and their increments is governed by (2.27). By contrast, the group

C_{12} describes translation of the window index; Θ_{108} preserves the step calendar. This distinction specifies what is transformed and what is observed when a cycle is completed.

In coordinates $\theta = 9b + r$, with $0 \leq r < 9$, advancing 9 steps preserves r and increases b by one. Addition of phases introduces the quotient

$$c_0(r, r') = \left\lfloor \frac{r + r'}{9} \right\rfloor,$$

whose cocycle identity ensures associativity of composition. The finite calendar preserves $[b]_{12}$; a record of turns preserves $b \in \mathbb{Z}$, or $b \in \mathbb{N}_0$ for a history with fixed origin. Section 4 proves the nonsplit exact sequence (4.5). Its function is to describe the relation between these coordinates before they are used to construct the signed state.

The signed record receives phase and memory data through the composition developed in Section 9:

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \quad (2.28)$$

Its domain is the terminal state with memory; its output belongs to $\mathbb{Z}^{12}$. The reversible transformation between this signed state and K , together with the rational reading of K , is proved in the corresponding chapter. Record operations precede these transformations. Reconstructing a record from its cyclic differences verifies its recoverability; its dynamic provenance must be preserved through the events that produced it.

For the encoded trace (d_t) retained by the record, the event reader counts codes $\{2, 4, 6, 8\}$ in each window with orientation $(+, +, +, -, -, -, +, +, +, -, -, -)$. These codes and the cardinal symbols of $\mathcal{D}$ are different record coordinates. Chapter 9 develops this reader, the integral decomposition $1 + 5 + 6$, the congruences characterizing its image, and its exact inverse. The inversion proof establishes which data this chart preserves; the definition of $\mathcal{R}_{12}$ specifies the oriented history on which it acts.

2.3.6 Phase incidence and internal channel construction

The selector determines three subsets of D_9 :

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}.$$

Let $H_{\text{act}} = H_+ \sqcup H_-$ be the set of active phases and $O_{\text{ph}} = D_9 \setminus \{9\}$ the chart preceding the marked return. Denote by $P_2(H_{\text{act}})$ the set of unordered pairs of active phases. Internal incidence is given by

$$\mathcal{F}_6^{\text{ph}} = H_{\text{act}} \times P_2(H_{\text{act}}), \quad \mathcal{F}_8^{\text{ph}} = O_{\text{ph}} \times P_2(H_{\text{act}}), \quad \Theta_{\text{ph}} = D_9 \times P_2(H_{\text{act}}). \quad (2.29)$$

The indices describe the number of positions in the first component. The construction uses the phases and calendar origin already fixed; subsequent exceptional realizations will transport these incidences to their own domains.

Proposition 2.21 (Cardinalities and oriented phase incidence). *The sets in (2.29) have cardinalities 90, 120, 135, respectively. If*

$$\partial_{\text{ph}} = \Theta_{\text{ph}} \times \{-1, +1\}, \quad E_{\text{ph}} = \partial_{\text{ph}} \sqcup \{\infty\}, \quad L_{\text{ph}} = H_0 \times P_2(H_{\text{act}}),$$

then

$$|\partial_{\text{ph}}| = 270, \quad |E_{\text{ph}}| = 271, \quad |L_{\text{ph}}| = 45, \quad |\partial_{\text{ph}} \sqcup L_{\text{ph}}| = 315.$$

Simultaneously translating the phase origin and marked return preserves all these cardinalities and inclusions.

Proof. There are six active phases, so $|P_2(H_{\text{act}})| = \binom{6}{2} = 15$. Multiplying by 6, 8, 9 gives 90, 120, 135. Orientation doubles 135; the return mark adds one element; the three neutral phases give $3 \cdot 15 = 45$. A calendar translation transports each factor by a bijection and carries the marked point to its image. Products and disjoint unions thus preserve the incidence. $\square$

Here the cardinality 271 has a phase-incidence realization. The cubic completion in Section 4 has its own construction of the corona 1000 – 729. The correspondence between the two realizations must preserve the maps and marks identifying them, in addition to their cardinality. Likewise, the selector ε_9 , the incidence (90, 120), and the extractor $\mathcal{E}_{108}^{90,120}$ retain the domains and operations just distinguished.

2.3.7 Trajectory reversal and conjugate prolongations

Bidirectionality begins with an operation on trajectories before scalar values are assigned to them. For $\gamma = (x_0; d_1, \dots, d_n)$, with endpoint x_n , define

$$\text{rev } \gamma = (x_n; jd_n, \dots, jd_1).$$

The reversed trajectory starts at the endpoint of the original. Composition satisfies

$$\text{rev}^2 \gamma = \gamma, \quad \text{rev}(\gamma_2 \circ \gamma_1) = \text{rev } \gamma_1 \circ \text{rev } \gamma_2,$$

and integer displacement changes sign. These identities follow by reversing the order of steps and replacing each cardinal vector with its opposite. For a loop, it follows that $\text{wind}(\text{rev } \gamma) = -\text{wind}(\gamma)$.

On records, reversal also requires transporting the order of components and the orientation of events. Route reversal and signature conjugation are distinguishable operations: their composition is specified when applied to a concrete record. In particular, the regional involution interchanging closure and propagation components preserves the self-scaling axis; its formula is given in Section 6.17. This regional operation acts on a different domain from cardinal trajectory reversal.

Extensions over a route $r : q \rightarrow q'$ are expressed by relations between fibers of compatible states. If $\mathcal{F}_q$ denotes the fiber of complete states over q , we require

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \Delta_{\mathcal{F}_q} \subseteq \text{Ext}_{\text{id}_q},$$

$$\text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

Their composition preserves the evaluations of both sheets, orientation, quotients, prefix, and boundary. Route reversal does not by itself imply that every extension relation is a bijection. Conjugate prolongation is defined on the domain where transport of these data satisfies the corresponding compatibilities.

2.3.8 Nonadic holonomy and continuity of refinement

A phase turn composes transitions on the state with memory:

$$\Gamma_{9,k} = \mathcal{U}_{k+8} \circ \dots \circ \mathcal{U}_k. \quad (2.30)$$

On an individualized history, this functional composition realizes the holonomy relation; on the full domain we retain $\Gamma_{9,k} \subseteq \widehat{X}_k \times \widehat{X}_{k+9}$, with the admissible prolongations developed later. The cyclic base has nine phases, while the fiber preserves trajectory, quotients, and boundary. Monodromy is the action of one turn of this base on the fibers; holonomy $\Gamma_{9,k}$ expresses its realization through the effective TPK transitions.

If g observes phase and M the number of retained turns,

$$g(\Gamma_{9,k}x) = g(x), \quad M(\Gamma_{9,k}x) = M(x) + 1.$$

The first identity expresses phase closure and the second identifies the new temporal depth. Coinductive prolongation also preserves its cylinders and prefixes. Restrictions between depths compose an inverse system; a compatible section gathers its antecedents without replacing them by the latest observation. This arithmetic of histories and its unity are developed in 4.8–4.10; the regions and their readers are constructed in Section 6.

Phase transfer on emissions already constructed has a different domain. Let $\mathcal{A}_+$ and $\mathcal{A}_\times$ be the additive and multiplicative signature alphabets, with cardinalities 18 and 26. Their product identifies the catalog $\mathcal{U}_6$. For $f : \mathcal{A}_+ \times \mathcal{A}_\times \rightarrow \mathbb{R}$, the sheet averages are

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s, e').$$

These emissions and their cardinalities are constructed through the recurrence in Section 4. Once they have been obtained, the transfer operator is defined on $\mathcal{U}_6 \times C_9$ by

$$P_+ = E_+, \quad P_- = E_\times, \quad P_0 = I, \quad \mathcal{K}_{\text{ph}}((u, r), (v, r+1)) = P_{M_{\text{ph}}(r)}(u, v),$$

with zero weight for all other phase transitions. The index r uses the positive representative of the cyclic class. The phase order $(+1, -1, 0)^3$ then produces the renewal

$$\mathcal{K}_{\text{ph}}^9|_r = E_+ E_\times.$$

Indeed, both averages are idempotent, commute, and average different factors; their composition assigns weight $1/(18 \cdot 26) = 1/468$ to each emission. This equality characterizes a subsequent transfer observable. History transport remains given by (2.30), with its memory and boundary.

The resulting hierarchy establishes the function of the following chapters. Finite emission constructs the domain and regions; extension relations preserve their prefixes; the nonadic connection prolongs them; the dodecaphase record determines the data entering K and the closure of α . Numerical and incidence realizations receive this preceding structure. This order allows every consequence to be presented in its own chapter while preserving, from the formal kernel onwards, the objects that make it possible.

2.4 Accumulated memory and transport resolvents

Calendar return does not remove increments produced during the journey. To express this distinction in operator form, an accumulated trajectory-event record is constructed first, followed by its generating series. Reduction of memory variables will be performed on this transport, also preserving the source and output reader.

2.4.1 Oriented events and record update

Let (d_t) be the sequence of integer direction codes retained by a TPK trajectory. The code d_t and the cursor's cardinal direction are different coordinates; no identification of their alphabets is assumed. We number the windows starting from zero:

$$I_m = \{9m + 1, \dots, 9m + 9\}, \quad m \geq 0.$$

In the first cycle, I_m is window E_{m+1} of the dodecaphase record. Its orientation is

$$\sigma = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1).$$

For a prolonged history, $\sigma_m \in \{-1, 1\}$ is the sign retained in the corresponding window. The signed event is

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9. \quad (2.31)$$

Each summand is incorporated when its transition occurs. This definition counts trace events; it does not select directions through a value that is to be obtained later.

Let $\mathbf{e}_0, \dots, \mathbf{e}_{11}$ be the basis of $\mathbb{Z}^{12}$ and $\mathbf{S}\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$. Write $R = S^{-1}$. The accumulated record in fixed coordinates satisfies

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}.$$

A prefix with no accumulated history starts at $z_0 = 0$; at a different origin, z_0 retains the preceding record. The frame accompanying the calendar is $y_m = S^{-m} z_m$.

Proposition 2.22 (Update and return with memory). *The record in the moving frame satisfies*

$$y_{m+1} = Ry_m + \eta_m, \quad \eta_m = a_m R \mathbf{e}_0. \quad (2.32)$$

For every $n \geq 1$,

$$y_n = R^n y_0 + \sum_{j=0}^{n-1} R^{n-1-j} \eta_j. \quad (2.33)$$

In particular,

$$y_{12} = y_0 + \sum_{j=0}^{11} a_j \mathbf{e}_j. \quad (2.34)$$

Proof. Multiplying the update of z_m by $S^{-(m+1)}$ gives $y_{m+1} = S^{-1}y_m + a_m S^{-1} \mathbf{e}_0$, since $S^{-m} \mathbf{e}_{m \bmod 12} = \mathbf{e}_0$. Formula (2.33) follows by induction. For $n = 12$, $R^{12} = I$ and $R^{11-j} \eta_j = a_j R^{12-j} \mathbf{e}_0 = a_j \mathbf{e}_j$, which proves (2.34). $\square$

Thus twelve windows restore the frame and preserve the event vector. The count alone does not recover the order of all steps within each window; that order remains in the recorded trajectory.

To specify the balance, define

$$D_3 = I - S^3, \quad D_4 = I - S^4, \quad B = D_3^\top D_3 + D_4^\top D_4,$$

$$\mathcal{M}(y) = \frac{1}{2} y^\top B y, \quad q(y) = \sum_{j=0}^{11} y_j.$$

The first is a quadratic form of the record; the second records the sum of its coordinates. They are not identified here with physical energy or charge.

Proposition 2.23 (Event-induced balance). *Update (2.32) satisfies*

$$\begin{aligned} \mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) &= a_m (B y_m)_0 + 2a_m^2, \\ q(y_{m+1}) - q(y_m) &= a_m. \end{aligned} \quad (2.35)$$

Proof. Expanding the products gives $B = 4I - S^3 - S^{-3} - S^4 - S^{-4}$; hence $R^\top B R = B$ and $B_{00} = 4$. Since $y_{m+1} = R(y_m + a_m \mathbf{e}_0)$, the quadratic difference is $a_m \mathbf{e}_0^\top B y_m + \frac{1}{2} a_m^2 B_{00}$. The coordinate sum is invariant under permutation R , giving the second equality. $\square$

2.4.2 Generating series and uniform memory control

The series preserves all increments, not only their sum at the end of a cycle. With a formal variable u , let

$$Y(u) = \sum_{m \geq 0} y_m u^m, \quad H_\eta(u) = \sum_{m \geq 0} \eta_m u^m.$$

Proposition 2.24 (Resolvent and tail bound). *The formal identity*

$$Y(u) = (I - uR)^{-1}(y_0 + uH_\eta(u)) \quad (2.36)$$

holds. Both series converge for $|u| < 1$. If $\ell : \mathbb{R}^{12} \rightarrow \mathbb{R}$ is linear, $L = \|\ell\|_{1 \rightarrow \mathbb{R}}$, $M_0 = \|y_0\|_1$, and $0 < r < 1$, then, for every integer $N \geq 0$ and $|u| \leq r$,

$$\left| \sum_{m > N} \ell(y_m) u^m \right| \leq Lr^{N+1} \left(\frac{M_0 + 9(N+1)}{1-r} + \frac{9r}{(1-r)^2} \right). \quad (2.37)$$

The derivative series converges uniformly on every disk $|u| \leq r < 1$.

Proof. Multiplying (2.32) by u^{m+1} and summing gives $Y - y_0 = uRY + uH_\eta$. The constant term of $I - uR$ is invertible, so its formal inverse is $\sum_{k \geq 0} u^k R^k$. Moreover, R preserves the ℓ^1 norm and $\|\eta_m\|_1 \leq 9$, whence $\|y_m\|_1 \leq M_0 + 9m$. For the tail, sum the majorants $L(M_0 + 9m)r^m$, using

$$\sum_{m > N} r^m = \frac{r^{N+1}}{1-r}, \quad \sum_{m > N} mr^m = r^{N+1} \left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2} \right).$$

Finally, $\sum_{m \geq 1} Lm(M_0 + 9m)r^{m-1} < \infty$ majorizes the derivative series and proves its uniform convergence. $\square$

The variable u counts windows of nine transitions. Identifying it with another reader's variable requires preserving that clock and the map relating the two readings.

2.4.3 Memory elimination: operator, source, and reader

On a domain where the update is fixed as $U : X \rightarrow X$, the free vector space $V = \mathbb{Q}^{(X)}$ on the states admits the operator $T\delta_x = \delta_{U(x)}$. The clock is included in x if the transition depends on it. This linear extension preserves distinct states even when they share an observable phase.

Consider a decomposition $V = V_0 \oplus V_1$, where V_1 is the auxiliary sector to be eliminated, and write

$$T = \begin{pmatrix} A & B_1 \\ C & D \end{pmatrix}, \quad v = \begin{pmatrix} v_0 \\ v_1 \end{pmatrix}, \quad \ell = (\ell_0, \ell_1).$$

Here $A : V_0 \rightarrow V_0$, $B_1 : V_1 \rightarrow V_0$, $C : V_0 \rightarrow V_1$, and $D : V_1 \rightarrow V_1$. The following identities are formal; they also hold wherever the resolvents converge.

Proposition 2.25 (Schur reduction with source and reader). *Let*

$$D_u = (I - uD)^{-1}, \quad \mathcal{F}(u) = I - uA - u^2 B_1 D_u C.$$

The retained component of $(I - uT)^{-1}v$ is

$$w_0 = \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1), \quad w_1 = D_u(v_1 + uCw_0). \quad (2.38)$$

Its complete reading satisfies

$$\begin{aligned} \ell(I - uT)^{-1}v &= (\ell_0 + u\ell_1 D_u C) \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1) \\ &\quad + \ell_1 D_u v_1. \end{aligned} \quad (2.39)$$

Proof. The two equations of $(I - uT)w = v$ are $w_0 = v_0 + uAw_0 + uB_1w_1$ and $w_1 = v_1 + uCw_0 + uDw_1$. Solving the second and substituting it into the first gives (2.38); applying ℓ_0 and ℓ_1 to the two components gives (2.39). All formal inverses exist because their constant terms are identities. $\square$

The term $u^2B_1D_uC = \sum_{k \geq 0} u^{k+2}B_1D^kC$ describes departure into memory, k steps within it, and a return. By contrast, $u\ell_1D_uC$ reads memory before that return. Thus reducing only the operator does not necessarily preserve the observation: the source and reader must also be transformed.

The distinction is visible in the cycle already constructed. If P retains $\mathbf{e}_0$, then $PRP = 0$, but

$$\langle \mathbf{e}_0, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{1}{1 - u^{12}}, \quad \langle \mathbf{e}_{11}, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{u}{1 - u^{12}}. \quad (2.40)$$

Indeed, $R^n\mathbf{e}_0 = \mathbf{e}_{-n \bmod 12}$: the two series collect, respectively, $n \equiv 0$ and $n \equiv 1 \pmod{12}$. The resolvent of the compression is I , whereas compression of the resolvent preserves the omitted returns. This example concerns the twelve-window record, not the entire TPK state.

2.4.4 Segment composition and coherence of reduction

Let three composable affine transports be $F_i(y) = R_iy + b_i$, with compatible domains and codomains. The memory produced by the complete journey is

$$F_3F_2F_1(y) = R_3R_2R_1y + b_3 + R_3b_2 + R_3R_2b_1. \quad (2.41)$$

The proof is successive substitution of the three affine formulas. Grouping the first two segments first, or the last two, produces the same expression; the factors transport each increment from its point of origin. In (2.32), $R_i = R$ and b_i is the event of its window.

Eliminating several layers preserves a similar coherence. To make it explicit, now decompose $V = V_0 \oplus V_1 \oplus V_2$ and write

$$T = \begin{pmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{pmatrix}, \quad Q_2 = (I - uD_{22})^{-1}.$$

Eliminating the third component leaves the four blocks

$$\begin{aligned} A' &= A + uB_2Q_2C_2, & B'_1 &= B_1 + uB_2Q_2D_{21}, \\ C'_1 &= C_1 + uD_{12}Q_2C_2, & D'_{11} &= D_{11} + uD_{12}Q_2D_{21}. \end{aligned} \quad (2.42)$$

Proposition 2.26 (Transitivity of elimination). *If $D_* = (D_{ij})_{i,j=1}^2$, the complement obtained in two stages coincides with that obtained by eliminating both memories together:*

$$\begin{aligned} I - uA' - u^2B'_1(I - uD'_{11})^{-1}C'_1 &= I - uA \\ &\quad - u^2(B_1 \ B_2)(I - uD_*)^{-1} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \end{aligned} \quad (2.43)$$

Proof. Solving the third equation of $(I - uT)w = v$ and substituting it into the first two produces (2.42). Eliminating the second component then gives the left-hand side of (2.43). Solving the two memory equations together gives the right-hand side, by Proposition 2.25. The formal solution is unique because $I - uT$ has constant term I . Both procedures also transform the source and reader according to (2.38)–(2.39). $\square$

Balance (2.35) describes record increments. The preceding identities allow them to be transported when memory is reused in dodecaphase closure; they do not by themselves determine additional analytic coefficients.

3 Topological Phase Kernel and generation of the seed catalogue

The two APP evaluations acquire generative efficacy once their activation, evaluation position, positional transport and the information preserved by each transition have been specified. The *Topological Phase Kernel*, abbreviated TPK, combines these operations into a dynamics on enriched states. Its definition comprises selection, transport, arithmetic updating and preservation of the record. The finite construction developed in this section provides the seeds on which the regional lifts and coinductive prolongation subsequently act.

The principal result of this stage is a generated and classified catalogue: the initial conditions of the two cursors produce integer emissions; their ternary reading determines oriented words; and the dihedral action organises these words into orbits. Each step preserves its preimages and the relations that permit recovery of the preceding description. The chain of cardinalities

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729, \quad 243 \longrightarrow 43,$$

abbreviates distinct objects. Each object is constructed and each count justified below, before any of them is used as data for regional selection.

3.1 Phase selection, transport and enriched state

3.1.1 Tritic selector of operative phase

The nonadic phase takes its values in $D_9 = \{1, \dots, 9\}$. The tritic selector is the function

$$M_{\text{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \quad M_{\text{ph}}(g) = \begin{cases} +1, & g \in \{1, 4, 7\}, \\ -1, & g \in \{2, 5, 8\}, \\ 0, & g \in \{3, 6, 9\}. \end{cases} \quad (3.1)$$

In the emitter realisation, the value $+1$ activates the additive evaluation, -1 activates the multiplicative evaluation, and 0 records the neutral event. The cardinal orientation of the cursor is another state coordinate. This distinction specifies both the operation being read and the direction in which its evaluation point is transported.

For transitions indexed by $t \geq 1$, the phase preceding the reading is

$$g_t = 1 + [(t-1)]_9, \quad \epsilon_t = M_{\text{ph}}(g_t), \quad (3.2)$$

where $[n]_b$ denotes the representative of n modulo b in $\{0, \dots, b-1\}$. A window of nine transitions contains the sequence $(+1, -1, 0, +1, -1, 0, +1, -1, 0)$.

3.1.2 Oriented cursors and cardinal transport

The indices $I_9 = \{0, \dots, 8\}$ are used for positions, and the positive labels $(a, b) = (i+1, j+1)$ for APP evaluations. A finite cursor is

$$c = (i, j; d) \in \mathcal{S} := I_9^2 \times \mathcal{D}, \quad \mathcal{D} = \{N, E, S, O\}.$$

The four directions act by

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The cardinal map corresponding to d is

$$T_d(i, j) = ([i + (v_d)_1]_9, [j + (v_d)_2]_9).$$

Thus M_{ph} and T_d have distinct domains, codomains and actions. Tritic activation selects one of the two cursors; cardinal transport changes its position in the direction already retained by that cursor.

The integer lift of the cursor is $\tilde{c} = (x, y; d) \in \mathbb{Z}^2 \times \mathcal{D}$, with finite position $([x]_9, [y]_9)$. Its winding counts are recovered from

$$x = 9[x/9] + [x]_9, \quad y = 9[y/9] + [y]_9. \quad (3.3)$$

These two divisions concern spatial transport. The division of an additive or multiplicative evaluation constitutes a distinct arithmetic record, which is preserved together with them.

3.1.3 Composition of the transition

The state of the finite realisation includes the two lifted cursors, their directions, the phase, the accumulators of the active window, the ordered list of blocks already emitted and the transition record. Each local record contains the preceding positions, the integer evaluations of both sheets, their residues and quotients, the active regime and the orientations. Prefix and boundary fields arising in a prolongation are incorporated into the same state with their compatibility rules.

The transition is ordered as

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (3.4)$$

Selection determines the regime and preserves the sample taken before any displacement. Transport applies the active movement and the cardinal reversal prescribed by the schedule. Updating incorporates that sample into the accumulators and the record; at the end of a window, it emits the corresponding block. The composition therefore uses a sample taken before transport, although its definitive storage occurs at the end of the transition.

3.2 Initial conditions and reversible enumeration

The additive and multiplicative sheets have independent cursors. The domain of initial conditions is the ordered product

$$\Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}| = 9^2 \cdot 4 = 324.$$

Consequently,

$$|\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (3.5)$$

The number 81 counts positions; 324 counts the oriented positions of one cursor; 104 976 counts ordered pairs of cursors. Choosing a position in APP retains only part of an initial condition of the emitter.

Fix the numbering

$$\text{ind}(N) = 0, \quad \text{ind}(E) = 1, \quad \text{ind}(S) = 2, \quad \text{ind}(O) = 3,$$

and define

$$\nu(i, j; d) = 4(9i + j) + \text{ind}(d). \quad (3.6)$$

Proposition 3.1 (Exact enumeration of the initial conditions). *The map ν is a bijection from $\mathcal{S}$ onto $\{0, \dots, 323\}$. The pair of indices is reversibly encoded by $324\nu_+ + \nu_\times \in \{0, \dots, 104\,975\}$.*

Proof. Division by four recovers

$$\text{ind}(d) = [\nu]_4, \quad j = [\lfloor \nu/4 \rfloor]_9, \quad i = \lfloor \nu/36 \rfloor.$$

The bounds on the domain ensure that these three quantities belong to their respective index sets and reconstruct ν through (3.6). For the pair, Euclidean division by 324 recovers the quotient ν_+ and the residue $\nu_\times$. $\square$

The enumeration traverses the complete domain before any regional grouping. The indices locate the preimages of an emission while keeping explicit the condition that produced each trajectory.

3.3 Sheet evaluation and arithmetic record

At a positive position (a, b) , the integer evaluations are $A_+(a, b) = a + b$ and $A_\times(a, b) = ab$. For $n > 0$, retain

$$\rho_9(n) = 1 + [n - 1]_9, \quad q_9(n) = \lfloor (n - 1)/9 \rfloor, \quad n = \rho_9(n) + 9q_9(n). \quad (3.7)$$

Additive activation incorporates $\rho_9(A_+)$ into the corresponding accumulator. Multiplicative activation uses an observable of the residue $\rho_9(A_\times)$, defined next.

3.3.1 Discrete exponent in the units modulo nine

The powers of 2 modulo 9 traverse

$$2^0, 2^1, \dots, 2^5 \equiv 1, 2, 4, 8, 7, 5 \pmod{9}, \quad 2^6 \equiv 1 \pmod{9}.$$

The discrete logarithm in this group of six units is represented by

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5. \quad (3.8)$$

The emission law extends the observable by $\ell_9(3) = \ell_9(6) = \ell_9(9) = 0$. To preserve its provenance, the recorded sample includes the original residue and the indicator of membership in the group of units. An observed value of zero can therefore originate from the unit 1 or from one of the three non-invertible residues, with its origin remaining recoverable.

3.3.2 Gauges of the multiplicative observable and generator normalization

The choice of discrete exponent admits a finite comparison in which the APP support, the initial conditions of both cursors, the tritic selector, the reversal calendar, and the emission rule remain fixed. The variation is confined to the observable on the multiplicative sheet. Let ℓ_2 and ℓ_5 be the logarithms of $(\mathbb{Z}/9\mathbb{Z})^\times$ with generators 2 and 5, respectively, extended by zero on $\{3, 6, 9\}$. They are compared with v_3 , evaluated on the positive representatives $1, \dots, 9$, and with the identity reading on those same representatives.

Proposition 3.2 (Normalization of the multiplicative logarithm). *The two isomorphisms from the unit group modulo nine to $\mathbb{Z}/6\mathbb{Z}$, normalized by sending a generator to 1, are ℓ_2 and ℓ_5 . They satisfy*

$$\ell_5(x) \equiv -\ell_2(x) \pmod{6} \quad (x \in (\mathbb{Z}/9\mathbb{Z})^\times).$$

Choosing the least positive generator fixes $\ell_2 = \ell_9$, with the values in (3.8).

Proof. The powers of 2 traverse 1, 2, 4, 8, 7, 5 and close at the sixth step. The generators of a cyclic group of order six correspond to exponents coprime to six: 1 and 5. Their representatives are 2 and $2^5 \equiv 5 \pmod{9}$. Since $5 \equiv 2^{-1} \pmod{9}$, expressing a unit as a power of 5 changes the sign of its exponent in base 2. Normalization by the least positive representative selects 2. $\square$

The homomorphism condition alone also admits maps with proper image. The proposition therefore distinguishes a faithful representation of the unit group from its quotients. Extension by zero also specifies the treatment of the three noninvertible residues; the provenance record retains which residue was observed.

The comparison of the four observables traverses the same 104 976 initial conditions and groups their emissions over six windows. Writing E_g for the emitter with multiplicative observable g , one obtains the following image census.

Both logarithms satisfy the additive exponent law. The valuation v_3 vanishes on every unit but takes the values 1, 1, 2 on 3, 6, 9; this accounts for the difference between its two control columns. The identity retains the positive residue and fails the homomorphism law into $\mathbb{Z}/6\mathbb{Z}$; for example, at the unit 1, the required equality would be $1 \equiv 2 \pmod{6}$. The census of 468 emissions is shared by

Observable	$ \text{im } E_g $	Zero roots	Additive law	Normalization
ℓ_2	468	yes	yes	least generator
ℓ_5	468	yes	yes	inverse generator
v_3	144	no	yes	zero unit image
id	684	no	no	positive residue

Table 1: Comparison of four gauges on the same transport and the same domain of initial conditions.

both logarithms and therefore retains the inversion ambiguity. The least-generator criterion resolves that ambiguity before any regional reading of constants. The cardinalities in the table describe the exhaustive evaluation of these four observables on the fixed domain; their role is to test the emitter's operational normalization.

3.3.3 Order of reading and movement

Evaluations are performed at the positions preceding the transition. If $\epsilon_t = +1$, the additive reading is accumulated and its cursor is displaced. If $\epsilon_t = -1$, the multiplicative discrete exponent is accumulated and the second cursor is displaced. If $\epsilon_t = 0$, the neutral counter is incremented and both positions are preserved.

At the end of steps 27 and 54, both directions are replaced by their opposites. The windows are concatenated with positions and orientations preserved. At the start of each window, only its three local accumulators are reset to zero; the preceding record remains available.

Proposition 3.3 (Inversion of the lifted transport). *Let $a, b \in \{0, 1\}$ be the activation and reversal indicators of a cursor. If opp interchanges $N \leftrightarrow S$ and $E \leftrightarrow O$, the transformation*

$$F_{a,b}(z, d) = (z + av_d, \text{opp}^b(d))$$

has inverse

$$F_{a,b}^{-1}(z', d') = (z' - av_{\text{opp}^b(d')}, \text{opp}^b(d')).$$

Spatial reduction modulo nine intertwines this transformation with the finite transport.

Proof. Applying opposition twice recovers the initial direction. Once that direction has been recovered, subtracting av_d reverses the displacement. For the spatial projection, it suffices to use $[z + av_d]_9 = [[z]_9 + av_d]_9$ componentwise. Directional reversal preserves this equality. $\square$

The phase determines the indicators used by the inverse. The spatial and arithmetic records can therefore be preserved jointly, including when the boundary of the representative I_9^2 is crossed.

3.4 Six windows and decimal block emission

3.4.1 Accumulation within a nonadic window

Window m , with $1 \leq m \leq 6$, comprises steps $9m - 8, \dots, 9m$. Let S_m be the sum of its active positive additive residues, E_m the sum of its active discrete exponents, and Z_m its number of neutral events.

Lemma 3.4 (Activation multiplicities). *Each window contains three additive activations, three multiplicative activations and three neutral activations. In particular, $Z_m = 3$.*

Proof. The phases of a window traverse exactly $1, \dots, 9$. Each of the three sets in (3.1) has cardinality three. $\square$

The six windows comprise 54 transitions and produce six blocks. This length belongs to the initial emitter defined here. Continuation in depth subsequently acts on the words and their compatible records.

3.4.2 Decimal encoding rule

The emission law specifies the digits

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (3.9)$$

The weights 100, 10, 1 are the hundreds, tens and units positions. The subscript 10 denotes the decimal residue, and each block satisfies $0 \leq U_m \leq 999$. The emitted word is the ordered sequence $\mathbf{U} = (U_1, \dots, U_6)$, with a fixed width of three digits per block.

The coefficients 3, 5, 7 constitute the explicit weights of this emission law. Once the rule has been fixed, each digit is calculated from the accumulators, and the results of this section follow from that rule. The number 1 appearing in its reduced form has a further specific provenance: $7Z_m = 21 \equiv 1 \pmod{10}$.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, one obtains

$$G(s_m, e_m) = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10} = U_m. \quad (3.10)$$

Here the letter e_m denotes a component of the discrete-exponent signature. Its definition belongs to the finite emitter.

Proposition 3.5 (Recovery of the two signatures). *The map $G : \{0, \dots, 9\}^2 \rightarrow \{0, \dots, 999\}$ is injective. For a block $U = 100a + 10b + c$ in its image, the inverse is*

$$s = a, \quad e = [b - a]_{10}, \quad c = [3s + 5e + 1]_{10}.$$

Consequently, the word $\mathbf{U}$ determines the two six-component signatures $\mathbf{s}$ and $\mathbf{e}$.

Proof. The last two summands of (3.10) belong to $\{0, \dots, 99\}$, so the hundreds digit recovers s . The tens digit is $[s + e]_{10}$; subtracting s and reducing modulo ten recovers e . The units digit verifies the third congruence. The reconstruction applies independently to the six positions of the word. $\square$

The integer totals S_m, E_m are recovered from the recorded readings; the hundreds and tens digits recover their residues. For example, a total $S_m = 15$ produces $s_m = 5$. This distinction identifies precisely which information each level of encoding preserves.

3.4.3 Decimal alphabet and check-digit congruence of the emitter

Retain the emission law (3.9), with the APP readings and observable ℓ_9 of (3.8), the cursors of Section 3.1, and the six-window calendar of Section 3.4. The accumulators are S_m, E_m, Z_m , with $Z_m = 3$, and the output digits are denoted $d_{m,1}, d_{m,2}, d_{m,3}$. Proposition 3.5 already recovers both residual signatures. Combining this recovery with the actual accumulator images determines the following alphabet.

Proposition 3.6 (Check-digit congruence for the third digit). *Every emitted block satisfies*

$$d_{m,3} = [5d_{m,2} - 2d_{m,1} + 1]_{10}.$$

Proof. The inverse in Proposition 3.5 gives $[E_m]_{10} = [d_{m,2} - d_{m,1}]_{10}$. Substituting $E_m \equiv d_{m,2} - d_{m,1} \pmod{10}$ and $Z_m = 3$ into the third digit gives $3d_{m,1} + 5(d_{m,2} - d_{m,1}) + 21 \equiv 5d_{m,2} - 2d_{m,1} + 1 \pmod{10}$. $\square$

Proposition 3.7 (Decimal alphabet of forty-two blocks). *The per-window image of the emitter consists exactly of the blocks*

$[E_m]_{10}$	U_m						
0	114	227	443	556	669	885	998
1	129	232	458	561	674	890	903
3	149	252	478	581	694	810	923
5	169	272	498	501	614	830	943
7	189	292	418	521	634	850	963
9	109	212	438	541	654	870	983

Proof. A cardinal trajectory reads three consecutive terms of the additive cycle

$$(2, 3, 4, 5, 6, 7, 8, 9, 1).$$

Their sums are (9, 12, 15, 18, 21, 24, 18, 12, 6); reversing direction preserves the set of sums. Hence

$$\text{Im } S_m = \{6, 9, 12, 15, 18, 21, 24\}, \quad \text{Im}[S_m]_{10} = \{1, 2, 4, 5, 6, 8, 9\}.$$

In the multiplicative channel, a fixed factor divisible by three contributes zero. For invertible factors, the cyclic sums of three readings are

fixed factor	possible sums
1, 8	{1, 3, 7, 9}
2, 7	{3, 5, 9}
4, 5	{1, 5, 7}

Thus $\text{Im } E_m = \{0, 1, 3, 5, 7, 9\}$. Attainability can already be verified in the first window: every cursor in the following two tables has direction E .

S_1	6	9	12	15	18	21	24
(i_+, j_+)	(0, 8)	(0, 0)	(0, 1)	(0, 2)	(0, 3)	(0, 4)	(0, 5)

E_1	0	1	3	5	7	9
(i_x, j_x)	(2, 0)	(0, 0)	(0, 1)	(1, 1)	(0, 2)	(0, 4)

Independence of the two cursors realizes every one of the $7 \cdot 6$ pairs. The first two digits recover the residual pair, so their 42 images are distinct. The check-digit congruence fixes the third digit and yields exactly the stated table. $\square$

The decimal block therefore preserves both residual signatures through an injective encoding and a check digit. This result describes the finite emitter alphabet; continuation of the regional readings additionally retains the phase, orientation and memory conditions of the enriched state.

The per-window image therefore has 42 blocks. Compatible concatenation of six windows preserves the census of 468 emissions in Proposition 3.9, whose ternary reduction yields 243 words within the 729-word ambient space. Each cardinality counts the object of its stage and retains the provenance of the same emitter.

3.4.4 Positional encoding of the six-block word

The six blocks admit a fixed-width integer encoding,

$$\mathcal{N}(\mathbf{U}) = \sum_{m=1}^6 U_m 1000^{6-m}, \quad 0 \leq \mathcal{N}(\mathbf{U}) < 1000^6. \quad (3.11)$$

The first block occupies the position of greatest weight, and the sixth the base-thousand units position. A block 058 occupies a complete position with integer value 58; its three-digit width belongs to the decimal presentation of that position.

Proposition 3.8 (Positional recovery of the emissions). *The encoding (3.11) determines the complete word by*

$$U_m = \left[\left[\frac{\mathcal{N}(\mathbf{U})}{1000^{6-m}} \right] \right]_{1000}, \quad 1 \leq m \leq 6.$$

Proof. On division by 1000^{6-m} , the preceding blocks yield integer multiples of 1000, block m yields U_m , and the contribution of the subsequent blocks belongs to $[0, 1)$. Taking the integer part eliminates the latter contribution, and the residue modulo 1000 eliminates the former. $\square$

This encoding preserves the six emissions. Its domain has ordered base-thousand positions, whereas the APP domain has spatial positions modulo nine. The composition retains both indices: m identifies an emission window, and (i, j) identifies the cell visited during a transition in that window.

3.5 Complete calculation of an emission beginning with 501

Fix the initial conditions

$$c_{+,0} = (0, 4; N), \quad c_{\times,0} = (2, 3; N), \quad \nu_+ = 16, \quad \nu_\times = 84.$$

The initial positive positions are $(1, 5)$ and $(3, 4)$. The preceding chronological index is $t = 0$, the first phase read is $g_1 = 1$, and the record is empty. These coordinates are the inputs to the calculation; the blocks are obtained by executing the recurrence.

Table 2: First window: preceding readings and active accumulation.

t	ϵ_t	position +	position $\times$	ΔS	ΔE	(S, E, Z)
1	+1	(1, 5)	(3, 4)	6	0	(6, 0, 0)
2	-1	(9, 5)	(3, 4)	0	0	(6, 0, 0)
3	0	(9, 5)	(2, 4)	0	0	(6, 0, 1)
4	+1	(9, 5)	(2, 4)	5	0	(11, 0, 1)
5	-1	(8, 5)	(2, 4)	0	3	(11, 3, 1)
6	0	(8, 5)	(1, 4)	0	0	(11, 3, 2)
7	+1	(8, 5)	(1, 4)	4	0	(15, 3, 2)
8	-1	(7, 5)	(1, 4)	0	2	(15, 5, 2)
9	0	(7, 5)	(9, 4)	0	0	(15, 5, 3)

Note. Positions are positive labels of the cells preceding movement. Neutral events increment only Z .

The three active additive evaluations are

$$1 + 5 = 6, \quad 9 + 5 = 14 = 5 + 9, \quad 8 + 5 = 13 = 4 + 9,$$

so $S_1 = 6 + 5 + 4 = 15$. The three active multiplicative evaluations are $3 \cdot 4 = 12$, $2 \cdot 4 = 8$ and $1 \cdot 4 = 4$. Their positive residues are 3, 8, 4, with observables 0, 3, 2; hence $E_1 = 5$. Finally, $Z_1 = 3$. The decimal rule gives

$$d_{1,1} = [15]_{10} = 5, \quad d_{1,2} = [15 + 5]_{10} = 0, \quad d_{1,3} = [45 + 25 + 21]_{10} = 1,$$

and one obtains $U_1 = 501$.

The second additive reading above has arithmetic quotient 1, whereas its lifted cursor is at $(-1, 4; N)$, with vertical winding count -1 . These are distinct data in the same record. Both participate in the complete recovery of the evaluation and its position.

The resulting word and its signatures are

$$\begin{aligned} \mathbf{U} &= (501, 614, 498, 169, 272, 272), \\ \mathbf{s} &= (5, 6, 4, 1, 2, 2), \end{aligned} \quad \mathbf{e} = (5, 5, 5, 5, 5, 5).$$

Cardinal reversal after step 27 changes the trajectories of the final three windows. The repeated block 272 corresponds to two successive windows with their own positions and chronological record.

Table 3: The six windows of initial condition (16, 84).

m	steps	S_m	E_m	Z_m	s_m	U_m	$[-U_m]_3$
1	1–9	15	5	3	5	501	0
2	10–18	6	5	3	6	614	1
3	19–27	24	5	3	4	498	0
4	28–36	21	5	3	1	169	2
5	37–45	12	5	3	2	272	1
6	46–54	12	5	3	2	272	1

As a second case, the minimal initial condition $(\nu_+, \nu_-) = (0, 0)$ produces

$$\mathbf{U} = (252, 109, 252, 903, 850, 850), \quad \mathbf{s} = (2, 1, 2, 9, 8, 8), \quad \mathbf{e} = (3, 9, 3, 1, 7, 7).$$

In this case, both cursors return to their initial positions and orientations after 54 transitions. The nonadic phase completes six cycles, the chronological counter advances by 54 units, and the record contains 54 samples. The positional reading of the return and the recorded history describe distinct properties of the execution.

3.6 Factorisation of the finite census

The signatures $f_+(\nu) = \mathbf{s}$ and $f_-(\nu) = \mathbf{e}$ are calculated by executing the corresponding cursor through the six windows. The independence of positions and directions between the two cursors gives

$$T_{\text{em}}(\nu_+, \nu_-) = G^{(6)}(f_+(\nu_+), f_-(\nu_-)), \quad (3.12)$$

where $G^{(6)}$ applies (3.10) at each position. This factorisation first permits enumeration of $324 + 324$ trajectories and then combination of the distinct signatures, preserving the multiplicities of their preimages.

3.6.1 Images of the two signatures

The following tables include every signature. Its multiplicity is the number of initial conditions producing it; the minimal index locates one such condition. The complete preimage is defined by

$$f_{\sigma}^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_{\sigma}(\nu) = a\}.$$

The recurrence of the preceding sections determines every element of this set by a finite integer calculation.

Table 4: The eighteen additive signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
122564	18	17	564122	18	16
122898	18	24	645212	18	4
212645	18	5	654889	18	33
212988	18	0	889221	18	13
221456	18	29	889654	18	32
221889	18	12	898122	18	25
456221	18	28	898465	18	20
465898	18	21	988212	18	1
546988	18	9	988546	18	8

Table 5: The twenty-six multiplicative signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
000000	108	8	555555	24	84
177393	8	1	555717	8	14
177555	8	54	555771	8	18
177771	8	11	555933	8	24
339555	8	6	717555	8	12
339771	8	15	717717	8	21
339933	8	59	717933	8	19
393177	8	0	771177	8	9
393393	8	45	771339	8	13
393555	8	40	771555	8	16
555177	8	52	933339	8	57
555339	8	4	933555	8	26
555393	8	41	933717	8	17

Proposition 3.9 (Image and multiplicities of the emitter). *The image $\mathcal{U}_6 = \text{im } T_{\text{em}}$ contains 468 distinct words. Their preimages are distributed into 432 fibres of cardinality 144, 18 of cardinality 432 and 18 of cardinality 1944.*

Proof. Exhaustive evaluation of the algorithm in Section 3.9 on the indices $0, \dots, 323$ gives the two tables above. The first partition has 18 classes of cardinality 18; the second has 24 classes of cardinality 8, one of cardinality 24 and one of cardinality 108. The sums $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$ verify the total mass of both partitions. Proposition 3.5 makes the coupling of their images injective; there are therefore $18 \cdot 26 = 468$ emissions. By (3.12), the preimage of a pair of signatures is the product of their two preimages. This gives $18 \cdot 24 = 432$ products of cardinality $18 \cdot 8 = 144$, 18 of cardinality $18 \cdot 24 = 432$, and 18 of cardinality $18 \cdot 108 = 1944$. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The enumerative part is an exact finite check of the recurrence; the factorisation explains why this count covers every pair. $\square$

The emission in Example 3.5 has a fibre of cardinality $18 \cdot 24 = 432$. The example with indices $(0, 0)$ has a fibre of cardinality $18 \cdot 8 = 144$. In both cases, the emitted word recovers the signatures, whereas identification of a particular initial condition uses its index or its trajectory record.

3.7 Oriented ternary reading and preservation of fibres

Oriented reduction acts separately on each block:

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \quad \Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (3.13)$$

Its sign fixes an orientation of the catalogue. The balanced identification $0 \mapsto 0, 1 \mapsto +1, 2 \mapsto -1$ is then applied componentwise.

Modulo nine organises APP evaluations and positions; residues modulo ten form each digit; base thousand combines three decimal positions; and modulo three determines the ternary signature of each block. These four uses have defined objects and functions. In particular, Ψ reads six blocks and produces six trits; a positional conversion of the integer formed by eighteen digits would be a different operation.

For the two conditions already calculated, one obtains

$$\begin{aligned} \Psi(501, 614, 498, 169, 272, 272) &= (0, 1, 0, 2, 1, 1), \\ \Psi(252, 109, 252, 903, 850, 850) &= (0, 2, 0, 0, 2, 2). \end{aligned}$$

The decimal emitter is defined before this reduction. The function of Ψ is to produce the ternary object on which orbital symmetry and regional selectors act. This function explains its position in the genealogy: it connects an already calculated emission with a structural classification.

The image $\mathcal{W}_6 := \Psi(\mathcal{U}_6)$ contains exactly 243 words within the ambient set of $3^6 = 729$ words. Enumeration by the final algorithm gives the histogram

$$\begin{array}{c|cccccc} |\Psi^{-1}(w)| & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline \text{number of words } w & 132 & 66 & 18 & 3 & 6 & 18. \end{array} \quad (3.14)$$

The sums for the two readings are

$$132 + 66 + 18 + 3 + 6 + 18 = 243, \quad 132 + 2 \cdot 66 + 3 \cdot 18 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 18 = 468.$$

The first counts ternary words; the second recovers the number of emissions distributed among their fibres.

Example 3.10 (Two emissions with the same ternary signature). The emissions

$$(498, 501, 614, 252, 252, 109), \quad (870, 810, 923, 252, 252, 109)$$

are distinct and both have image $(0, 0, 1, 0, 0, 2)$ under Ψ . Each has 144 initial conditions. The ternary word preserves a reading class; the emissions and their preimages distinguish the data that converge in that class.

Preservation of fibres is expressed concretely by

$$\mathcal{F}_w = \{\mathbf{U} \in \mathcal{U}_6 : \Psi(\mathbf{U}) = w\}, \quad \Omega_{\mathbf{U}} = T_{\text{em}}^{-1}(\mathbf{U}).$$

The complete preimage of w in the initial domain is therefore the disjoint union of the sets $\Omega_{\mathbf{U}}$, with $\mathbf{U} \in \mathcal{F}_w$. The enriched record can retain the emission, its signatures and the initial condition together with its ternary reading. The exact inverse then concerns the recorded datum with its type; the isolated six-trit word retains the precise scope of (3.13).

3.8 Dihedral action and the forty-three orbits

Define the following permutations on six coordinates:

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4), \quad (3.15)$$

$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3). \quad (3.16)$$

The first rotates the two groups of three coordinates in conjugate directions; the second interchanges the groups. Their composition satisfies $r^3 = s^2 = 1$ and $srs = r^{-1}$. They therefore generate an action of the six-element dihedral group D_3 .

Proposition 3.11 (Equivariance and stability of the catalogue). *Reduction by Ψ commutes with the action of D_3 . The sets $\mathcal{U}_6$ and $\mathcal{W}_6$ are stable under its generators.*

Proof. The equality $\Psi(g\mathbf{U}) = g\Psi(\mathbf{U})$ holds at every coordinate because g permutes positions and Ψ applies the same reduction at each position. Stability of the catalogue is a further check: generating its 468 elements by (3.12), one verifies that $r\mathbf{U}$ and $s\mathbf{U}$ again belong to that set. The algorithm in Section 3.9 performs these membership checks. Equivariance then transfers stability to the ternary image. $\square$

Proposition 3.12 (Complete orbital decomposition). *The set $\mathcal{W}_6$ decomposes into 43 orbits: 38 of cardinality six and 5 of cardinality three.*

Proof. For each $w \in \mathcal{W}_6$, form

$$\mathcal{O}(w) = \{w, rw, r^2w, sw, srw, sr^2w\}.$$

Choose the lexicographically least word in this set as representative. Two words have the same representative exactly when they belong to the same orbit. The algorithm generates the list of representatives in Table 6; their cardinalities sum to $38 \cdot 6 + 5 \cdot 3 = 243$. Since every word in the catalogue participates in this operation and the catalogue is stable, the list covers the entire image. $\square$

Table 6: Minimal representatives and cardinalities of the forty-three orbits.

representative	cardinality	representative	cardinality	representative	cardinality
001002	6	002112	6	012222	6
001011	6	002211	6	021022	6
001012	6	002220	6	021121	6
001021	6	011021	6	021202	6
001022	6	011112	6	022022	3
001100	3	011120	6	022112	6
001101	6	011201	6	022121	6
001110	6	011211	6	022202	3
001112	6	011220	6	022211	6
001202	6	011222	6	022222	6
001210	6	012112	6	112112	3
001211	6	012121	6	112211	3
001220	6	012202	6	112222	6
001222	6	012210	6		
002012	6	012220	6		

The orbits inherit integer invariants from their fibres. For each word, $n(w) = |\mathcal{F}_w|$ counts emissions, whereas

$$M(w) = \sum_{\mathbf{U} \in \mathcal{F}_w} |\Omega_{\mathbf{U}}|$$

counts initial conditions. The symbol census $c(w) = (n_0, n_1, n_2)$ records the number of occurrences of each ternary class. Together with the orientation and the support of the trajectories, these data provide the predicates for subsequent regional selection. The classification thus preserves more information than the form of a representative word.

3.9 Self-contained enumeration and verification algorithm

The enumeration uses integers, finite lists and the operations already defined. The following procedure specifies its execution; σ takes the values $+$ and $\times$.

1. For each $v = 0, \dots, 323$, recover $(i, j; d)$ using the inverse of (3.6). Initialise an empty signature list.
2. For each window $m = 1, \dots, 6$, initialise the accumulator $A = 0$. Traverse $t = 9m - 8, \dots, 9m$, calculating ϵ_t by (3.2).
3. If $\sigma = +$ and $\epsilon_t = +1$, add $\rho_9(i + j + 2)$ to A and update $(i, j) \leftarrow T_d(i, j)$. If $\sigma = \times$ and $\epsilon_t = -1$, add $\ell_9(\rho_9((i + 1)(j + 1)))$ and perform the same transport.
4. At the end of $t = 27$ or $t = 54$, replace d by its opposite. At the end of the window, append $[A]_{10}$ to the signature list. Preserve the position and direction for the next window.
5. Group the 324 indices according to equality of their six-component lists. These two groupings give f_+ , $f_\times$ and all their preimages.
6. For each pair of distinct signatures, apply $G^{(6)}$. Assign to the emission the product multiplicity of the two preimages. Verify signature recovery in each block and the sum of multiplicities 104 976.
7. Apply Ψ to each emission and group by equality of words. Count the 243 values and the multiplicities in (3.14).
8. Apply r, s to each emission and verify membership in the catalogue. For each ternary word, form its six dihedral images, remove repetitions and record its lexicographic minimum. Count the 43 orbits and the two sizes in Table 6.

The execution terminates: the first stage evaluates 648 trajectories of 54 transitions; the second combines 468 pairs of signatures; the third groups a set of 243 words. An additional direct check can execute the 104 976 pairs and compare the result with the factorisation (3.12); its scope is the same finite domain.

3.10 Preservation of the record and the function of the seeds

Let $\text{Sample}(q_t)$ be the sample preceding the transition, and $\mathcal{M}_t$ the accumulated record. The memory update is

$$\mathcal{M}_{t+1} = \mathcal{M}_t \parallel \text{Sample}(q_t), \quad (3.17)$$

where $\parallel$ denotes ordered concatenation. Induction gives

$$\mathcal{M}_n = \mathcal{M}_0 \parallel (\text{Sample}(q_0), \dots, \text{Sample}(q_{n-1})).$$

The accumulators are reconstructed by summing the active contributions of each segment of nine records. Their quotients, the winding counts of the cursors and the orientations remain associated with the same transitions. Execution determines that record through the recurrence; storage preserves the specific provenance of the readings.

3.11 Cyclic correlation and preservation of path order

The record (3.17) preserves the ordered concatenation of samples. The following example isolates order information extracted by two distinct readers from a twelve-position word, retaining the domain of each reading.

Let $\tau = (\tau_1, \dots, \tau_{12}) \in \{-1, 0, 1\}^{12}$ be an ordered word, with indices taken modulo 12. Define

$$\begin{aligned} G_a &= \{a, a + 4, a + 8\} \quad (1 \leq a \leq 4), \\ v_{120}(\tau) &= \sum_{a=1}^4 \text{sgn} \left(\sum_{m \in G_a} \tau_m \right), \\ v_{270}(\tau) &= \sum_{m=1}^{12} \tau_m \tau_{m+3}. \end{aligned}$$

The marginal histogram is $p_j(\tau) = \#\{m : \tau_m = j\}/12$ for $j \in \{-1, 0, 1\}$, and its entropy is $H(\tau) = -\sum_j p_j(\tau) \log p_j(\tau)$, with $0 \log 0 = 0$.

Proposition 3.13 (Order information beyond the marginal distribution). *There exist words with identical histograms and distinct correlations ν_{270} , even when ν_{120} has the same value.*

Proof. Consider

$$\tau^{(a)} = (1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0), \quad \tau^{(b)} = (1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0).$$

Both have $p_1 = 1/6$, $p_0 = 5/6$ and $p_{-1} = 0$, so their marginal entropies coincide. In each case, two of the sets G_a contain a unit and the other two contain only zeros: $\nu_{120} = 2$. In $\tau^{(a)}$, no pair of units has cyclic separation three, whereas in $\tau^{(b)}$ precisely the term $m = 1$ contributes. Thus $\nu_{270}(\tau^{(a)}) = 0$ and $\nu_{270}(\tau^{(b)}) = 1$. $\square$

This separation identifies the information retained by the ordered record: incidences between positions in addition to symbol frequencies. A reduced state retaining only the histogram assigns the same output to both words and loses this correlation. Marginal entropy, uncertainty between bases and entanglement entropy are defined on different objects; relating them requires retaining the distribution, the change of basis, and the state together with its partition, respectively.

Proposition 3.14 (Parity of the readers under oriented reversal). *With cyclic indices in $\mathbb{Z}/12\mathbb{Z}$, let $(C_M\tau)_m = -\tau_{-m}$ and $\text{sgn}(0) = 0$. The two readers satisfy*

$$\nu_{120}(C_M\tau) = -\nu_{120}(\tau), \quad \nu_{270}(C_M\tau) = \nu_{270}(\tau).$$

Proof. The reflection $m \mapsto -m$ permutes the four position classes modulo four. Reversal changes the sign of each class sum and hence, by the oddness of sgn , the sign of ν_{120} . In the second reader, the signs of the two factors cancel:

$$\sum_m (C_M\tau)_m (C_M\tau)_{m+3} = \sum_m \tau_{-m} \tau_{-m-3} = \sum_k \tau_{k+3} \tau_k = \nu_{270}(\tau).$$

$\square$

Orientation thus distinguishes an odd reading from an even correlation. Both depend on the arrangement of samples in the record; their transformation laws specify which information persists when the route is reversed.

At the end of this stage, the complete domain of initial conditions, decimal emission, its two recoverable signatures, the ternary image, its fibres and its orbital classification have been constructed. The counts 104 976, 468, 243, 729 and 43 now have a domain and a mechanism of derivation. Regional selections receive these structures with their invariants, and the lifts act on oriented words with preserved provenance. This continuity makes the finite catalogue a constructive antecedent of the coinductive development.

4 Development of the formal kernel: initial conditions, geometry, and refinement

The eighty-one-position support does not coincide with the domain of initial conditions of the dynamics. An initial condition must also specify an orientation; and the additive and multiplicative sheets have independent cursors. This distinction makes it possible to reconstruct the emission exhaustively without selecting a numerical region in advance.

4.1 Initial states and the emission recurrence

For enumeration we use indices $I_9 = \{0, \dots, 8\}$, related to the positive labels by $i \mapsto i + 1$. Let

$$\mathcal{S} = I_9^2 \times \{N, E, S, O\}, \quad \Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times.$$

Each initial condition $\omega = (s_+, s_\times)$ determines two positions and two directions. The domain comprises all ordered pairs, so

$$|\mathcal{S}| = 324, \quad |\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (4.1)$$

Appendix A.1 gives a bijective numbering of these conditions, the recurrence, and the complete signature tables. The rows of an emission table are fibers of this domain of initial conditions.

The finite realization uses six windows of nine transitions. At each time t , the phase signature is $M_{\text{ph}}(1 + ((t-1) \bmod 9))$. If it equals $+1$, the positive residue $\Sigma(i+1, j+1) = \rho_9(i+j+2)$ of the first cursor is read, and that cursor is then moved; if it equals -1 , the positive residue $\Pi(i+1, j+1) = \rho_9((i+1)(j+1))$ of the second cursor is read, and the second cursor is then moved. The value zero increments the neutral counter. At the end of transitions twenty-seven and fifty-four, the cardinal orientations are reversed. Integer evaluations and their quotients are retained in the register, but the emitter uses the residue readouts specified here.

In the unit sector, the multiplicative readout uses the base-two discrete logarithm in $(\mathbb{Z}/9\mathbb{Z})^\times$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

For the finite emission it is extended by zero to the nonunit sector. This extension is an observable, not an identification of the elements 3, 6, 9 with the multiplicative identity. The sheet and its membership in the radical sector remain in the trajectory register.

Let S_m be the sum of the three additive positive residues Σ in window m , E_m the sum of the three discrete exponents of the multiplicative residues Π , and $Z_m = 3$ the number of neutral events. Thus S_m does not sum the integer evaluations $S = i + j$, whose quotients remain in the register. With $[a]_{10} \in \{0, \dots, 9\}$, the decimal emission rule is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (4.2)$$

The integer coefficients and calendar of this recurrence are an explicit part of the declared emission law. No values of π, φ, e or α are received. The distinction between an emission rule and a constant recognized from its output prevents the genealogy from being hidden in a table of decimals.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, the same rule reads

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \quad (4.3)$$

The final one is the residue of $7Z_m = 21$ modulo ten. The oriented ternary reduction of the catalogue adopts the convention

$$\Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (4.4)$$

The sign of this chart is explicit: changing it modifies the words and requires transporting subsequent orientations. The balanced identification of each component of $\mathbb{F}_3$ with the TRIT is made after (4.4).

Proposition 4.1 (Injective factorization of the emission). *The word $U = (U_1, \dots, U_6)$ determines the two signatures $s = (s_1, \dots, s_6)$ and $e = (e_1, \dots, e_6)$. The emission image has $18 \cdot 26 = 468$ elements. The ternary projection has 243 distinct values in the ambient space of 729 words.*

Proof. The hundreds digits of U_m recover s_m . If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit checks the third congruence in (4.3). Thus the coupling is injective on the product of signatures.

Enumerating the 324 conditions of a cursor using the preceding recurrence produces eighteen additive signatures, each with eighteen preimages; and twenty-six multiplicative signatures, of which twenty-four have eight preimages, one has twenty-four, and one has one hundred and eight. Tables 13 and 14 preserve the signatures, their multiplicities, and a witness for each fiber; the recurrence in Appendix A.1 determines all their preimages. Injectivity of the coupling gives $18 \cdot 26$ emissions and multiplicities

$$\begin{array}{ccc} 432 & 18 & 18 \\ \hline 18 \cdot 24 & \text{fibers of } 144, & \text{fibers of } 432, \quad \text{fibers of } 1944. \end{array}$$

Their sum is $432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976$. Applying (4.4) to the 468 emissions produces exactly 243 words. This last number is the cardinality of the image, not of the space $\mathbb{F}_3^6$. $\square$

This factorization explains the difference between finite exhaustiveness and unlimited depth. The domain (4.1) is complete for the declared support and emitter; it does not already contain every prefix of every prolongation. Depth belongs to the system of compatible extensions constructed over those conditions.

4.2 Calendar closure and memory advancement

The calendar of one hundred and eight transitions consists of twelve nonadic windows. Its group structure is more precise than an informal product of two periods:

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \quad \iota([b]) = [9b], \quad p([t]) = [t]_9. \quad (4.5)$$

If $t = 9b + r$ with $0 \leq r < 9$, adding two phases produces the carry

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor, [r + r']_9 \right),$$

where the first coordinate is reduced modulo twelve. Composition preserves the term linking the two coordinates.

Proposition 4.2 (Nonsplitting of the calendar). *The sequence (4.5) is exact and admits no section that is a group homomorphism.*

Proof. The kernel of reduction modulo nine is the subgroup of multiples of nine, the image of ι . If the extension split, commutativity would give $C_{108} \cong C_{12} \times C_9$. The first group has exponent 108, whereas the second has exponent $\text{lcm}(12, 9) = 36$, a contradiction. $\square$

Nine transitions restore the nonadic phase and advance one window; one hundred and eight restore the finite calendar. Neither fact requires erasing the trajectory or restoring the prefix of a prolongation. The nonadic holonomy Γ_9 denotes the transport of one winding on states with memory, so that

$$g(\Gamma_9 x) = g(x), \quad M(\Gamma_9 x) = M(x) + 1. \quad (4.6)$$

Equality of phase and increment of memory are compatible and constitute two observations of the same transport.

4.2.1 Temporal parametrization of compatibility and tritic orientation

The temporal interpretation of TRIT is established on transitions of the complete state. Let $x \xrightarrow{\text{adm}} y$ be the admissibility relation determined by TPK operations on $\hat{X}$. A trajectory parametrized by a discretely ordered set $\mathcal{T}$ is a map

$$\gamma : \mathcal{T} \longrightarrow \hat{X}, \quad \gamma(t) \xrightarrow{\text{adm}} \gamma(t^+),$$

where t^+ is the successor of t . The compatibility relation precedes the choice of parameter: the order allows its transitions to be expressed as an evolution. When the term section is used, the projection $p : E \rightarrow \mathcal{T}$ and condition $p \circ \gamma = \text{id}_{\mathcal{T}}$ are also declared. The state space, temporal order, and transported information are thereby specified.

The signature τ selects a quadratic regime and admits a temporal reading relative to this parametrization. Type +1, with $J_{+1}^2 = -I$, represents return and memory; type 0, with $J_0^2 = 0$, represents the transition threshold; type -1, with $J_{-1}^2 = I$, represents opening in two eigendirections. In the model's temporal terminology, these functions correspond, respectively, to past, present, and future. The correspondence concerns the operational position of a transition within a history: retained antecedent, update, and admissible prolongation.

This interpretation acquires content through three distinct operations. Restriction of a trajectory recovers the prefix already constructed. The update $\mathcal{U}_t$ composes selection, transport, and recording in the observed transition. Prolongation considers continuations that preserve the boundary conditions of that prefix. If $\mathcal{P}_n(x)$ denotes the set of admissible continuations of length n from x , deleting the last step defines

$$r_n : \mathcal{P}_{n+1}(x) \longrightarrow \mathcal{P}_n(x).$$

An unlimited continuation is a collection $(\gamma_n)_{n \geq 1}$ satisfying $r_n(\gamma_{n+1}) = \gamma_n$. The inverse system thus preserves the restrictions each depth imposes on preceding depths. Existence of a compatible collection is established in the corresponding prolongation domain; merely specifying the sets $\mathcal{P}_n(x)$ does not replace that proof.

Bidirectionality therefore combines reconstruction of antecedents with compatibility of continuations. On the signature sequence, the involution C_{rev} reverses order and changes the sign of each component. The trajectory is also subject to the transformations of position, sheet, and boundary specified by transport. The reversed signature is an observation of the conjugate trajectory, not a replacement for its other coordinates. The threshold remains distinguished: the visible value 0 retains its transition function even when its quotient or historical record is nonzero.

The relation with exponential geometries is equally precise. In each fixed regime, $\exp(tJ_\tau)$ composes parameters and preserves the algebraic norm. Elliptic closure belongs to that local representation. Phase return of Γ_0 , by contrast, acts on the trajectory with memory, according to (4.6). A trajectory can recover its phase while retaining a new antecedent: historical data have changed although the phase observation is the same. The parabolic threshold retains a linear displacement; hyperbolic opening separates two eigendirections of expansion and contraction. These three realizations provide composition laws, whereas the TPK determines their selection and effective concatenation.

The aperiodic suspension in Section 6 will make this distinction explicit through a finite phase factor, an irrational rotation, and a memory degree. The link between TRIT and temporality is thus expressed by operations on trajectories and their quadratic representations. Its content exceeds a classification of signs: it preserves the regime, the direction of transport, and the relation between antecedent, transition, and continuation.

The monodromic representation of this return is developed in Section 6.14. There, the holonomy of the state is distinguished from its representation on phase and counter. The symbolic suspension in Section 6.16 defines the aperiodic time-crystal model, and Section 6.18 composes this temporality with the three Euler realizations.

4.3 Observable compression and orthogonal conservation

An isometric realization makes it possible to quantify what a reduced observation loses. Let $U : \mathcal{H} \rightarrow \mathcal{H}$ be an isometric realization of the transport of one winding, $U^*U = I$, and $J : \mathcal{H}_{\text{vis}} \rightarrow \mathcal{H}$ an isometric inclusion. Define

$$T = J^*UJ, \quad D = (I - JJ^*)UJ.$$

Then

$$UJ = JT + D, \quad J^*D = 0, \quad I - T^*T = D^*D. \quad (4.7)$$

The last equality follows by taking the adjoint product of the first and using orthogonality. Contraction of the visible component does not by itself constitute destruction of information in the total state.

Proposition 4.3 (Finite-horizon conservation). *If $I - T^*T = D^*D$, then, for every $n \geq 1$,*

$$I = \sum_{r=0}^{n-1} T^{*r}D^*DT^r + T^{*n}T^n. \quad (4.8)$$

Proof. Multiplying $D^*D = I - T^*T$ by T^{*r} and T^r and summing produces a telescoping sum. The interior terms cancel, leaving $I - T^{*n}T^n$. $\square$

The last summand preserves the terminal state. Removing it before justifying a limit would alter the equality. This distinction is the operational content of preservation of the unit under compression: the norm is distributed among the visible component, memory, and terminal, without being replaced by a single residue.

4.4 Local development: commutation, Lorentzian signature, and discrete curvature

The adjacent diagonal block with equal diagonal entries modulo nine is unique. Indeed,

$$k^2 \equiv (k+1)^2 \pmod{9} \iff 2k+1 \equiv 0 \pmod{9}$$

gives $k = 4$. Its visible matrix is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I.$$

With $P_{\pm} = (I \pm R)/2$, one obtains $B_c = 9P_+ + 5P_-$. The unit of stochastic normalization and preservation of an indefinite form are two realizations of this block:

$$\frac{B_c}{9} = P_+ + \frac{5}{9}P_-, \quad \frac{B_c}{\sqrt{45}} = \exp(\eta R), \quad \eta = \frac{1}{2} \log \frac{9}{5}.$$

For $Q = \text{diag}(1, -1)$, the second matrix satisfies $B_c^T Q B_c = 45Q$. This gives a local realization in $SO^+(1, 1)$; no prior introduction of a four-dimensional spacetime is needed to prove it.

Uniform averaging over the classes $C_1 = \{1, 4, 7\}$, $C_2 = \{2, 5, 8\}$, $C_0 = \{3, 6, 9\}$ gives

$$M_{\Pi} = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \quad M_{\Sigma} = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

The characteristic polynomial of M_{Π} is

$$(\lambda + 1)((\lambda - 9)^2 - 72),$$

so its quadratic form has signature $(2, 1)$. The local spectral gap $9 - 5 = 4$, the coefficient 2, the aggregate difference $51 - 45 = 6$, and the integrated defect $9 \cdot 6 = 54$ describe different levels of the same arithmetic comparison. They are not interchangeable indices of the same operator.

4.4.1 Invariant sublattice and quadratic unit of the aggregated operator

Aggregation by residue classes preserves an integral structure that specifies the preceding spectral decomposition. The initial operator is M_{Π} , obtained by averaging the multiplicative sheet of APP over the blocks $C_a \times C_b$. Its integral domain is $\mathbb{Z}^3$, with coordinates ordered by C_1, C_2, C_0 .

Proposition 4.4 (Decomposition on an index-two sublattice). *Let*

$$v_- = (1, -1, 0)^T, \quad v_+ = (1, 1, 0)^T, \quad v_0 = (0, 0, 1)^T,$$

and let $L = \mathbb{Z}v_- \oplus \mathbb{Z}v_+ \oplus \mathbb{Z}v_0$. Then

$$L = \{(x, y, z) \in \mathbb{Z}^3 : x \equiv y \pmod{2}\}, \quad [\mathbb{Z}^3 : L] = 2. \quad (4.9)$$

The sublattice L is invariant under M_{Π} . In the ordered basis (v_-, v_+, v_0) , the restriction has matrix

$$[M_{\Pi}|_L] = (-1) \oplus 3H, \quad H = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}, \quad \det H = 1. \quad (4.10)$$

Proof. The combination $av_- + bv_+ + cv_0 = (a + b, -a + b, c)$ has first and second coordinates of equal parity. Conversely, for a vector with this parity, its coordinates in the stated basis are $a = (x - y)/2$, $b = (x + y)/2$ and $c = z$, all integers. The change-of-basis matrix and its rational inverse are

$$P = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1/2 & -1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

For this column order, $\det P = 2$, giving the index in (4.9). The action on the three generators is

$$M_{\Pi}v_- = -v_-, \quad M_{\Pi}v_+ = 9v_+ + 12v_0, \quad M_{\Pi}v_0 = 6v_+ + 9v_0.$$

These equalities prove both invariance and the identity

$$P^{-1}M_{\Pi}P = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 9 & 6 \\ 0 & 12 & 9 \end{pmatrix} = (-1) \oplus 3H.$$

Finally, $\det H = 9 - 8 = 1$. □

Corollary 4.5 (Pell iteration on the rank-two submodule). *For $n \geq 0$, there are integers a_n, b_n such that*

$$H^n = \begin{pmatrix} a_n & b_n \\ 2b_n & a_n \end{pmatrix}, \quad a_n + b_n\sqrt{2} = (3 + 2\sqrt{2})^n, \quad a_n^2 - 2b_n^2 = 1. \quad (4.11)$$

In particular, $(a_0, b_0) = (1, 0)$, $(a_1, b_1) = (3, 2)$ and $(a_2, b_2) = (17, 12)$. The eigenvalues of H are the conjugate quadratic units $3 \pm 2\sqrt{2}$.

Proof. Multiplication of matrices of the stated form corresponds to

$$(a, b)(c, d) = (ac + 2bd, ad + bc),$$

the multiplication law for $a + b\sqrt{2}$. Hence the powers of H satisfy $a_{n+1} = 3a_n + 4b_n$, $b_{n+1} = 2a_n + 3b_n$, with the preceding initial data. The multiplicative norm of $3 + 2\sqrt{2}$ is one; thus $a_n^2 - 2b_n^2 = 1$ at every degree. The characteristic polynomial of H , $\lambda^2 - 6\lambda + 1$, gives the two conjugate units. The identity $H^T \text{diag}(2, -1)H = \text{diag}(2, -1)$ expresses the same conservation as an integral bilinear form. □

The integral splitting therefore resides in L ; the same matrix P gives a complete decomposition over $\mathbb{R}^3$. The factor 3 in the restriction preserves the scale of the APP average: its rank-two iteration is $3^n H^n$. The central block $B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$, in contrast, resides in two adjacent positions of the table and satisfies $B_c^\top \text{diag}(1, -1) B_c = 45 \text{diag}(1, -1)$. Compression into three classes and central restriction to two positions thus retain their respective domains, bases and normalizations. The Pell structure proceeds from the first operation through (4.10).

Discrete curvature can be described directly before this spectral realization. In residues, let $S(x, y) = x + y$, $P(x, y) = xy$, and Δ_x, Δ_y be the forward differences. The link variables are potential differences, $A_x^T = \Delta_x T$, $A_y^T = \Delta_y T$, and we set

$$F(T_x, T_y) = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Since $\Delta_x S = \Delta_y S = 1$, $\Delta_x P = y$, and $\Delta_y P = x$,

$$F(S, S) = F(P, P) = 0, \quad F(S, P) = +1, \quad F(P, S) = -1 \pmod{9}. \quad (4.12)$$

Sum and product, taken separately, induce flat connections; their ordered mixing changes the sign of plaquette curvature. Summing over a rectangle with sides a, b , the interior edges cancel and the oriented boundary sum equals $\pm ab$ modulo nine. An exact relation thus appears between composition order and boundary orientation.

4.5 Cubic completion and two notions of boundary

4.5.1 Threshold difference and composition of shells

The enlargement of a discrete cell preserves every stratum appearing when its size increases by one unit. The threshold difference expresses this operation on the cardinality reader and tracks its behaviour under addition and multiplication before a metric realization is introduced.

Definition 4.6 (Threshold difference). Let R be a commutative ring with identity and let $f : \mathbb{Z}_{\geq 0} \rightarrow R$. Define

$$(D_{\#}f)(n) = f(n+1) - f(n). \quad (4.13)$$

For an integer reader of nested finite cells, this difference records the cardinality of the shell added in the step $n \rightarrow n+1$.

Proposition 4.7 (Additivity and exact product law). For $f, g : \mathbb{Z}_{\geq 0} \rightarrow R$,

$$D_{\#}(f+g)(n) = D_{\#}f(n) + D_{\#}g(n), \quad (4.14)$$

$$D_{\#}(fg)(n) = f(n+1)D_{\#}g(n) + g(n)D_{\#}f(n) \quad (4.15)$$

$$= f(n)D_{\#}g(n) + g(n)D_{\#}f(n) + D_{\#}f(n)D_{\#}g(n). \quad (4.16)$$

Proof. The difference of a sum separates term by term. For the product, adding and subtracting $f(n+1)g(n)$ gives

$$\begin{aligned} f(n+1)g(n+1) - f(n)g(n) &= f(n+1)(g(n+1) - g(n)) \\ &\quad + g(n)(f(n+1) - f(n)). \end{aligned}$$

This proves (4.15). Substitution of $f(n+1) = f(n) + D_{\#}f(n)$ yields (4.16), including its complete bilinear term. $\square$

If $A_n \subseteq A_{n+1}$ and $B_n \subseteq B_{n+1}$ are finite sets, the shell of the product decomposes into three disjoint parts:

$$\begin{aligned} (A_{n+1} \times B_{n+1}) \setminus (A_n \times B_n) &= (A_n \times (B_{n+1} \setminus B_n)) \\ &\sqcup ((A_{n+1} \setminus A_n) \times B_n) \\ &\sqcup ((A_{n+1} \setminus A_n) \times (B_{n+1} \setminus B_n)). \end{aligned}$$

Taking cardinalities gives exactly (4.16). The bilinear term counts positions new in both factors; it resides in the intersection of the two enlargements, with incidence preserved.

Proposition 4.8 (Binomial law for the dimensional shell). *For $d \geq 1$, let $f_d(n) = n^d$. Then*

$$D_{\#}f_d(n) = (n+1)^d - n^d = \sum_{k=0}^{d-1} \binom{d}{k} n^k = \sum_{r=1}^d \binom{d}{r} n^{d-r}. \quad (4.17)$$

Proof. The binomial expansion of $(n+1)^d$ contains n^d and the remaining d terms in the first sum. Substituting $r = d - k$ gives the second sum. In the support $\{1, \dots, n+1\}^d$, each added position has a nonempty set of coordinates equal to $n+1$. If that set has r elements, there are $\binom{d}{r}$ choices of positions and n^{d-r} choices of previous coordinate values. These strata are disjoint and cover the shell, also proving the formula by finite incidence. $\square$

For $d = 3$, the three strata have cardinalities $3n^2$, $3n$ and 1 . At the nonadic–decimal step, $n = 9$, the operation gives $243 + 27 + 1 = 271$; excluding the final vertex leaves 270 positions. The following cubic completion makes explicit the union of this shell with the 729 previous positions. The threshold law thus preserves both the total and its distribution by the number of new coordinates. Subsequent lattice realizations transport their incidences through the maps specific to each construction.

Enlarging the cubic support of side nine to that of side ten admits an exact positional stratification:

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1. \quad (4.18)$$

The first term corresponds to three interior coordinates; the second to one new coordinate; the third to two; and the last to the vertex with three new coordinates. The enlargement shell contains 271 positions. By contrast, the internal boundary of the discrete cube of side nine contains $9^3 - 7^3 = 386$. The difference between these counts is a difference of domain, not a conflict of values.

Dividing (4.18) by 1000 gives a normalized partition of unity. Cube incidence distinguishes six faces, twelve edges, and eight vertices. The subsequent link with Steiner hexads and octads requires the corresponding combinatorial maps; the cardinalities of the cube do not replace them. This completion here provides the relation between the bases 729 and 1000, while the exceptional prolongation will be developed after numerical generation, the dodecaphase record, and the constructions of α .

4.5.2 Stratification of the completion and restoration of the apex

The completion admits a set-theoretic description making its operation explicit. Let E_9 be the set of nine residue classes, represented positively by $1, \dots, 9$, and let $*$ be an additional element. The symbol $*$ differs from the zero class, already represented by 9. Consequently, extending E_9 to $\hat{E} = E_9 \sqcup \{*\}$ increases the cardinality from nine to ten without changing the internal modular identification. The cubic completion is $\hat{C} = \hat{E}^3$, and the initial cube is $C = E_9^3$.

Definition 4.9 (Strata of combinatorial codimension). For $0 \leq r \leq 3$, define

$$S_r = \{(x_1, x_2, x_3) \in \hat{E}^3 : \#\{j : x_j = *\} = r\}.$$

The apex of the completion is $a_* = (*, *, *)$, and the punctured completion is $\hat{C}^\circ = \hat{C} \setminus \{a_*\}$.

Proposition 4.10 (Exact decomposition of the completion). *The strata are disjoint and satisfy*

$$\hat{C} = S_0 \sqcup S_1 \sqcup S_2 \sqcup S_3, \quad |S_r| = \binom{3}{r} 9^{3-r}, \quad (4.19)$$

$$|\hat{C}^\circ| = 729 + 243 + 27 = 999, \quad |\hat{C}| = 999 + 1 = 1000. \quad (4.20)$$

The punctured expansion contains 270 elements in addition to the initial cube; the complete expansion contains 271.

Proof. An element of S_r is obtained by choosing the r coordinates containing $*$, in $\binom{3}{r}$ ways, and assigning one of the nine classes of E_9 to each remaining coordinate. Classification by the number of additional coordinates is exhaustive and its classes are disjoint. The sole element of S_3 is a_* . Removing it subtracts one from the total cardinality; restoring it recovers the complete Cartesian product. $\square$

The expression *holographic extreme and mean ratio*, abbreviated EMRH, denotes this articulation of the initial body, expansion strata, and unit restoration. Its immediate content is the exact partition of a finite product and its normalization. The number 999 represents a definite object, the punctured completion; 1000 represents its closure through restoration of the apex. The added unit therefore has an explicit incidence location.

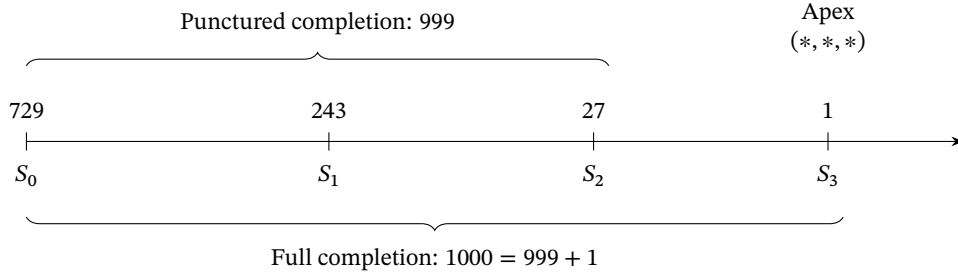


Figure 3: Strata of the cubic completion. The arrangement represents the set-theoretic decomposition, not a metric scale. The 243 elements of S_1 form three coordinate strata of 81 elements; the 27 elements of S_2 form three strata of 9. The apex completes the partition.

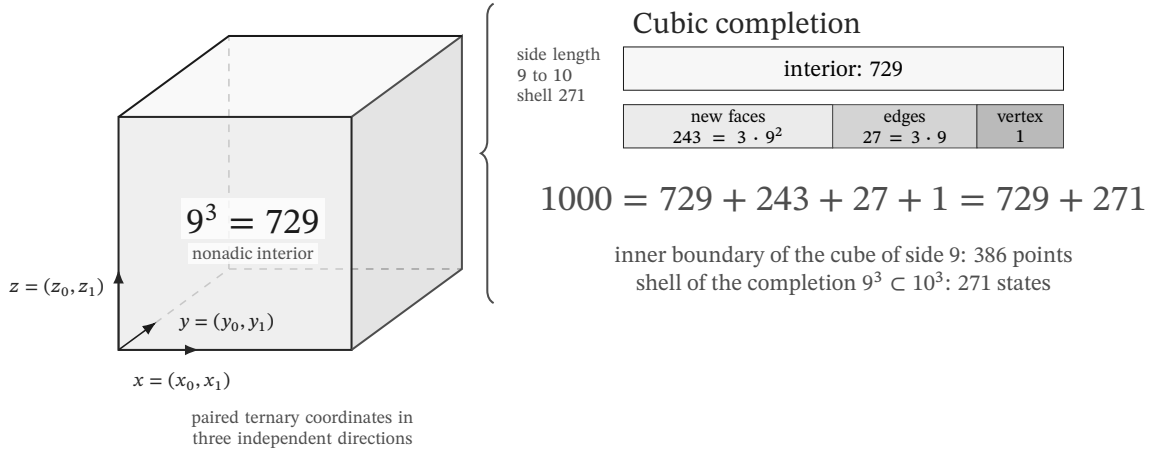


Figure 4: Cubic representation of the expansion from nine to ten classes per coordinate. Strata S_1, S_2, S_3 locate the added incidences on faces, edges, and the apex; their cardinalities form the 271-element shell. The 386-element boundary belongs to the initial cube. This geometric perspective complements the partition in Figure 3; graphical spacing is schematic.

4.5.3 Conservation of unity and dimensional compatibility

Normalization is not limited to dividing an isolated identity. In dimension d , the product $\hat{E}^d$ carries the uniform measure $\mu_d(\{x\}) = 10^{-d}$. The coordinate projection $p_d : \hat{E}^{d+1} \rightarrow \hat{E}^d$ deletes the last coordinate.

Proposition 4.11 (Projective compatibility of normalized measures). *For every $d \geq 1$,*

$$(p_d)_* \mu_{d+1} = \mu_d, \quad \mu_d(\hat{E}^d) = 1.$$

The mass of the stratum with r additional coordinates is $\binom{d}{r} 9^{d-r}/10^d$, and the masses of all strata sum to one.

Proof. Each point of $\widehat{E}^d$ has ten preimages. Their total mass is $10 \cdot 10^{-(d+1)} = 10^{-d}$. Equality for any subset follows by summing over its points. The stratified formula follows from the same count as (4.19); the binomial $(9 + 1)^d$ gives total normalization. $\square$

In dimension three, removing the apex leaves mass 999/1000, and restoring it adds 1/1000. Renormalizing before restoration would define a different measure: the distinction is operationally relevant. In particular, conservation of probability mass, conservation of transport records, and thermodynamic dissipation concern different objects. The proposition proves the first and provides the compatible support for studying the second; the third requires a specified dynamic and energetic realization.

The internal geometric boundary also remains distinct. On the Cartesian representative $\{1, \dots, 9\}^3$, removing $\{2, \dots, 8\}^3$ leaves 386 points. This boundary belongs to the initial cube, whereas $\widehat{C} \setminus C$ belongs to its expansion. Cubic completion, base-thousand positional expansion, and exceptional-incidence reading share support data; their respective maps determine which information from these data each representation uses.

4.6 Cubic incidence and Lorentzian realization

Cubic completion supplies a relation between interior and boundary. Its geometry can be developed through incidence of the six faces with the eight vertices. Label faces by (i, ε) , $i \in \{1, 2, 3\}$, $\varepsilon \in \{+1, -1\}$, and vertices by $\nu \in \{+1, -1\}^3$. The incidence matrix is

$$M_{(i,\varepsilon),\nu} = \mathbf{1}_{\{\nu_i=\varepsilon\}}.$$

Let $u = (1, \dots, 1)^T \in \mathbb{R}^6$, let R_F denote face opposition, and let $P_u = uu^T/6$, $P_- = (I - R_F)/2$.

Proposition 4.12 (Four-dimensional incidence quotient). *The incidence Gram matrix satisfies*

$$MM^T = 12P_u + 4P_-.$$

Its eigenvalues 12, 4, 0 have multiplicities 1, 3, 2. Consequently,

$$Q_{\square} = \mathbb{R}^6 / \ker(MM^T) \simeq \mathbb{R}u \oplus E_-, \quad E_- = \text{im } P_-.$$

Proof. A face contains four vertices; two opposite faces have empty intersection, and two distinct non-opposite faces share two vertices. The matrix $12P_u + 4P_-$ has exactly these entries. The uniform subspace has dimension one; differences between opposite faces generate a three-dimensional space orthogonal to it. Its complement has dimension two and belongs to the kernel. $\square$

The decomposition $1 + 3$ comes from incidence. The temporal choice is declared through the bilinear form

$$B_{\square}([x], [y]) = \langle x, (P_- - P_u)y \rangle.$$

It is well defined on the quotient because both projectors annihilate the kernel. In the basis

$$t = [u]/\sqrt{6}, \quad d_i = [e_{i,+} - e_{i,-}]/\sqrt{2},$$

its matrix is $\text{diag}(-1, 1, 1, 1)$. The domain, decomposition, and sign convention producing the Minkowski realization are thus explicit. The positive Gram matrix determines the quotient; the temporal assignment determines the indefinite form on it.

The operator

$$W_{\square} = \sqrt{6}P_u + \sqrt{2}P_-, \quad W_{\square}e_{i,\varepsilon} = t + \varepsilon d_i$$

sends the six face labels to null directions, since $B_{\square}(t + \varepsilon d_i, t + \varepsilon d_i) = -1 + 1 = 0$. Direction reversal and face opposition thus acquire a concrete geometric realization.

4.6.1 Soldering, reversal, and subdivision

The cellular realization uses an invertible transport $U_m(a)$, preserving the Lorentzian form, between the endpoint fibers of an oriented dual edge a , a face label $\lambda_m(a)$, and an orientation $\sigma_m(a)$. With $\ell_m = \ell_0/9^m$, the soldering map is

$$\beta_m(a) = \ell_m \sigma_m(a) W_{\square, s(a)} e_{\lambda_m(a)}.$$

Compatible reversal satisfies

$$\beta_m(\bar{a}) = -U_m(a) \beta_m(a).$$

Identify the source fibers at two consecutive levels through the declared refinement isometry. If the nine subsegments $a_1, \dots, a_9$, at level $m+1$, preserve label and orientation when transported to that common fiber, then

$$\beta_m(a) = \sum_{j=1}^9 U_{m+1}(a_1 \cdots a_{j-1})^{-1} \beta_{m+1}(a_j). \quad (4.21)$$

Each transported term equals $\ell_m/9$ times the same null direction, proving the identity. The compatibility hypothesis determines which subdivisions realize this refinement. The equation brings together direction, scale, and transport, rather than limiting geometric correspondence to a dimension count.

With orientation and the Lorentzian form also fixed, Hodge duality on $\Lambda^2 Q_{\square}^*$ satisfies $\star^2 = -I$. In dimension four, the sign is $(-1)^{2(4-2)+1} = -1$. This yields a real complex structure on two-forms and, after complexification, the eigenspaces with eigenvalues $+i$ and $-i$. The electromagnetic realization will use this domain after constructing the electronic section and declaring its coupling.

4.7 Compatibility of modular and positional reductions

The block-decimal base enters through Euclidean division preserving residue and quotient simultaneously:

$$n = 1000q + r, \quad 0 \leq r < 1000.$$

Since $1000 \equiv 1 \pmod{9}$ and $1000 \equiv 1 \pmod{3}$,

$$n \equiv q + r \pmod{9}, \quad n \equiv q + r \pmod{3}.$$

Reduction of a block therefore requires the quotient's contribution to recover the integer's class. The numbers 0 and 1000 have the same remainder modulo 1000, but different classes modulo 9 and 3. This is an operational reason for preserving memory through a change of resolution.

Ternary refinement and determination of a decimal prefix can be coordinated through a single integer remainder. If $N/3^L$ is a cylinder endpoint and $D/1000^r$ a decimal endpoint, define

$$A = 1000^r N - D 3^L.$$

On adding six ternary symbols, $N' = 729N + \nu$, $L' = L + 6$, with $0 \leq \nu < 729$. If the decimal prefix remains fixed, one obtains

$$A' = 729A + \nu 1000^r.$$

If a decimal block δ is also incorporated, then $D' = 1000D + \delta$, $r' = r + 1$, and

$$A' = 729000A + \nu 1000^{r+1} - 729\delta 3^L. \quad (4.22)$$

Both identities follow by substitution of the definitions. Comparing endpoints is thus reduced to integer operations retaining each prolongation's contribution. This compatibility makes it possible to refine a region before evaluating its real coordinate.

4.8 Number as a compatible section and value as a reading

Distinguishing state from observation specifies what object is prolonged as resolution increases. At depth n , let X_n be the set of admissible states produced by APP-TRIT-TPK. Their data include position, sum and product sheets, residues and quotients, regime, orientation, carry, phase, route, boundary, memory, and survival. These are not independent coordinates: earlier arithmetic evaluations and transports impose their relations. A prolongation preserves these relations in each antecedent, even while adding new steps and memory data. The compatible section (4.23) preserves these antecedents; the operations in Sections 4.9 and 4.10 specify their composition and refinement.

Let $r_{m,n} : X_m \rightarrow X_n$, for $m \geq n$, be depth restrictions, with $r_{n,n} = \text{id}$ and $r_{\ell,n} = r_{m,n} \circ r_{\ell,m}$. Each restriction preserves the prefix and the information already determined in it.

Definition 4.13 (Compatible numerical section). An HMT number is a compatible history of the state system:

$$x = (x_n)_{n \geq 0} \in X_\infty := \varprojlim_n X_n, \quad r_{m,n}(x_m) = x_n. \quad (4.23)$$

A reader is an additional operation on a specified domain of these histories. A digit is a finite observation; a scalar value is an evaluation of the reader. The pair consisting of a section and its reader constitutes a read presentation, not a redefinition of the section's identity.

On the Archimedean domain X_∞^{Arch} , the reader associates a closed cylinder with each prefix, preserves nesting, and makes its diameter tend to zero. It thus determines

$$\text{ev}_\mathbb{R} : X_\infty^{\text{Arch}} \longrightarrow \mathbb{R}, \quad \{\text{ev}_\mathbb{R}(x)\} = \bigcap_n I_n(x). \quad (4.24)$$

The concrete construction of these cylinders and the determination of their boundaries are developed in Section 6. Definition (4.23) does not by itself assert that the image of (4.24) is the entire line: a coverage claim requires its own theorem. Nor does equality of values establish equality of histories without an injectivity result. This separation allows precision to be discussed as depth while preserving the distinction between the arithmetic antecedent and its scalar realization.

4.9 Composition of histories and memory multiplier

Fix a sector of states at depth n and a category $\mathcal{C}$ whose arrows are its admissible histories. Composition is written $vu = v \circ u$: first traverse u , then v . Elementary steps come from TPK selection, transport, and updating. A local tritic fiber admits the notation $\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x)$, where $\tau(x) \in \{+1, 0, -1\}$. Regime and orientation belong to the transported state; a transition between regimes does not amount to identifying their quadratic algebras.

The following formulation specifies an algebraic sector of this composition. For each object x , a unital associative real algebra A_x , possibly noncommutative, is given. For each $u : x \rightarrow y$, a unital homomorphism $\rho_u : A_x \rightarrow A_y$ is given. Assume a strict action:

$$\rho_{\text{id}_x} = \text{id}_{A_x}, \quad \rho_v \rho_u = \rho_{vu}. \quad (4.25)$$

For each composable pair $u : x \rightarrow y, v : y \rightarrow z$, fix an invertible central multiplier $\Omega(v, u) \in Z(A_z)^\times$. Require the normalization $\Omega(\text{id}_y, u) = \Omega(u, \text{id}_x) = 1_y$ and, for every composable triple, the identity

$$\rho_w(\Omega(v, u))\Omega(w, vu) = \Omega(w, v)\Omega(wv, u). \quad (4.26)$$

These are explicit hypotheses on transports and their coefficients, not a consequence of calling the dynamics holonomic. Applying this presentation to a TPK sector requires identifying its ρ_u and $\Omega(v, u)$. Transitions expressed through bimodules require preserving that structure and are not replaced by (4.25).

The space of finite sums

$$\mathcal{H}(\mathcal{C}, A, \rho, \Omega) = \bigoplus_{u \in \text{Mor}(\mathcal{C})} A_{I(u)} T_u$$

receives the bilinear product

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v, u)T_{vu}, \quad (4.27)$$

when the arrows are composable, and zero otherwise. Here $t(u)$ denotes the terminal state. Addition gathers histories with coefficients; multiplication concatenates histories while preserving the record multiplier.

Proposition 4.14 (Sectorial associativity with a central multiplier). *Under (4.25) and (4.26), product (4.27) is associative. If the object set is finite, its identity is $\sum_x 1_x T_{\text{id}_x}$.*

Proof. For $u : x \rightarrow y, v : y \rightarrow z, w : z \rightarrow t$, let $a \in A_y, b \in A_z, c \in A_t$. The two coefficients of T_{wvu} are

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: c\rho_w(b)\Omega(w, v)\rho_{wv}(a)\Omega(wv, u), \\ (cT_w)((bT_v)(aT_u)) &: c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v, u))\Omega(w, vu). \end{aligned}$$

The strict action identifies $\rho_w\rho_v(a)$ with $\rho_{wv}(a)$. Centrality of $\Omega(w, v)$ allows it to move to the right of that factor; (4.26) then equates the remaining products. The coefficients a, b, c are not permuted. If the triple is not composable, both parenthesizations are zero because of their initial and terminal states. Bilinearity completes the proof. Local identities and normalization of Ω verify the stated identity element. $\square$

Calendar carry provides an effective example. In the category with one object and arrows C_9 , take coefficients $A = \mathbb{R}[z, z^{-1}]$, the identity action, and

$$\Omega(b, a) = z^{c_0(a, b)}, \quad c_0(a, b) = \left\lfloor \frac{a + b}{9} \right\rfloor, \quad 0 \leq a, b < 9.$$

The two carry sums of a triple are $\lfloor (a + b + c)/9 \rfloor$; hence (4.26) holds. The formal variable z records the turn without assigning it a numerical value. Subsequently imposing $z^{12} = 1$ recovers the dodecaphase record of calendar (4.5); retaining z without that relation distinguishes successive turns. The example realizes this calendar's memory without identifying it with all TPK memory.

4.10 Double variance of refinement and conservation of unity

Restricting states prolongs observables in the opposite direction. Consider the function algebras, with pointwise product, $D_n = \text{Fun}(X_n, \mathbb{R})$. Precomposition defines

$$r_{m,n}^* : D_n \longrightarrow D_m, \quad r_{m,n}^* f = f \circ r_{m,n}. \quad (4.28)$$

These are unital morphisms; they are also injective when the corresponding restrictions are surjective. The direct limit $D_{\text{cyl}} = \varinjlim_n (D_n, r_{n+1,n}^*)$ gathers finite-depth observations. This algebra is not identified here with the whole route algebra: prolonging transport operators also requires their compatibility with refinement.

Lemma 4.15 (Identity and naturality of evaluation). *If e_x is the indicator of $x \in X_n$, then*

$$r_{m,n}^* e_x = \sum_{y: r_{m,n}(y)=x} e_y, \quad r_{m,n}^* 1_n = 1_m, \quad (4.29)$$

and, for every $f \in D_n, y \in X_m$,

$$\langle r_{m,n}^* f, y \rangle = \langle f, r_{m,n} y \rangle.$$

Each section $x \in X_\infty$ determines a well-defined evaluation $D_{\text{cyl}} \rightarrow \mathbb{R}$, given by $[f] \mapsto f(x_n)$ for $f \in D_n$.

Proof. The fibers of $r_{m,n}$ partition X_m . Their indicators are orthogonal and sum to 1_m , proving (4.29). Naturality is the definition of precomposition. Finally, $(r_{m,n}^* f)(x_m) = f(x_n)$, so two compatible representatives give the same evaluation in the direct limit. $\square$

The global identity is the constant function one on all possibilities; an individual section selects a compatible history. Preserving the former does not mean immobilizing the latter. Refinement may increase the number of extensions and the depth of each history while preserving the identity and all evaluations of its antecedents.

4.11 Phase return, Weyl representation, and two readings

Law (4.6) acquires an operator realization in the bilateral suspension constructed in Section 6. There, T is one-step advance, V_9 observes the nonadic phase, W_δ the rotation phase, and D_M the memory degree. Writing $\hat{\Gamma}_9 = T^9$ for the turn in this representation, the Weyl relations (6.28) imply, on finitely supported vectors,

$$\begin{aligned} V_9 \hat{\Gamma}_9 &= \hat{\Gamma}_9 V_9, & W_\delta \hat{\Gamma}_9 &= e^{2\pi i 9\delta} \hat{\Gamma}_9 W_\delta, \\ [D_M, \hat{\Gamma}_9] &= \hat{\Gamma}_9. \end{aligned} \tag{4.30}$$

The first equality follows from $\zeta_9^9 = 1$; the second from nine applications of the covariance of W_δ ; the third expresses the unit memory increment. The finite phase returns while irrational rotation and memory distinguish the next state. Characters and parameter δ are realized subsequently from the numerical construction specified there: they do not select the seeds retrospectively. This representation describes suspension observables and translations; it does not replace all TPK coordinates or operators.

The double reading extends the same distinction between history and observable. On its Archimedean domain, (4.24) applies; on the calibrated incidence domain, the maps Q_n of Section 11.1 preserve words and boundary frames. The equality proved there, $r_{n,m} Q_n = Q_m p_{n,m}$, determines their prolongation Q_∞ . On the common domain of both readings, $\text{ev}_\mathbb{R}(x)$ and $Q_\infty(x)$ can be considered jointly: the former retains the value; the latter, the transported incidence preserved by its codomain. The section precedes both. This articulation makes it possible to pass from the arithmetic of histories to constants and exceptional reading without turning their realizations into new initial data.

5 Correlative construction of the continuum: histories, transport, incidence and terminal

The arithmetic expressed by a normal form preserves a precise readout of the state producing it. To situate that readout within HMT, the histories, sheets and operations acting on them must also be maintained. This chapter brings together that common substrate. Its different constructions are obtained from the same APP-TRIT-TPK execution tree; expository divisions allow their properties to be studied without turning them into independent domains chosen by a subsequent application.

The starting point is the formal kernel presented earlier. An admissible emission contains the APP operation, TRIT orientation, sheet change, residue and quotient, together with the effective TPK memory update. When only a residue is expressed, the execution history still belongs to the starting object. Two executions with identical final readouts are not identified on that ground alone.

5.1 Admissible histories and memory by composition

Let X_k be the set of admissible histories of length k . Its elements are constructed by successive emissions from the kernel's initial domain. Write $h\varepsilon$ for prolongation of h by an admissible emission ε , and $\rho_k(h\varepsilon) = h$ for deletion of the last emission. The emission label is preserved, even when different labels produce the same expressed state.

On the two arithmetic sheets, local data satisfy the exact divisions

$$\delta_\diamond = r_\diamond + 9q_\diamond, \quad 0 \leq r_\diamond < 9, \quad \diamond \in \{\Sigma, \Pi\}.$$

The oriented TRIT coordinate likewise admits an expression $\Delta = o_r + 3\kappa$, with the oriented representative declared by the kernel. Preservation of the quotient allows these representations to be composed without replacing an emission by its isolated residue.

In a memory chart, an emission's update is written

$$U_\varepsilon(m) = C_\varepsilon m + \kappa_\varepsilon.$$

For a trajectory $\alpha = (\varepsilon_1, \dots, \varepsilon_k)$, the effective operation is the ordered composition of these updates. Its linear part and translation are

$$C_\alpha = C_{\varepsilon_k} \cdots C_{\varepsilon_1}, \quad \kappa_\alpha = \sum_{j=1}^k C_{\varepsilon_k} \cdots C_{\varepsilon_{j+1}} \kappa_{\varepsilon_j}.$$

The empty product in the last sum is the identity.

Proposition 5.1 (Affine compatibility of concatenation). *If α precedes β , then*

$$C_{\beta\alpha} = C_\beta C_\alpha, \quad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta, \quad U_{\beta\alpha} = U_\beta U_\alpha.$$

Proof. Substituting $U_\alpha(m)$ into U_β produces $C_\beta C_\alpha m + C_\beta \kappa_\alpha + \kappa_\beta$. Decomposition into linear part and translation gives the three identities. Associativity of composition proves that the result does not depend on an auxiliary subdivision of the trajectory. $\square$

This cocycle identity is the precise content of accumulated memory. The quantity transported by a second trajectory depends on the chart update it performs; it is not, in general, the untransported sum of two registers.

Nonadic return preserves a second distinction. In phase and winding coordinates, with $\bar{r} \in \{0, \dots, 8\}$, its representation is

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + 1 \pmod{9} \right), \quad \sigma_9^9(w, r) = (w + 1, r).$$

During nine steps, exactly one crossing from 8 to 0 occurs. Phase returns and the winding coordinate increases. Enriched holonomy also retains the memory updates accompanying this route.

5.2 Tree refinement and measure of histories

Fix the finite nonempty initial set of admissible seeds, with strictly positive initial masses summing to 1. In this subsection each history has a finite positive number of prolongations. Deletion of emissions defines the refinement maps. Kernel admissibility supplies the prolongations; when a residence uses a neutral emission, that emission explicitly gives a section of ρ_k . Existence of that section concerns the tree including this emission, not every arbitrary restriction of routes.

Let $d(h) \geq 1$ be the number of admissible emissions at h . Equiprobable choice among its successors defines

$$\mu(h\varepsilon) = \frac{\mu(h)}{d(h)}.$$

More generally, a positive local distribution $p(\varepsilon | h)$, summing to 1, defines $\mu(h\varepsilon) = \mu(h)p(\varepsilon | h)$. This second formulation preserves the readout's weights when numbers of prolongations are not uniform.

Proposition 5.2 (Cylindrical compatibility and conditional expectation). *At each level one has*

$$\sum_{\rho_k(y)=x} \mu(y) = \mu(x).$$

On the spaces $L^2(X_k, \mu_k)$, the maps

$$(J_k f)(y) = f(\rho_k(y)), \quad (E_k g)(x) = \sum_{\rho_k(y)=x} \frac{\mu(y)}{\mu(x)} g(y)$$

satisfy $E_k = J_k^*$, $E_k J_k = I$ and $\|J_k f\| = \|f\|$.

Proof. The first identity is normalization of local probabilities. Applying it to $|f(x)|^2$ gives the isometry. Grouping by predecessors the sum defining $\langle J_k f, g \rangle$ yields the formula for E_k ; on a function constant on each fibre, that formula returns its value. $\square$

Compatible masses define a probability measure on the space of infinite histories: the cylinder of a finite history receives its mass $\mu(h)$. The measure extends from the cylinder algebra, and uniqueness follows because the cylinders generate the sigma-algebra. If the tree is finitely branching and every history has a prolongation, the inverse limit of its levels under emission deletion is nonempty. In the cylindrical topology, every nonempty cylinder has positive measure.

This measure distinguishes histories converging to the same representation. The uniform measure on terminal values would be a different readout, since it would remove the multiplicities just preserved by the construction.

5.3 Sheetwise transport and exact accounting of outputs

First consider the space whose orthonormal basis consists of histories at level k . Transport preserving the label of each prolongation is

$$\mathcal{U}_k e_h = \sum_{\varepsilon \text{ admissible at } h} \sqrt{p(\varepsilon | h)} e_{h\varepsilon}.$$

The successor sets of distinct predecessors are disjoint. Normalization of p therefore proves $\mathcal{U}_k^* \mathcal{U}_k = I$. The representation using history masses is related to this one by the diagonal basis change incorporating $\sqrt{\mu(h)}$.

The sheet label belongs to a history's complete register. Write

$$\hat{h} = ((h_0, s_0), \dots, (h_k, s_k)), \quad s_j \in \{\Sigma, \Pi\},$$

where s_j is the active sheet at step j . Prolongation adds the emission and sheet s_{k+1} determined by the TPK operational selector, preserving the entire prefix, including the initial sheet. In the basis of these lifted histories,

$$\mathcal{U}_k e_{\hat{h}} = \sum_{\varepsilon \text{ admissible at } \hat{h}} \sqrt{p(\varepsilon | \hat{h})} e_{\hat{h}\varepsilon}, \quad P_k e_{\hat{h}} = \mathbf{1}_{s_k=\Sigma} e_{\hat{h}}, \quad Q_k = I - P_k.$$

Thus, P_k reads the current sheet, while the distinction between histories is maintained by their complete prefixes. Two initial histories reaching the same active sheet still have distinct successors. The isometry proof using disjoint collections of prolongations applies verbatim to this basis. Tric orientation is another register coordinate: the value 0 preserves the threshold readout on each sheet and does not add a third sheet to this decomposition. We now separate sheet preservation from sheet interchange:

$$A_k = P_{k+1} \mathcal{U}_k P_k + Q_{k+1} \mathcal{U}_k Q_k, \quad \eta_k = Q_{k+1} \mathcal{U}_k P_k + P_{k+1} \mathcal{U}_k Q_k.$$

Proposition 5.3 (Sheetwise Gram identity). *For a linear transport U , whether or not it is isometric, the preceding decomposition satisfies*

$$A^* A + \eta^* \eta = P U^* U P + Q U^* U Q.$$

In particular, for $U = \mathcal{U}_k$,

$$A_k^* A_k + \eta_k^* \eta_k = I.$$

Proof. In $A^* A$ and $\eta^* \eta$, mixed terms containing $P_{k+1} Q_{k+1}$ vanish. Summing the remaining terms yields $P_k U^* (P_{k+1} + Q_{k+1}) U P_k$ and its analogue with Q_k . The first identity follows from complementarity of the projectors; the second additionally uses $U^* U = I$. $\square$

The first identity is the general formula. Replacing its right-hand side by $U^* U$ requires the cross-blocks of that Gram operator to vanish. Isometric preservation by history transport provides this condition here; it must not be attributed automatically to a subsequent compression.

Write $F_{0,0} = I$, $F_{0,n} = A_{n-1} \cdots A_0$ and $T_N = F_{0,N}^* F_{0,N}$.

Theorem 5.4 (Finite decomposition with terminal residual). *For each horizon N ,*

$$I = \sum_{n=0}^{N-1} F_{0,n}^* \eta_n^* \eta_n F_{0,n} + T_N.$$

The sequence T_N is positive and decreasing and has a positive strong limit T_∞ . The sum of outputs converges strongly to $I - T_\infty$.

Proof. Proposition 5.3 gives $T_n - T_{n+1} = F_{0,n}^* \eta_n^* \eta_n F_{0,n}$. Telescoping summation proves the finite equality and monotonicity. For each vector, the quadratic forms decrease to a limit; polarization defines a bounded positive form and, by the representation theorem for bounded forms, an operator T_∞ . Convergence is strong because, for a positive operator $S \leq I$, $\|Sv\|^2 \leq \langle v, Sv \rangle$; apply this to $S = T_N - T_\infty$. $\square$

The terminal residual is preserved until its extinction is proved in the domain considered. This distinction prevents phase return alone from becoming a hypothesis that all histories are exhausted.

5.4 Oriented balance and cancellation under refinement

A history carries its accumulated orientation $o(h)$ and the counts of its visits to the two sheets. The corresponding local balance is

$$b(h) = o(h)(N_\Sigma(h) - N_\Pi(h)), \quad b^+ = \frac{|b| + b}{2}, \quad b^- = \frac{|b| - b}{2}.$$

When a readout expresses the average over prolongations, define $b_c = Eb_f$. Orientation remains in the argument of b_f ; it is not removed before averaging.

Proposition 5.5 (Exact cancellation defect). *The quantity*

$$\kappa_{\text{can}} = \frac{E|b_f| - |Eb_f|}{2}$$

is nonnegative and satisfies

$$E(b_f^+) = (b_c)^+ + \kappa_{\text{can}}, \quad E(b_f^-) = (b_c)^- + \kappa_{\text{can}}.$$

Proof. The weighted triangle inequality gives $|Eb_f| \leq E|b_f|$. Writing the positive and negative parts as $(|b| \pm b)/2$ yields both equalities. $\square$

The information omitted when only the balance is expressed is the common cancellation of opposite contributions. This quantity is determined from the same prolongations used by the preceding transport. The factor $1/9$ appears only when the fibre contains nine equally weighted successors; conditional expectation is the rule that remains valid for general branching and weights.

5.5 Ternary sheet incidence and rectangular complex

The two sheets determine the module $V = \mathbb{F}_3 e_\Sigma \oplus \mathbb{F}_3 e_\Pi$. Its four projective directions are the lines generated by $e_\Sigma, e_\Pi, e_\Sigma + e_\Pi, e_\Sigma - e_\Pi$. They yield six unordered pairs. The eight nonzero vectors are their oriented representatives. Thus the counts 4, 6, 8 belong to distinct operations on a single sheet module.

On the six pair positions, define

$$(Ba)_{\{L, L'\}} = a_L + a_{L'}, \quad s(q) = \sum_{\{L, L'\}} q_{\{L, L'\}}.$$

Each direction occurs in three pairs; hence $sB = 0$ in $\mathbb{F}_3$. The readout $W(q) = \mathbf{1}_{s(q)=0} - 1/3$ is invariant under $q \mapsto q + Ba$. On the uniform ambient space $\mathbb{F}_3^6$, its moments are

$$\mathbb{E}W = 0, \quad \mathbb{E}W^2 = \frac{2}{9}, \quad \mathbb{E}W^3 = \frac{2}{27}.$$

Indeed, s is surjective and each fibre contains 3^5 elements. The readout equals $2/3$ with probability $1/3$ and $-1/3$ with probability $2/3$. These averages describe the uniform ambient space; the distribution induced by a subset of histories must be computed with its own multiplicities, already preserved in X_k .

To fix the ports without losing multiplicities, at depth k retain the labelled readouts $\lambda_\Sigma(h) = (h, \Sigma)$ and $\lambda_\Pi(h) = (h, \Pi)$. Their sets are

$$I_k = \{\lambda_\Sigma(h) : h \in X_k\}, \quad J_k = \{\lambda_\Pi(h) : h \in X_k\}.$$

They are finite and nonempty because X_k is. Values, quotients and residues are expressed on these ports; equality of values does not identify their labels. The product $I_k \times J_k$ is the rectangular-readout domain, within which the support of admissible incidences is recorded.

With this level fixed, write $I = I_k, J = J_k$ and $A_N = \mathbb{Z}/3^N\mathbb{Z}$. Choose reference ports i_0, j_0 . This choice merely provides coordinates for reconstruction formulas. The maps

$$\kappa(u, v)_{ij} = u_i - v_j, \quad (\delta w)_{ij} = w_{ij} - w_{ij_0} - w_{i_0j} + w_{i_0j_0}$$

define the following sequence, valid over the finite ring A_N :

$$0 \longrightarrow A_N \longrightarrow A_N^I \oplus A_N^J \xrightarrow{\kappa} A_N^{I \times J} \xrightarrow{\delta} A_N^{(I \setminus \{i_0\}) \times (J \setminus \{j_0\})} \longrightarrow 0.$$

Proposition 5.6 (Exactness of rectangular incidence). *The preceding sequence is exact, with first arrow given by the constant diagonal.*

Proof. The equality $u_i - v_j = 0$ for every pair forces all coordinates of u and v to be the same constant. Direct substitution gives $\delta\kappa = 0$. If $\delta w = 0$, take $u_i = w_{ij_0}$ and $v_j = w_{i_0j_0} - w_{i_0j}$. Then $u_i - v_j = w_{ij}$. Finally, any matrix in the last term extends with zero reference row and column; its image under δ is the initial matrix. The proof uses only addition and subtraction, so it requires no division by a ring unit. $\square$

Reduction from A_{N+1} to A_N , with k, I_k, J_k, i_0, j_0 fixed, commutes with both maps. Reconstruction formulas are the same at each modular depth. This is the arithmetic compatibility proved in this subsection. Prolongation of histories $k \rightarrow k + 1$ preserves prefix labels; its maps between port sets are distinct data from the ring change $N \rightarrow N + 1$.

These maps are obtained from emission deletion:

$$\rho_k^\Sigma(h\varepsilon, \Sigma) = (h, \Sigma), \quad \rho_k^\Pi(h\varepsilon, \Pi) = (h, \Pi).$$

Fix a reference history with compatible prolongations, whose existence was proved for the tree without terminal leaves. Its two labels define compatible reference ports at every level. Coordinate transport is precomposition:

$$(L_k u)_{i'} = u_{\rho_k^\Sigma(i')}, \quad (L_k v)_{j'} = v_{\rho_k^\Pi(j')}, \quad (L_k w)_{i'j'} = w_{\rho_k^\Sigma(i'), \rho_k^\Pi(j')}.$$

For the sequence's last module, first extend the matrix by zero in the reference row and column, apply this same formula and restrict to the new complement of the reference ports. This rule defines L_k even when a new port has the marked port as its predecessor.

Proposition 5.7 (Naturality of the rectangular complex under prolongation). *The preceding maps satisfy $L_k \kappa_k = \kappa_{k+1} L_k$ and $L_k \delta_k = \delta_{k+1} L_k$. They also commute with coefficient reduction $A_{N+1} \rightarrow A_N$ and compose according to prefix deletion.*

Proof. The first identity follows from $u_{\rho_k^\Sigma(i')} - v_{\rho_k^\Pi(j')}$. In the second, each of the four terms defining δ is transported by the same precomposition. Compatibility of the marked ports identifies their rows and columns; when a predecessor is the marked port, the four terms cancel and agree with extension by zero. All formulas consist of additions and subtractions with integer coefficients, so they commute with modular reduction. Composition of precompositions is precomposition with the composite prefix, proving the last statement. $\square$

5.6 Memory complex and compatibility of incidences

The same accumulated memory coordinate $c_y \in \mathbb{Z}$, at a residence y , defines a free integer complex. Its reduction to A_N uses the same integer register before reduction; successive depths are therefore compatible. Its generators are f_y in degree 2, e_y, b_y, r_y in degree 1, and g_y in degree 0:

$$\partial f_y = 3c_y e_y + b_y, \quad \partial e_y = g_y, \quad \partial b_y = -3c_y g_y, \quad \partial r_y = 0.$$

The identity $\partial^2 = 0$ is verified on f_y through $3c_y g_y - 3c_y g_y = 0$, and is immediate on the other generators.

If a trajectory $\alpha : y \rightarrow y'$ updates memory, its transport in the complex sends the identically named generators to their new representatives, except that

$$\Psi_\alpha(b_y) = b_{y'} + 3(c_{y'} - c_y)e_{y'}.$$

Proposition 5.8 (Naturality of the memory complex). *One has $\partial\Psi_\alpha = \Psi_\alpha\partial$ and $\Psi_{\beta\alpha} = \Psi_\beta\Psi_\alpha$. The maps extend linearly over the chosen coefficient ring.*

Proof. On f_y , the image of its boundary is $3c_y e_{y'} + b_{y'} + 3(c_{y'} - c_y)e_{y'} = 3c_{y'} e_{y'} + b_{y'}$. On b_y , the boundary of its image is $-3c_{y'} g_{y'} + 3(c_{y'} - c_y)g_{y'} = -3c_y g_{y'}$. The other two cases are immediate. In a concatenation, memory differences sum telescopically, proving the second identity. $\square$

The relation between two representatives of the same residence can also be written as an explicit boundary. For an integer coordinate v , reduced in the same ring when appropriate, let

$$z^- = r_y, \quad z^+ = r_y + 3c_y v e_y + v b_y, \quad h_* = -v f_y.$$

Then $\partial h_* = z^- - z^+$. Under a transport also updating v , the correction $K_\alpha = (v_{y'} - v_y)f_{y'}$ satisfies $K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha$. The symbols K_α of this local homotopy do not denote the dodecaphasic seal K : they belong to another domain and their role is specified by the preceding identity. That role can also be checked in the equality

$$z_{y'}^+ - \Psi_\alpha z_y^+ = \partial K_\alpha.$$

Both sides equal $(v_{y'} - v_y)(3c_{y'} e_{y'} + b_{y'})$.

Proposition 5.9 (Homology and memory of the local complex). *Over $\mathbb{Z}$, or after reduction to A_N , one has*

$$H_0(\Gamma) = H_2(\Gamma) = 0, \quad H_1(\Gamma) \cong A_N[r_y]$$

in the modular realization, and $H_1(\Gamma) \cong \mathbb{Z}[r_y]$ in the integer realization.

Proof. Define the degree-1 homotopy by $h(g_y) = e_y$, $h(b_y) = f_y$, and zero on the other generators. Let p be projection onto the summand generated by r_y , and ι its inclusion. Checking on each generator gives

$$\partial h + h\partial = I - \iota p.$$

In particular, on b_y the terms $3c_y e_y$ and $-3c_y e_y$ cancel. The complex homotopy retracts onto the module generated by r_y , concentrated in degree 1, proving the assertion. $\square$

The transport Ψ_α is a determinant-1 shear. Thus this complex explicitly preserves how c_y changes at chain level, but its local homology does not distinguish values of that coordinate. Complete memory remains in the labels and transports of the source state; it is not recovered from H_1 alone. A memory-dependent numerical torsion requires the specified assembly and determinantal readout in addition to this local complex.

5.7 Assembly with terminal paths

For a finite horizon, residences and their prolongations form a path category. To each residence ν , assign the free module $W(\nu) = A_N[\text{Path}(\nu, \text{Term})]$, preserving each terminal continuation as a generator. An initial trajectory acts on W by precomposition. The complex $\Gamma(\nu)$ of the preceding subsection acts covariantly through Ψ .

Assembly uses both transports in the same bigraded module:

$$\mathcal{B}_{q,r} = \bigoplus_{\nu_0 \rightarrow \dots \rightarrow \nu_q} W(\nu_q) \otimes_{A_N} \Gamma_r(\nu_0).$$

The normalized path complex is used: degenerate chains containing identities are omitted. The finite horizon then bounds the number of strict prolongations and the horizontal degree. The horizontal boundary deletes an intermediate residence by composing adjacent arrows. At the first endpoint it transports Γ ; at the last it precomposes W . Alternating signs give the path boundary d_{bar} . The total differential is

$$D = d_{\text{bar}} + (-1)^q \partial_\Gamma.$$

The face identities cancel the terms of d_{bar}^2 in pairs. Naturality from Proposition 5.8 implies that d_{bar} commutes with ∂_Γ ; the factor $(-1)^q$ cancels the cross-terms of the square. Thus $D^2 = 0$. Compatibility is proved on path generators and is not introduced through a label asserting membership in a complex.

Terminal evaluation does not require choosing a particular history either. For each continuation h , the effective composition of its updates is applied. The result preserves the pair consisting of h and its readout, with its multiplicity. If $k < \ell < N$, continuations decompose as a disjoint union of prefixes up to ℓ and their continuations up to N . Proposition 5.1 proves that direct evaluation and evaluation of these two parts produce the same readout. This gives the terminal naturality square.

Path modules, incidences, memory transport and their differentials arise from the same prolongations. Their assembly does not consist of adding five coordinates to a state that would then be preserved by definition: their relations are constructed through sums, boundaries and effective compositions, and verified on their generators.

5.8 Determinants, limit and arithmetic readout

At each residence, the free local complex Γ admits its graded determinant line

$$\text{Det } \Gamma = \bigotimes_r \left(\bigwedge^{\text{rk } \Gamma_r} \Gamma_r \right)^{(-1)^r}.$$

The total assembly of the preceding subsection, in turn, has modules $\text{Tot}_d \mathcal{B} = \bigoplus_{q+r=d} \mathcal{B}_{q,r}$. At finite horizon and over the same ring, these are free of finite rank and only finitely many are nonzero. Its line is

$$\text{Det}(\text{Tot } \mathcal{B}) = \bigotimes_{q,r} \left(\bigwedge^{\text{rk } \mathcal{B}_{q,r}} \mathcal{B}_{q,r} \right)^{(-1)^{q+r}}.$$

This formula follows from direct-sum decomposition in each degree. It distinguishes the local complex's line from that of assembly with all paths; their bases are ordered through the preserved labels. Modular reduction commutes with these exterior powers because the modules and their bases arise from the same free integer modules at fixed horizon. The line is defined by the modules and their degrees, both over $\mathbb{Z}$ and after reduction to A_N ; a negative exponent denotes the dual module. A nonzero numerical torsion additionally requires specifying the complex over a field, its bases and the relevant homology identifications. To compute Smith factors, the integer matrix produced before reduction is used, not an arbitrary lift of a modular matrix. A square integer block with nonzero determinant is invertible over $\mathbb{Q}$; its reduction is invertible over A_N only when that determinant is a unit, that is, when it is not divisible by 3. Refinement of memories and incidences allows stability of the relevant invariant factors to be formulated and checked; these conditions do not follow from existence of the determinant line.

Passage to the limit preserves compatible inverse systems of residues, histories and their multiplicities. The squares proved in the preceding subsections remain commutative because their maps are given by the same formulas at every level. For an Archimedean coordinate, the additional step is to express nonempty cylindrical intervals, compatible by inclusion and with diameter tending to zero. Their nested

compact closures have a one-point intersection: existence follows from compactness and uniqueness from the diameter bound. This is the value of the limit readout. If the intervals are half-open, the point may belong only to their closures; the boundary representation rule then determines its positional expression. The value does not replace the history producing it. The other representation preserves the incidences of the same state. The two destinations share their origin without thereby receiving the same information.

In the following chapters, the coinductive regions, dodecaphasic register and exceptional incidence are expressed from that substrate through their respective maps. The Hilbert-space realization subsequently preserves phase labels and specifies its own unital inclusion; duality and the screens act on the codomains constructed there. The joint operations brought together here transmit histories, sheets, orientation and memory to these representations. Identifying a concrete functional with an external analytic form in turn requires writing its action on generators and verifying its inner products: the conservation identities of this chapter do not replace that composition.

5.9 Provenance, verification and scope

The five operations are recognized by their action on the same histories and by the relations proved in the preceding subsections. The affine memory transport U_α and linear history transport $\mathcal{U}_k$ have the respective domains of Sections 5.1 and 5.3. The sheet grading is $\sigma_k = R_k - Q_k$. For a history $h_j = (\varepsilon_1, \dots, \varepsilon_j)$, accumulation of the balance is made explicit by

$$M_0^\pm = 0, \quad M_k^\pm(h_k) = \sum_{j=1}^k b^\pm(h_j), \quad M_{k+1}^\pm(h_k \varepsilon) = M_k^\pm(h_k) + b^\pm(h_k \varepsilon).$$

Each summand is the signed readout of the corresponding prefix, defined in Section 5.4. Separating the last summand proves the update; the identities $M_k^+ - M_k^- = \sum_{j=1}^k b(h_j)$ and $M_k^+ + M_k^- = \sum_{j=1}^k |b(h_j)|$ follow from the definitions of the positive and negative parts. Accumulation and conditional cancellation are thus distinct operations on the same contributions.

Joint operation	Objects and effective action	Development and proof
Sheet transport and grading	$\mathcal{U}_k : \ell^2(\widehat{X}_k) \rightarrow \ell^2(\widehat{X}_{k+1})$, with $\widehat{X}_k$ consisting of lifted histories; $\sigma_k = R_k - Q_k$, A_k and η_k preserve or interchange the current sheet.	Section 5.3; Proposition 5.3 and Theorem 5.4.
Balance, cancellation and memory	$b : X_k \rightarrow \mathbb{Z}$, $b^\pm : X_k \rightarrow \mathbb{N}_0$; accumulation $M_k^\pm$ over prefixes and cancellation $\kappa_{\text{can}} = (E b_f - Eb_f)/2$ when publishing the conditional average.	Section 5.4; Proposition 5.5.
Ternary sheet incidence	$B : \mathbb{F}_3^4 \rightarrow \mathbb{F}_3^6$, $(Ba)_{\{L, L'\}} = a_L + a_{L'}$; the full ambient space of q is preserved, with $sB = 0$ and $W(q + Ba) = W(q)$.	First part of Section 5.5.
Ports and rectangular complex	$\kappa : A_N^{I_k} \oplus A_N^{J_k} \rightarrow A_N^{I_k \times J_k}$, $\kappa(u, v)_{ij} = u_i - v_j$; the difference δ and precomposition L_k preserve labels and their multiplicities.	Propositions 5.6 and 5.7.
Cellular memory, filling and determinant line	$\Psi_\alpha : \Gamma_y \rightarrow \Gamma_{y'}$, $K_\alpha \in \Gamma_{y', 2}$, $\partial h_* = z^- - z^+$; the lines $\text{Det } \Gamma$ and $\text{Det}(\text{Tot } \mathcal{B})$ are formed on the corresponding based complexes.	Sections 5.6–5.8; Propositions 5.8 and 5.9.

The table identifies five actions of one construction: each prefix preserves its sheets, orientation, residues, quotients, memory and terminal prolongations. Port precomposition takes coordinates from one level to its refinement; coefficient reduction takes A_{N+1} to A_N . Their directions and domains are distinct, and their commutative squares are made explicit in Proposition 5.7. The assembly in Section 5.7 simultaneously uses terminal continuations and transport of the memory complex; its composition formulas bring together the table's actions on the same history system. Reading a determinant line as a scalar preserves the conditions on bases, homology and nonsingularity in Section 5.8.

This development brings together the constructions of state, tree and common measure with the correlative intrinsic operations. The balance in 5.4, the sequence in 5.6, the memory transport in

5.8 and the assembly in 5.7 provide the conservation and composition identities used here. The five constructions of the continuum remain operations on a single tree. Each specialization retains its explicit conditions: isometric transport, uniform weights, terminal extinction and nonsingularity of a determinantal block.

The accompanying checker compares exact instances of the cocycles, weighted cancellation, rectangular sequence and memory complex. These checks accompany the text's general proofs and verify the operators defined here in their respective domains. The subsequent passage towards exceptional representations and the Hilbert-space realization preserves their actions on generators: Section 12 specifies the phase fibre, Weyl operators and inclusions determining their closures. Documentary preservation and finite checks retain their function of traceability and comparison for these proofs.

6 Coinductive generation of π, φ, e to arbitrary depth

The six-block emission is a finite stage of the construction, not a complete decimal expansion. Its function is to produce regions with multiplicity, orientation, and transport rules. Prolongation must preserve these data and determine a compatible collection of prefixes of arbitrary length. The word *generation* designates this operation; *genealogy* designates the reconstruction of its dependencies. Both notions enter the discussion, but perform distinct functions.

6.1 Orbits, regions, and characters of closure, propagation, and self-scaling

On a word $w = (w_1, \dots, w_6)$, we define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \quad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

We have $r^3 = s^2 = I$ and $srs = r^{-1}$; these permutations therefore realize the dihedral group D_3 . The action preserves the catalog and multiplicities and commutes with Ψ . The quotient of the 243 visible words has 43 orbits: thirty-eight of cardinality six and five of cardinality three.

The closure class is recognized by a fiber predicate: its six orientations have five emissions per word, total multiplicity 1008, and distribution $432 + 4 \cdot 144$. There is a single orbit with these properties in the catalog. To specify the other two selections, we write $f(w) = \#\Psi^{-1}(w)$, $m(w) = \sum_{U \in \Psi^{-1}(w)} \text{mult}(U)$, and $\nu(w) = (n_0, n_1, n_2)$. The additive diagonal class of an emission is $d(U) = [i + j]_9$ in indices from zero to eight; its additive signature preserves this class and the orientation *ES* or *NO*. Let $\mathcal{D}(\mathcal{O})$ be the set of these classes for all emissions over an orbit.

Among the orbits of size six, the joint predicate

$$f(w) = 1, \quad m(w) = 144 \quad (w \in \mathcal{O}), \quad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\}$$

leaves exactly six orbits. In this sector, the census $\nu = (2, 3, 1)$, equal to that of closure, selects propagation uniquely; the census $\nu = (2, 2, 2)$ selects self-scaling uniquely. The diagonal support cannot be omitted: without it there are four balanced singleton-fiber orbits of multiplicity 144, not one. Finally, the existence of an emission with $d = 6$ and orientation *ES* selects a representative of each orbit and leads to

$$w^{\text{cl}} = 010211, \quad w^{\text{pr}} = 201101, \quad w^{\text{au}} = 121200. \quad (6.1)$$

These symbols first designate regions of the catalog. Scalar identification will be proved through their readers; they are not chosen because they coincide with digits of a known constant.

The closure word 010211 has the following five preimages:

Six-block emission	Multiplicative signature	Role	Multiplicity
501, 614, 498, 169, 272, 272	555555	transversal	432
810, 923, 870, 169, 272, 272	339555	positive orientation	144
810, 983, 810, 169, 272, 272	393555	positive orientation	144
870, 923, 810, 169, 272, 272	933555	positive orientation	144
870, 923, 810, 418, 674, 521	933717	oriented return	144

The multiplicity 432 and constant signature 555555 characterize the transversal region. They do not identify the central position (5, 5) of APP. Confusing a six-window signature with a position of the support would eliminate the distinction between numerical region and electronic structure.

The microfiber also has a precise bipartite morphology. Each emission is a product of an additive-cursor class and a multiplicative class. The three positive regions share the same additive class of eighteen conditions; together they contain three multiplicative classes of eight conditions each. The transversal region has a product 18×24 ; the three positive regions together, another 18×24 ; and the return, a product 18×8 . The decomposition by emissions has five parts, whereas the decomposition by connected components of the bipartite graph has three. Both preserve the same 1008 seeds without reducing form to a cardinality.

6.2 Incidences between halves of regional emissions

Let $\mathcal{U}_6$ be the image of six consecutive windows of the emitter of Section 3.4. For $u = (u_1, \dots, u_6)$, define $A(u) = (u_1, u_2, u_3)$ and $B(u) = (u_4, u_5, u_6)$. The half-emission graph has these triples as vertices and an undirected edge between $A(u)$ and $B(u)$ for each emission.

Proposition 6.1 (Cycle of regional junctions). *The vertices*

$$\begin{aligned} A &= (830, 943, 830), & B &= (109, 252, 252), \\ C &= (810, 923, 870), & D &= (169, 272, 272), \\ E &= (850, 963, 890), & F &= (149, 252, 212) \end{aligned}$$

form the cycle $A - B - C - D - E - F - A$.

Proof. Write each seed as (i, j, d) , with residual positions between 0 and 8 and directions $N = (-1, 0)$, $E = (0, 1)$ and $S = (1, 0)$. Under the 54-step recurrence, the following seed pairs produce the six emissions shown:

emission	s_+	$s_\times$
$A \mid B$	$(0, 6, E)$	$(0, 6, N)$
$B \mid C$	$(0, 6, N)$	$(1, 5, E)$
$C \mid D$	$(0, 6, E)$	$(0, 1, S)$
$D \mid E$	$(0, 6, N)$	$(0, 4, S)$
$E \mid F$	$(0, 6, E)$	$(0, 3, E)$
$F \mid A$	$(0, 6, N)$	$(0, 1, S)$

Each row witnesses membership in $\mathcal{U}_6$ by successive evaluation of the two accumulators. Splitting each emission into its two halves yields exactly the six edges of the cycle. $\square$

In the classification (6.1), $A \mid B$, $C \mid D$ and $E \mid F$ respectively produce the words w^{au} , w^{cl} and w^{pr} . The regional readers subsequently identify these outputs with φ , one region of π and e . The other three emissions exhibit their junctions in the graph. The result describes incidences between emitter outputs. Temporal realization of an incidence additionally uses the orientation, phase and memory of the enriched state. Including the APP anchor and all five regions of π imposes further path conditions; the cycle above retains its role as a local junction witness.

The six-edge cycle belongs to this undirected half-emission graph. An enriched temporal path additionally retains the anchoring, orientation and closure data of its construction. Incidence between two halves and transport of a state thus have explicit domains and composition conditions.

6.2.1 Block multigraph and five minimal regional closures

The 468 emissions in $\mathcal{U}_6$, produced by the APP-TRIT-TPK emitter, determine a block multigraph. Identifying its regional anchors makes it possible to solve a joint edge-covering problem: to traverse the APP, propagation, autoscale, and each closure region's incidences within a single closed walk. The local six-junction cycle of Section 6.2 and these covers retain their respective traversal conditions.

Block graph and exact phase quotient. For $U = (U_1, \dots, U_6) \in \mathcal{U}_6$, define

$$A(U) = (U_1, U_2, U_3), \quad B(U) = (U_4, U_5, U_6). \quad (6.2)$$

The undirected multigraph $\tilde{G}$ has as vertices the triples occurring as $A(U)$ or $B(U)$, and one edge labelled by each U , incident at those two triples. All 468 labels are retained: distinct emissions are not identified merely because they share endpoints or because one interchanges the halves of another.

Internal phase acts on each triple by cyclic rotation:

$$(a, b, c) \sim (b, c, a) \sim (c, a, b), \quad \text{can}(a, b, c) = \min_{\text{lex}}\{(a, b, c), (b, c, a), (c, a, b)\}. \quad (6.3)$$

The graph G replaces the endpoints of edge U by $\text{can } A(U)$ and $\text{can } B(U)$, retaining its label. The equivalence acts precisely by the three cyclic rotations of each vertex; every emission retains its identity as an edge.

Enumeration from the same emitter and component searches give

	$ V $	component sizes
$\bar{G}$	96	(28, 28, 28, 4, 4, 4)
G	32	(28, 4)

(6.4)

These data are obtained by collecting distinct endpoints, adding each incidence in both directions, and traversing each unvisited component by breadth-first search. Adjacency may be deduplicated when counting components, but parallel labels are retained when searching for closures with prescribed edges.

APP anchor and regional numbering. The three common edges are

$$\begin{aligned}
 U_{\text{APP}} &= 903|850|850|252|109|252, \\
 U_e &= 850|963|890|149|252|212, \\
 U_\varphi &= 830|943|830|109|252|252.
 \end{aligned}
 \tag{6.5}$$

Fix the endpoint $v_{\text{APP}} = \text{can } A(U_{\text{APP}}) = 850|850|903$. For the five already constructed regions, use the following order, which places the transverse region in the fifth row:

i	$U_\pi^{(i)}$	ρ_i	m_i
1	810 923 870 169 272 272	++	144
2	810 983 810 169 272 272	++	144
3	870 923 810 169 272 272	++	144
4	870 923 810 418 674 521	--	144
5	501 614 498 169 272 272	$\perp$	432

(6.6)

This numbering distinguishes the three ++ regions, the mutated return --, and the transverse region $\perp$. Accordingly, $\rho = (++, ++, ++, --, \perp)$ and $m = (144, 144, 144, 144, 432)$ in this order; their sum remains 1008.

Five complete witnesses with traversal directions. In each table, row j means traversing the edge labelled e_j from v_{j-1} to v_j . Endpoints are canonical phase representatives; since G is undirected, the pair (v_{j-1}, v_j) specifies the traversal direction of the labelled edge. Each table contains exactly eight edges and satisfies $v_0 = v_8 = v_{\text{APP}}$.

The breadth-first search described in the proof yields the following witnesses, one for each set of four target edges. Minimality concerns length; several walks may attain that length.

Witness $i = 1$.

j	v_{j-1}	e_j	v_j
1	850 850 903	109 252 252 850 903 850	109 252 252
2	109 252 252	109 252 252 810 923 870	810 923 870
3	810 923 870	810 923 870 169 272 272	169 272 272
4	169 272 272	169 272 272 850 963 890	850 963 890
5	850 963 890	850 963 890 149 252 212	149 252 212
6	149 252 212	149 252 212 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Witness $i = 2$.

j	v_{j-1}	e_j	v_j
1	850 850 903	169 272 272 850 903 850	169 272 272
2	169 272 272	169 272 272 810 983 810	810 810 983
3	810 810 983	810 983 810 169 272 272	169 272 272
4	169 272 272	169 272 272 850 963 890	850 963 890
5	850 963 890	850 963 890 149 252 212	149 252 212
6	149 252 212	149 252 212 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Witness $i = 3$.

j	v_{j-1}	e_j	v_j
1	850 850 903	169 272 272 850 903 850	169 272 272
2	169 272 272	169 272 272 850 963 890	850 963 890
3	850 963 890	850 963 890 149 252 212	149 252 212
4	149 252 212	149 252 212 870 923 810	810 870 923
5	810 870 923	870 923 810 169 272 272	169 272 272
6	169 272 272	169 272 272 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Witness $i = 4$.

j	v_{j-1}	e_j	v_j
1	850 850 903	169 272 272 850 903 850	169 272 272
2	169 272 272	169 272 272 850 963 890	850 963 890
3	850 963 890	850 963 890 149 252 212	149 252 212
4	149 252 212	149 252 212 870 923 810	810 870 923
5	810 870 923	870 923 810 418 674 521	418 674 521
6	418 674 521	418 674 521 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Witness $i = 5$.

j	v_{j-1}	e_j	v_j
1	850 850 903	109 252 252 850 903 850	109 252 252
2	109 252 252	109 252 252 501 614 498	498 501 614
3	498 501 614	501 614 498 169 272 272	169 272 272
4	169 272 272	169 272 272 850 963 890	850 963 890
5	850 963 890	850 963 890 149 252 212	149 252 212
6	149 252 212	149 252 212 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Theorem 6.2 (Minimality of the five regional edge covers). *For each $i \in \{1, \dots, 5\}$, among closed walks in G traversing every edge of*

$$E_i^* = \{U_{\text{APP}}, U_e, U_\varphi, U_\pi^{(i)}\}$$

at least once, the minimum length, counting traversals with multiplicity, is eight. Vertices and edges may be repeated.

Proof. Every closed walk traversing U_{APP} passes through v_{APP} . Cyclic rotation of its step list allows it to start at that vertex without changing length or coverage. It therefore suffices to solve the problem with a fixed starting vertex.

The search space is the finite set

$$\mathcal{S}_i = V(G) \times \mathcal{P}(E_i^*), \quad |\mathcal{S}_i| \leq 32 \cdot 16 = 512.$$

The initial state is $s_0 = (v_{\text{APP}}, \emptyset)$, and the terminal state is $s_* = (v_{\text{APP}}, E_i^*)$. Traversing an edge e incident at v towards its other endpoint v' updates the state by

$$(v, M) \mapsto (v', M \cup (E_i^* \cap \{e\})). \quad (6.7)$$

The mask records edge labels, not just endpoints; distinct parallel edges must therefore not be collapsed.

Every transition has unit cost. A FIFO queue initialized at s_0 extracts states in increasing distance, marks each state on its first visit, and stores its predecessor and traversed label. Equivalently, with $\mathcal{F}_0 = \{s_0\}$, the new layers are

$$\mathcal{F}_{k+1} = \text{Succ}(\mathcal{F}_k) \setminus \bigcup_{j=0}^k \mathcal{F}_j.$$

Induction shows that $\mathcal{F}_k$ contains exactly the states whose minimum distance from s_0 is k . Finite enumeration over the specified emitter gives, for each of the five regions,

$$s_* \notin \bigcup_{k=0}^7 \mathcal{F}_k, \quad s_* \in \mathcal{F}_8, \quad (\ell_1, \ell_2, \ell_3, \ell_4, \ell_5) = (8, 8, 8, 8, 8).$$

The tables exhibit the five upper bounds, including all four required labels in each case. Exhaustive exploration of the preceding layers proves the lower bound. Together they establish minimality for the declared problem. $\square$

Length eight characterizes the joint coverage of four incidences in each closure region. The quotient retains emission labels and their junctions between phase classes; enriched temporal evolution also preserves orientation and memory. This description links all five regions to the same propagation, autoscale, and APP anchors and complements the local six-junction cycle. Coinductive prolongation retains its operators and cylinders together with these orientation and memory data.

6.3 Finite transport and the prolongation boundary

The endomorphisms L_0, L_1 of the ternary-word space act before Archimedean realization. Their determination begins with transitions of the arithmetic state, not with the prescription of two matrices. If U_i is the composite step of selection, transport, and updating, the observed block satisfies

$$b_j^\chi = \Pi_6(x_j^\chi), \quad x_{j+1}^\chi = U_{9j+9} \cdots U_{9j+1}(x_j^\chi), \quad \chi \in \{\text{cl}, \text{pr}, \text{au}\}.$$

Six transitions are formed for each regime: two consecutive transitions in each of the three regions. The pairs of source and target matrices obtained in the record are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

Each six-digit string represents a row in $\mathbb{F}_3^6$. Since $\det X_0 = 1$ and $\det X_1 = -1$ in $\mathbb{F}_3$, the equations $X_a L_a = Y_a$ uniquely determine $L_a = X_a^{-1} Y_a$. In that row-vector chart, the result is

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix} \pmod{3}.$$

The calendar (L_0, L_0, L_1, L_1) produces four new blocks of six components. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate accumulated length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions have been fixed; it does not replace the production of those transitions by U_t . The provenance of their record and the examination of the sixteen binary calendars are documented in the technical supplement. The indicated calendar is the only one of those sixteen that jointly reproduces the three sequences of five blocks.

6.4 Correlated construction and ordering of the readers

The common antecedent of the three regions is the arithmetic emission with its trajectory record. APP provides the sum and product evaluations; TRIT provides the oriented signature; transition (2.20) composes mode selection, cardinal transport, and memory update. The census of that emission produces the catalogue and its fibres. The multiplicity, diagonal-support, and orientation predicates select the representatives in (6.1). Regional correlation preserves this joint provenance, together with the incidences and restrictions between its records.

The transports L_0, L_1 act on the selected representatives in the declared row chart. The calendar and concatenation distinguish prefix length, visible phase, and accumulated memory. The deep coordinate and its quotients are made explicit below through (6.8) and (6.9). Restrictions of histories are preserved in the sense of 4.8. The concrete oriented selection of R_{36} has its incidence proof in 11.2.1, after the construction of A_W ; that placement preserves the antecedents used by its proof.

Each regional reader adds a typed operation to the record it receives. For closure, the symmetric reader of the penta fibre fixes the primitive slope, and the composition law determines the compensator in (6.11). For propagation, formal composition, the unit, and the oriented generator determine recurrence (6.12). For autoscaling, the mode calendar produces incidence (6.13), whose positive ray fixes the normalized coordinate. The following proofs retain the normalizations and domains of these three operations.

Rational bounds and cylinder separation subsequently evaluate the resulting characters. Effective positional emission and its compatibility are proved in 7.2 and 7.3. The dodecaphasic register retains its own construction in 8; its coordinates and the three regional readings subsequently enter the integer closure in 10. Thus regional selection, transport, character construction, and prefix evaluation retain their domains and order within the same APP–TRIT–TPK genealogy.

6.5 Deep memory and compatibility of the cylinders

Prolongation compares states that simultaneously retain residues, quotients, and boundaries. In the six-coordinate linear chart of the record, take the integer representative of L_0 whose entries were displayed above:

$$A_H = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad \det A_H = 1.$$

The choice of this representative is part of the chart: reduction modulo three retains L_0 , whereas the integer representative fixes the operation on the deep quotients. With row vectors, this gives

$$\begin{aligned} \text{Div}_H : \mathbb{Z}^6 &\longrightarrow \{0, 1, 2\}^6 \times \mathbb{Z}^6, \\ x_H &\longmapsto (u_H, x_H^+), \\ y_H = x_H A_H, \quad u_H &= [y_H]_3, \quad x_H^+ = (y_H - u_H)/3. \end{aligned} \quad (6.8)$$

The pair (u_H, x_H^+) retains $y_H = u_H + 3x_H^+$. Since A_H is unimodular, its inverse has integer entries and

$$x_H = (u_H + 3x_H^+)A_H^{-1}. \quad (6.9)$$

This identity recovers the deep coordinate from its residue and quotient. The other state fields retain their own records. The ternary orientation is subsequently read through $0 \mapsto 0$, $1 \mapsto +1$, $2 \mapsto -1$, retaining the residue $u_H \in \{0, 1, 2\}^6$ and the quotient in (6.8).

The cylinder chart relates a ternary prefix of length L , with integer index N , to k_{dec} decimal triads, with index D . The corresponding half-open cylinders are

$$I_3(N, L) = \left[\frac{N}{3^L}, \frac{N+1}{3^L} \right), \quad I_{1000}(D, k_{\text{dec}}) = \left[\frac{D}{1000^{k_{\text{dec}}}}, \frac{D+1}{1000^{k_{\text{dec}}}} \right).$$

Define the boundary integers

$$a_{\text{cyl}} = N1000^{k_{\text{dec}}} - D3^L, \quad b_{\text{cyl}} = (N+1)1000^{k_{\text{dec}}} - D3^L.$$

For a block of six residues, fix its integer encoding as

$$\nu(u_H) = \sum_{j=1}^6 (u_H)_j 3^{6-j} \in \{0, \dots, 728\}.$$

Lemma 6.3 (Integer update of compatible cylinders). *The preceding cylinders intersect if and only if $a_{\text{cyl}} < 3^L$ and $b_{\text{cyl}} > 0$. Appending a ternary block with index $\nu \in \{0, \dots, 728\}$ gives $N' = 729N + \nu$ and $L' = L + 6$. For $\Delta k_{\text{dec}} \in \{0, 1\}$, the boundaries update according to*

$$\begin{aligned} \Delta k_{\text{dec}} = 0 : \quad a'_{\text{cyl}} &= 729a_{\text{cyl}} + \nu 1000^{k_{\text{dec}}}, \\ \Delta k_{\text{dec}} = 1 : \quad D' &= 1000D + \delta, \\ a'_{\text{cyl}} &= 729000a_{\text{cyl}} + \nu 1000^{k_{\text{dec}}+1} - \delta 729 \cdot 3^L, \end{aligned}$$

where $\delta \in \{0, \dots, 999\}$ is the triad of the compatible prolongation. In both cases,

$$b'_{\text{cyl}} = a'_{\text{cyl}} + 1000^{k_{\text{dec}}+\Delta k_{\text{dec}}}.$$

Proof. Two half-open intervals intersect precisely when each left endpoint is smaller than the right endpoint of the other interval. Multiplication of these two inequalities by $3^L 1000^{k_{\text{dec}}}$ gives $a_{\text{cyl}} < 3^L$ and $b_{\text{cyl}} > 0$. Substitution of N' , L' , and D' into

$$a'_{\text{cyl}} = N'1000^{k_{\text{dec}}+\Delta k_{\text{dec}}} - D'3^{L'}$$

gives the two recurrences. Subtracting the definitions of b'_{cyl} and a'_{cyl} yields the final identity. $\square$

These operations act on integer prefixes and their compatible restrictions. The Archimedean reading appears when the resulting system of cylinders is evaluated. The prolongation domain also incorporates orientation, carry, the joint signature, and the boundary data of the state $\hat{X}$ in (2.20); interval comparison is one component of this joint compatibility.

6.6 Five closure regions and angular compensation

The closure region contains a transversal component and four companions. The symmetric reader acts on the space of its five positions through

$$P_5 = \frac{1}{5}\mathbf{1}\mathbf{1}^\top, \quad \mathbf{1} = (1, 1, 1, 1, 1)^\top.$$

The invariance $P_5\lambda = \lambda$ and conservation $\mathbf{1}^\top\lambda = 1$ force $\lambda = \mathbf{1}/5$. This normalization distributes unity among reader positions; it does not replace the seed multiplicities 432, 144, 144, 144 by uniform frequencies. The reader's elliptic chart realizes each coordinate through $1 + \lambda_j J$; its slope is λ_j , so the primitive slope is $1/5$. This makes explicit the rule passing from a normalized coordinate to a slope, instead of identifying the two notions without a map. The reader composes the four companion positions by the law

$$a \oplus b = \frac{a + b}{1 - ab}. \quad (6.10)$$

This law is obtained by multiplying $(1 + aJ)(1 + bJ)$ in the regime $J^2 = -I$ and dividing by its scalar component. Slope composition is therefore a realization of the quadratic structure of TRIT, not an ordinary sum of four fractions.

Proposition 6.4 (Closure compensator). *The composition of four slopes $1/5$ is $120/119$. The condition of return to slope one determines a unique positive compensator of the form $1/q$, with $q > 1$, and gives $q = 239$.*

Proof. The first doubling gives $(1/5) \oplus (1/5) = 5/12$; the second gives $(5/12) \oplus (5/12) = 120/119$. The condition

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1$$

is equivalent to $(120 - 119)q = 120 + 119$, hence $q = 239$. $\square$

The resulting closure character has realization

$$\Pi_{\text{cl}} = 4 \left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \right). \quad (6.11)$$

Machin's angular identity recognizes $\Pi_{\text{cl}} = \pi$. The meaning of the construction is precise: the slope and number of compositions are declared by the pentafiber reader; the compensation equation forces 239. Once this character has been produced, its evaluation by alternating series does not select the seed orbit anew.

For $0 < x < 1$, the alternating sums

$$A_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{2j+1}$$

enclose $\arctan x$ between two consecutive truncations, with difference $x^{2n+3}/(2n+3)$. Applied to the two slopes in (6.11), they provide rational endpoints with explicit error. The numerical value is thus calculated as a consequence of the exact character, without introducing a target prefix.

6.7 Propagation reader and normalization of unity

The propagation region is realized through a formal collection $P_t \in \mathcal{A}[[t]]$, where $\mathcal{A}$ is a unital commutative algebra over $\mathbb{Q}$. Its composition satisfies $P_{t+s} = P_t P_s$, with unit $P_0 = 1$, and oriented elementary transport fixes the scalar generator $P'_0 = +1$. The condition fixes its sign, not only its magnitude. Writing

$$P_t = \sum_{n \geq 0} a_n t^n,$$

normalization determines $a_0 = a_1 = 1$. Comparing the coefficient of $s^n t$ in $P_{s+t} = P_s P_t$ gives $(n+1)a_{n+1} = a_n a_1$; equivalently, $P'_t = P_t$. Thus,

$$(n+1)a_{n+1} = a_n, \quad a_n = \frac{1}{n!}. \quad (6.12)$$

The character at one unit of the parameter is $\Pi_{\text{pr}} = P_1$, subsequently recognized as e .

The normalization of the generator matters. Without that datum, the semigroup law would admit $P_t = e^{ct}$ for different c ; the condition fixing the parameter unit in the reader is therefore not omitted. This condition precedes decimal comparison and belongs to the exposition of the operational rule, not to the choice of an experimental value.

Proposition 6.5 (Rational propagation bounds). *For $n \geq 1$, let $E_n = \sum_{j=0}^n 1/j!$. Then*

$$E_n < \Pi_{\text{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

Proof. The first omitted term is $1/(n+1)!$. Each subsequent ratio is at most $1/(n+2)$. The geometric upper bound for the tail is

$$\frac{1}{(n+1)!} \frac{1}{1 - 1/(n+2)} = \frac{n+2}{(n+1)(n+1)!}.$$

The strict inequality follows from the decrease in successive ratios. $\square$

6.8 Modal incidence and generation of self-scaling

The tritic selector produces the cycle $(+1, -1, 0)$. In the modes Σ, Π , the rule is $+1 : m \mapsto \Sigma, -1 : m \mapsto \Pi, 0 : m \mapsto m$. With cyclic closure, the observed mode is (Σ, Π, Π) . Its edges are $\Sigma \rightarrow \Pi, \Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Its incidence, in that order of modes, is therefore

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (6.13)$$

The four entries are obtained from the calendar, not from an equation previously supplied with the value of φ . Repetition of the calendar gives counts $3F_{\text{av}}, 9F_{\text{av}}, 36F_{\text{av}}$ at lengths nine, twenty-seven, and one hundred eight. These counts are not the powers of the matrix.

Theorem 6.6 (Positive self-scaling character). *The incidence (6.13) has a unique strictly positive eigenray. When normalized as $(1, x)^T$, its coordinate x satisfies $x^2 - x - 1 = 0$, $x > 0$, and is $\varphi = (1 + \sqrt{5})/2$.*

Proof. The eigenvalue equation gives $x = \lambda$ and $1 + x = \lambda x$. Eliminating λ produces the polynomial. Only one of its two roots is positive. Since F_{av}^2 has all entries positive, the positive ray is unique and simple. The radical expression recognizes the coordinate obtained; it does not participate in producing F_{av} . $\square$

With $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, the incidence induces, for $n \geq 1$,

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

Consecutive ratios F_{n+1}/F_n alternate around φ . The determinant identity

$$F_{n+1}^2 - F_n F_{n+2} = (-1)^n$$

gives an exact width $1/(F_n F_{n+1})$ between two consecutive ratios. Their growth makes this rational bracket arbitrarily small.

The measure on all compatible histories does not coincide with the mode frequency of the three-event calendar. The chronological frequency is $(1/3, 2/3)$. The measure of maximal entropy of the symbolic envelope defined by F_{av} is constructed from its eigenvectors and has weight ratio φ^{-2} . The latter enters the areal-volumetric reading; it is not obtained by simply substituting one counting frequency for another.

6.8.1 Trajectory space and invariant measure between sheets

The toroidal topology of APP fixes the support of cardinal translations and their winding classes. The correspondence between sheets arises from another operation on that same support: temporal selection of the additive or multiplicative evaluation. Its geometric typing is

$$\eta_{\text{av}}(\Sigma) = \mathcal{H}_{\text{ar}}, \quad \eta_{\text{av}}(\Pi) = \mathcal{H}_{\text{vol}}.$$

The first sheet corresponds to the additive reading of the boundary; the second, to the multiplicative reading of the interior. This map transports mode labels and preserves the incidence (6.13). The square of the self-scaling ratio is determined through the trajectories of that incidence.

Definition 6.7 (Memory-one symbolic envelope). Let

$$X_{\text{av}} = \{(s_k)_{k \in \mathbb{Z}} \in \{\mathcal{H}_{\text{ar}}, \mathcal{H}_{\text{vol}}\}^{\mathbb{Z}} : (F_{\text{av}})_{s_k, s_{k+1}} = 1 \text{ for every } k\}.$$

Index shift acts on this space. It is the smallest memory-one symbolic system containing the modal sequences and preserving their consecutive pairs: it admits the two cross-transitions and volumetric self-retention. Phase, cardinal orientation, and the trajectory records remain in the TPK state from which this envelope is obtained.

Proposition 6.8 (Weighting trajectories between sheets). *In areal–volumetric order, the invariant measure of maximal entropy of X_{av} has transition matrix and stationary distribution*

$$P_{\text{av}} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}, \quad (\mu_{\text{ar}}, \mu_{\text{vol}}) = \frac{(1, \varphi^2)}{1 + \varphi^2}. \quad (6.14)$$

In particular, $\mu_{\text{ar}}/\mu_{\text{vol}} = \varphi^{-2}$.

Proof. The already constructed incidence has positive eigenvector $r = (1, \varphi)^\top$ and eigenvalue φ . The normalization

$$(P_{\text{av}})_{ij} = \frac{(F_{\text{av}})_{ij} r_j}{\varphi r_i}$$

gives the stated matrix. Its rows sum to one because $\varphi^{-2} + \varphi^{-1} = 1$. The symmetry of F_{av} implies that weights proportional to r_i^2 satisfy detailed balance; therefore, $\mu P_{\text{av}} = \mu$. Irreducibility and volumetric self-retention determine a unique stationary distribution.

To verify maximality, every allowed transition satisfies $\log(P_{\text{av}})_{ij} = \log r_j - \log r_i - \log \varphi$. Under any invariant measure, the endpoint terms cancel on average. If v_i are its vertex weights and $q_i(j) = \nu(s_1 = j \mid s_0 = i)$, for $v_i > 0$, then

$$h_\nu \leq H_\nu(s_1 \mid s_0) = \log \varphi - \sum_{i: v_i > 0} v_i D(q_i \parallel R_i) \leq \log \varphi,$$

where $D(q \parallel p) = \sum_j q_j \log(q_j/p_j)$ is the relative entropy on allowed pairs and $0 \log 0 = 0$ is adopted. The Markov measure in (6.14) attains the bound. Equality in the first inequality requires the Markov property; equality in the second requires $q_i = R_i$. Stationarity and irreducibility then fix μ , establishing uniqueness. This is the Parry measure of the defined envelope. $\square$

The frequency $(1/3, 2/3)$ counts the instants of the primitive calendar; μ weights the trajectories of the envelope. Its quadratic ratio expresses the product of the incoming and outgoing incidence amplitudes. TPK memory and phase restrictions retain their own domain; the entropy of the envelope corresponds to the symbolic system specified in this definition.

6.8.2 Closure operator and limit of marked returns

The closure coordinate π , generated by the regional reader, enters after the construction of modal incidence. Let $e_{\text{ar}} = (1, 0)^T$, $e_{\text{vol}} = (0, 1)^T$, and $P_i = e_i e_i^T$ be the sheet projectors. The reader's conditions are preservation of both sheets, unit volumetric normalization, and an areal mark equal to the closure coordinate. These conditions determine the positive operator

$$D_\pi = \pi P_{\text{ar}} + P_{\text{vol}} = \begin{pmatrix} \pi & 0 \\ 0 & 1 \end{pmatrix}. \quad (6.15)$$

Diagonality expresses preservation of sheets, and the two entries express their normalizations; this makes explicit the uniqueness of the operator for the fixed reader.

Proposition 6.9 (Ratio of boundary returns). *For $n \geq 1$, the marked evaluation*

$$\Theta_n = \frac{\langle e_{\text{ar}}, D_\pi F_{\text{av}}^n e_{\text{ar}} \rangle}{\langle e_{\text{vol}}, F_{\text{av}}^n e_{\text{vol}} \rangle} = \pi \frac{F_{n-1}}{F_{n+1}} \quad (6.16)$$

satisfies

$$\lim_{n \rightarrow \infty} \Theta_n = \pi \frac{\mu_{\text{ar}}}{\mu_{\text{vol}}} = \frac{\pi}{\varphi^2}. \quad (6.17)$$

Proof. The diagonal entries of F_{av}^n count the paths of length n returning to their initial sheet. The already proved power formula provides F_{n-1} and F_{n+1} . The operator D_π multiplies the first count by π . Finally, F_{n-1}/F_{n+1} is the product of two consecutive ratios converging to φ^{-1} ; its limit is φ^{-2} . The identity with the weight ratio follows from (6.14). $\square$

The closure operator appears only once, in the terminal evaluation of the return. The transfer F_{av} remains determined by the calendar. Information from two readers of the same state is thus composed: modal incidence determines the quadratic weighting, and the closure region contributes its coordinate to the boundary.

The composition also preserves refinement. For positive cylinders $I = [a, b]$ and $J = [c, d]$, the image under the reader $\mathcal{R}(x, y) = x/y^2$ is the interval

$$\mathcal{Q}(I, J) = \left[\frac{a}{d^2}, \frac{b}{c^2} \right].$$

Monotonicity in each argument proves that the nested closure and self-scaling cylinders produce nested intervals for the ratio; positivity of their limits and contraction of their diameters determine π/φ^2 . The scalar expression thus retains a chain of operations compatible with depth, which is applied next in its two orientations.

After the characters have been generated, the two orientations of the reader

$$R(x, y) = \frac{x}{y^2}$$

provide

$$\rho_A = \frac{\pi}{\varphi^2}, \quad \rho_E = \frac{\varphi}{\pi^2}, \quad \varphi^3 \rho_A^2 \rho_E = 1. \quad (6.18)$$

The first expression is also obtained as the limit of the marked reading $\pi F_{n-1}/F_{n+1}$, according to (6.17). Exchanging the arguments relates the two readings. This arithmetic relationship does not turn a reparameterization of the electronic equation into a second independent derivation of all its coefficients.

6.9 Iteration of the ratio reader and phase recovery

The reader $R(x, y) = x/y^2$, applied to the positive representations of closure and self-scaling, admits the oriented composition

$$\mathcal{Q}_{\text{rat}}(x, y) = (R(x, y), R(y, x)) = (x/y^2, y/x^2), \quad x, y > 0.$$

This composition receives coordinates already produced by APP-TRIT-TPK. Its study retains both the product $P = xy$ and the oriented ratio $O = x/y$, whose transformations are

$$P \circ \mathcal{Q}_{\text{rat}} = P^{-1}, \quad O \circ \mathcal{Q}_{\text{rat}} = O^3.$$

The notation $\mathcal{Q}_{\text{rat}}$ distinguishes this map from the interval extension of the scalar reader.

Proposition 6.10 (Positive iteration and complex extension). *For $n \geq 0$, let*

$$a_n = \frac{(-1)^n + 3^n}{2}, \quad b_n = \frac{(-1)^n - 3^n}{2}.$$

Then

$$\mathcal{Q}_{\text{rat}}^n(x, y) = (x^{a_n} y^{b_n}, x^{b_n} y^{a_n}). \quad (6.19)$$

The same Laurent expression defines a map on $(\mathbb{C}^\times)^2$. For $n \geq 1$, this map is an unramified covering of degree 3^n , with kernel

$$\ker \mathcal{Q}_{\text{rat}}^n = \{(\zeta, \zeta^{-1}) : \zeta^{3^n} = 1\}. \quad (6.20)$$

Its restriction to the positive quadrant has a unique positive preimage for every positive pair.

Proof. In positive logarithmic coordinates, the matrix is

$$B = \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}.$$

The vectors $(1, 1)$ and $(1, -1)$ have eigenvalues -1 and 3 . Decomposition along these two directions gives $B^n = \begin{pmatrix} a_n & b_n \\ b_n & a_n \end{pmatrix}$ and proves (6.19). Since a_n, b_n are integers, the identity extends as a Laurent identity.

Write $m = 3^n$, $\epsilon = (-1)^n$. For an image $(u, v) \in (\mathbb{C}^\times)^2$, the product of a preimage is fixed by $p = xy = (uv)^\epsilon$. The equations are equivalent to

$$x^m = u p^{-b_n}, \quad y = p/x.$$

The first has exactly m distinct roots. The second determines y ; moreover, the product of the two images is $p^\epsilon = uv$, so the second original equation also holds. Every fibre contains exactly m points. For $(u, v) = (1, 1)$, this gives (6.20); the other fibres are its translates in the multiplicative group.

The differential satisfies

$$\det D\mathcal{Q}_{\text{rat}}^n = (-3)^n (xy)^{\epsilon-1} \neq 0.$$

The local description by roots of a nonzero number provides the m local branches and proves the covering assertion. In the positive quadrant only one of these roots is positive. In particular, for one step,

$$x = (uv^2)^{-1/3}, \quad y = (u^2v)^{-1/3},$$

with the positive real roots. □

In the complex extension, the condition $xy = (uv)^{-1}$ couples the two cube roots in the one-step inverse. The coordinates (P, O) also identify (x, y) and $(-x, -y)$; their map has kernel $\{(1, 1), (-1, -1)\}$. The preceding proof retains (x, y) and therefore the distinction between this coordinate identification and the fibres of $\mathcal{Q}_{\text{rat}}^n$.

Proposition 6.11 (Reconstruction from the ternary register). *On the subgroup*

$$(x, y) = (e^{2\pi i \theta}, e^{-2\pi i \theta}),$$

the map induces $\theta \mapsto 3\theta \pmod{1}$. With representatives $\theta_j \in [0, 1)$, let

$$d_j = \lfloor 3\theta_j \rfloor, \quad \theta_{j+1} = 3\theta_j - d_j, \quad C_n = \sum_{j=0}^{n-1} d_j 3^{n-1-j}.$$

Exact recovery is given by

$$\theta_0 = \frac{\theta_n + C_n}{3^n}. \quad (6.21)$$

Proof. Substitution into the Laurent expression gives phase tripling. The recurrence $C_{n+1} = 3C_n + d_n$, with $C_0 = 0$, gives $\theta_n = 3^n \theta_0 - C_n$ by induction. This proves (6.21). Each d_j belongs to $\{0, 1, 2\}$, so n ternary digits retain the choice among the 3^n preimages of the reduced phase. The branches $B_d(\theta) = (\theta + d)/3$ satisfy

$$B_a \circ B_b(\theta) = \frac{\theta + b + 3a}{9}.$$

Thus two steps retain the ordered digit $3d_0 + d_1$. □

The product returns after two iterations, whereas the oriented ratio is raised to the ninth power. The recovery proved here has this reader and its phase register as its domain. Its insertion into a representation of the complete enriched state additionally retains the maps of sheets, orientation, memory and survival belonging to that representation.

6.10 Exponential realizations of the generated coordinates

The regional coordinates already determined allow recognition of concrete realizations of the TRIT quadratic collection. The order is substantive: construction of the regions precedes evaluation of an exponential at parameters π or $2 \log \varphi$. These expressions identify the role of the coordinates in operators constructed from APP and TPK; they do not participate in selecting the prefixes that produce them.

6.10.1 Ternary cycle, nilpotent transport, and sheet incidence

Multiplication $a(r) = 4r$ in $\mathbb{Z}/9\mathbb{Z}$ has order 3. On the units it determines the orbits $\{1, 4, 7\}$ and $\{2, 8, 5\}$. In the augmentation module of either orbit, consisting of integer combinations whose coefficients sum to zero, the basis $(e_r - e_{4r}, e_{4r} - e_{7r})$ gives

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^2 + C + I = 0.$$

The restricted form has Gram matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. On residue evaluations modulo 9, the action distinguishes the sheets: $S(ax, ay) = aS(x, y)$, whereas $P(ax, ay) = a^{-1}P(x, y)$. The cyclic realization therefore preserves a contragredient orientation between sum and product. Quotients of the integer lift are retained in the corresponding arithmetic record.

Elementary parabolic transport is represented by

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The areal–volumetric incidence already constructed in (6.13) gives

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = F_{\text{av}}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The three operators act on declared real spaces: the augmentation module tensored with $\mathbb{R}$, the nilpotent-transport plane, and the modal-incidence plane. Sharing a matrix dimension does not identify these spaces or their coordinates.

Proposition 6.12 (Concrete generators of the three types). *The operators*

$$J_e = \frac{2C + I}{\sqrt{3}}, \quad J_p = N, \quad J_h = \frac{2A - 3I}{\sqrt{5}}$$

satisfy, respectively,

$$J_e^2 = -I, \quad J_p^2 = 0, \quad J_h^2 = I.$$

Proof. The relation for C implies $(2C + I)^2 = -3I$. Direct multiplication gives $N^2 = 0$. Finally, A has trace 3 and determinant 1, so $A^2 - 3A + I = 0$ and $(2A - 3I)^2 = 5I$. $\square$

Proposition 6.13 (Three evaluations of the exponential identity). *In the preceding realizations,*

$$\exp(\pi J_e) + I = 0, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{\text{av}}^2. \quad (6.22)$$

Proof. The elliptic specialization of the tritic exponential gives $\cos \pi I + \sin \pi J_e = -I$. Nilpotency removes exactly the terms of degree at least 2 in the second expression. For the third, $\varphi^2 = \varphi + 1$ implies

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2}.$$

Thus the exponential equals $\frac{3}{2}I + \frac{\sqrt{5}}{2}J_h = A$. $\square$

The first equality represents the elliptic half-turn; the complete turn satisfies $\exp(2\pi J_e) = I$. The second retains a nonzero linear term: the parabolic regime expresses transport, not absence of evolution. The third realizes a unimodular hyperbolic advance. Its eigenvalues are φ^2 and φ^{-2} ; one direction expands and the other contracts while preserving the determinant. The three evaluations use different parameters of one common exponential collection.

In this collection, e enters through the scalar evaluation $\exp(I) = eI$ and the law $\exp(sI)\exp(tI) = \exp((s+t)I)$. Its role is not restricted to the parabolic type. The coordinate e , previously obtained from its region, is recognized here as the unit evaluation of the exponential; this identity does not replace its coinductive genealogy.

6.10.2 Polynomial resolution of the regimes

The three realizations admit a compact presentation on the direct sum of their spaces. Define

$$\mathbb{J} = J_e \oplus J_p \oplus J_h, \quad Q = \mathbb{J}^2.$$

The preceding relations imply

$$\mathbb{J}^6 = \mathbb{J}^2, \quad Q^3 = Q.$$

Using Q keeps this operator square distinct from the dodecaphase record K .

Proposition 6.14 (Polynomial regime projectors). *The maps*

$$p_e = \frac{Q^2 - Q}{2}, \quad p_p = I - Q^2, \quad p_h = \frac{Q^2 + Q}{2}$$

are mutually orthogonal idempotents, sum to I , and recover the three summands of the representation.

Proof. The three polynomials take values $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ at the points $-1, 0, +1$, respectively. The differences between their squares and themselves, as well as their cross products, are multiples of $X^3 - X$. Substituting Q , which annihilates that polynomial, proves the identities. On the three blocks, Q acts as $-I, 0, I$. $\square$

One thus obtains

$$\begin{aligned} \exp(t\mathbb{J}) &= p_e(\cos tI + \sin t\mathbb{J}) + p_p(I + t\mathbb{J}) \\ &\quad + p_h(\cosh tI + \sinh t\mathbb{J}). \end{aligned} \quad (6.23)$$

Each summand preserves its quadratic relation, and conjugation reverses its evolution. The total representation gathers the three regimes, but does not by itself define a transition operator between them. Such a transition requires the TPK transport maps with their domains, orientation, and memory. The exponential articulation allows recognition of their types without replacing those dynamics by an identification of fibers.

6.11 Compatible cylinders and arbitrary depth

The realization of a ternary block $b = (b_1, \dots, b_6)$ is the integer

$$\nu(b) = \sum_{j=1}^6 b_j 3^{6-j} \in \{0, \dots, 728\}.$$

A sequence of blocks defines integer prefixes by $N_{n+1} = 729N_n + \nu(b_{n+1})$, starting at $N_0 = m$, where m is the integer part determined by the reader. The preceding brackets place the closure, propagation, and self-scaling characters in (3, 4), (2, 3), and (1, 2), respectively; their initial values are therefore $m = 3, 2, 1$. Subsequent blocks refine the fractional part. The corresponding cylinder is

$$I_n = \left[\frac{N_n}{729^n}, \frac{N_n + 1}{729^n} \right], \quad |I_n| = 729^{-n}. \quad (6.24)$$

The half-open convention is used to select a representative when a value falls on a boundary. For passage to the limit, the closures of these intervals are used and the two equivalent boundary expansions are identified. This specification prevents the assumption that every intersection of nested half-open intervals is automatically nonempty.

Theorem 6.15 (Determination of scalar realization). *A compatible block history determines a unique limit of the left endpoints of (6.24). If an exact character has rational brackets of arbitrarily small width and is separated from the boundary of each decimal cylinder examined, each decimal prefix of finite length is determined after a finite computation. The prefixes thus obtained are mutually compatible.*

Proof. Extending a block preserves the preceding prefix by Euclidean division. The closed cylinders are nested and their diameter tends to zero; their left endpoints are Cauchy and have a unique limit. For a fixed decimal length, a value separated from the boundaries of the partition has positive distance from them. A sufficiently narrow bracket lies within a single cell and determines its prefix. Uniqueness of the cell and nesting of the partitions prove compatibility under truncation. At points with a terminating expansion, the exact boundary decision and its representation convention must be added; that decision does not follow from numerical width alone. $\square$

Corollary 6.16 (Finite computation and compatibility of regional prefixes). *For each of the realizations π, φ, e and for every finite decimal length, the constructed brackets determine the corresponding prefix by a finite computation. Prefixes obtained at different depths are compatible under truncation.*

Proof. The closure, propagation, and self-scaling characters have the preceding brackets. After their recognition, the irrationality of π, e, φ guarantees their separation from the rational boundaries of each finite decimal partition. The preceding theorem applies to each character. $\square$

Arbitrary depth expresses the quantifier of the corollary: for every finite horizon there exists a computation determining its prefix. The mathematical object is the complete compatible collection; each execution obtains a finite projection of it.

6.12 Specular involution and synchronization of the regions

The nonadic connection transports the three characters jointly. In the ninth-phase chart, the self-scaling axis and the ternary sum of closure and propagation remain fixed under specular exchange. Helical inversion changes the direction of transport; the mirror of the two channels changes their relative orientation. They are different involutions, although both act on the same system of trajectories.

Coincidences in the census illustrate this coordination:

Phase	N_π	N_e	N_φ	Equality of counts
4	207	207	122	$N_\pi = N_e$
5	39	151	151	$N_e = N_\varphi$
6	110	110	69	$N_\pi = N_e$
7	81	29	81	$N_\varphi = N_\pi$
8	59	59	59	triple synchronization
9	43	43	43	triple synchronization

These integers count compatible coordinates of the regions. They do not assert equality among π , e , and φ as real numbers. If

$$\partial(a, b, c) = (a - b, b - c, c - a),$$

its image belongs to the lattice $A_2 = \{(u, v, w) \in \mathbb{Z}^3 : u + v + w = 0\}$. At phases eight and nine the relative component vanishes, but the diagonal component changes from (59, 59, 59) to (43, 43, 43). Relative synchronization coexists with a shift of -16 per channel. The closure leading to α must preserve both pieces of information: relative coincidence and the evolution of memory.

6.13 Sturmian helical suspension

The relation between bases nine and ten of the completion defines the dimensionless parameter

$$\delta = \frac{\log(10/9)}{\log 10}. \quad (6.25)$$

It is not obtained from a target physical constant. It is the relative logarithmic scale of two bases already present in the construction. Its use as a symbolic reading of refinement has an exact consequence.

Lemma 6.17 (Irrationality of the slope). *The number δ is irrational.*

Proof. If $\delta = p/q$, $q > 0$, then $(10/9)^q = 10^p$ and $9^q = 10^{q-p}$. The factorization of the left-hand side contains the prime three and that of the right-hand side only the primes two and five. Equality is impossible. $\square$

The lower mechanical word, with intercept θ , is

$$z_n(\theta) = \lfloor (n+1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor \in \{0, 1\}.$$

It is obtained by observing the rotation $\theta \mapsto \theta + \delta \pmod{1}$ through the interval $[1 - \delta, 1)$. The event sequence has density δ , and the sum over any window of length L equals $\lfloor L\delta \rfloor$ or $\lceil L\delta \rceil$. Since $21 < 1/\delta < 22$, the gaps between events are twenty-one or twenty-two. The word indicating which intervals are short is another mechanical word of slope $\sigma = 22 - 1/\delta$; its events are separated by six or seven primary intervals. The second scale is induced by the first and is not a second free parameter.

6.13.1 Capacity index, vacancies, and induction of the tritic regime

The vacancy sequence is linked to refinement through an increment identity. To exhibit it, fix the phase origin $\theta = 0$. Let

$$\rho = \log_{10} 9 = 1 - \delta, \quad \mathcal{C}_t = \lfloor (t+1)\rho \rfloor \quad (t \geq 0).$$

The index $\mathcal{C}_t$ expresses the decimal capacity of refinement in the normalization comparing a ternary block of 6 positions, with 729 possibilities, and a decimal block of 3 positions, with 1000 possibilities. Indeed, $\log_{1000} 729 = \log_{10} 9 = \rho$. The letter K is reserved for the dodecaphase record constructed later.

Proposition 6.18 (Complementarity of capacity and vacancy). *For $t \geq 1$,*

$$\mathcal{C}_t - \mathcal{C}_{t-1} = 1 - z_t(0). \quad (6.26)$$

Consequently, if $0 \leq a < b$,

$$\mathcal{C}_b - \mathcal{C}_a + \sum_{t=a+1}^b z_t(0) = b - a.$$

Proof. Irrationality of δ gives

$$\mathcal{C}_t = t + 1 - \lfloor (t+1)\delta \rfloor = t - \lfloor (t+1)\delta \rfloor.$$

Subtracting consecutive terms produces (6.26). Summation over the interval telescopes. $\square$

Each transition increases the block counter by one. If $z_t = 0$, capacity also increases; if $z_t = 1$, capacity remains unchanged while a transition is incorporated into the history. Here a vacancy is a capacity-retention event, with a determined position and antecedent. The preceding balance preserves length exactly between acquired capacity and recorded vacancies. The observed phase $g_t^{\text{cap}} = [\mathcal{C}_t]_9$ satisfies

$$g_t^{\text{cap}} - g_{t-1}^{\text{cap}} \equiv 1 - z_t \pmod{9}.$$

This capacity phase and chronological phase $[t]_9$ are two different observations. Local capacity retention and a complete holonomy turn have distinct mechanisms, although both allow the phase observation to be preserved while the state changes.

The relation with TRIT can be followed in the vacancy positions themselves. For $j \geq 1$, define

$$p_j = \left\lfloor \frac{j}{\delta} \right\rfloor, \quad v_j = \mathcal{C}_{p_j}, \quad h_j = p_{j+1} - p_j.$$

The inequality $p_j\delta < j < (p_j + 1)\delta$ identifies p_j as the position of event j . Since $0 < \delta < 1$, it also gives $\lfloor (p_j + 1)\delta \rfloor = j$, and hence

$$v_j = p_j - j, \quad v_{j+1} - v_j = h_j - 1.$$

If $s_j = 22 - h_j$, then $s_j \in \{0, 1\}$ and

$$[v_{j+1}]_3 = [v_j - s_j]_3. \quad (6.27)$$

An interval of length 22 preserves the class; one of length 21 shifts it by one in the negative direction. The canonical selector is applied to the positive phase $\bar{r}_j = 1 + [v_j]_9$:

$$\tau_j = M_{\text{ph}}(\bar{r}_j).$$

A tritic regime is thus determined by the capacity index retained at each event, in addition to the vacancy's binary encoding.

The positions of the short intervals form the induced mechanical coding of slope $22 - 1/\delta$. Their separations 6 and 7 determine the lengths of plateaus of the class $[v_j]_3$, once the origin and endpoint convention are fixed. The first classes are 2 for 6 events, 1 for 7, and 0 for 7; the first segment of 6 is the initial restriction of a plateau. Their signatures traverse 0, -1 , $+1$. Equations (6.26) and (6.27) make explicit the link between refinement, vacancies, and regime selection. The quadratic reading of

TRIT subsequently realizes these types through their respective generators; the temporal event and its geometric representation thus retain an identifiable dependence.

Inverting the monotone index supplies another useful observation:

$$\mathcal{T}(k) = \min\{t : \mathcal{C}_t = k\} = \left\lceil \frac{k}{\rho} \right\rceil - 1 \quad (k \geq 1).$$

If $\sigma = \rho^{-1} - 1$, the number of blocks needed to gain m units of capacity is

$$\mathcal{T}(k+m) - \mathcal{T}(k) = m + [(k+m)\sigma] - [k\sigma] \in \{m + \lfloor m\sigma \rfloor, m + \lceil m\sigma \rceil\}.$$

In particular, 9 units of capacity require 9 or 10 blocks. The period of a finite phase remains exact, whereas return time measured by the other index admits two values. This distinction belongs to refinement dynamics and is not an a posteriori correction of its numerical values.

Theorem 6.19 (Complexity of phase readings). *For $m \geq 1$, the mechanical word has factor complexity $p(m) = m + 1$. When the phase $n \bmod 9$ is retained, the complexity is $p_9(m) = 9(m + 1)$; when only its ternary class is retained, it is $p_3(m) = 3(m + 1)$. All three topological entropies are zero.*

Proof. The predecessors of the two endpoints of the coding interval form, for factors of length m , $m + 1$ distinct points on the circle: coincidence between shifted endpoints eliminates duplications and irrationality prevents additional coincidences. These points determine $m + 1$ distinct intervals of constant itinerary. Every irrational orbit visits each interval, whence $p(m) = m + 1$.

For a fixed phase modulo nine, positions of that phase follow a rotation of step 9δ , also irrational. By density, all factors occur in every phase, and the first coordinate distinguishes their nine copies. The same argument with step 3δ gives three copies after ternary reduction. Finally, $m^{-1} \log(c(m + 1)) \rightarrow 0$ for $c = 1, 3, 9$. $\square$

The phase product $C_9 \times \mathbb{T}$, translated by $(1, \delta)$, has characters of frequency

$$\frac{a}{9} + k\delta \pmod{1}, \quad a \in \mathbb{Z}/9\mathbb{Z}, k \in \mathbb{Z}.$$

The spectral module is not $\log 10$: it is the additive module generated by $1/9$ and δ , modulo the integers. The factor $\log 10$ belongs to the normalization in (6.25).

An operator presentation makes memory explicit. In $\ell^2(\mathbb{Z})$, let $T|n\rangle = |n + 1\rangle$, $V_9|n\rangle = \zeta_9^n|n\rangle$, $W_\delta|n\rangle = e^{2\pi i n \delta}|n\rangle$, and $D_M|n\rangle = \lfloor n/9 \rfloor |n\rangle$. On finitely supported vectors,

$$V_9 T = \zeta_9 T V_9, \quad W_\delta T = e^{2\pi i \delta} T W_\delta, \quad [D_M, T^9] = T^9. \quad (6.28)$$

The first two relations are Weyl covariances; the third expresses that a nonadic turn increases the memory degree. The base rotation is abelian, whereas the algebra of observables and translations is not. The inversion $|n\rangle \mapsto |-n\rangle$ conjugates advance with retreat. This realization provides an aperiodic helical suspension with finite phase return, not a periodic word of length nine.

For this count we fix the intercept $\theta = 0$ and the accumulated endpoints $(N_0, \dots, N_4) = (0, 729, 972, 999, 1000)$, corresponding to the cubic order $(729, 243, 27, 1)$. The lower counts are $\lfloor N_j \delta \rfloor - \lfloor N_{j-1} \delta \rfloor$, and give $(33, 11, 1, 0)$. The upper counts are $\lceil N_j \delta \rceil - \lceil N_{j-1} \delta \rceil$, with the same initial boundary zero, and give $(34, 11, 1, 0)$. These values depend on the intercept and the ordering of the strata; they are not asserted for an arbitrary initial phase. Their totals are forty-five and forty-six. The increment corresponds to a single boundary unit, located in the first stratum of this ordered partition. The numbers thirty-four and eleven thus appear in a single count preceding any chart of units; identifying them with physical exponents additionally requires the corresponding dimensional derivation.

The term *aperiodic crystal* is used here for this symbolic order, with its complexity, spectrum, and return specified. Zero topological entropy does not by itself amount to zero physical dissipation: a thermodynamic realization requires a defined energy dynamics and erasure operation. Likewise, this Sturmian reading does not replace all TPK operators or by itself prove periodicity or aperiodicity of the complete output of α . Its function is to make explicit an order structure not described by the simple return of the calendar.

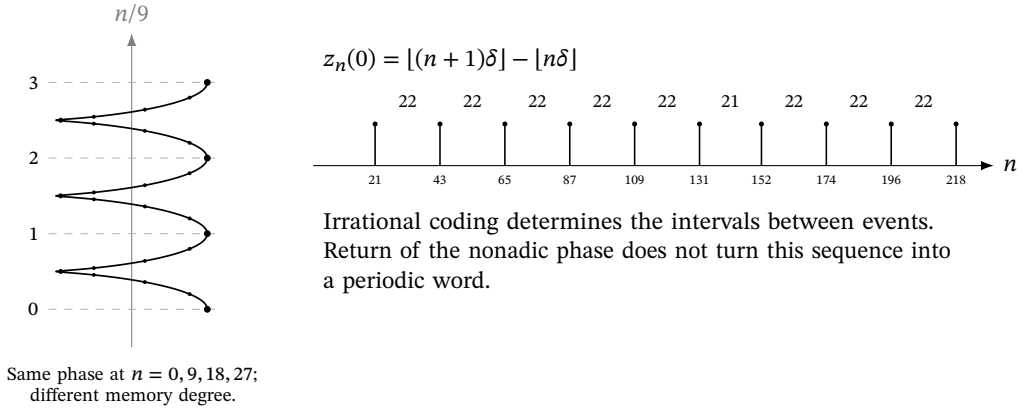


Figure 5: Complementary representations of transport. Left: projection of the helix $(\cos(2\pi n/9), \sin(2\pi n/9), n/9)$: advancing nine steps recovers the phase and increases the height. Right: the first ten events of the lower word for $\delta = \log_{10}(10/9)$, with exact separations 21 or 22. The helical representation introduces no additional physical metric.

6.14 Nonadic connection, holonomy, and monodromic representation

The already constructed regional connection is now studied through three objects: transport of the complete state, the representation of its turns, and the suspension of its itineraries. The APP-TRIT-TPK operations and the initial catalog retain their preceding definitions. The necessary distinction here concerns what each representation retains of the same history. Definition 6.20 preserves the admissible extensions; Propositions 6.21 and 6.22 distinguish their iteration and the bilateral representation of the index.

At prolongation index k , a state chart preserves

$$\hat{x}_k = (B_k, C_k, p_k, c_k, \Sigma_k, W_k, g_k, m_k).$$

Here B_k brings together the three regional blocks, C_k is the cylinder, p_k the prefix, c_k the carry record, Σ_k the boundary signature, and W_k the incidence information. The phase is $g_k = 1 + [(k-1)]_9$; m_k counts complete turns. The reduction $\beta_k = [m_k]_{12}$ preserves the dodecaphase position, whereas the integer m_k distinguishes its successive turns.

Definition 6.20 (Transport through one nonadic turn). Let $\mathcal{R}_k \subseteq \hat{X}_k \times \hat{X}_{k+1}$ be the admissible-extension relations of the TPK. The realization of holonomy at level k is the relational composition

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \cdots \circ \mathcal{R}_k.$$

For $s \geq 1$, its prolongation to s turns is

$$\Gamma_{9s,k} = \Gamma_{9,k+9(s-1)} \circ \cdots \circ \Gamma_{9,k}.$$

The relational formulation preserves admissible continuations. On an individualized history, each arrow leads to the state actually prolonged; the isolated visible block may still admit more than one continuation. This difference is precisely the function of memory and boundary conditions in determining the trajectory.

Proposition 6.21 (Iteration of return and accumulated memory). *In every admissible composition of s turns,*

$$g_{k+9s} = g_k, \quad m_{k+9s} = m_k + s, \quad \beta_{k+9s} = \beta_k + s \pmod{12}.$$

Proof. The one-turn rule preserves phase and increases the counter by one unit. Applying it to the already transported state repeats exactly those two operations. Induction gives the first two equalities; reducing the second modulo 12 gives the third. $\square$

The counter thus provides an integer witness to the difference between turns. After 12 turns the pair of finite phases is restored, but the counter increases by 12. The nonsplit extension (4.5) expresses the composition of the calendar through its cocycle; the trajectory additionally retains the integer lift of that datum. Conservation here means transport of previous contributions, not immobility of every coordinate.

6.14.1 Bilateral representation of the return index

The shift T in (6.28) admits an expression in phase and turn coordinates. We number phases from 0 to 8, by shifting the origin relative to g_k , and write $k = 9m + r$, with $0 \leq r < 9$, on the integer line covering the phase cycle. On $\mathcal{H}_{\text{ind}} = \ell^2(\mathbb{Z})$, the basis $|k\rangle = |r, m\rangle$ expresses that same unitary shift

$$T|r, m\rangle = \begin{cases} |r+1, m\rangle, & r < 8, \\ |0, m+1\rangle, & r = 8. \end{cases}$$

The inverse reduces r by one unit when $r > 0$, and sends $|0, m\rangle$ to $|8, m-1\rangle$. Passage through the phase origin therefore incorporates the exact quotient unit.

Proposition 6.22 (Representation of the monodromy of the cycle). *Let $S = T^9$. Then*

$$S|r, m\rangle = |r, m+1\rangle, \quad \mathbb{Z} \longrightarrow \mathcal{U}(\mathcal{H}_{\text{ind}}), \quad s \longmapsto S^s$$

is a faithful unitary representation of the turn group of the cycle.

Proof. Nine shifts preserve the remainder of division by 9 and increase the quotient. The powers satisfy $S^{s+t} = S^s S^t$, and $S^s|r, m\rangle = |r, m+s\rangle$ distinguishes every integer s . A permutation of an orthonormal basis is unitary. $\square$

This representation corresponds to the covering of the cycle, whose turn group is $\mathbb{Z}$, and represents phase together with its counter. The TPK state additionally contains the trajectory coordinates already described. The bilateral extension admits negative indices as coordinates of the covering; a trajectory starting from an initial condition uses its admissible half-axis. This provides an operator inverse of the index without turning every extension relation into a bijection of the complete state.

With $D_M|k\rangle = [k/9]|k\rangle$, the identity $[D_M, S] = S$ expresses the memory increment on the dense domain of finitely supported vectors. If $\mathcal{I}|k\rangle = |-k\rangle$ and P_0 projects onto indices divisible by 9, one additionally obtains

$$\mathcal{I}T\mathcal{I} = T^{-1}, \quad \mathcal{I}D_M\mathcal{I} = -D_M - (I - P_0). \quad (6.29)$$

Indeed, for $k = 9m + r$, $[-k/9] = -m$ if $r = 0$ and equals $-m-1$ if $r > 0$. Inversion thus preserves the boundary term of Euclidean division.

6.15 Block clock and capacity clock

The index of an application of refinement and the number of available decimal positions have related but different laws. To express them, we reserve n for the ternary-block index and define

$$\rho = \log_{1000} 729 = \log_{10} 9, \quad \delta = 1 - \rho, \quad \mathcal{C}_n = \lfloor (n+1)\rho \rfloor, \quad n \geq 0.$$

The letter $\mathcal{C}_n$ here designates capacity, not the dodecaphase record K . At the canonical phase origin, the preceding mechanical word provides

$$Z_n = \sum_{j=0}^n z_j(0) = \lfloor (n+1)\delta \rfloor.$$

Proposition 6.23 (Exact balance between capacity and vacancies). *For $n \geq 0$ and, in the second identity, $n \geq 1$,*

$$\mathcal{C}_n = n - Z_n, \quad \mathcal{C}_n - \mathcal{C}_{n-1} = 1 - z_n(0).$$

Proof. The irrationality of δ implies $[(n+1)\delta] = [(n+1)\delta] + 1$. Then

$$[(n+1)(1-\delta)] = n+1 - [(n+1)\delta] = n - Z_n.$$

Subtracting two consecutive identities completes the proof. $\square$

An event $z_n = 1$ adds a block to the history while maintaining capacity $\mathcal{C}_n$; an event $z_n = 0$ increases both counters. In any interval, capacity increments and vacancies sum to exactly its length. Reducing modulo q , for $q = 9$ or $q = 108$, gives

$$[\mathcal{C}_n]_q = [n]_q - [Z_n]_q.$$

The uniform block phase and the capacity phase are related through the integrated memory of vacancies. In particular, a retention of capacity and a complete turn of the connection have different mechanisms, although both may preserve a phase observation.

Monotone inversion of the clock gives a first-arrival time:

$$T_{\text{cap}}(a) = \min\{n : \mathcal{C}_n = a\} = \left\lfloor \frac{a}{\rho} \right\rfloor - 1, \quad a \geq 1.$$

With $\sigma = \rho^{-1} - 1$, the time taken to gain b units of capacity is

$$T_{\text{cap}}(a+b) - T_{\text{cap}}(a) = b + [(a+b)\sigma] - [a\sigma].$$

The difference of ceilings takes one of the two integers adjacent to $b\sigma$. Thus 9 or 10 blocks are obtained to gain 9 units of capacity, and 113 or 114 to gain 108. These are times of the inverse clock, measured from a first arrival; the turn counter is always interpreted in the index to which it belongs.

6.16 Symbolic suspension and aperiodic time crystal

Rotation coding combines phase periodicity, local recurrence, and global aperiodicity. Let $\mathcal{X}_\delta \subseteq \{0, 1\}^{\mathbb{Z}}$ be the closure, in the product topology, of the translates of the bilateral word $z_n(\theta)$. Denote by σ_{sh} the shift of its sequences. Retaining a nonadic phase, the step is

$$F(r, z) = ([r+1]_9, \sigma_{\text{sh}}z), \quad (r, z) \in C_9 \times \mathcal{X}_\delta.$$

Definition 6.24 (Symbolic aperiodic time crystal). The unit-roof suspension of the preceding system is the space

$$\mathcal{S}_\delta = ((C_9 \times \mathcal{X}_\delta) \times \mathbb{R}) / \sim, \quad (y, t+1) \sim (Fy, t),$$

with flow $\Phi_s[y, t] = [y, t+s]$. This model of temporal order, together with its vacancy coding and representation of the turn counter, is called a *symbolic aperiodic time crystal*.

The adjective temporal designates the action of the flow and its return section; aperiodicity characterizes its itineraries. The frequency of ones in every sequence is δ : the uniform discrepancy bound of less than one, obtained by telescoping in any window, persists on taking the closure. A periodic sequence would have rational frequency. Therefore, $\mathcal{X}_\delta$ contains no periodic orbits. The suspension also has no closed orbits: a return of the flow would require an integer time and periodicity of the itinerary.

Finite words do recur. Itineraries of length m arise from a partition of the circle by $m+1$ points; each interval is visited recurrently by the irrational rotation. Even when times are restricted to a fixed phase, the step 9δ remains irrational. This is why every word appears in all 9 phases, and explains the already established complexities

$$p(m) = m+1, \quad p_9(m) = 9(m+1), \quad p_3(m) = 3(m+1).$$

Recurrence of local configurations coexists with the absence of a global period. Its topological entropy is zero because these complexities grow linearly.

Realization through the observables V_9, W_9 and shift T expresses the same order through the covariances (6.28). Frequencies of the uniform clock belong to $\frac{1}{9}\mathbb{Z} + \delta\mathbb{Z}$, modulo 1. The capacity clock additionally preserves the relation $[C_n]_9 = [n]_9 - [Z_n]_9$; it is not replaced by the uniform phase when interpreting returns. The term *time crystal* is thus defined through a mathematical dynamical system. Its physical realization requires specification of an energy dynamics, its temporal observables, and its stability, data distinct from the present symbolic construction.

6.17 Specular symmetry of the regions and double realization

Specular exchange acts on the regional block $B = (u^\pi, u^e, u^\varphi) \in (\mathbb{F}_3^6)^3$ by

$$cB = (u^e, u^\pi, u^\varphi).$$

Its fixed axis and aggregate are, respectively, u^φ and $s = u^\pi + u^e$. This fixation refers to the involution within a fiber; both coordinates may evolve during nonadic transport. With $q = u^\pi + u^e + u^\varphi$, $a = u^\pi + u^e - u^\varphi$, and $d = u^\pi - u^e$, one recovers

$$u^\varphi = 2(q - a), \quad s = q - u^\varphi, \quad u^\pi = 2(s + d), \quad u^e = 2(s - d).$$

The equalities hold componentwise in $\mathbb{F}_3$, where $2^{-1} = 2$. The mirror preserves q, a, s and reverses d ; this antisymmetric coordinate individualizes the distribution that the aggregate preserves without distinguishing it. The inversion $\mathcal{I}$ of the temporal index and the channel involution c consequently act on different data.

Phase return has an explicit witness:

$$\begin{aligned} B_1 &= (012222 \mid 121221 \mid 112202), \\ B_{10} &= (101222 \mid 112221 \mid 112002). \end{aligned}$$

Both positions have phase 1; the block, axis, and memory have changed. Likewise, synchronization of the censuses at phases 8 and 9 cancels their relative differences, while the diagonal component passes from $(59, 59, 59)$ to $(43, 43, 43)$. Equality of a relative observable does not exhaust the state determining it.

Double realization preserves that architecture. In Section 11.1, the same prefix determines nested cylinders on the one hand and a sequence of coded words with their frames on the other. The first map passes to the limit by contraction of diameters; the second satisfies $r_{n,m}Q_n = Q_m p_{n,m}$ and defines Q_∞ . Figure 6 anticipates these two maps. The subsequent incidence development constructs codes, lattices, and the realization of Moonshine in their own domains; the automorphism group acts on that realization, not on the digits through an implicit identification.

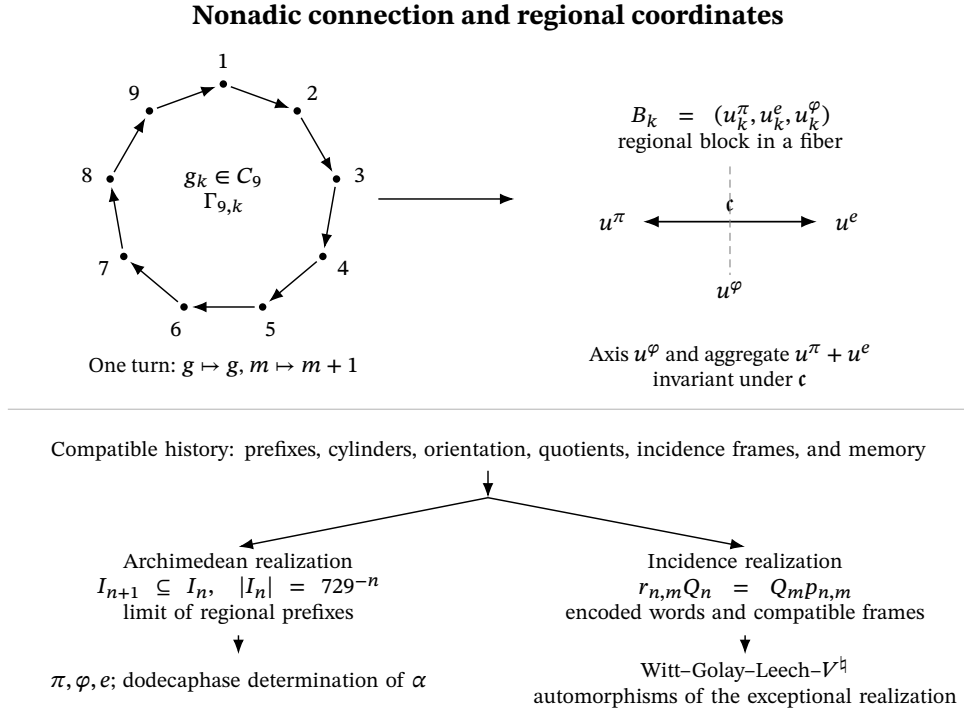


Figure 6: Nine phases of a connection and two realizations of a compatible history. The cycle represents phase; each turn increments the counter. The regional diagram represents the involution $\mathfrak{c}$, which interchanges closure and propagation and fixes the self-scaling axis in a fiber. This invariance does not require the axis to remain constant during transport. The two lower maps use the same prefix: one determines its numerical limit, while the other preserves words and incidence frames. The maps of the second realization are specified in Section 11.1; the exceptional construction is developed afterwards.

6.17.1 Rectilinear representation of phases and return

The representation on the segment $[0, 10]$ preserves the positional origin of APP and orders its nine interior marks. In this diagram, the integers $1, \dots, 9$ are phase representatives. Horizontal position indicates the order of the charts; upper labels specify regional operations. Figure 7 thus brings together the initial line, fixation of the self-scaling axis, the specular distribution of the two channels, and return with memory. The axis and aggregate relations and the regional census are made explicit below; return is governed by Proposition 6.21.

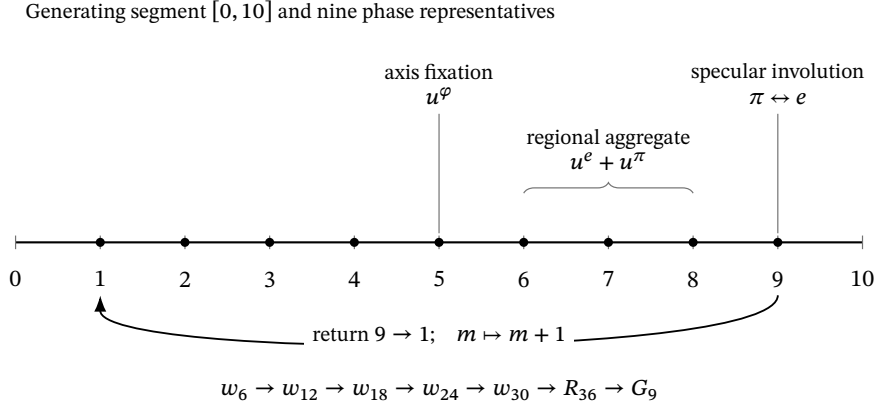


Figure 7: Rectilinear representation of the nonadic connection. The mark 5 locates the first strong saturation and the regional self-scaling axis; the brace 6–8 identifies the ternary aggregate fixed within each fiber; phase 9 selects the closure orientation. Return preserves phase and increases memory. The abscissae number operational positions, while the labels u^π, u^e, u^φ designate blocks of $\mathbb{F}_3^6$.

Fixation of the axis is formulated in the fiber of a given boundary. If $q = u^\pi + u^e + u^\varphi$ and $a = u^\pi + u^e - u^\varphi$, then

$$u^\varphi = 2(q - a), \quad s = u^\pi + u^e = q - u^\varphi \quad \text{in } \mathbb{F}_3^6.$$

The data (q, a) determine the axis and aggregate; the ordered distribution of s retains the specular freedom resolved by prolongation. The fifth phase has a local solution and an extension, with boundary datum $c = 111111$ and residual signature $\rho_W = 000$. This is the meaning of its first strong saturation. The symbol φ placed above the mark 5 identifies the regional axis entering that operation.

The intermediate phases preserve this decomposition within each fiber, while their boundary data evolve. The complete census of local multiplicities and extensions of the distinguished branch is

phase	1	2	3	4	5	6	7	8	9
N_{loc}	3	2	5	4	1	1	3	3	2
N_{ext}	2	5	4	1	1	3	3	2	1

Phase 6 combines local uniqueness and three prolongations; phase 7 resolves a triple specular fiber; phase 8 synchronizes the three regional censuses. Phase 9 preserves two local distributions with the same axis and aggregate, one of which prolongs. These counts express different functions within a single connection and justify the marks on the line.

In particular, for phases 6, 7, 8, the regional sums are respectively 012100, 101001, and 112000. The conservation represented by the brace is an invariance under the specular distribution within each fiber. Subsequent real evaluation applies the readers of compatible cylinders to complete histories; the internal sum represented in the figure belongs to the ternary domain.

Finally, the arc $9 \rightarrow 1$ represents passage from one turn to the next. The formulas of Section 6.14 preserve the phase remainder and increase the turn counter. The rectilinear diagram and the cyclic diagram in Figure 6 therefore express two complementary aspects of the same holonomy: the order of positions and composition of the return on the state with memory.

6.18 Temporal transport and tritic exponential composition

The Euler relations of Section 6.10 describe local realizations of the three regimes. For a fixed signature τ , the generator satisfies $J_\tau^2 = -\tau I$; algebraic conjugation sends J_τ to $-J_\tau$. Consequently,

$$\exp(sJ_\tau)\exp(tJ_\tau) = \exp((s+t)J_\tau), \quad \overline{\exp(tJ_\tau)} = \exp(-tJ_\tau),$$

and the product of an evolution with its conjugate is unity. In the elliptic type this is realized through $\cos^2 t + \sin^2 t = 1$; in the hyperbolic type, through $\cosh^2 t - \sinh^2 t = 1$; in the parabolic type, $(I + tN)(I - tN) = I$, since $N^2 = 0$. The same conservation law admits three quadratic expressions.

The elliptic realization closes a turn; the nilpotent one preserves a linear displacement; the hyperbolic one separates expansion and contraction with unit determinant. At the already generated regional parameters,

$$\exp(\pi J_e) = -I, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2.$$

The evaluation $\exp(I) = eI$ recognizes unity in the common exponential law. The number e is not assigned exclusively to the parabolic regime.

Transport of a local realization by an isomorphism U preserves these relations: if $J' = UJU^{-1}$, the exponential series gives $U\exp(tJ) = \exp(tJ')U$. The identity transports the regime and its norm. A transition between different quadratic types belongs, by contrast, to the TPK selection of sheets and regimes; it is not a conjugation changing the value of J^2 . The temporal product therefore preserves the order of transitions and the information at their endpoints.

This articulation distinguishes closure of a local representation from holonomy of a history. The identity $\exp(2\pi J_e) = I$ may hold at the same time as $S|r, m\rangle = |r, m+1\rangle$. The first equality restores the orientation of the regime; the second records another turn. The discrete structure preserves both: local return and the accumulated provenance of the state realizing it.

6.19 Multiplicative irreducibility and return periods

The distinction between structure, evaluation, and phase also appears in the multiplicative sector. APP's local product contains both residue and quotient; iterating it constructs normal forms and factorizations before their decimal periods are examined. This development places the expression *prime mode* in a precise domain and relates irreducibility to the operations already defined, rather than to an isolated coincidence of periods. The multiplicative recurrence below and its evaluation identity (6.30) distinguish these two properties.

Let

$$\mathcal{W}_9 = \{\emptyset\} \sqcup \bigsqcup_{\ell \geq 1} \{1, \dots, 9\}^\ell.$$

Sequences are ordered from the least significant digit. The evaluation $\text{ev}_9(u) = \sum_j u_j 9^j$, with $\text{ev}_9(\emptyset) = 0$, is bijective onto the nonnegative integers: for a positive integer, division of $n-1$ by 9 uniquely determines the first digit and continuation. This is bijective nonadic numeration; it retains the boundary representative 9.

Addition $\boxplus_\#$ is obtained by applying APP's additive sheet to the digits at each position and propagating their quotients. If there is only one digit, it is the provisional residue; a nonzero incoming quotient is normalized through a second additive operation. The sum of quotients is transported to the next position. To multiply a sequence by a digit d , the multiplicative cell produces $u_j d = 9q_j + r_j$, and the incoming quotient c_j is incorporated by

$$(w_j, c_{j+1}) = \begin{cases} (r_j, q_j), & c_j = 0, \\ (R^+(r_j, c_j), q_j + q^+(r_j, c_j)), & c_j > 0. \end{cases}$$

Start with $c_0 = 0$ and incorporate the final quotient when nonzero. During addition, $c_j \leq 2$; during multiplication by a digit, $c_j \leq 9$. Local operations thus receive arguments within their domain. Denote this second procedure by $d \odot_\# u$, with $d \odot_\# \emptyset = \emptyset$.

For $v = (v_0, \dots, v_{m-1})$, the complete product is constructed through the recurrence

$$a_m = \emptyset, \quad a_j = (9 \odot_{\#} a_{j+1}) \boxplus_{\#} (v_j \odot_{\#} u), \quad u \boxtimes_{\#} v = a_0.$$

The rule composes both local sheets. Its evaluation satisfies

$$\text{ev}_9(u \boxtimes_{\#} v) = \text{ev}_9(u) \text{ev}_9(v). \quad (6.30)$$

Indeed, each transition preserves the partial sum plus the pending quotient weighted by the power of 9. The recurrence then gives $\text{ev}_9(a_j) = 9 \text{ev}_9(a_{j+1}) + v_j \text{ev}_9(u)$, whose iteration proves (6.30). Uniqueness of normal form transports associativity and commutativity to the constructed product; its identity is (1).

Definition 6.25 (Reduced multiplicative coproduct). On the finitely supported vector space generated by nonempty normal forms, define

$$\tilde{\Delta}_{\#} e_w = \sum_{\substack{u \boxtimes_{\#} v = w \\ u, v \neq (1)}} e_u \otimes e_v.$$

A non-unit normal form is irreducible when its image under $\tilde{\Delta}_{\#}$ is zero.

Factorization fibers are finite by (6.30). In the completion $\mathcal{H}_{\#} = \ell^2(\mathcal{W}_9 \setminus \{\emptyset\})$, images of distinct forms have disjoint supports. If $d_{\#}(w)$ counts ordered non-unit factorizations, the domain of the closure is

$$\text{Dom}(\overline{\tilde{\Delta}_{\#}}) = \left\{ x \in \mathcal{H}_{\#} : \sum_w d_{\#}(w) |x_w|^2 < \infty \right\}.$$

Indeed, the squared norm of the image is this weighted sum. Hence the operator

$$A_{\#} = (\overline{\tilde{\Delta}_{\#}})^* \overline{\tilde{\Delta}_{\#}}$$

is diagonal, positive, and self-adjoint: its eigenvalue at e_w is the number of ordered non-unit factorizations of w . The projection

$$P_{\text{irr}} = (I - |e_{(1)}\rangle\langle e_{(1)}|) \mathbf{1}_{\{0\}}(A_{\#})$$

selects exactly the irreducible forms. Through multiplicative evaluation, these correspond to ordinary primes. Selection depends on factorization structure and keeps the unit state separate.

For a value $n \geq 2$ coprime to 10, decimal division induces the residue permutation $r \mapsto 10r \pmod{n}$. Return of residue 1 has period

$$\tau_n = \text{ord}_n(10).$$

If $n = \prod_p p^{a_p}$, the Chinese remainder theorem gives

$$\tau_n = \text{lcm}_{p^{a_p} \parallel n} \text{ord}_{p^{a_p}}(10).$$

Composite return synchronizes the returns of prime powers. For a prime $p \notin \{2, 3, 5\}$,

$$\tau_p \mid p-1, \quad M_p = \frac{10^{\tau_p} - 1}{p} \in \mathbb{N}, \quad M_p \equiv 0 \pmod{9}.$$

The last congruence expresses nonadic closure of decimal repetition: $10^{\tau_p} - 1$ is a multiple of 9, and p is invertible modulo 9. The same congruence holds for every n coprime to 90. In particular, 7, 13, 77, 91, 143 have decimal period 6, although only the first two are irreducible. The coproduct and the period therefore record different properties of the same constructed value.

This differentiation specifies the relation with aperiodic refinement. The vacancy sequence determines increments of the capacity index; factorization determines multiplicative modes; decimal division determines their return times. The nonadic representative preserves phase closure, while quotient and factorization retain information that this closure does not distinguish. The connection to the treatise's spectral constructions resides in these operators and their intertwining. Proof of their subsequent analytic results retains its own development; it is not replaced by periodicity of a residue observation.

6.19.1 Vacancy discrepancy and Dirichlet readings

The connection also admits an exact analytic expression. Retain the same coding $z_n = z_n(0)$, for $n \geq 1$, and consider two subsequent observables: the series over all indices and the product over indices selected by irreducibility. For $\text{Re } s > 1$,

$$Z_{\text{vac}}^+(s) = \sum_{n \geq 1} \frac{z_n}{n^s}, \quad Z_{\text{vac}}^\times(s) = \prod_p (1 - p^{-s})^{-z_p}.$$

Both converge absolutely in this half-plane. Their common factor is density δ , but their discrepancies differ.

Proposition 6.26 (Separation of vacancy density). *We have*

$$Z_{\text{vac}}^+(s) = \delta \zeta(s) + H_{\text{vac}}(s), \quad H_{\text{vac}} \in \mathcal{O}(\text{Re } s > 0).$$

In $\text{Re } s > 1$, with logarithms defined by absolutely convergent Euler series,

$$Z_{\text{vac}}^\times(s) = \zeta(s)^\delta G_{\text{vac}}(s), \quad \log G_{\text{vac}}(s) = \sum_p (z_p - \delta) \log(1 - p^{-s})^{-1}.$$

Proof. The centered partial sums satisfy

$$A_N := \sum_{n=1}^N (z_n - \delta) = \lfloor (N+1)\delta \rfloor - N\delta, \quad |A_N| < 1.$$

Abel summation represents the centered term by $s \int_1^\infty A_{\lfloor x \rfloor} x^{-s-1} dx$. The integral converges locally uniformly in $\text{Re } s > 0$, proving holomorphy. The second identity follows by separating $z_p = \delta + (z_p - \delta)$ in the logarithm of the product, within its domain of absolute convergence. $\square$

The decomposition reveals what the reading on primes adds: it examines the same vacancy sequence on a subset determined by factorization. Controlling discrepancy over all integers gives the first holomorphic continuation; the second observable preserves its own discrepancy on primes. These are two applications of the same discrete structure, with explicit analytic domains. This articulation explains their relevance to the arithmetic kernel without transferring the proof of the treatise's terminal results to this article.

7 Integral compatibility of regional generation, cylindrical refinement, and incidence

Regional prolongation produces ordered sequences of blocks together with their reading depths. The Archimedean projection and ternary incidence are calculated on these same sequences. The connection between the two realizations can be formulated entirely through integer operations: selection by rational intervals, Euclidean division, lifting of six trits, carry reading, and the Witt transformation. This formulation makes it possible to trace each coefficient from its generating block to its subsequent use.

The construction uses the three regional readers already obtained from the APP-TRIT-TPK state. We denote them by the indices $c \in \{P, E, F\}$, corresponding, respectively, to the closure, propagation, and autoscaling regions. Their real realization will be written as v_c . The subsequent identifications $v_P = \pi_{\text{HMT}}$, $v_E = e_{\text{HMT}}$, and $v_F = \varphi_{\text{HMT}}$ preserve this causal order. The operation that emits a block inspects the rational intervals of the regional reader; the real value enters the proof of correctness of that operation.

7.1 Positional emission through rational separation of cells

For each region, two rational sequences $\ell_c(j), u_c(j)$, constructed by its reader, are available and satisfy

$$\ell_c(j) \leq v_c \leq u_c(j).$$

The cell-separation property proved for these readers states that, for each integer $s > 0$, there exists J such that, for every $j \geq J$,

$$\lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1 = \lfloor sv_c \rfloor. \quad (7.1)$$

Avoidance of rational boundaries by the regional values is the hypothesis of the preceding theorems that guarantees this separation. Here we use their operational conclusion together with the readers and intervals that produce it.

Definition 7.1 (Separation index and emitted prefix). For $s > 0$, let

$$j_c(s) = \min\{j \in \mathbb{N} : \lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1\}.$$

For an integer base $b \geq 2$ and a depth $n \geq 0$, define

$$P_c(b, n) = \lfloor b^n \ell_c(j_c(b^n)) \rfloor. \quad (7.2)$$

The prefix contains the integer part and the first n positions in base b ; consequently, $P_c(b, 0)$ is the regional integer part.

Theorem 7.2 (Termination and correctness of emission). *The index $j_c(s)$ exists for every $s > 0$. The emitted prefixes satisfy*

$$P_c(b, n) = \lfloor b^n v_c \rfloor, \quad P_c(b, n) \leq b^n v_c < P_c(b, n) + 1. \quad (7.3)$$

For each scale s , a single sufficiency index $J(s)$ can also be chosen that is valid simultaneously for all three regions.

Proof. The eventual equality in (7.1) makes the set defining $j_c(s)$ nonempty. Its minimum exists by the well-ordering of $\mathbb{N}$. At any index where the equality holds, the rational lower bound gives $\lfloor s\ell_c(j) \rfloor \leq \lfloor sv_c \rfloor$. The upper bound, together with avoidance of a cell boundary, gives $\lfloor sv_c \rfloor \leq \lfloor su_c(j) \rfloor - 1$. When the two integer endpoints coincide, the intervening integer coincides with them. This proves (7.3). For the joint statement, it suffices to take the maximum of the three sufficiency indices obtained at the same scale. The joint index controls all three emissions simultaneously, while each region retains its own intervals. $\square$

Proposition 7.3 (Exact compatibility of prefixes). *For $n, m \geq 0$,*

$$\left\lfloor \frac{P_c(b, n+m)}{b^m} \right\rfloor = P_c(b, n). \quad (7.4)$$

In particular, the next block $d_c(b, n) = P_c(b, n+1) \bmod b$ satisfies

$$P_c(b, n+1) = bP_c(b, n) + d_c(b, n), \quad 0 \leq d_c(b, n) < b. \quad (7.5)$$

Proof. Write $b^n v_c = N + \theta$, with $N = P_c(b, n)$ and $0 \leq \theta < 1$. Then $P_c(b, n+m) = b^m N + \lfloor b^m \theta \rfloor$, and the second summand belongs to $\{0, \dots, b^m - 1\}$. Euclidean division by b^m recovers N . The specialization $m = 1$ yields (7.5) and its residual bound. $\square$

7.2 Regional blocks of six trits at arbitrary depth

The integer equality $729 = 3^6$ determines the grouping of six ternary positions into a block. For $t \geq 0$, define

$$u_c(t) = P_c(729, t+1) \bmod 729, \quad B_{c,j}(t) = \left\lfloor \frac{u_c(t)}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6. \quad (7.6)$$

Thus $B_c(t)$ is an ordered word of six trits, and

$$u_c(t) = \sum_{j=0}^5 B_{c,j}(t) 3^{5-j}. \quad (7.7)$$

The word and its integer lift are mutually recoverable representations of the same emitted block.

Theorem 7.4 (Stability of extraction at any depth). *For every n ,*

$$P_c(729, n) = P_c(3, 6n).$$

If $t < n$, then

$$u_c(t) = \left\lfloor \frac{P_c(729, n)}{729^{n-t-1}} \right\rfloor \bmod 729. \quad (7.8)$$

Consequently, every extension of the prefix preserves all previously emitted blocks and all signatures calculated from them.

Proof. The first equality follows because both prefixes are separated at the same scale, $729^n = 3^{6n}$, and from Theorem 7.2. For the second, apply (7.4) at depths n and $t+1$, and then take the residue modulo 729. The ternary extraction in (7.6) is deterministic. Any explicit function of those eighteen coordinates therefore retains the same result when the prefix is extended. $\square$

The construction is quantified over $t \in \mathbb{N}$. A table materializing one hundred blocks is a finite evaluation of this construction; formula (7.8) determines its compatibility with every subsequent extension exactly.

Example 7.5 (First joint block). Evaluation of the selected regional readers gives the words

$$B_P(0) = 010211, \quad B_E(0) = 201101, \quad B_F(0) = 121200.$$

Their lifting by (7.7) produces

$$u_P(0) = 103, \quad u_E(0) = 523, \quad u_F(0) = 450.$$

For example, $201101_3 = 2 \cdot 243 + 27 + 9 + 1 = 523$. All six positions, including leading zeros, are retained in the band representation. The leading zero in 010211 has a defined position even though the decimal notation for 103 does not contain it.

7.3 Transition registers and reconstruction of linear lifts

Over $\mathbb{F}_3$, collect two consecutive emissions from each region in the matrix

$$R(t) = \begin{pmatrix} B_P(t) \\ B_P(t+1) \\ B_E(t) \\ B_E(t+1) \\ B_F(t) \\ B_F(t+1) \end{pmatrix}. \quad (7.9)$$

This definition fixes both the regional order and the temporal order. The first four registers are

$$X_0 = R(0), \quad Y_0 = R(1), \quad X_1 = R(2), \quad Y_1 = R(3).$$

Their values, obtained by extraction from the regional prefixes, are

$$\begin{aligned} X_0 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, & Y_0 &= \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Theorem 7.6 (Unique reconstruction of the two initial links). *The equations $X_0 L_0 = Y_0$ and $X_1 L_1 = Y_1$ each have a unique solution in $M_6(\mathbb{F}_3)$. These solutions are*

$$\begin{aligned} L_0 &= \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, & L_1 &= \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}. \end{aligned}$$

Both matrices are invertible.

Proof. The matrices

$$\begin{aligned} A_0 &= \begin{pmatrix} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 & 2 & 2 \\ 2 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}, & A_1 &= \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix} \end{aligned}$$

satisfy $A_i X_i = X_i A_i = I$, by multiplication in $\mathbb{F}_3$. The solutions are therefore forced by $L_i = A_i Y_i$; multiplication produces the two displayed matrices. Their inverses are

$$L_0^{-1} = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}, \quad L_1^{-1} = \begin{pmatrix} 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

The bilateral verification $L_i L_i^{-1} = L_i^{-1} L_i = I$ proves exact recovery of each transformed word. All products use residues 0, 1, 2 with their interpretation in $\mathbb{F}_3$. $\square$

The causal reading of this result is precise: the regional emissions produce $R(t)$; the registers $R(0), \dots, R(3)$ determine the two lifts above; their inverses recover the transformed words. Continuity for every t belongs to the coinductive readers and Theorem 7.4. The matrix reconstruction proved here has the two links specified in its statement as its domain.

7.4 Simultaneous refinement in ternary and decimal bases

A ternary emission of six trits and a decimal emission of three digits are readings in different bases. To preserve their relationship, record the state

$$s = (n, k, N, D),$$

where N is a prefix at depth n in base 729, and D is a prefix at depth k in base 1000. Their Archimedean cylinders are

$$I_{729}(s) = \left[\frac{N}{729^n}, \frac{N+1}{729^n} \right), \quad I_{1000}(s) = \left[\frac{D}{1000^k}, \frac{D+1}{1000^k} \right). \quad (7.10)$$

The half-open upper endpoint assigns each value to a unique cell at a given base and depth.

Definition 7.7 (Integer compatibility defects). Define

$$A(s) = N1000^k - D729^n, \quad (7.11)$$

$$B(s) = (N+1)1000^k - D729^n = A(s) + 1000^k. \quad (7.12)$$

The state is compatible when

$$A(s) < 729^n, \quad B(s) > 0. \quad (7.13)$$

Lemma 7.8 (Integer criterion for intersection). *The cylinders in (7.10) have a nonempty intersection if and only if (7.13) holds.*

Proof. Two half-open intervals $[a, b)$ and $[c, d)$, with $a < b$ and $c < d$, intersect exactly when $a < d$ and $c < b$. Applying this condition to (7.10) and multiplying by the positive denominators gives $N1000^k < (D+1)729^n$ and $D729^n < (N+1)1000^k$. These are the two inequalities in (7.13). $\square$

Definition 7.9 (Update of depths and prefixes). For $0 \leq u < 729$ and $0 \leq d < 1000$, define

$$T_u(n, k, N, D) = (n+1, k, 729N+u, D),$$

$$T_{u,d}(n, k, N, D) = (n+1, k+1, 729N+u, 1000D+d).$$

The first operation refines only the ternary cylinder. The second refines both readings and preserves both depths in the state.

Proposition 7.10 (Exact boundary recurrences). *The updates above satisfy*

$$A(T_u s) = 729A(s) + u1000^k, \quad (7.14)$$

$$A(T_{u,d} s) = 729000A(s) + u1000^{k+1} - d729^{n+1}. \quad (7.15)$$

Proof. For the first identity,

$$(729N + u)1000^k - D729^{n+1} = 729(N1000^k - D729^n) + u1000^k.$$

For the second,

$$\begin{aligned} (729N + u)1000^{k+1} - (1000D + d)729^{n+1} \\ = 729000(N1000^k - D729^n) + u1000^{k+1} - d729^{n+1}. \end{aligned}$$

Both are integer identities preceding any real approximation. □

Theorem 7.11 (Refinement of the generated regional cylinders). *Let*

$$s_c(n, k) = (n, k, P_c(729, n), P_c(1000, k)).$$

Then $s_c(n, k)$ is compatible for all n, k . Moreover,

$$\begin{aligned} T_{u_c(n)}s_c(n, k) &= s_c(n + 1, k), \\ T_{u_c(n), d_c(1000, k)}s_c(n, k) &= s_c(n + 1, k + 1). \end{aligned}$$

Euclidean division of the refined prefixes recovers the preceding prefixes exactly. For every $a, b \geq 0$,

$$\frac{P_c(729, n + a) - P_c(729, n + a) \bmod 729^a}{729^a} = P_c(729, n),$$

and the analogous identity holds in base 1000.

Proof. Theorem 7.2 places v_c simultaneously in both intervals in (7.10); the intersection criterion gives compatibility. The update identities are (7.5) in each base. Recovery follows from (7.4). Thus the defect recurrence and prefix recovery are consequences of the same update. □

Example 7.12 (First pair of regional cylinders). For $n = k = 1$, evaluation of the three regions gives

c	N	D	A	B
P	2290	3141	211	1211
E	1981	2718	-422	578
F	1179	1618	-522	478

In all three rows, $A < 729$ and $B > 0$. The first prefix column decomposes as $2290 = 3 \cdot 729 + 103$, $1981 = 2 \cdot 729 + 523$, and $1179 = 729 + 450$: exactly the previously emitted six-trit blocks reappear. The different signs of A express the relative positions of the two lower endpoints without altering the regional value's membership in both intervals.

7.5 Selection of the next block by the regional reader

Compatibility of two intervals expresses a relationship between readings. Determining the next block additionally uses the rational interval produced by the region. This operation can be formulated without discarding the depth history.

Definition 7.13 (Admissibility of regional prolongation). For fixed c, n , let $s = 729^{n+1}$ and $j = j_c(s)$. A residue $u \in \{0, \dots, 728\}$ is admissible when

$$[s\ell_c(j)] \leq 729P_c(729, n) + u \leq [su_c(j)] - 1. \quad (7.16)$$

Theorem 7.14 (Uniqueness of the prolongation compatible with the reader). *Condition (7.16) is equivalent to $u = u_c(n)$. There is therefore exactly one admissible prolongation of each region at each depth.*

Proof. At the separation index, both integer endpoints in (7.16) coincide with $P_c(729, n + 1)$. The condition is therefore equivalent to $729P_c(729, n) + u = P_c(729, n + 1)$. By (7.5), the right-hand side is $729P_c(729, n) + u_c(n)$. Cancelling the common term gives equality of the residues. Conversely, this equality satisfies the condition. $\square$

The effective dependence of this selection is fully specified: selected region, rational reader, depth, separation index, preceding prefix, and next residue. The defect A summarizes a boundary relationship; the state (n, k, N, D) and the regional intervals retain the data used by the update.

7.6 Ternary signature, carry, Witt incidence, and reading clock

At a fixed depth t , write $B_{c,j} = B_{c,j}(t)$. For each column j , calculate the following integers and residues:

$$S_j = B_{P,j} + B_{E,j} + B_{F,j}, \quad q_j = S_j \bmod 3, \quad c_j = \lfloor S_j/3 \rfloor, \quad (7.17)$$

$$a_j = (B_{P,j} + B_{E,j} - B_{F,j}) \bmod 3, \quad w_j = \sum_c \mathbf{1}_{B_{c,j} \neq 0}, \quad \bar{w}_j = w_j \bmod 3. \quad (7.18)$$

The residue of the axial expression is taken in the integers. An implementation using natural numbers calculates it as $(B_{P,j} + B_{E,j} + 3 - B_{F,j}) \bmod 3$, which preserves that residue exactly.

The ternary incidence matrix used by the reader is

$$W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \quad (7.19)$$

Two residual charges are preserved for each region:

$$v_c^{\text{res}} = \sum_j B_{c,j} \bmod 3, \quad v_c^W = \sum_j (B_c W)_j \bmod 3. \quad (7.20)$$

The reader's complete signature is $\mathcal{S}(B) = (q, a, c, w, \bar{w}, v^{\text{res}}, v^W)$. Its structure separately preserves the column residue, carry, axial contrast, support weight, and the two charge readings.

Theorem 7.15 (Exact recovery and bounds of the signature). *For each column and each region,*

$$S_j = q_j + 3c_j, \quad 0 \leq q_j, a_j < 3, \quad 0 \leq c_j \leq 2, \quad (7.21)$$

$$B_{F,j} = 2(q_j + 3 - a_j) \bmod 3, \quad 0 \leq w_j \leq 3, \quad 0 \leq \bar{w}_j < 3. \quad (7.22)$$

Moreover,

$$v_c^W = \left(\sum_j ((B_c W)_j \bmod 3) \right) \bmod 3. \quad (7.23)$$

The stated recoveries determine each column's total and its autoscaling coordinate; the regional register additionally preserves the three ordered bands.

Proof. Identity (7.21) is Euclidean division of S_j by 3. Each summand of S_j lies between 0 and 2, so $0 \leq S_j \leq 6$ and $c_j \leq 2$. In $\mathbb{F}_3$, subtracting the two congruences defining q_j and a_j gives $q_j - a_j = 2B_{F,j}$. Since the inverse of 2 is 2, this yields (7.22); the summand 3 permits an expression using natural numbers. The bounds on w_j follow by summing three indicators. Finally, reducing an integer sum modulo 3 is equivalent to reducing each summand and then reducing their sum, which proves (7.23). $\square$

Example 7.16 (Signature of the first joint block). The three initial bands produce

$$\begin{aligned} q &= (0, 0, 2, 2, 1, 2), & a &= (1, 2, 0, 1, 1, 2), \\ c &= (1, 1, 0, 1, 0, 0), & w &= (2, 2, 2, 3, 1, 2), \\ v^{\text{res}} &= (2, 2, 0), & v^W &= (2, 1, 1). \end{aligned}$$

The fourth column has total $2 + 1 + 2 = 5$; its signature records $q_3 = 2$ and $c_3 = 1$, so $q_3 + 3c_3 = 5$. Axial recovery gives $2(2 + 3 - 1) \bmod 3 = 2$, precisely the autoscaling coordinate. The Witt images, reduced componentwise, are

$$(2, 0, 2, 1, 1, 2), \quad (0, 0, 0, 2, 0, 2), \quad (2, 1, 1, 2, 1, 0).$$

Their residual sums by row yield the three charges $(2, 1, 1)$.

7.7 The carry cocycle and the integer lift of the regional signature

The regional signature retains the residue and quotient simultaneously. The following construction makes the algebraic law of the latter explicit. APP supplies integer evaluations on its two sheets; TRIT retains orientation and regime; TPK transport produces the regional bands and updates their memory. In a column of these bands, Euclidean division determines the carry. The construction uses the representatives $\{0, 1, 2\}$ of $\mathbb{F}_3$; the oriented TRIT representation separately retains its sign and transport quotient.

Let $(p, e, f) \in \{0, 1, 2\}^3$ be the entries of a regional column. In this subsection, e denotes the ternary entry of the propagation region. Define

$$\begin{aligned} q &= [p + e + f]_3, & c &= (p + e + f - q)/3, \\ a &= [p + e - f]_3, & h &= (p + e - f - a)/3, \end{aligned} \tag{7.24}$$

where $[z]_3 \in \{0, 1, 2\}$ is the remainder of any integer z . The variable a is the axial coordinate of this column. The constant α_{HMT} retains its subsequent dodecaphasic construction.

Proposition 7.17 (Local composition of carries). *For $x, y, z \in \{0, 1, 2\}$, let*

$$\kappa_3(x, y) = \frac{x + y - [x + y]_3}{3}, \quad \beta_3(x, y) = \frac{x - y - [x - y]_3}{3}.$$

With $r = [p + e]_3$, the two laws are

$$c = \kappa_3(p, e) + \kappa_3(r, f), \quad h = \kappa_3(p, e) + \beta_3(r, f).$$

Moreover,

$$\kappa_3(x, y) + \kappa_3([x + y]_3, z) = \kappa_3(y, z) + \kappa_3(x, [y + z]_3). \tag{7.25}$$

Proof. Adding $p + e = r + 3\kappa_3(p, e)$ to $r + f = q + 3\kappa_3(r, f)$ gives the first formula. Subtracting f and using $r - f = a + 3\beta_3(r, f)$ gives the second. Both sides of (7.25) equal $(x + y + z - [x + y + z]_3)/3$. Composition therefore retains the same quotient under either association of the sum. $\square$

If $s_3 : \mathbb{F}_3 \rightarrow \mathbb{Z}$ is the representative section, then

$$s_3(x) + s_3(y) - s_3(x + y) = 3\kappa_3(x, y).$$

This is the cocycle of the extension $0 \rightarrow \mathbb{Z} \xrightarrow{\times 3} \mathbb{Z} \rightarrow \mathbb{F}_3 \rightarrow 0$. It arises from lifting a residue operation to integer values. The visible coordinate and the memory of the multiple of three are related by an exact identity.

Table 7: Complete law of the 27 regional columns. The values of h are displayed as integers.

p	e	f	q	c	a	h
0	0	0	0	0	0	0
0	0	1	1	0	2	-1
0	0	2	2	0	1	-1
0	1	0	1	0	1	0
0	1	1	2	0	0	0
0	1	2	0	1	2	-1
0	2	0	2	0	2	0
0	2	1	0	1	1	0
0	2	2	1	1	0	0
1	0	0	1	0	1	0
1	0	1	2	0	0	0
1	0	2	0	1	2	-1
1	1	0	2	0	2	0
1	1	1	0	1	1	0
1	1	2	1	1	0	0
1	2	0	0	1	0	1
1	2	1	1	1	2	0
1	2	2	2	1	1	0
2	0	0	2	0	2	0
2	0	1	0	1	1	0
2	0	2	1	1	0	0
2	1	0	0	1	0	1
2	1	1	1	1	2	0
2	1	2	2	1	1	0
2	2	0	1	1	1	1
2	2	1	2	1	0	1
2	2	2	0	2	2	0

Theorem 7.18 (The additional algebraic component of carry). *Over $\mathbb{F}_3$, the functions q, a are linear in (p, e, f) . The reductions of c, h have polynomial representations*

$$\begin{aligned} \bar{c} = & 2ef + 2ef^2 + 2e^2f + 2pf + 2pf^2 \\ & + 2pe + pef + 2pe^2 + 2p^2f + 2p^2e, \end{aligned} \quad (7.26)$$

$$\begin{aligned} \bar{h} = & 2f^2 + ef + 2ef^2 + e^2f + pf + 2pf^2 \\ & + 2pe + 2pef + 2pe^2 + p^2f + 2p^2e. \end{aligned} \quad (7.27)$$

If V is the evaluation matrix of $1, p, e, f$ on the 27 columns, then

$$\text{rank}_{\mathbb{F}_3} V = 4, \quad \text{rank}_{\mathbb{F}_3}(V \mid \bar{c}) = 5, \quad \text{rank}_{\mathbb{F}_3}(V \mid \bar{h}) = 5.$$

Proof. Substituting the 27 triples from the table into the two polynomials reproduces $\bar{c}, \bar{h}$. The representation of degree at most two in each variable is unique: a polynomial of degree at most two in one variable that vanishes at all three elements of $\mathbb{F}_3$ is zero. Applying this argument successively to the three variables proves independence of the 27 monomials.

Evaluation at $0, (1, 0, 0), (0, 1, 0), (0, 0, 1)$ proves independence of $1, p, e, f$. The function $\bar{c}$ vanishes at these four inputs and equals one at $(1, 1, 1)$; it therefore lies outside their affine evaluation space. For $\bar{h}$, the first three inputs force the constant, p , and e coefficients to zero, while $\bar{h}(0, 0, 1) = 2$ forces the f coefficient to two. That affine function would equal one at $(0, 0, 2)$, whereas $\bar{h}(0, 0, 2) = 2$. Each additional column therefore increases the rank by one. $\square$

The proof specifies the information supplied by the quotient. Residual superposition produces q, a ; integer evaluation also incorporates c, h . For $c \in \{0, 1, 2\}$, the residue determines its integer value; for $h \in \{-1, 0, 1\}$, it does so together with this declared choice of representatives. The polynomials describe the reductions; complete memory also retains these lifts.

7.7.1 Transport of the regional quotients

Let M be an integer transport matrix already fixed at a stage of the construction, with row states x_t^χ for $\chi \in \{\pi, e, \varphi\}$. Perform the following division coordinatewise:

$$y_t^\chi = x_t^\chi M, \quad u_t^\chi = [y_t^\chi]_3, \quad x_{t+1}^\chi = (y_t^\chi - u_t^\chi)/3.$$

Define $X_t^+ = x_t^\pi + x_t^e + x_t^\varphi$ and $X_t^\sigma = x_t^\pi + x_t^e - x_t^\varphi$. Applying the preceding laws columnwise gives

$$\begin{aligned} X_{t+1}^+ &= \frac{X_t^+ M - q_t}{3} - c_t, \\ X_{t+1}^\sigma &= \frac{X_t^\sigma M - a_t}{3} - h_t. \end{aligned} \tag{7.28}$$

Indeed, the sum of the three residues is $q_t + 3c_t$; their signed combination is $a_t + 3h_t$. Substitution of both identities into division of the states proves (7.28). The statement applies to the declared integer transports. Regional emissions and their continuity retain the coinductive readers previously constructed.

As a complete evaluation on a six-column band,

$$u^\pi = 021020, \quad u^e = 001220, \quad u^\varphi = 100210$$

give $q = 122120$, $c = 000110$, $a = 222000$, and $h = (-1, 0, 0, 1, 0)$. The orientation $\sigma = (1, 1, -1)$ has doubled residue $2\sigma = (2, 2, 1)$ in $\mathbb{F}_3^3$. This last equality records the sign convention and is distinguished from the six-column bands.

7.7.2 Boundary memory and interval reading

The local carry law couples to the previously established cylinder refinement. One example exhibits the role of the complete interval:

$$\left[\frac{1}{729}, \frac{2}{729} \right) \cap \left[\frac{2}{1000}, \frac{3}{1000} \right) = \left[\frac{2}{1000}, \frac{2}{729} \right) \neq \emptyset.$$

By contrast, $\lfloor 1000/729 \rfloor = 1$. The lower-endpoint reading returns block 1, whereas interval compatibility also admits cell 2. The boundary state retains both bounds needed to decide that compatibility.

For one column, fixing q, a fixes

$$f = 2(q - a), \quad p + e = q - f \quad \text{in } \mathbb{F}_3.$$

Each of the nine fibres of (q, a) contains exactly three triples: choosing p determines e and leaves f fixed. On six columns, before additional constraints are imposed, the freedom is the product of these six fibres and has cardinality $3^6 = 729$. Fixing the coordinate refers to alternatives at one boundary. As depth increases, the register and its memory change under transport. Local carry, cylinder bounds, and regional incidence therefore act jointly on the same arithmetic history.

7.8 Integer clock of decimal reading

Definition 7.19 (Integer clock of decimal reading). The transition index t and decimal representation depth are related by

$$\kappa(t) = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\}. \quad (7.29)$$

Proposition 7.20 (Characterization and monotonicity of the clock). *The integer $\kappa(t)$ is uniquely determined by*

$$1000^{\kappa(t)} \leq 729^{t+1} < 1000^{\kappa(t)+1}. \quad (7.30)$$

The function κ is monotone and satisfies $\kappa(20) = \kappa(21) = 20$.

Proof. The powers of 1000 are strictly increasing and unbounded; there is therefore a unique interval between consecutive powers containing 729^{t+1} . Monotonicity of the latter expression proves that of κ . The final two evaluations are the integer comparisons

$$1000^{20} \leq 729^{21} < 1000^{21}, \quad 1000^{20} \leq 729^{22} < 1000^{21}.$$

These comparisons use exact integer powers. □

Equality of decimal depth at two distinct transitions explains why both indices must be retained: the reading clock can remain constant while the ternary band advances and its signature is updated. Chronological information resides in the ordered register of transitions.

7.9 Composition theorem for the block, cylinder, and signature

Theorem 7.21 (Joint compatibility of regional readings). *For each depth n , there exists a unique triple of blocks (u_P, u_E, u_F) satisfying regional admissibility (7.16) in all three regions. This triple is $(u_P(n), u_E(n), u_F(n))$. Simultaneously:*

1. *each block updates its cylinder through recurrences (7.14) and (7.15);*
2. *its six trits produce the signature $\mathcal{S}(B_P(n), B_E(n), B_F(n))$;*
3. *the column totals are recovered as $q + 3c$, and the Witt charges are obtained from the same bands through (7.23);*
4. *every deeper prefix returns exactly these blocks, truncated cylinders, and signatures.*

Proof. Apply Theorem 7.14 to each of the three regional indices. Coordinatewise uniqueness gives joint uniqueness. Substitute those same residues into the operations $T_u, T_{u,d}$, which allows Theorem 7.11 to be applied. The deterministic application of (7.6) produces the bands used by (7.17)–(7.20). The recovery identities are those of Theorem 7.15. Finally, Theorem 7.4 and truncation in each base prove stability at greater depth. At every step the identity of the emitted block has been preserved; neither reading requires the selection of a different block. □

The conclusion is an operational compatibility between cylindrical contraction and preservation of incidence. Both use the same generated register, retain the depths, and admit integer recovery. This result is the explicit form of the connection

$$\text{emitted regional block} \longrightarrow \begin{cases} \text{cylinder and integer boundary,} \\ \text{band, signature, carry, and Witt charge.} \end{cases}$$

The subsequent development of incidence and fields uses the second reading while preserving its common provenance with the first.

7.10 Calendar census and simultaneous compatibility of the regional biographies

The preceding rows are assembled, without yet applying any decimal reader, into the three block biographies produced by the transitions:

$$\begin{aligned} B^{\text{cl}} &= 010211|012222|010211|002111|110221, \\ B^{\text{pr}} &= 201101|121221|102011|012222|102011, \\ B^{\text{au}} &= 121200|112202|121020|010210|010200. \end{aligned} \tag{7.31}$$

Indeed, the consecutive pairs of the first two segments are the rows of X_0, Y_0 , and those of the last two are the rows of X_1, Y_1 . Thus, the biographies are another representation of the record of twelve transitions, not a collection of conventional prefixes.

To verify the operative word of the calendar, let $c = c_1c_2c_3c_4 \in \{0, 1\}^4$ and, for each regime $\chi \in \{\text{cl}, \text{pr}, \text{au}\}$, define

$$u_0^\chi = b_0^\chi, \quad u_j^\chi = u_{j-1}^\chi L_{c_j} \quad (1 \leq j \leq 4).$$

To condense the census, denote by

$$N_{\text{ar}}(c) := \#\{(j, \chi) : u_j^\chi = b_j^\chi, 1 \leq j \leq 4\}, \quad N_{\text{bio}}(c) := \#\{\chi : (u_0^\chi, \dots, u_4^\chi) = B^\chi\}.$$

Exact multiplication in $\mathbb{F}_3$ for the sixteen calendars gives:

c	$N_{\text{ar}}(c)$	$N_{\text{bio}}(c)$
0000, 0001	6	0
0010	9	0
0011	12	3
0100, 0101, 0110, 0111	3	0
1000, 1001, 1010, 1011	0	0
1100, 1101, 1110, 1111	0	0

(D)

The table exhausts the $2^4 = 16$ possibilities, and each of its entries is obtained by four multiplications of a row by one of the two displayed matrices. Consequently, 0011 is the unique calendar that simultaneously reconstructs the three complete biographies. In operator notation,

$$(L_0, L_0, L_1, L_1) \tag{7.32}$$

is not an independent prescription: it is the unique binary reading compatible with the twelve edges already produced by U_t .

The calendar (7.32) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate cumulative length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace their production by U_t . Conversely, (D) proves within the article the uniqueness of the ordering of the four transports. The causal chain is thereby closed: the APP census produces the initial regions, TRIT fixes regime and orientation, TPK produces the transition record, and that record determines L_0, L_1 and 0011 before any Archimedean realization.

8 Regional determination of the dodecaphasic descriptor and selection of the coupling register

The determination of the coupling register combines two constructions that should be developed in their actual order. The first produces the inputs to the selector from the regional representations of the TPK: six-trit bands, column signatures, a conversion calendar, an oriented descriptor, and a functional panel. The second operates on these inputs and on the terminal eight-block repertoire, enumerates its admissible orderings, and selects a unique register through two simultaneous conditions: exceptional incidence and reading heterotypy. The value of the register appears at the end of this composition.

The present development presupposes the regional representations obtained in the preceding chapters from APP-TRIT-TPK. Its purpose is to make explicit every intermediate operation leading from those representations to the finite selector. The realization of the same register through signed incidences, Hadamard inversion, and the representation of the fine-structure constant are then incorporated. The three constructions retain their respective domains: selection of the register, recovery from its channels, and arithmetic evaluation of its representations.

8.1 Regional bands, encoding, and integer signatures

8.1.1 Domains and encoding of the regional bands

The channel order is fixed as

$$\chi \in \{\pi, e, \varphi\}, \quad B_\chi(t) \in \mathbb{F}_3^6, \quad t \in \mathbb{N}.$$

The channel symbols identify the regions already generated. Each $B_\chi(t)$ retains its place in its sequence of representations; the index t counts ternary blocks. The natural-number evaluation of a band $b = (b_0, \dots, b_5)$ is

$$\text{enc}_6(b) = 243b_0 + 81b_1 + 27b_2 + 9b_3 + 3b_4 + b_5, \quad 0 \leq \text{enc}_6(b) < 729. \quad (8.1)$$

Its fixed-width inverse is

$$\text{dig}_j(n) = \left\lfloor \frac{n}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6.$$

Leading zeros are thus retained: the words 002001 and 2001 could share an abbreviated integer notation, but the type of the former contains six positions. All subsequent concatenations use this width.

Within the hundred-block window, the six-hundred-trit representation of channel χ determines an integer $P_\chi^{(600)}$. Extraction is performed by

$$b_\chi(t) = \left\lfloor \frac{P_\chi^{(600)}}{729^{99-t}} \right\rfloor \bmod 729, \quad 0 \leq t < 100, \quad B_\chi(t) = \text{enc}_6^{-1}(b_\chi(t)). \quad (8.2)$$

The six-hundred-trit integer is a representation of the regional producer; the hundred-row table is the evaluation of (8.2). This separation allows a historical table to be replaced by its producer without changing the selector that consumes it.

8.1.2 Residues, quotients, occupancy, and orientation

For each column j and time t , introduce

$$T_j(t) = B_\pi(t)_j + B_e(t)_j + B_\varphi(t)_j, \quad (8.3)$$

$$q_j(t) = T_j(t) \bmod 3, \quad c_j(t) = \lfloor T_j(t)/3 \rfloor, \quad (8.4)$$

$$a_j(t) = (B_\pi(t)_j + B_e(t)_j - B_\varphi(t)_j) \bmod 3, \quad (8.5)$$

$$w_j(t) = \mathbf{1}_{B_\pi(t)_j \neq 0} + \mathbf{1}_{B_e(t)_j \neq 0} + \mathbf{1}_{B_\varphi(t)_j \neq 0}, \quad \bar{w}_j(t) = w_j(t) \bmod 3. \quad (8.6)$$

The subtraction in the third expression is performed in the integers and then reduced. In an implementation over natural numbers, the equivalent expression is $(B_\pi + B_e + 3 - B_\varphi) \bmod 3$; truncated subtraction would change the reader.

Proposition 8.1 (Reconstruction of the column sum). *For all admissible bands,*

$$T_j(t) = q_j(t) + 3c_j(t), \quad 0 \leq q_j(t) < 3, \quad 0 \leq c_j(t) \leq 2, \quad 0 \leq w_j(t) \leq 3.$$

Proof. Each column sums three representatives from $\{0, 1, 2\}$, so that $0 \leq T_j(t) \leq 6$. Euclidean division by three yields the stated residue and quotient. Occupancy counts three binary conditions; its bound follows directly from that definition. $\square$

This identity preserves the quotient of the local sum. Reconstructing the three individual bands requires the bands themselves or an additional reader that distinguishes them: a column sum has fewer coordinates than the triple that produces it. Accordingly, the temporal row used below jointly retains the three bands and the four fields $q, a, c, \bar{w}$.

8.2 Conversion clock and first zero advance

8.2.1 Ternary capacity and decimal depth

A representation of $t + 1$ bands contains $6(t + 1)$ trits. The associated conversion calendar is defined by the integer

$$d_t = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\} = \max\{k \in \mathbb{N} : 1000^k \leq 729^{t+1}\}. \quad (8.7)$$

The definition compares integer capacities. The counter t and the depth d_t record different quantities: a transition adds one six-trit band; the clock counts complete powers of the decimal base one thousand contained in the accumulated ternary capacity.

Lemma 8.2 (Integer characterization of the clock). *For $t, k \in \mathbb{N}$,*

$$d_t = k \iff 1000^k \leq 729^{t+1} < 1000^{k+1}.$$

Moreover, $d_t \leq d_{t+1}$.

Proof. The powers of one thousand form a strictly increasing, unbounded sequence. The power 729^{t+1} lies between two consecutive terms; the lower exponent is precisely the maximum in (8.7). The inequality $729^{t+1} < 729^{t+2}$ implies monotonicity of the maximum. $\square$

Proposition 8.3 (First zero advance). *The least time satisfying $d_{t+1} = d_t$ is*

$$t_* = 20.$$

In particular,

$$d_0 = 0, \quad d_{19} = 19, \quad d_{20} = 20, \quad d_{21} = 20, \quad d_{22} = 21.$$

Proof. The exact integer comparisons

$$1000^{20} \leq 729^{21}, \quad 729^{22} < 1000^{21}, \quad 1000^{21} \leq 729^{23} < 1000^{22}$$

give the values at times twenty, twenty-one, and twenty-two. For $0 \leq t \leq 20$, the expression $729^{t+1}/1000^t = 729(729/1000)^t$ is decreasing and its value at twenty is greater than or equal to one. Hence, $1000^t \leq 729^{t+1} < 1000^{t+1}$ and $d_t = t$. No time less than twenty can satisfy $d_{t+1} = d_t$; at twenty it does hold. All comparisons used involve integer powers. $\square$

The number twenty is therefore an output of the minimality criterion. The rule selecting the instant is “the first zero advance of the clock”; the algorithm and its proof can use this rule without receiving twenty as an external parameter.

8.2.2 Nonadic return and temporal antecedent

The length of the phase return determines

$$r = t_* - 9 = 11. \quad (8.8)$$

The descriptor queries are made at $t_* + 1$, $r + 1$, and r . This distribution preserves the relation between the pause instant, the nonadic translation, and the immediately associated phase reading. Times eleven, twelve, and twenty-one designate positions in the regional calendar; their temporal difference remains recorded even when two times produce the same decimal depth.

8.3 Oriented construction of the twenty-four-trit descriptor

8.3.1 Opposite half-band defects

Take the oriented defects of the regional chart

$$\delta_+ = (0, 0, 0, 1, 1, 1), \quad \delta_- = (0, 0, 0, 2, 2, 2) \in \mathbb{F}_3^6.$$

Their opposition is verified componentwise:

$$\delta_+ + \delta_- = 0 \quad \text{in } \mathbb{F}_3^6.$$

The representative 2 realizes -1 in the ternary chart. Concatenating the three-position tails in negative-positive order produces the word 222111. Adding the defects and concatenating their tails are different operations: the former has codomain $\mathbb{F}_3^6$, whereas the latter preserves the order of two segments of length three.

8.3.2 Dual charge and phase selection

In the regional chart, the six-coordinate transformation uses

$$A_{\text{reg}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

With row vectors, the dual charge of $b \in \mathbb{F}_3^6$ is the sum of the coordinates of bA_{reg} . The integer row sums of A_{reg} are $(5, 7, 7, 7, 7, 7)$; this gives the local expression

$$\eta^\vee(b) = 2b_0 + b_1 + b_2 + b_3 + b_4 + b_5 \pmod{3}. \quad (8.9)$$

The reading phase at instant r is $\theta_r = (r - 1) \bmod 3$. For each channel, define

$$\varepsilon_\chi(r) = \begin{cases} 1, & \eta^\vee(B_\chi(r)) = \theta_r, \\ 2, & \eta^\vee(B_\chi(r)) \neq \theta_r. \end{cases}$$

The phase-charge representation is

$$D(r) = \varepsilon_\pi(r)B_\pi(r) + \varepsilon_e(r)B_e(r) + \varepsilon_\varphi(r)B_\varphi(r) \quad \text{in } \mathbb{F}_3^6. \quad (8.10)$$

The coefficient is determined by an equality of charges before evaluating the word that will result from the combination.

8.3.3 Composition of the descriptor

Let $\parallel$ denote the concatenation of fixed-width words. The complete construction is

$$\Omega_{24} = (B_e(t_* + 1) + \delta_-) \parallel (\delta_-|_{3,4,5} \parallel \delta_+|_{3,4,5}) \parallel (B_\pi(r + 1) + B_e(r + 1)) \parallel D(r). \quad (8.11)$$

Each length-six summand is evaluated in $\mathbb{F}_3^6$; the final concatenation has length $6 + 6 + 6 + 6 = 24$. This construction distinguishes the descriptor Ω_{24} from the regional stages w_{24}^χ : the two objects share a length, but their producers and their places in the composition differ.

Proposition 8.4 (Regional evaluation of the descriptor). *The regional representations and the first-zero-advance criterion determine*

$$\Omega_{24} = 020022\ 222111\ 211201\ 021101, \quad [\Omega_{24}]_3 = 66241910521.$$

Proof. The required bands, preserving the order (π, e, φ) , are

t	$B_\pi(t)$	$B_e(t)$	$B_\varphi(t)$
11	022212	110001	100021
12	012012	202222	000120
21	221022	020100	211021

and come from extraction (8.2). At time twenty-one, $020100 + 000222 = 020022$ in $\mathbb{F}_3^6$. The defect tails provide 222111. At time twelve, $012012 + 202222 = 211201$.

At time eleven, the dual charges are 0, 1, 2, respectively; the phase is $(11 - 1) \bmod 3 = 1$. The resulting coefficients are (2, 1, 2) and

$$2(022212) + (110001) + 2(100021) = 021101 \quad \text{in } \mathbb{F}_3^6.$$

Concatenating these four results gives the stated word. Its integer evaluation is obtained by Horner's method in base three, or by three successive concatenations in base 729. $\square$

The preceding table serves a verification purpose: each of its entries is obtained from the regional producer. Construction rule (8.11) makes it possible to reproduce the descriptor without introducing the final twenty-four-trit string as an input.

8.4 Functional panel and terminal repertoire

8.4.1 Nine regional positions

The functional panel retains the first three bands of each of the three channels, in the declared regional order:

$$\mathcal{P}_9 = (b_\pi(0), b_\pi(1), b_\pi(2), b_e(0), b_e(1), b_e(2), b_\varphi(0), b_\varphi(1), b_\varphi(2)). \quad (8.12)$$

Its evaluation is

$$\mathcal{P}_9 = (103, 161, 103, 523, 457, 301, 450, 398, 438).$$

The nine positions retain their regional provenance, although the value 103 occurs twice. The panel has nine ordered incidences and eight distinct values. The subsequent enumeration uses the incidences; candidate deduplication takes place at a later stage.

8.4.2 Unordered eight-block repertoire

The terminal repertoire involved in this construction is

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\} \subset \mathbb{F}_3^6. \quad (8.13)$$

Its eight elements are distinct. The integer encoding, sorted by value solely to fix a computational representation, is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

The documentary provenance of this repertoire is the terminal segmentation of the seventy-two-trit word and its position-wise recovery through incidences of the phase-charge and affine readers. The selector developed below receives the repertoire without its historical ordering and tests its 8! orderings.

The exact formulation of the reassembled implementation maintains an explicit interface here: the regional producers materially replace the descriptor, the panel, and the temporal rows; repertoire (8.13) retains its earlier documentary producer. This distinction identifies where each input enters. Membership of eight words in a temporal reader repertoire, selection of those eight words, and determination of their terminal order are three different statements. The terminal-selection proof establishes the third for the declared repertoire.

Documentary recovery by position can be expressed exactly. Let $\mathcal{I}$ be the phase-charge incidence register and let $\mathcal{J}$ be the register of the affine extension. Each incidence retains the position ℓ in the seventy-two-trit word and the emitted word u . For $5 \leq \ell \leq 12$, form

$$\mathcal{S}_\ell = \{u : (\ell, u) \in \mathcal{I} \cup \mathcal{J}\}.$$

The recovery procedure checks that all eight positions occur and that each $\mathcal{S}_\ell$ is a singleton. If s_ℓ is its unique element, the output delivered to the selector is

$$\mathcal{S}_{\text{term}} = \{s_5, s_6, \dots, s_{12}\}.$$

The singleton proof for each position preserves the coherence of the historical witnesses; the subsequent export retains the eight words and removes only the historical ordering. The selector reconstructs the order through complete enumeration and the terminal conditions, instead of receiving it from the provenance register.

8.4.3 Temporal incidences of the terminal repertoire and recovery by position

The representation of the terminal repertoire uses the same regional blocks, the same carries, and the same calendar that produce the initial twenty-four-trit descriptor. Two operations in this composition should be made explicit separately. The first recovers eight words from recorded temporal incidences. The second presents those eight words to the dodecaphasic selector as an unordered set. The selector re-establishes the order through its incidence, cylinder, and closure conditions. The distinction preserves the provenance witnesses without turning the order of a historical segmentation into an input to the selector.

Let $B_t = (b_{t,0}, b_{t,1}, b_{t,2}) \in (\mathbb{F}_3^6)^3$ be the regional boundary already generated at time t , with $0 \leq t < 100$. In the chart used by the historical register, the incidence matrix is

$$A_{\text{hist}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

The words are regarded as row vectors. Their two charges and the chronological phase are

$$r_{t,i} = \sum_{j=1}^6 b_{t,i,j}, \quad d_{t,i} = \sum_{j=1}^6 (b_{t,i} A_{\text{hist}})_j, \quad \vartheta_t = t - 1 \pmod{3}.$$

All operations in these three expressions take place in $\mathbb{F}_3$. The historical subscript identifies a specific chart of the Witt incidence; it keeps the coordinate transformation explicit when another presentation of that same incidence is used.

For $c \in \{r_t, d_t\}$, $\delta \in \mathbb{F}_3$ and $\epsilon \in \{1, -1\}$, define

$$\eta_i(c, \delta, \epsilon) = \begin{cases} \epsilon, & c_i = \vartheta_t + \delta, \\ -\epsilon, & c_i \neq \vartheta_t + \delta, \end{cases} \quad L_{t,c,\delta,\epsilon} = \sum_{i=0}^2 \eta_i(c, \delta, \epsilon) b_{t,i}.$$

The orientation -1 is written as 2 when printing the ternary coefficients. Alongside these twelve comparative readers per boundary, the register has the affine reader

$$L_t^{\text{af}} = \sum_{i=0}^2 (d_{t,i} + \vartheta_t) b_{t,i}.$$

Adding the phase in the affine reader and comparing charges in the preceding reader are different operations, both acting on data already produced by the TPK boundary.

t	$b_{t,0}$	$b_{t,1}$	$b_{t,2}$	r_t	d_t
8	200121	212212	110110	011	202
11	022212	110001	100021	001	012
30	210112	110202	211110	100	012
34	121022	000011	112110	220	021
37	120121	011202	112221	100	201
49	110212	200212	100122	110	201
58	111121	022121	122201	122	220
67	010012	001220	011120	122	122
75	220000	020111	100002	120	021
81	122201	020202	201122	202	001
90	022112	022110	121010	202	200

Table 8: Temporal boundaries realizing the printed incidences of the terminal repertoire. Each six-digit entry preserves the order of the six coordinates of the regional boundary.

The following table displays all comparative incidences of the register on the twelve blocks of the segmentation under consideration. The letters r and d denote the visible charge and the dual charge, respectively. Column j retains the position index of the historical witness; the operation delivering the data to the selector forgets this index after forming the set of words.

The tenth block has an affine witness. At $t = 30$, $d_{30} = (0, 1, 2)$ and $\vartheta_{30} = 2$, whence

$$(d_{30,0} + \vartheta_{30}, d_{30,1} + \vartheta_{30}, d_{30,2} + \vartheta_{30}) = (2, 0, 1)$$

and, using the bands in Table 8,

$$L_{30}^{\text{af}} = 2(210112) + 0(110202) + (211110) = (001001) \quad \text{in } \mathbb{F}_3^6.$$

The equality holds coordinatewise: before reduction modulo three, the sum vector is $(6, 3, 1, 3, 3, 4)$.

Proposition 8.5 (Complete recovery of the repertoire through incidences). *Consider the comparative-incidence register in Table 9, augmented by the affine witness $(j, t, L) = (10, 30, 001001)$. For each position $5 \leq j \leq 12$, the set of output words of its incidences is a singleton. Extracting these eight words and subsequently forgetting their order produces exactly*

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\}.$$

Their encoding as base-three integers is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

j	t	charge	δ	ϵ	$(\eta_0\eta_1\eta_2)$	$L_{t,c,\delta,\epsilon}$
4	11	d	0	1	212	021101
5	67	d	1	-1	211	002001
5	67	d	2	1	211	002001
5	67	r	1	-1	211	002001
5	67	r	2	1	211	002001
6	81	d	0	1	222	020111
6	81	r	2	1	222	020111
7	34	r	1	-1	111	200110
7	75	d	1	-1	211	200110
7	75	r	2	-1	211	200110
8	90	r	0	1	121	121012
8	90	r	1	-1	121	121012
9	49	d	0	-1	121	010122
11	11	d	2	-1	211	221110
11	37	d	0	-1	121	221110
12	8	d	0	-1	111	222110
12	8	r	1	-1	111	222110
12	58	d	1	-1	111	222110
12	58	r	0	-1	111	222110

Table 9: Complete phase-charge incidences on the terminal segmentation within the hundred boundaries of the register. Word matches between rows retain their times, orientations, and charge types.

Proof. The incidence multiplicities at positions $j = 5, 6, 7, 8, 9, 10, 11, 12$ are, respectively,

$$(4, 2, 3, 2, 1, 1, 2, 4).$$

At each position, all rows print the same word. Each row is verified by substituting its three bands and three coefficients into the formula for L ; the tenth case has been calculated explicitly through L_{30}^{af} . For example, at $t = 67$ and with coefficients $(2, 1, 1)$, one obtains

$$2(010012) + (001220) + (011120) = (002001) \pmod{3}.$$

The eight results are distinct as words of length six. Their encoding $\sum_{k=1}^6 w_k 3^{6-k}$ gives, in position order, $(55, 175, 498, 437, 98, 28, 714, 687)$. Forgetting the order yields the stated set of integers. Thus, extraction preserves each complete word, including its zero positions, and delivers eight distinct elements to the selector. $\square$

The preceding proof is an exact recovery from identified incidences. Its input comprises the incidence register with its positions and witnesses. The subsequent orbital selection receives only $\mathcal{S}_{\text{term}}$ and the initial regional descriptor; it traverses the $8!$ orderings and determines their compatibility with the functional boundaries and the oriented incidences of the dodecaphasic wheel. Both operations are thus presented contiguously, with explicit domains: temporal provenance of the eight words and constructive determination of their order.

The comparative table contains the nineteen incidences that its formula produces on the twelve-block segmentation over the hundred declared times. This finite exhaustiveness is checked by traversing $100 \cdot 2 \cdot 3 \cdot 2 = 1200$ evaluations. The affine reader extends coverage of the terminal positions from seven to eight. This count characterizes the local reader repertoire just defined; the other TPK operations retain the domains and constructive functions established in their respective derivations.

8.5 Comparative readers, affine reader, and temporal incidence

8.5.1 Witt transformation of the selector

The selector uses the six-coordinate chart

$$A_{\text{ter}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Define $W(b) = bA_{\text{ter}}$ in $\mathbb{F}_3^6$, together with the charges

$$\eta(b) = \sum_{j=0}^5 b_j \pmod{3}, \quad \eta_{\text{ter}}^{\vee}(b) = \eta(W(b)).$$

The descriptor chart and the selector chart are related by the permutation $\sigma = (0, 1, 2, 4, 5, 3)$, written as a list of images:

$$(A_{\text{ter}})_{ij} = (A_{\text{reg}})_{\sigma(i), \sigma(j)}. \quad (8.14)$$

Lemma 8.6 (Compatibility of the charge functional). *For every band b ,*

$$\eta_{\text{ter}}^{\vee}(b) = \eta^{\vee}(b).$$

Proof. The integer sum of each corresponding row of the two matrices agrees: it is five in the first row and seven in the other five. By distributivity,

$$\sum_j \sum_i b_i (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{reg}})_{ij}.$$

Reduction modulo three gives the equality of charges. Matrix identity (8.14) is verified by substituting its thirty-six entries and also preserves the coordinate change. $\square$

Equality of the total charge functional allows its representations to be compared. The vector transformations remain identified by their matrices and the explicit permutation; the preceding proof preserves that coordinate information.

8.5.2 Twelve comparative readers per instant

Define the temporal phase

$$\theta(t) = (t + 2) \pmod{3}.$$

This expression realizes $(t - 1) \pmod{3}$ also at $t = 0$, where the phase equals two. For each charge type $\nu \in \{\eta, \eta_{\text{ter}}^{\vee}\}$, offset $o \in \{0, 1, 2\}$ and orientation $s \in \{1, 2\}$, take

$$\epsilon_{\chi}^{\nu, o, s}(t) = s \begin{cases} 1, & \nu(B_{\chi}(t)) = \theta(t) + o \pmod{3}, \\ 2, & \nu(B_{\chi}(t)) \neq \theta(t) + o \pmod{3}. \end{cases}$$

The corresponding comparative word is

$$C_{\nu, o, s}(t) = \sum_{\chi \in \{\pi, e, \varphi\}} \epsilon_{\chi}^{\nu, o, s}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (8.15)$$

The parameters generate twelve labelled representations per row. Two labels may produce the same word; the temporal witnesses and labels remain available in the complete calendar.

8.5.3 Affine phase–charge reader

The affine reader uses coefficients

$$a_{\chi}^{\text{af}}(t) = \eta_{\text{ter}}^{\vee}(B_{\chi}(t)) + \theta(t) \pmod{3},$$

and produces

$$F(t) = \sum_{\chi \in \{\pi, e, \varphi\}} a_{\chi}^{\text{af}}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (8.16)$$

The comparative rule distinguishes agreement and difference of charges; the affine rule adds charge and phase before combining the bands. This operational difference is what heterotypic selection uses.

8.5.4 Word–residue incidence

Let R be the cyclic rotation of the six positions of a band. In the row at time t , form the residual repertoire

$$\mathcal{R}(t) = \{R^j v : v \in \{q(t), a(t), c(t), \bar{w}(t)\}, 0 \leq j < 6\}.$$

The temporal incidence relations are

$$\mathcal{A}_{\text{cmp}} = \{(C_{\nu, o, s}(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t), \nu, o, s \text{ admissible}\}, \quad (8.17)$$

$$\mathcal{A}_{\text{af}} = \{(F(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t)\}. \quad (8.18)$$

Each pair relates a read word to a residue of the same row. Time is quantified jointly across both coordinates of the pair. In the calendar used, times $0, \dots, 99$ are distinct; identifying the same row and identifying the same time are therefore equivalent operations.

Proposition 8.7 (Existential compression of the temporal repertoire). *The function*

$$M(v, u) = \mathbf{1}_{(v, u) \in \mathcal{A}_{\text{cmp}}} + 2\mathbf{1}_{(v, u) \in \mathcal{A}_{\text{af}}} \in \{0, 1, 2, 3\}$$

preserves exactly the existence of each reading type for the pair (v, u) .

Proof. The first bit equals one exactly when there exist a time, a residual field, a rotation, and a comparative label that produce the pair. The second bit expresses the corresponding condition for the affine reader. Union by bitwise disjunction preserves each existence condition as all rows are traversed. A pair may admit both types, in which case the mask equals three. $\square$

The function M is an existence reader. The witness times, their multiplicities, and the individual labels remain in (8.18) and in the calendar that produces them. The finite implementation uses M to decide heterotypy, whereas the genealogical archive retains the richer provenance data.

8.6 Enumeration of cylinders and decimal candidates

8.6.1 Concatenation of seventy-two and seventy-eight trits

For an ordering $\tau = (s_1, \dots, s_8)$ of $\mathcal{S}_{\text{term}}$, define

$$P_{72}(\tau) = [\Omega_{24} \| s_1 \| \cdots \| s_8 \|]_3.$$

Recursively, start from $p_0 = [\Omega_{24}]_3$ and compute $p_i = 729p_{i-1} + [s_i]_3$ for $1 \leq i \leq 8$. This recurrence preserves the leading zeros of each block through explicit multiplication by 729. For each position b of the functional panel,

$$P_{78}(\tau, b) = 729P_{72}(\tau) + b.$$

The lengths are $24 + 8 \cdot 6 = 72$ and $72 + 6 = 78$, respectively. The corresponding ternary cylinder is

$$I_{\tau, b} = \left[\frac{P_{78}(\tau, b)}{3^{78}}, \frac{P_{78}(\tau, b) + 1}{3^{78}} \right). \quad (8.19)$$

8.6.2 Exact intersection with the twelve-triad grid

The decimal reading uses twelve triads, that is, the grid $N/10^{36}$. Write $D = 10^{36}$ and $Q = 3^{78}$ for brevity.

Lemma 8.8 (Integer endpoints of the cylinder). *For an integer prefix p of length seventy-eight, the numerators of the grid points belonging to its cylinder are*

$$\left\{ N \in \mathbb{Z} : \left\lceil \frac{pD}{Q} \right\rceil \leq N < \left\lceil \frac{(p+1)D}{Q} \right\rceil \right\}. \quad (8.20)$$

Each cylinder contains at most one point of this grid.

Proof. The condition $N/D \in [p/Q, (p+1)/Q]$ is equivalent to $pD/Q \leq N < (p+1)D/Q$. For integer N , the first inequality is equivalent to $\lceil pD/Q \rceil \leq N$; the second is equivalent to $N < \lceil (p+1)D/Q \rceil$. The interval length in coordinate N is $D/Q < 1$, since $10^{36} < 3^{78}$. It can therefore contain at most one integer. $\square$

Ceilings are evaluated by integer division: $\lceil a/Q \rceil = \lfloor (a+Q-1)/Q \rfloor$ for $a \geq 0$. The entire enumeration uses exact integers; the choice of boundary points is fixed by the half-open right endpoint.

First retain the indexed list of incidences

$$\mathcal{C}_{\text{ind}} = \{(\tau, j, N) : N/D \in I_{\tau, \mathcal{P}_9(j)}\}, \quad 1 \leq j \leq 9,$$

and then obtain the set of numerators

$$\mathcal{C} = \{N : \exists \tau, j (\tau, j, N) \in \mathcal{C}_{\text{ind}}\}.$$

Passing to $\mathcal{C}$ removes repetitions of a numerator, retained in $\mathcal{C}_{\text{ind}}$ together with their provenance.

Lemma 8.9 (Recovery of the ordering from the cylinder point). *If $(\tau, j, N) \in \mathcal{C}_{\text{ind}}$, block number i of the seventy-eight-trit word is recovered by*

$$\left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13.$$

In particular, blocks five through twelve recover τ .

Proof. Membership in the cylinder means $p \leq QN/D < p+1$, where $p = P_8(\tau, \mathcal{P}_9(j))$. Hence, $\lfloor QN/D \rfloor = p$. Extraction of the earlier blocks agrees with successive division of the prefix p by powers of 729: truncating a positional expansion before extracting a block preserves that block. The displayed formula performs exactly this extraction. The first four blocks form Ω_{24} and the next eight are the ordering τ . $\square$

Proposition 8.10 (Exact candidate count). *For Ω_{24} , $\mathcal{S}_{\text{term}}$ and $\mathcal{P}_9$ defined above,*

$$\# \text{Ord}(\mathcal{S}_{\text{term}}) = 40320, \quad \# \mathcal{C}_{\text{ind}} = 21902, \quad \# \mathcal{C} = 19446.$$

Proof. The eight blocks are distinct, so their orderings are the $8! = 40320$ permutations. For each permutation and each of the nine panel positions, compute $p = P_8(\tau, b)$, apply endpoints (8.20), and add the unique integer when the integer interval is nonempty. The sum of the cardinalities of these intervals is 21902. The union of their integers contains 19446 distinct elements. The exhaustive evaluation of these operations is certified by the theorems `indexed_cylinder_count` and `candidate_count` of the finite selector. The algorithm performs exactly the enumeration described: permutations, nine panel incidences, bounds by integer division, and duplicate removal. $\square$

The numbers 21902 and 19446 count different objects. The former counts indexed incidences containing a decimal point; the latter counts distinct numerators. The total number of cylinder queries is $40320 \cdot 9 = 362880$, including cylinders whose intersection with the grid is empty.

8.7 Exceptional incidence and heterotypy condition

8.7.1 Exceptional face produced by the panel

A band is lifted to twelve coordinates by

$$\Gamma(b) = (b, W(b)) \in \mathbb{F}_3^{12}.$$

The support of a word is the set of positions of its nonzero coordinates, with human-readable indices $1, \dots, 12$. The first regional bands of the panel produce

$$\mathcal{B} = \text{supp } \Gamma(B_\pi(0)) \cap \text{supp } \Gamma(B_e(0)) = \{4, 6, 9, 10\}. \quad (8.21)$$

The operation uses the regional words 103 and 523 and the selector's Witt matrix. The face is computed before querying the candidate coordinates.

For $N \in \mathcal{C}$, its twelve triads are

$$K_i(N) = \left\lfloor \frac{N}{1000^{12-i}} \right\rfloor \bmod 1000, \quad 1 \leq i \leq 12. \quad (8.22)$$

The crown support is

$$\mathcal{H}(N) = \{i : K_i(N) \geq 729\}, \quad \mathcal{C}_{\mathcal{B}} = \{N \in \mathcal{C} : \mathcal{H}(N) = \mathcal{B}\}.$$

The boundary $729 = 3^6$ relates the six-trit band to the decimal triad. Exhaustive evaluation of the equality of supports gives

$$\#\mathcal{C}_{\mathcal{B}} = 79. \quad (8.23)$$

The support preserves positions, while the complete register also preserves the twelve magnitudes. The next stage uses this positional information together with relations between magnitudes.

8.7.2 Ternary words and oriented differences of the candidate

The i th ternary band of the point N/D is obtained by

$$T_i(N) = \left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13. \quad (8.24)$$

The oriented edge $a \rightarrow b$ has residue

$$R_{a,b}(N) = (K_a(N) - K_b(N)) \bmod 729. \quad (8.25)$$

Subtraction is performed in $\mathbb{Z}$ before reduction. For this edge, the pair

$$(\text{enc}_6^{-1}(T_b(N)), \text{enc}_6^{-1}(R_{a,b}(N))).$$

is queried in relations (8.18). It is therefore a joint query of the destination word and the oriented difference of the triads.

Write $\text{Cmp}(a, b; N)$ when the pair belongs to $\mathcal{A}_{\text{cmp}}$ and $\text{Af}(a, b; N)$ when it belongs to $\mathcal{A}_{\text{af}}$.

8.7.3 Determination of the transverse orbit

The twenty-four-trit descriptor determines the first three decimal triads before the terminal filtrations. Indeed, its cylinder satisfies the exact inclusion

$$\left[\frac{66241910521}{3^{24}}, \frac{66241910522}{3^{24}} \right) \subseteq \left[\frac{234543140}{10^9}, \frac{234543141}{10^9} \right).$$

The two endpoint inequalities are verified by integer cross-products. Thus, every candidate shares the three reference positions 1, 2, 3 and their representations (234, 543, 140).

The step-four translation on $\mathbb{Z}/12\mathbb{Z}$ has the four orbits

$$(1, 5, 9), \quad (2, 6, 10), \quad (3, 7, 11), \quad (4, 8, 12).$$

The first three each contain one of the reference positions; the fourth is the only one disjoint from $\{1, 2, 3\}$. The oriented forest of steps three and four is fixed by the edges

$$\begin{aligned} 1 &\rightarrow 4, & 1 &\rightarrow 5, & 2 &\rightarrow 6, \\ 3 &\rightarrow 7, & 4 &\rightarrow 8, & 5 &\rightarrow 9, \\ 6 &\rightarrow 10, & 7 &\rightarrow 11, & 8 &\rightarrow 12. \end{aligned}$$

Its first edge has step three and the remaining edges have step four. The two internal edges of the distinguished orbit are precisely $4 \rightarrow 8$ and $8 \rightarrow 12$. The location of the reading condition is thus fixed by the geometry of the positions and by the initial prefix, before examining the candidates of the exceptional face.

8.7.4 Terminal heterotypy condition

The terminal condition is

$$\begin{aligned} \mathcal{J}(N) = & [\text{Cmp}(4, 8; N) \wedge \text{Af}(8, 12; N)] \\ & \vee [\text{Af}(4, 8; N) \wedge \text{Cmp}(8, 12; N)]. \end{aligned} \tag{8.26}$$

Both assignments of types to the two edges are admitted. The selection requires the existence of a realization with distinct types; the individual labels and their times remain available in the incidence witnesses.

Theorem 8.11 (Terminal selection of the coupling register). *On the candidate domain produced by Ω_{24} , $\mathcal{S}_{\text{term}}$, $\mathcal{P}_3$ and the hundred-row regional calendar, there exists a unique numerator simultaneously satisfying exceptional incidence and heterotypy:*

$$\{N \in \mathcal{C}_{\mathcal{B}} : \mathcal{J}(N)\} = \{234543140729659824621058914794146601\}.$$

Its twelve-triad register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \tag{8.27}$$

Proof. The preceding count produces the 79 elements of $\mathcal{C}_{\mathcal{B}}$. For each one, its twelve coordinates are extracted by (8.22); the destination bands T_8 , T_{12} and residues $R_{4,8}$, $R_{8,12}$ are computed; the two masks of the temporal repertoire are queried; and disjunction (8.26) is applied. The resulting list contains exactly the stated integer, according to the exhaustive evaluation certified by `terminal_selection`. Equality of the list with a singleton implies existence and uniqueness in the declared domain. Base-one-thousand extraction from that integer gives (8.27).

The selection procedure receives the enumerated producers and rules. The final integer appears in the statement of the verified equality after selection has been executed. This logical position allows the result to be checked without using it as a selection criterion. $\square$

8.7.5 Arithmetic witness of membership in a cylinder

One of the incidences of the selected numerator uses the ordering

$$(002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110)$$

and the boundary $438 = 121020_3$. For this incidence,

$$P_2 = 5283881584882673136514102926090957,$$

$$P_{78} = 3851949675379468716518781033120308091.$$

The integer-endpoint formula yields

$$\left\lceil \frac{P_{78} 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146601,$$

$$\left\lceil \frac{(P_{78} + 1) 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146602.$$

The unique integer in the interval is therefore N_K . The thirteen ternary bands of its decimal point are

020022	222111	211201	021101
002001	020111	200110	121012
010122	001001	221110	222110
121020			

This calculation exhibits a concrete witness of membership in the enumerated domain. Uniqueness of the register follows from the exhaustive examination of all candidates, in addition to this particular witness.

8.8 Regional composition and continuity of the realizations

8.8.1 Composition of the regional generators

Let $\text{Sel}(\Omega, \mathcal{S}, \mathcal{P}, \mathcal{N})$ be the selector defined by the enumeration and the two preceding filters. The regional producers satisfy the equalities

$$\Omega_{\text{reg}} = \Omega_{24}, \quad \mathcal{P}_{\text{reg}} = \mathcal{P}_9, \quad \mathcal{N}_{\text{reg}} = \mathcal{N}_{100},$$

where $\mathcal{N}_{\text{reg}}$ is computed by band extraction and signature reading. By congruence of functions,

$$\begin{aligned} \text{Sel}(\Omega_{\text{reg}}, \mathcal{S}_{\text{term}}, \mathcal{P}_{\text{reg}}, \mathcal{N}_{\text{reg}}) \\ = \text{Sel}(\Omega_{24}, \mathcal{S}_{\text{term}}, \mathcal{P}_9, \mathcal{N}_{100}) = \{N_K\}. \end{aligned} \tag{8.28}$$

The equality replaces three documentary interfaces with their actual regional construction. The terminal repertoire retains the provenance specified in (8.13).

The register is then formed by the twelve Euclidean divisions in (8.22). Each coordinate belongs to $\{0, \dots, 999\}$ and the length is twelve by construction. The implementation that takes the first element of the selected list is justified by equality with the singleton: the default value of a computational operation on empty lists never enters this result.

8.8.2 Relation to signed incidences and inversion

The present construction provides a prospective selection in an explicit finite domain. The subsequent development of the dodecaphasic register produces its oriented channels from the enriched terminal state and proves an integral inversion through Hadamard. That inversion recovers a register from compatible channels; here a register is selected from its regional and incidence-based inputs. Equality of their results allows the two realizations to be composed while preserving the origin of each operation.

Uniqueness of selection and uniqueness of inversion are distinct properties. The former quantifies over the candidates constructed in this chapter. The latter quantifies over the registers sharing a compatible signature. Their conjunction justifies transporting the selected register to the signed realization and recovering it from that realization, without turning a retrospective recovery into the producer of the register.

8.8.3 Relation between the terminal counts

The sequence

$$40320 \longrightarrow 21902 \longrightarrow 19446 \longrightarrow 79 \longrightarrow 1$$

records, successively, orderings, indexed incidences with the grid, distinct numerators, candidates of the exceptional face, and heterotypic survivors. The arrow notation indicates the order of the operations; the second number counts a different class of objects from the first.

The regional-catalogue count $196 \cdot 197 \cdot 196 \longrightarrow 331 \longrightarrow 2 \longrightarrow 1$, developed in its corresponding place, starts from another population: regional blocks from three catalogues and terminal conditions of Gram, occupancy, distance, and orientation. Both counts retain their domains. Their convergence to the same representation is expressed through the identification of their readers and outputs; the intermediate numbers describe different operations and remain documented separately.

8.8.4 Regional composition of the fine-structure constant

The output K subsequently enters the ordered regional composition and the base-one-thousand carry. Its later use preserves the twelve positions and the reading boundary. Coupling to the analytic root and representations at arbitrary depth uses that same output, together with the regional coordinates already generated and the rules of the analytic chart. This dependency prepares the chapter devoted to the two correlated realizations of the fine-structure constant.

The scope of this chapter is precise: production of the descriptor and its regional inputs, determination of the finite domain, unique selection on that domain, and material composition with the producers. The analytic recurrence at all orders and the compatibility of infinite carries are developed subsequently with their own proofs.

8.9 Executable certification and traceability of the operations

The formal development preserves the correspondence between the mathematical definitions and their executable realizations. The signature and clock proofs are found in `N69RegionalSignature.lean`; band extraction and reproduction of the hundred rows, in `GeneratedN69Rows.lean`; the minimality criterion and construction of Ω_{24} , in `RegionalW24.lean`. Compatibility of the two Witt matrices is assembled in `WittReaderCompatibility.lean`.

The temporal readers, rotations, oriented residues, and heterotypic condition are formalized in `TerminalReaders.lean`. Enumeration and counts are formalized in `TerminalOrbitalSelection.lean`. The regional panel is linked through `RegionalPanelBridge.lean`, and composition (8.28) is expressed in `SelectedFromRegionalInputs.lean`.

The finite enumeration checks use Lean’s native evaluation; their reports record `Lean.ofReduceBool`, in addition to the library’s customary logical axioms. The regional reconstruction of the descriptor uses kernel proofs and ordinary reduction. This distinction describes the verification mechanisms actually employed. The compilation receipts attest to their declared modules and dependencies; the mathematical proofs retain the domains, producers, and hypotheses made explicit in the body of the chapter.

9 Dodecaphase record and the rational phase constant

Generation of the regional coordinates does not exhaust the information in the APP-TRIT-TPK state. One Archimedean readout determines a value through compatible cylinders; another readout preserves the opposition of the sheets, the orientation of the route, and the record of the incidences traversed. The dodecaphase record belongs to this second readout before entering the closure of alpha. It is therefore not defined by subtracting a physical constant from a combination of previously known values. It is reconstructed from a signed coordinate of the same arithmetic state with trajectory memory that produces the regional coordinates.

The distinction is necessary even when two objects have twelve coordinates. A twelve-block catalogue word, a list of twelve oriented events, a signed integer vector, and a base-thousand dodecaphase record are different objects. Comparing them requires the maps that relate them. Equality of length does not provide these maps, and loss of signs is not merely a simplification of notation.

This chapter preserves the causal cutoff established in the preceding chapters: the nine APP positions on the horizontal and vertical axes produce the arithmetic sheets; TRIT types regime and orientation; TPK selects, transports, updates, and prolongs, preserving residue, quotient, carry, sheet, route, boundary, and memory. The coordinates used here are obtained from the joint discrete structure of the continuum. The chain $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ is not replaced by a finite schedule. Nor is the order $U_t \rightarrow \Gamma_9 \rightarrow K_{\text{ph}}$ reversed: the last map is a reduced-memory readout, not the generator of the signed record.

Reversible coordinatization and incidence determination. The result organizing this chapter is the equivalence between three coordinatizations of a record previously produced by APP-TRIT-TPK. The Hadamard transformation relates the signed record to the integer record; the difference extractor relates the latter to its transverse variations and its uniform component:

$$U \xrightleftharpoons[\mathcal{H}_{12}]{\mathcal{H}_{12}/4} K \xrightleftharpoons{(D_3, D_4, Q)} (D_3 K, D_4 K, Q(K)).$$

The first equivalence is established on the integral sublattice $\mathcal{L}_H$; the second, on the image of the extractor. With base 1000, the 12 positions, and the phase origin fixed, the evaluation $K \mapsto \kappa_{\text{per}}$ adds a fourth reversible coordinatization: multiplication by $1000^{12} - 1$ followed by Euclidean division recovers the complete record. The proofs in §9.8, §9.9, and §9.11 establish these equivalences in their respective domains.

Two compositions subsequently act on this record. The positional composition combines its components with the closure, propagation, and self-scaling coordinates, and its normalization determines α_A . The incidence composition applies the cubic threshold and yields

$$K \longmapsto B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The complete record allows both compositions to be reconstructed. The support B_K specifically preserves membership in the class selected by the threshold. The identity $B_K = H_e \cap P_{\pi}$ in (11.23) links this integer extraction to the incidence of the regional codes. The negative support of their balance with K completes the flag $B_K \subset H_{\alpha}^-$ developed in the exceptional extension. Thus, the function of K is defined through an invertible transformation and precise subsequent maps; the additional information on depth and transport remains in the state from which the record originates.

9.1 Cyclic extension and memory grading

The observable schedule contains one hundred and eight positions, organized into twelve nonadic windows. With positive indices, let

$$E_m = \{9(m-1) + 1, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The ordered support $E_1 \sqcup \dots \sqcup E_{12}$ preserves position and window membership. The group $C_{108} = \mathbb{Z}/108\mathbb{Z}$, by contrast, preserves the law of cyclic translation. For the representative $0 \leq \tilde{\theta} < 108$, the coordinates

$$\beta(\theta) = \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad r(\theta) = 1 + (\tilde{\theta} \bmod 9)$$

separate the dodecaphase sector from the interior nonadic position. Neither by itself preserves the total number of turns.

Proposition 9.1 (The carry of the schedule). *The maps*

$$\iota : C_{12} \longrightarrow C_{108}, \quad [b] \longmapsto [9b], \quad p : C_{108} \longrightarrow C_9, \quad [\theta] \longmapsto [\theta]_9$$

form a nonsplit exact sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0.$$

For the set-theoretic section choosing the representatives $0, \dots, 8$, the failure of additivity is the carry

$$c(r, r') = \left[\left[\frac{\tilde{r} + \tilde{r}'}{9} \right] \right]_{12}.$$

Proof. The kernel of p consists of the multiples of nine and coincides with the image of ι . Reduction is surjective, and multiplication by nine is injective on the stated domain. If the sequence split, the middle group would be isomorphic to $C_{12} \times C_9$, whose exponent is $\text{lcm}(12, 9) = 36$; the exponent of C_{108} is 108. This contradiction proves nonsplitting. Finally, Euclidean division of $\tilde{r} + \tilde{r}'$ by nine gives the stated carry and verifies the identity for the section. $\square$

The result explains a distinction that will reappear in alpha: returning to the phase is not equivalent to returning to the state. If q_{mem} counts nonadic turns, the schedule preserves only $\beta = [q_{\text{mem}}]_{12}$. After one hundred and eight steps, β returns, but q_{mem} increases by twelve. The record also preserves the cylinder, the incidence ledger, and the vertical memory. Closure of a finite chart does not turn this history into a memoryless cycle.

9.2 The oriented record and the integral 1 + 5 + 6 chart

For a history produced by TPK, let τ_m be the subroute over E_m , σ_m its orientation, κ_m the boundary carry, and Λ_m the local incidence ledger. The record is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}.$$

Its domain is the space of arithmetic histories with trajectory memory, not the set of phase labels. Projection onto the schedule forgets precisely some of the data required by the signed readout.

An initial chart of the record can be written in terms of oriented events. In this chart, the event-code set is $\{2, 4, 6, 8\}$, and the sector orientation is

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

If d_t is the direction code at step t , this gives

$$e_i = \Sigma_{12}(i+1) \# \{t \in E_{i+1} : d_t \in \{2, 4, 6, 8\}\}, \quad 0 \leq i < 12.$$

The formula keeps the nonnegative event count separate from orientation. It does not identify a cardinal direction with a TRIT sign. The vector e , whose components are signed local counts, is also not yet the vector U that will appear below.

Opposite positions determine, for $0 \leq i < 6$,

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}.$$

With

$$s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5),$$

the integral chart is defined by

$$\mathcal{L}_{12}(e) = (s_C, M_0, \dots, M_4, a_0, \dots, a_5).$$

The decomposition $1 + 5 + 6$ is not an arbitrary allocation of coordinates: one coordinate preserves the charge, five compare the pairs with a reference pair, and six preserve the opposition of the half-crowns.

Lemma 9.2 (Inversion of the integral chart). *An integer tuple $(s_C, M_0, \dots, M_4, a_0, \dots, a_5)$ belongs to the image of $\mathcal{L}_{12}$ if and only if*

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6),$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad p_i = p_5 + M_i \quad (i < 5).$$

In that case its preimage is unique and is given by

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}.$$

Proof. The sum of the five margins is $s_C - 6p_5$, which gives the division by six. Once the p_i are reconstructed, each pair of equations $p_i = e_i + e_{i+6}$, $a_i = e_i - e_{i+6}$ has the indicated solution. This solution is integral exactly when the parity congruence holds. All operations are inverted, so no residual freedom remains. $\square$

This reversibility shows what information is preserved; it does not authorize discarding the rest of the ledger. The complete record has a larger domain than this count chart. Extraction of the channels feeding U uses that complete record and cannot be deduced from the list (e_i) by identifying names.

9.3 Incidence evaluation preceding the dodecaphasic record

Recovering a record from its cyclic differences and total charge is distinct from evaluating those coordinates on the incidence ledger. The first problem is solved through the transverse extractor of Proposition 9.12. For the second, a historical presentation of the record provides three integer counts in each window. It is useful to make their composition explicit and determine exactly the information their evaluation requires.

Let $\mathcal{L}$ be a previously generated ledger of visits and oriented traces. Each visit preserves the route, APP position, sheet, instant, multiplicity, and boundary incidence. Arithmetic evaluation produces the integer and its decomposition

$$n_t = r_t + 9q_t, \quad 1 \leq r_t \leq 9.$$

The integer n_t is computed from the sum or product sheet; its remainder is not received as an independent value. The distinction between corona incidence and interior incidence remains explicit for visits with remainder 9.

In window E_m , the count presentation considers

$$A_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{1,3,5,7\}}(r(v)), \quad (9.1)$$

$$C_m = \sum_{j \geq 0} (\nu_{120,m}^-(j) - \nu_{120,m}^+(j)), \quad (9.2)$$

$$V_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{9\}}(r(v)) \epsilon_\delta(v). \quad (9.3)$$

Here $w(v)$ is the incidence multiplicity, and $\epsilon_\delta(v) = 1$ on the corona, -1 in the interior. The traces counted by $\nu^\pm$ must be enumerated through the corresponding ledger rule. The integer sum requires finite support or an alternative construction determining its meaning. The number of chronological incidences and the reduced length $120(1 + 3j)$ have different types; their relation requires the reader's length unit.

Define $[x]_{10} \in \{0, \dots, 9\}$ as the Euclidean representative and retain the quotients

$$x = [x]_{10} + 10[x/10].$$

The incidence block is

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}. \quad (9.4)$$

Composition yields the map

$$\mathcal{E}_{\text{cnt}}(\mathcal{L}) = \left((z_m - z_{m+3})_{m=1}^{12}, (z_m - z_{m+4})_{m=1}^{12}, \sum_{m=1}^{12} z_m \right). \quad (9.5)$$

Indices are read cyclically modulo 12. In this subsection, $(D_j z)_m = z_m - z_{m+j}$, for $j = 3, 4$, and $Q(z) = \sum_{m=1}^{12} z_m$; thus the preceding composition is $(D_3 z, D_4 z, Q(z))$. This convention is retained in §9.9, where complete integral reconstruction is proved. This formula produces its coordinates from the counts. It does not use K , U , α , or the expected transverse coordinates as arguments. Its identification with $\mathcal{E}_{108}^{90,120}$ requires composition with the effective collection and incidences of the terminal state; it does not follow solely from invertibility of the transverse system.

Proposition 9.3 (Sufficient and necessary arithmetic information). *Let $N, N' \in \mathbb{Z}^{12 \times 3}$, ordered by the counts (A, C, V) . For the map defined by (9.4)–(9.5),*

$$\mathcal{E}_{\text{cnt}}(N) = \mathcal{E}_{\text{cnt}}(N') \iff N - N' \in 10\mathbb{Z}^{36}.$$

In particular, the 36 sector residues determine exactly the 25-coordinate output. Preservation of quotients and provenance constitutes additional information in the record.

Proof. Equality modulo 10 produces the same blocks and therefore the same transverse coordinates. Conversely, if the transverse coordinates agree, $D_3(z - z') = D_4(z - z') = 0$. Steps 3 and 4 generate the cycle of length 12, so $z - z'$ is constant. Equality of total charges annihilates that constant. Finally, the decimal representation of an integer between 0 and 999 has three unique digits. Thus the three residues of every window are recovered. The statement is an algebraic identity for all N, N' , not an inference obtained from finite tests. $\square$

9.4 Conjugation of counts and boundary terms

Let $\rho(m) = 13 - m$. Consider an incidence conjugation that reverses the window order, preserves the arithmetic and boundary classes of visits, and exchanges the trace channels. Then

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\rho(m)}, -C_{\rho(m)}, V_{\rho(m)}).$$

If $c_m = [C_m]_{10}$, decimal reduction produces the identity

$$z_m^\dagger = z_{\rho(m)} + 100 \mathbf{1}_{\{c_{\rho(m)} \neq 0\}} - 20c_{\rho(m)}. \quad (9.6)$$

Indeed, $[-C]_{10} = 0$ when $[C]_{10} = 0$, and $[-C]_{10} = 10 - [C]_{10}$ otherwise. The formula preserves exactly this difference. Negating the channel is not equivalent to negating the entire decimal block.

Applying this identity to a concrete TPK involution requires checking its action on incidences. For an occupation reader evaluated before advancement, reversing a route $\gamma = (x_0, \dots, x_n)$ satisfies

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0).$$

Endpoint terms are incorporated before reduction modulo 10. On closed routes they vanish; in open windows they can alter the residues. This correction, already present in the bidirectional transport development, specifies what the connection between route and counts must preserve.

Proposition 9.4 (Necessity of the record's nonperiodic information). *Suppose the observable of a fixed route configuration satisfies $o(t + 54) = o(t)$, multiplicity is preserved under that return, and the same local reader acts in windows of 9 instants. Then its counts and blocks satisfy*

$$(A, C, V)_{m+6} = (A, C, V)_m, \quad z_{m+6} = z_m.$$

Proof. Translation $t \mapsto t + 54$ bijectively sends E_m to E_{m+6} , preserving each counted incidence and its multiplicity. Equality passes to the sum and decimal reduction. $\square$

Thus a presentation with 12 distinct blocks requires the reader to preserve some information that does not factor through that period-54 observable: incidence selection, effective orientation, endpoints, multiplicity, or memory. The proposition delimits an inadmissible state reduction; it does not assert that the full dynamics has period 54 or itself determine its terminal reader.

9.5 Complete cylinder census and terminal-boundary selection

The terminal reading uses two kinds of information jointly: compatibility of the regional cylinders and an incidence profile with labelled channels and positions. This subsection constructs the complete catalogues, enumerates their triples, and proves the effect of each reader condition. The starting point consists of the regional outputs already produced in Section 6, not values introduced to define APP-TRIT-TPK transport. Reduction to a matrix belongs to a subsequent reading of those outputs and preserves the order of the three channels.

The result is a census theorem with an explicit starting structure. The Gram, weight, distance, and margin profile is specified as a datum of the terminal reader. Exhaustive enumeration proves what this profile selects from the fixed catalogues; it does not, by itself, prove that the profile is a universal law or that it can be reconstructed from the unique triple that ultimately satisfies it. This distinction allows the complete finite proof to be incorporated without reversing its genealogy.

9.5.1 From regional outputs to ternary catalogues

Let $x_\pi, x_e, x_\varphi \in [0, 1)$ be the fractional parts of the three regional coordinates already constructed. We use their expressed cylinders of one thousand decimal places,

$$J_\chi^{(1000)} = \left[\frac{d_\chi}{10^{1000}}, \frac{d_\chi + 1}{10^{1000}} \right), \quad \chi \in \{\pi, e, \varphi\}.$$

The integers d_χ are finite output readings. The length of one thousand is sufficient for the integers below, as verified from the two bounds of each interval; it is not a depth assumed to be infinite. Define

$$p = 30, \quad r = 1050, \quad L = p + r = 1080, \quad k = 171, \quad D = 3^L, \quad Q = 1000^k, \quad (9.7)$$

and

$$P_\chi = \lfloor 3^{30} x_\chi \rfloor, \quad T_\chi = \lfloor Q x_\chi \rfloor.$$

Here k counts decimal triads and does not denote the dodecaphase record K . We have $1000^{171} \leq 3^{1080} < 1000^{172}$.

Lemma 9.5 (Stable extraction of an integer). *For $d, m, h \in \mathbb{Z}$, with $m, h > 0$, the value of $\lfloor mx \rfloor$ is constant on $[d/h, (d+1)/h)$ if and only if*

$$\left\lfloor \frac{md}{h} \right\rfloor = \left\lfloor \frac{m(d+1)-1}{h} \right\rfloor. \quad (9.8)$$

When this holds, either side gives that value.

Proof. The minimum of mx is attained at the left endpoint. Its supremum is $m(d+1)/h$ and is not attained. The largest integer that $\lfloor mx \rfloor$ can attain is $\lfloor m(d+1)/h \rfloor - 1$, equal to the right-hand side because the numerator and denominator are integers. Equality of the endpoints of the interval of values proves the statement. $\square$

The stability test holds in all three channels for $m = 3^{30}, Q, 3^{1080}$. The last choice will be used solely for a subsequent comparison, not to select the profile. The first thirty recovered positions are precisely the five six-trit stages of regional transport:

Channel	Thirty-trit prefix	P_χ
π	010211012222010211002111110221	29152671743887
e	201101121221102011012222102011	147887858824447
φ	121200112202121020010210010200	127247717616687

The agreement preserves the extension already constructed; the decimal reading is not used to replace its transport matrices retrospectively.

For a fixed channel, a tail of r trits is represented by $0 \leq R < 3^r$. Its cylinder and the cylinder of the first k triads are

$$I_{\chi,R} = \left[\frac{P_\chi 3^r + R}{D}, \frac{P_\chi 3^r + R + 1}{D} \right), \quad J_{\chi,T} = \left[\frac{T_\chi}{Q}, \frac{T_\chi + 1}{Q} \right). \quad (9.9)$$

The catalogue includes every tail for which these two cylinders have non-empty intersection.

Proposition 9.6 (Catalogue by exact intersection). *The compatible tails are the integers in the interval $[a_\chi, b_\chi]$, where*

$$a_\chi = \max\left(0, \left\lfloor \frac{T_\chi D}{Q} \right\rfloor - P_\chi 3^r\right), \quad (9.10)$$

$$b_\chi = \min\left(3^r - 1, \left\lfloor \frac{(T_\chi + 1)D - 1}{Q} \right\rfloor - P_\chi 3^r\right). \quad (9.11)$$

For the fixed regional cylinders, neither truncation is active, giving

χ	$a_\chi \bmod 729$	$b_\chi \bmod 729$	$b_\chi - a_\chi + 1$
π	67	262	196
e	584	51	197
φ	117	312	196

(9.12)

Proof. Write $N = P_\chi 3^r + R$. The two half-open intervals in (9.9) intersect precisely when

$$NQ < (T_\chi + 1)D, \quad (N + 1)Q > T_\chi D.$$

The second inequality is equivalent to $N \geq \lfloor T_\chi D/Q \rfloor$; the first, to $N \leq \lfloor ((T_\chi + 1)D - 1)/Q \rfloor$. Subtracting $P_\chi 3^r$ and retaining $0 \leq R < 3^r$ proves the formulae. Evaluation by integer division of the T_χ extracted using (9.8) gives the table. Since $r \geq 6$, $P_\chi 3^r \equiv 0 \pmod{729}$, so the residue of the first endpoint is also obtained directly from $\lfloor T_\chi D/Q \rfloor \bmod 729$. $\square$

The last six trits of a tail form the word $\nu_6(R \bmod 729)$, where ν_6 is the ternary representation of length six, including leading zeros. By the table, each catalogue has fewer than 729 consecutive elements, so this reduction is injective on it. The three boundary catalogues are therefore described completely, without choosing the final triples:

$$\begin{aligned}\mathcal{B}_\pi &= \{\nu_6(67 + i) : 0 \leq i < 196\}, \\ \mathcal{B}_e &= \{\nu_6((584 + j) \bmod 729) : 0 \leq j < 197\}, \\ \mathcal{B}_\varphi &= \{\nu_6(117 + \ell) : 0 \leq \ell < 196\}.\end{aligned}\tag{9.13}$$

Requiring the left endpoint of $I_{\chi,R}$ to belong to $J_{\chi,T}$ would be a different condition: it would omit the first cylinder intersecting the boundary and yield 195, 196, 195. The census uses the intersection of the complete cylinders, in accordance with its definition.

9.5.2 Specified profile and compatibility indicators

For $u, v \in \mathbb{F}_3^6$, set $g(u, v) = \sum u_i v_i \in \mathbb{F}_3$, $n(u) = g(u, u)$, $w(u) = \#\{i : u_i \neq 0\}$, and $h(u, v) = \#\{i : u_i \neq v_i\}$. The last two functions are integer-valued; the first two take values in $\mathbb{F}_3$. The reader's local profile, in the order (π, e, φ) , is

$$\begin{aligned}(g_{\pi e}, g_{\pi \varphi}, g_{e \varphi}) &= (2, 2, 0), & (n_\pi, n_e, n_\varphi) &= (0, 0, 0), \\ (w_\pi, w_e, w_\varphi) &= (3, 3, 3), & (h_{\pi e}, h_{\pi \varphi}, h_{e \varphi}) &= (2, 5, 3).\end{aligned}\tag{9.14}$$

The words are ordered as in (9.13). We use $[E] \in \{0, 1\}$ for the indicator of a condition E . For each pair of channels χ, ψ , the rectangular matrix $A_{\chi\psi}$ has entry $[g(u, v) = g_{\chi\psi}]$. The matrix $A_{\chi\psi}^h$ multiplies that entry by $[h(u, v) = h_{\chi\psi}]$. Let D_χ^n be the diagonal matrix of $[n(u) = 0]$, and D_χ^{nw} the diagonal matrix of $[n(u) = 0][w(u) = 3]$.

Theorem 9.7 (Exhaustive enumeration by finite sums). *The product of the three catalogues contains 7 567 952 triples. The successive conditions in (9.14) leave*

$$7\,567\,952 \longrightarrow 275\,794 \longrightarrow 6235 \longrightarrow 3127 \longrightarrow 331.\tag{9.15}$$

These four counts are, respectively,

$$\begin{aligned}N_g &= \text{tr}(A_{\pi e} A_{e \varphi} A_{\varphi \pi}), \\ N_{gn} &= \text{tr}(D_\pi^n A_{\pi e} D_e^n A_{e \varphi} D_\varphi^n A_{\varphi \pi}), \\ N_{gnw} &= \text{tr}(D_\pi^{nw} A_{\pi e} D_e^{nw} A_{e \varphi} D_\varphi^{nw} A_{\varphi \pi}), \\ N_{gnwh} &= \text{tr}(D_\pi^{nw} A_{\pi e}^h D_e^{nw} A_{e \varphi}^h D_\varphi^{nw} A_{\varphi \pi}^h).\end{aligned}\tag{9.16}$$

All products and traces in this formula are computed over the integers, not over the field with three elements.

Proof. The size of the product is $196 \cdot 197 \cdot 196$. Expanding a trace in (9.16), its indices run over exactly one triple of words. The summand is one if it satisfies every represented condition and zero otherwise. Each triple is counted once. The diagonal matrices add the individual conditions; the rectangular factors add the pairwise conditions. This proves that the traces are precisely the stated cardinalities.

To make their evaluation explicit, fix the indices i, j of the first two channels and form the sets of indices ℓ of the third that satisfy each pairwise condition. Their indicators are intersected, the relevant individual indicators are added, and the resulting cardinality is summed. The final sum runs over $0 \leq i < 196, 0 \leq j < 197$. Substituting the words in (9.13) and the profile in (9.14) gives, in that order, 275794, 6235, 3127, 331. The supplement records the four contributions for each of the 196 indices i , together with the 331 complete triples; its program evaluates these sums using exact binary indicators and checks each survivor by scalar computation. No sampling or tolerance is involved. $\square$

9.5.3 Labelled incidence and orientation charge

For a surviving triple, let $B \in \mathbb{F}_3^{3 \times 6}$ be its matrix of ordered rows. The second profile requires

$$\text{rank}_{\mathbb{F}_3} B = 3, \quad w_{\text{col}}(B) = (1, 1, 2, 2, 3, 0), \quad q_{\partial}(B) = (1, 2, 2, 1, 2, 0), \quad (9.17)$$

where w_{col} counts the non-zero elements in each column and q_{∂} sums its entries modulo three. This is a profile of labelled columns; permuting them without transporting the profile defines a different reader. Write $\mathcal{F}$ for the set of 331 triples in the preceding theorem.

Proposition 9.8 (Reduction of the census to the terminal pair). *The indicator sums over $\mathcal{F}$, upon successively requiring rank, column weights, and column sums, are*

$$\sum_{B \in \mathcal{F}} [\text{rank } B = 3] = 331, \quad N_{\text{rango,pesos}} = 18, \quad N_{\text{rango,pesos,sumas}} = 2. \quad (9.18)$$

With catalogue indices starting at one, the two final triples are (120, 197, 157) and (129, 197, 148), and their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

Proof. For every triple in the already enumerated set, compute the rank by elimination over $\mathbb{F}_3$, count the non-zero elements in the six columns, and sum those columns modulo three. The products of the indicators in (9.17) give (9.18). Substitution of the two final index triples into (9.13) yields the displayed matrices. The complete list of survivors and the exact filters in the supplement allow the elimination to be reproduced without presupposing those two matrices as its domain. $\square$

The orientation $\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3$ selects the line $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$. The row charges of B_0, B_1 are $(0, 2, 0)$ and $(2, 2, 1)$; only the second belongs to that line. This recovers the final step of the selector in Section 9.6, now preceded by the construction and enumeration of its complete domain. The stable reading of $[3^{1080}x_{\chi}]$, performed after the filters, gives the boundaries 021020, 001220, 100210, with the same catalogue positions (129, 197, 148). This subsequent comparison neither defines the profile nor participates in the indicator sums.

9.5.4 Preserved information and scope of the result

The procedure preserves the thirty-trit prefix, the compatible tail, the regional order, the position of each trit, and the orientation. Reduction modulo 729 here preserves each candidate's identity because the interval of tails has length less than 729; this injectivity is not asserted for an arbitrary history. The census of 331 also proves that the local profile alone does not select a single matrix: the labelled margins and charge supply effective additional conditions.

A complete finite enumeration and its selector on output cylinders have therefore been recovered, with an explicit reading profile. Construction of the signed record additionally uses its memory reader and channels, as specified below; the visible matrix is not identified with that record merely because they have the same number of components. Nor does this count prove the universality of the profile, generate the regional coordinates anew, or decide a statement concerning alpha or the complete continuum.

9.6 Visible reading and signed record of the terminal state

Let x_{term} denote the state produced by the composition of selection, transport, and update up to the terminal cutoff. The operator expression is

$$x_{\text{term}} = (\mathcal{U}_{108} \circ \cdots \circ \mathcal{U}_1)(x_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Its two relevant readouts are

$$x_{\text{term}} \xrightarrow{(\pi_{\text{term}}, R_{\text{sgn}})} (B_{\text{term}}, U).$$

The matrix B_{term} retains the visible ternary readout; the vector U preserves another coordinate of the same history. No arrow $B_{\text{term}} \rightarrow U$ is inserted between them.

The selector of this terminal presentation acts on the constructed catalogs of sizes 196, 197, and 196. Their local signatures reduce the $196 \cdot 197 \cdot 196 = 7\,567\,952$ triples to 331. The rank, weight, and column-margin conditions $q_{\partial} = (1, 2, 2, 1, 2, 0)$, evaluated in Proposition 9.8 leave two matrices:

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row orientation is $\sigma = (1, 1, -1) = (1, 1, 2)$ in $\mathbb{F}_3^3$. The row charges of the candidates are $(0, 2, 0)$ and $(2, 2, 1)$. Only the second belongs to $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$; hence $B_{\text{term}} = B_1$. This check proves the final step of the selector from the census proved in Section 9.5, with the stated reader profile. It does not turn the column margin into a consequence of the row charge, nor replace the preceding enumeration with inspection of these two matrices.

The signed readout has the typed composition

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\text{TPK}}^{\text{term,mem}} \longrightarrow \mathbb{Z}^{12}. \quad (9.19)$$

The channel record determines

$$\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\text{term}})) = (b^{90}, b^{120}, Q_{\text{TPK}}),$$

with coordinates

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, \\ &\quad -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, \\ &\quad 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned}$$

These quantities are used as coordinates of the preceding record, not as parameters fitted through alpha. The reconstruction proved in the following subsections begins exactly at this typed output. The material trace of the preceding extractor and the scope of its self-contained transcription are preserved in the technical provenance note; the subsequent arithmetic is not presented as a substitute for that trace.

9.7 From the channel record to the signed vector

The harmonic readout Π_H can be written without using either K or α . To avoid confusing its orbit sums with the α blocks, we denote them by s_1, s_2, s_3 . Let

$$B_r = \sum_{j=0}^3 b_{r+3j}^{120}, \quad d_{r,j} = b_{r+3j}^{90}, \quad 1 \leq r \leq 3,$$

and

$$s_1 = \frac{Q_{\text{TPK}} + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2. \quad (9.20)$$

Division by three belongs to the readout of the three orbits. Its integrality is checked on compatible outputs of the record; it is not assumed for an arbitrary element of $\mathbb{Z}^{25}$.

The three signed blocks are

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \quad (9.21)$$

Integer substitution gives

$$(B_1, B_2, B_3) = (972, -1073, 101), \quad (s_1, s_2, s_3) = (2378, 1406, 2479),$$

and therefore

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns yields

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (9.22)$$

The presence of negative values and components greater than 999 shows that U is not a base-thousand digit word. Nor is it $U_6 \parallel U_6$, a catalogue concatenation of period six. The distinction concerns domain and preserved information, not merely values.

The constructive order thus far is

$$x_{\text{term}} \longrightarrow \mathcal{R}_{12} \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U.$$

The inversion that follows produces the dodecaphase record. The fact that the channels can subsequently be recovered from the dodecaphase record is a reversibility check; it does not reverse this causal order.

9.8 Orbitwise Hadamard transformation and integer reconstruction

In the twelve-position cycle, step three determines three orbits:

$$(1, 4, 7, 10), \quad (2, 5, 8, 11), \quad (3, 6, 9, 12).$$

The separation of four sectors in each orbit is what, once origin and orientation have been fixed, admits the ninety-degree angular readout. The transformation does not act on three consecutive tetrads in the natural ordering.

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

If P_{orb} reorders by the three indicated orbits, define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

Lemma 9.9 (Hadamard inversion and integrality). *We have*

$$H_4^2 = 4I_4, \quad H_4^{-1} = \frac{1}{4}H_4, \quad \det H_4 = -16.$$

For $u \in \mathbb{Z}^4$, the preimage $H_4^{-1}u$ is integral if and only if $H_4u \equiv 0 \pmod{4}$.

Proof. The rows are orthogonal and have squared norm four; the matrix is symmetric. This gives $H_4^2 = 4I_4$ and the inverse. Integer elimination, or expansion of the determinant, gives the negative sign and the value -16 . The integrality condition is exactly divisibility by four of all four coordinates of H_4u . $\square$

Definition 9.10 (Integral dodecaphase transformation). The sublattice of signed records and its integral coordinatization are

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}, \quad \mathcal{K}(u) = \frac{1}{4}\mathcal{H}_{12}u.$$

The map $\mathcal{K} : \mathcal{L}_H \rightarrow \mathbb{Z}^{12}$ is an isomorphism of abelian groups, with inverse $k \mapsto \mathcal{H}_{12}k$. The canonical record K is the image of the signed record generated by the preceding operations, with phase origin and orientation fixed.

Proof. The congruence defines precisely the integrality of $\mathcal{K}(u)$. The identity $\mathcal{H}_{12}^2 = 4I$ proves that both compositions are inverse. In particular, every $k \in \mathbb{Z}^{12}$ has preimage $\mathcal{H}_{12}k \in \mathcal{L}_H$. $\square$

Proposition 9.11 (Canonical integral record). *The inversion $K^{(r)} = H_4U^{(r)}/4$ of the blocks in (9.21) produces*

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 58, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

In the natural ordering, the dodecaphase record is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (9.23)$$

It is the unique rational preimage of U and belongs to $\{0, \dots, 999\}^{12}$.

Proof. The three products before division are

$$\begin{aligned} H_4U^{(1)} &= (936, 2916, 2484, 3176), \\ H_4U^{(2)} &= (2172, 2636, 232, 584), \\ H_4U^{(3)} &= (560, 3296, 3656, 2404). \end{aligned}$$

All coordinates are divisible by four. Dividing and undoing the orbit permutation gives (9.23). Uniqueness follows from rational invertibility, not from a search among candidates close to a physical value. $\square$

Integrality can also be formulated in lattice terms. The determinantal divisors of H_4 are 1, 2, 4, 16: the entries have greatest common divisor one; the minors of order two are even and one has absolute value two; those of order three are multiples of four and one has absolute value four. Hence

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4),$$

and for the global transformation,

$$\text{coker } \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3.$$

The index of the image is $16^3 = 4096$. Thus, not every signed integer vector admits an integer preimage; the vector obtained satisfies the precise congruences required by inversion. The factor $1/4$ is forced by the matrix and does not have the status of a free coefficient.

9.9 Cyclic differences and the uniform component

With cyclic indices modulo twelve, let

$$(D_3 z)_m = z_m - z_{m+3}, \quad (D_4 z)_m = z_m - z_{m+4}, \quad Q(z) = \sum_{m=1}^{12} z_m.$$

The first operator compares sectors separated by three positions; the second, by four. The channel names 90 and 120 refer here to these separations in the oriented cycle, not to identification with a physical operator from another domain.

Proposition 9.12 (Completeness of the transverse system). *The operator*

$$\mathcal{D} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and nonzero invariant factors

$$\text{diag}(1, \dots, 1, 12),$$

with eleven units. In particular, its outputs determine a unique dodecaphase record.

Proof. If $D_3 z = D_4 z = 0$, the vector is constant along both steps. Since 3 and 4 generate the cycle modulo twelve, $\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle$. The total charge eliminates this last freedom because $Q(\mathbf{1}) = 12$.

For the integral statement, the rows of (D_3, D_4) are oriented incidences of a connected graph. A spanning tree provides eleven unit invariant factors. Every minor of order twelve must contain the row Q . Adding the first eleven columns to the last makes the latter zero in the incidence rows and twelve in row Q . All maximal minors are divisible by twelve, and the tree produces one of absolute value exactly twelve. The final factor is therefore twelve. $\square$

Checking (9.23) gives

$$D_3 K = b^{90}, \quad D_4 K = b^{120}, \quad Q(K) = 6263.$$

Formulas (9.20) and (9.21), in turn, reconstruct U from those differences. This closes a triangle of exact representations of the same object: transverse record, signed vector, and dodecaphase record. Equality of results does not turn their codomains into a single one; it proves that the specified maps recover the relevant information.

9.10 Integral compatibility and determination of the transverse record

The domain of this construction is the representation of the record of an APP–TRIT–TPK history. Evaluation of record events, compatibility of its channels, and inversion of its coordinates are distinguished. The following result gives a complete integral criterion for the second level and an explicit factorization connecting the first to it. The proof uses the record operators and establishes their relations for every element of the indicated integer domains.

9.10.1 Characterization by adjacent increments

The indices belong to $\mathbb{Z}/12\mathbb{Z}$. Let

$$(Sz)_i = z_{i+1}, \quad D_3 = I - S^3, \quad D_4 = I - S^4, \quad Q(z) = \sum_{i=0}^{11} z_i.$$

Retain the transverse extractor

$$\mathcal{D} : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}, \quad k \longmapsto (D_3 k, D_4 k, Q(k)).$$

For a triple of integer coordinates (b, c, q) , write

$$w_i = c_i - b_{i+1}, \quad P_0 = 0, \quad P_i = \sum_{j=0}^{i-1} w_j \quad (1 \leq i \leq 12), \quad T(w) = \sum_{i=0}^{11} P_i. \quad (9.24)$$

Theorem 9.13 (Integral image of the transverse extractor). *A triple $(b, c, q) \in \mathbb{Z}^{25}$ belongs to $\text{im } \mathcal{D}$ if and only if it satisfies*

$$b_i = w_i + w_{i+1} + w_{i+2} \quad (0 \leq i < 12), \quad (9.25)$$

$$\sum_{i=0}^{11} w_i = 0, \quad (9.26)$$

$$q + T(w) \equiv 0 \pmod{12}. \quad (9.27)$$

Its unique integer preimage is

$$k_i = \frac{q + T(w)}{12} - P_i, \quad 0 \leq i < 12. \quad (9.28)$$

The twelve equations of (9.25) and equation (9.26) are linearly independent over $\mathbb{Q}$.

Proof. For an output of $\mathcal{D}$, one has

$$c_i - b_{i+1} = (k_i - k_{i+4}) - (k_{i+1} - k_{i+4}) = k_i - k_{i+1}.$$

The sum of three increments gives b_i ; their full cyclic sum is zero. Moreover, $P_i = k_0 - k_i$, so $q = 12k_0 - T(w)$. All conditions and the inverse formula follow.

Conversely, (9.27) makes formula (9.28) integral; (9.26) allows its cyclic prolongation. Its adjacent differences are w_i . Thus, $D_3 k = b$ and

$$(D_4 k)_i = w_i + w_{i+1} + w_{i+2} + w_{i+3} = w_i + b_{i+1} = c_i.$$

Summing the inverse gives $Q(k) = q$. The same formula proves uniqueness. For independence, the invertible integer change $(b, c) \mapsto (b, w = c - Sb)$ turns the first twelve relations into $b = (I + S + S^2)w$. Each solves for a different coordinate of b . The condition $\sum w_i = 0$ is an additional independent relation. $\square$

The rational image has dimension twelve. Its integral structure adds a congruence of order twelve: this is the constructive manifestation of the final Smith factor of $\mathcal{D}$. The eleven increments $w_0, \dots, w_{10}$ and the charge q are sufficient coordinates, with $w_{11} = -\sum_{i=0}^{10} w_i$ and congruence (9.27). The original twenty-five coordinates preserve these same twelve degrees of freedom through thirteen relations.

9.10.2 Commutativity with the harmonic reading

Let $\mathcal{H}_{12}$ be the record's Hadamard transformation, with orbits $(r, r+3, r+6, r+9)$ for $r = 0, 1, 2$. Its block is

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I_4.$$

For compatible (b, c, q) , define

$$B_r = \sum_{j=0}^3 c_{r+3j}, \quad A_0 = \frac{q + 2B_0 + B_1}{3}, \quad A_1 = A_0 - B_0, \quad A_2 = A_1 - B_1.$$

The reader Π_H of (9.21) has blocks

$$(\Pi_H(b, c, q))^{(r)} = (A_r, b_{r+3} - b_{r+9}, b_r + b_{r+6}, b_r - b_{r+6}).$$

Proposition 9.14 (Agreement of the two reconstructions). *On $\text{im } \mathcal{D}$,*

$$\Pi_H \mathcal{D} = \mathcal{H}_{12}, \quad \mathcal{D}^{-1} = \frac{1}{4} \mathcal{H}_{12} \Pi_H.$$

The image of Π_H restricted to that domain is exactly

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12} u \equiv 0 \pmod{4}\}.$$

Proof. For $k = \mathcal{D}^{-1}(b, c, q)$, let $a_r = \sum_{j=0}^3 k_{r+3j}$. Shifting four positions sends each orbit to the next, so $B_r = a_r - a_{r+1}$. Together with $q = a_0 + a_1 + a_2$, these identities give $A_r = a_r$. On an orbit, write (x_0, x_1, x_2, x_3) for the coordinates of k and $d_j = x_j - x_{j+1}$. Then

$$d_1 - d_3 = x_0 + x_1 - x_2 - x_3, \quad d_0 + d_2 = x_0 - x_1 + x_2 - x_3, \quad d_0 - d_2 = x_0 - x_1 - x_2 + x_3.$$

These are the remaining three rows of $H_4 k^{(r)}$. Inversion and the integral characterization follow from $\mathcal{H}_{12}^2 = 4I$. $\square$

The condition $\Pi_H(b, c, q) \in \mathcal{L}_H$ controls only the harmonic output. Membership of the twenty-five coordinates in $\text{im } \mathcal{D}$ additionally requires (9.25)–(9.27). For example, adding the vector $e_0 - e_3$ to c preserves the three orbital sums and leaves Π_H unchanged. Starting from $(0, 0, 0)$, one thus obtains a triple with $\Pi_H = 0$ that violates (9.25). This is an exact negative control of channel compatibility, independent of any terminal evaluation.

9.10.3 Determination and transport on the record's orbits

Theorem 9.13 determines k once its channels and charge have been produced. As a condition on outputs not yet individualized, the same image is a group isomorphic to $\mathbb{Z}^{12}$. Fixing only a charge q , the solutions form an affine class of

$$L_0 = \{v \in \mathbb{Z}^{12} : \sum v_i = 0\} = \bigoplus_{i=0}^{10} \mathbb{Z}(e_i - e_{11}).$$

For example, $100\mathbf{1} + e_0$ and $100\mathbf{1} + e_1$ are two different records of charge 1201, with components in $\{0, \dots, 999\}$. Each satisfies every image condition and every Hadamard congruence through its own channels. These witnesses compare compatibility equations; they are not attributed the status of terminal HMT histories.

Proposition 9.15 (Cyclic covariance and stabilizer of the record). *Let ρ be window translation on a record domain $\mathcal{R}$, and S the preceding translation on $\mathbb{Z}^{12}$. On the orbit of $R \in \mathcal{R}$, a covariant reader is determined by a vector*

$$k \in \text{Fix}(S^p),$$

where $p \mid 12$ is the least period of R under ρ . The charge condition $Q(k) = q$ admits integer solutions if and only if $12/p$ divides q ; when they exist, they form an affine class of rank $p - 1$. In particular, on a free orbit, covariance and charge leave eleven degrees of freedom.

Proof. The rule $F(\rho^j R) = S^j k$ is well defined exactly when $S^p k = k$. Such vectors repeat a word of length p exactly $12/p$ times. Their charge is $\frac{12}{p} \sum_{i=0}^{p-1} k_i$. Divisibility and rank follow from this formula. Channel covariance follows from $D_a S = S D_a$ and $Q S = Q$. On $\mathcal{L}_H$, the corresponding action is $\hat{S}_H = \mathcal{H}_{12} S \mathcal{H}_{12} / 4$; by Proposition 9.14, both reconstructions intertwine these actions. $\square$

If the domain carries a marked origin, the action transporting that mark as well must be used. The proposition concerns this declared cyclic action; it does not assume periodicity of the enriched history. Naturality under truncations after the 108 cutoff is obtained, for any already defined reader F_{108} , through $F_N = F_{108} \circ \tau_{N,108}$. This naturality preserves the initial evaluation rule; it does not by itself choose one of the possible rules on the first one hundred eight events.

9.10.4 Event evaluation and composition of contributions

The preceding characterization provides a factorization of extraction through two effective readings of the record:

$$\omega : \mathcal{R} \longrightarrow \{w \in \mathbb{Z}^{12} : \sum w_i = 0\}, \quad q_{\text{reg}} : \mathcal{R} \longrightarrow \mathbb{Z}, \quad q_{\text{reg}}(R) + T(\omega(R)) \equiv 0 \pmod{12}. \quad (9.29)$$

These maps must be evaluated on the events, orientation, incidence, and memory produced by APP-TRIT-TPK. Once their actions on these data are given, the composition

$$R \longmapsto (w, q) \longmapsto ((I + S + S^2)w, (I + S + S^2 + S^3)w, q) \xrightarrow{\Pi_H} u \xrightarrow{\mathcal{H}_{12/4}} k$$

is determined and satisfies all inverse checks. The formula $w = c - Sb$ also allows connection to a producer already delivering both channels: it is a check on its output, prior to any comparison with the terminal witness.

Proposition 9.16 (Accumulation of contributions and prospective uniqueness). *Suppose the initial state and each event a_t of the record produce, through previously fixed rules, contributions $(\delta w_t, \delta q_t)$ with*

$$\sum_i \delta w_{t,i} = 0, \quad \delta q_t + T(\delta w_t) \equiv 0 \pmod{12}.$$

With compatible initial conditions $(w^{(0)}, q^{(0)})$, the update

$$w^{(t+1)} = w^{(t)} + \delta w_t, \quad q^{(t+1)} = q^{(t)} + \delta q_t$$

produces a unique record at each stage. Its increment is

$$\delta k_{t,i} = \frac{\delta q_t + T(\delta w_t)}{12} - P_i(\delta w_t).$$

The construction respects concatenation of records, and the reading of a prefix remains unchanged under subsequent prolongations.

Proof. The compatibility conditions are preserved under addition. Formula (9.28) is additive in (w, q) and produces the indicated increment. Induction fixes each state successively. The sum over two segments is the concatenated sum, and a prefix receives exactly the same summands before and after prolonging the history. $\square$

The proposition provides a sufficient criterion, not a requirement of linearity for every TPK reader. If events are transported by nontrivial actions, their contributions can be expressed in the terminal chart through the corresponding affine cocycle. The cocycle equation then preserves the same independence from the partition of the trajectory. In both cases, event rules and the initial state constitute the constructive data that must come from the ledger: choosing them to reproduce an expected vector would replace that provenance with another task.

The result of this development is the exact determination condition (9.29), its proved inversion, and its transport identities. Its scope is the transverse record; the remaining HMT constructions retain their own domains and proofs.

9.11 Two rational readouts: κ_{36} and κ_{per}

Once K has been constructed, the base-thousand chart allows two different operations. The first reads a single turn; the second prolongs the dodecaphase record periodically. Let $B = 1000$ and define its concatenation integer

$$N_K = \sum_{m=1}^{12} K_m B^{12-m} = 234543140729659824621058914794146601. \quad (9.30)$$

The leading zeros of a block, as in 058, are part of its three-digit representation. They do not change its integer value, but they make it possible to concatenate the twelve blocks unambiguously.

Definition 9.17 (Finite and periodic readouts of the dodecaphase record). The one-turn readout is

$$\kappa_{36} = \sum_{m=1}^{12} K_m B^{-m} = \frac{N_K}{B^{12}}.$$

The periodic prolongation is

$$K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}, \quad \kappa_{\text{per}} = \sum_{j \geq 1} K_j^{\text{per}} B^{-j}.$$

Proposition 9.18 (Exact value and period of the prolonged dodecaphase record). *We have*

$$\kappa_{\text{per}} = \frac{N_K}{B^{12} - 1}, \quad \kappa_{\text{per}} - \kappa_{36} = \frac{N_K}{B^{12}(B^{12} - 1)} = B^{-12} \kappa_{\text{per}}. \quad (9.31)$$

The infinite word of the dodecaphase record has minimum period twelve in base thousand and minimum period thirty-six in base ten.

Proof. The sum of the first turn is $N_K B^{-12}$; each new turn multiplies its contribution by B^{-12} . The geometric series gives $N_K/(B^{12} - 1)$, and subtraction gives the second identity.

The twelve components of K are distinct. A proper period of the cycle would identify two of them; the minimum block period is therefore twelve. The decimal expansion is canonical, since the dodecaphase record has no tail of 999 blocks. If its minimum decimal period were d , it would divide 36. Grouping digits in threes would produce a block period of $d/\gcd(d, 3)$. No proper divisor of 36 gives twelve under this operation. Hence the minimum decimal period is 36. $\square$

Both readouts are rational, but they are not the same rational number. The first terminates after twelve blocks; the second repeats the turn. Moreover, $0 < \kappa_{\text{per}} - \kappa_{36} < B^{-12}$. Their proximity does not allow one to be substituted for the other in an exact identity. In particular, the finite closure of alpha uses κ_{36} , whereas the sum of the prolonged section uses κ_{per} .

9.12 Reversibility of the rational constant and phase change

The rational constant κ_{per} determines the entire record K when the base, length, and reading origin are retained. Its numerical and structural functions are related by an exact inversion.

Proposition 9.19 (Integral recovery and cyclic transformation). *Let $I(K) = \sum_{j=1}^{12} K_j 1000^{12-j}$. Then*

$$I(K) = (1000^{12} - 1) \kappa_{\text{per}}.$$

The integer expansion of $I(K)$ into twelve base-1000 blocks recovers every K_j , including the leading zeros in each block. If $\rho(K) = (K_2, \dots, K_{12}, K_1)$, then

$$\kappa(\rho K) = 1000 \kappa(K) - K_1. \quad (9.32)$$

Proof. The first identity inverts the definition of periodic evaluation. Iterated Euclidean division by 1000, with twelve fixed positions, recovers the vector. Furthermore,

$$I(\rho K) = 1000 I(K) - K_1 (1000^{12} - 1).$$

Dividing by $1000^{12} - 1$ yields (9.32). $\square$

Recovery therefore composes $\kappa \mapsto K \mapsto U$ and the already constructed transformations $D_3 K, D_4 K, Q$. Rational evaluation preserves the signed record and the cyclic differences that depend on it. The total depth and transport operators of a history remain in their own coordinates: the domain of the preceding inversion is the integral dodecaphasic record.

Write $\kappa_j = \kappa(\rho^j K)$, with indices modulo 12. The same identity gives

$$K_{j+1} = 1000\kappa_j - \kappa_{j+1}, \quad \sum_{j=0}^{11} \kappa_j = \frac{\sum_{j=1}^{12} K_j}{999} = \frac{6263}{999}.$$

Phase acts exactly on the constant: its value is fixed in the canonical reading, while its rotations follow an affine law. The orbit sum is invariant under the choice of origin.

The subsequent incidence also has an expression in these coordinates. The cubic threshold determines

$$B_K = \{j + 1 : 1000\kappa_j - \kappa_{j+1} \geq 729\} = \{4, 6, 9, 10\}.$$

The tetrad is recovered from the rational orbit because the latter first recovers the integer components. The signed support associated with α additionally uses its regional records and boundary normalization. The relationship between coordinate and incidence is thus expressed through identifiable operations.

This is an operational definition of the function of K : it provides a reversible integral coordinatization of the oriented record, whose periodic evaluation and incidence functions participate in subsequent constructions. The value, transformation, and domain are preserved together.

9.13 Periodicity and the cokernel of the transport operator

The periodicity just proved belongs to the dodecaphase record. It does not imply periodicity of the regional coordinates, of the closure carries, or of alpha. It is useful to make this separation precise through a second, purely integral quotient.

Let S be the cyclic shift of twelve coordinates, with fixed orientation, and let $b \geq 2$ be an integer base. The periodic carry operator is $\partial_b = S - bI$. If a periodic identity is written

$$A = P + E - \Phi - K + \partial_b C,$$

then the transformations

$$K \mapsto K + \partial_b h, \quad C \mapsto C + h$$

preserve its right-hand side. This invariance compares representatives of a carry class; it is not authorization to modify the dodecaphase record already selected by the record.

Proposition 9.20 (Periodic carry quotient). *With the preceding orientation,*

$$\text{coker}(S - bI) \cong \mathbb{Z}/(b^{12} - 1)\mathbb{Z}.$$

Proof. The cyclic module is presented as $\mathbb{Z}[t]/(t^{12} - 1)$. Imposing $S = bI$ amounts to imposing $t = b$, leaving the relation $b^{12} - 1 = 0$. Evaluating the coefficients at b , with the chosen concatenation order, represents the resulting class. $\square$

The denominator $b^{12} - 1$ of κ_{per} and this quotient come from the same cyclic length, but their objects remain different: one is a rational evaluation and the other a quotient of a carry module. The finite closure of alpha will be an oriented chain with a right boundary. Its prolongation will transport a compatible remote boundary, not a carry identified by decree between the first and thirteenth positions.

This prepares the next chapter. The dodecaphase record has been fixed before solving the alpha recurrence; its Hadamard reconstruction is integral and unique; its transverse differences recover the record; and its two rational evaluations have separate domains and uses. None of these facts uses alpha as an input, and none yet determines the arithmetic type of the output of the complete closure.

10 Derivation of the fine-structure constant

In this development, the fine-structure constant receives an arithmetic determination that brings together the three regional coordinates and the integral dodecaphase record. The operation combines their blocks with fixed orientation and transports the quotients of positional normalization. The resulting parameter expresses compatibility of that oriented difference with cylinder prolongation. Its value and the incidence data used in its construction will remain related throughout the exceptional development.

The internal order is important. The coordinates π , e , and φ , already constructed in Section 6, are prior coordinates of the same arithmetic state with trajectory memory and may legitimately act as inputs to a subsequent reader. They are not conventional constants injected to select alpha. Correlated generation of the four coordinates shares origin and coinductive compatibility without requiring that all four components be evaluated in a single step.

The two routes express different mathematical roles of the same parameter. The first realizes an integer closure by Euclidean division in base 1000. The second studies the analytic equilibrium of vacancies through the resolvent $E_{21/22}$ and its graded prolongations. One route determines blocks and quotients; the other determines a root through an equation and a uniqueness domain. Their identification is formulated afterward in the common cylinders at every depth. The exposition distinguishes these three results: positional construction, analytic determination, and compatibility of the two.

10.1 First route: positional determination by integer closure

10.1.1 Domain and orientation of integer closure

Let $P_j, E_j, \Phi_j \in \{0, \dots, 999\}$ be the canonical blocks of the three regional coordinates, and prolong the dodecaphase record by

$$K_j = K_{1+((j-1) \bmod 12)}.$$

The channel orientation is $(1, 1, -1)$. The contribution of the dodecaphase record is subtracted according to the already typed dodecaphase closure. For $B = 1000$, the local operation distinguishes the signed balance

$$Z_j := P_j + E_j - \Phi_j - K_j$$

from the sum incorporating the incoming quotient. The normalization is

$$s_j = Z_j + c_{j+1}, \quad A_j = s_j - B \left\lfloor \frac{s_j}{B} \right\rfloor, \quad c_j = \left\lfloor \frac{s_j}{B} \right\rfloor. \quad (10.1)$$

Thus $A_j \in \{0, \dots, B-1\}$, even when s_j is negative. The carry entering from the right belongs to the domain of the operation; it cannot be omitted because the table is printed from left to right.

Definition 10.1 (Finite-depth closure). For a depth N and an integer boundary t , the map $\mathcal{C}_N$ receives $(P_{1:N}, E_{1:N}, \Phi_{1:N}, K_{1:N}, c_{N+1} = t)$ and applies (10.1) in the order $N, N-1, \dots, 1$. Its output is the pair of words

$$(A_{1:N}, c_{1:N+1}).$$

Euclidean division has no residual sign choice: the remainder is fixed in $\{0, \dots, 999\}$, and the quotient is determined. For example, $-431 = 569 - 1000$ and $-1200 = 800 - 2 \cdot 1000$. The negative carries of closure are not anomalies; they are the integer information needed to maintain the oriented equality.

10.1.2 Dodecaphase window and carry conditions

The first twelve triads obtained are

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884), \\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497), \\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117). \end{aligned}$$

In the closure channel, the five regions determining coordinate P preserve their distinct antecedents. Sharing this reading does not identify them as regions or histories.

With boundary $c_{13} = 0$, the recurrence produces the following table. Each input, incoming carry, total before normalization, obtained block, and outgoing carry is shown. The table is a reproducible integer computation, not a list of blocks selected for their resemblance to a target.

j	R_j	E_j	Φ_j	K_j	c_{j+1}	s_j	A_j	c_j
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	-1	352	352	0
4	589	459	749	729	-1	-431	569	-1
5	793	045	894	659	-2	-717	283	-1
6	238	235	848	824	-1	-1200	800	-2
7	462	360	204	621	0	-3	997	-1
8	643	287	586	058	-1	285	285	0
9	383	471	834	914	-1	-895	105	-1
10	279	352	365	794	0	-528	472	-1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

Table 10: Initial closure with zero right boundary. The dependency propagates from the last row to the first.

The carry word is

$$(c_1, \dots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0),$$

and the window reading is the rational number

$$\alpha_{12}^{(0)} = \frac{7297352569283800997285105472380663}{10^{36}} = 0.007297352569283800997285105472380663. \quad (10.2)$$

The superscript recalls the chosen boundary. This window is exact in its finite domain; it is not simply identified with the truncation of the infinite section. The difference between the two readings will become visible when the carry of the remote tail is transported.

10.1.3 Telescoping and reversibility in a boundary fiber

The local form of the equality is

$$R_j + E_j - \Phi_j - K_j - A_j = Bc_j - c_{j+1}. \quad (10.3)$$

The carry term is a link between adjacent positions. Eliminating it coordinate by coordinate gives a false equality. Only an evaluation preserving its endpoints allows it to telescope.

Proposition 10.2 (Integer identity at arbitrary depth). *For $\iota_N(z) = \sum_{j=1}^N z_j B^{N-j}$, every execution of $\mathcal{C}_N$ satisfies*

$$\iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(K) - \iota_N(A) = B^N c_1 - c_{N+1}. \quad (10.4)$$

Proof. Multiplying (10.3) by B^{N-j} and summing produces on the right-hand side

$$\sum_{j=1}^N (Bc_j - c_{j+1}) B^{N-j} = B^N c_1 - c_{N+1}.$$

Each interior carry appears twice with opposite signs; only the two endpoints survive. $\square$

For Table 10, both endpoints are zero. If p_{12}, e_{12}, f_{12} are the three truncated fractional sums,

$$\alpha_{12}^{(0)} = p_{12} + e_{12} - f_{12} - \kappa_{36}.$$

The dodecaphase-record constant appearing here is κ_{36} , not κ_{per} . The domain of this identity contains truncations and a finite boundary; its form does not authorize erasing both conditions when passing to the complete section.

Local inversion is exact as well:

$$K_j = P_j + E_j - \Phi_j + c_{j+1} - A_j - Bc_j.$$

Given P, E, Φ , blocks A , and the complete carries, it recovers the dodecaphase record. Concatenation additionally recovers

$$\iota_N(K) = \iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(A) - B^N c_1 + c_{N+1}.$$

This reversibility is a property of the fiber with fixed data and boundaries. It must not be turned into a new causal recipe starting from alpha and manufacturing the record afterward. The direction of construction remains signed record, dodecaphase record, closure, and output.

10.1.4 Controls of orientation, boundary, and register position

Integer closure distinguishes three operational dependencies: the direction of carry propagation, the boundary condition, and the position of the dodecaphasic register. Their comparison uses the same twelve-block window as Table 10. The regional words P, E, Φ and the already generated register K are retained except for the modification identified in each case. The common reference is $\mathcal{C}_{12}$ with $c_{13} = 0$, whose output is (10.2).

Prospective production of the signed register and its inversion $K^{(r)} = H_4 U^{(r)}/4$ precede these controls. The normalization of Lemma 9.9 is retained, in particular $H_4^2 = 4I_4$ and $\det H_4 = -16$. Inversion determines the register once U has been produced; its operational selection remains in the construction preceding closure. The following variations of K examine the response of an already defined reader.

Proposition 10.3 (Separation of the dependencies of closure). *For the data in Table 10, the following four modifications have distinct effects:*

1. *Left-to-right quotient propagation, initialized with zero quotient, changes the third block from 352 to 353.*
2. *The condition $c_{13} = 1$, with the original orientation, changes the final block from 663 to 664 and preserves the preceding eleven blocks.*
3. *The rotation $K'_j = K_{j+1}$, with $K_{13} = K_1$, changes the output word while retaining $c_{13} = 0$.*
4. *The perturbation $K'_1 = K_1 + 1 = 235$, with the other eleven coordinates fixed, changes the first block from 007 to 006 and preserves the remaining blocks.*

Proof. For the first case, define the initial quotient $b_0 = 0$ and apply, in the order $j = 1, \dots, 12$,

$$\tilde{s}_j = P_j + E_j - \Phi_j - K_j + b_{j-1}, \quad \tilde{A}_j = [\tilde{s}_j]_{1000}, \quad b_j = \frac{\tilde{s}_j - \tilde{A}_j}{1000}.$$

This orientation produces, in two consecutive halves,

$$\begin{aligned} \tilde{A}_{1:6} &= (007, 297, 353, 570, 284, 800), \\ \tilde{A}_{7:12} &= (995, 285, 106, 471, 379, 663). \end{aligned}$$

The difference already appears at $j = 3$, where the quotient required by the original closure comes from the fourth position.

For the second case, the final sum changes from 663 to 664. Both have zero Euclidean quotient; hence $c_{12} = 0$ in both cases and the recurrence agrees from $j = 11$ to $j = 1$. The resulting word is

$$\begin{aligned} A_{1:6}^{(1)} &= (007, 297, 352, 569, 283, 800), \\ A_{7:12}^{(1)} &= (997, 285, 105, 472, 380, 664). \end{aligned}$$

Identity (10.4) records exactly the unit variation of the right endpoint.

The rotation in the third case, evaluated with the same zero boundary and the original recurrence, gives

$$\begin{aligned} A_{1:6}^{\text{rot}} &= (698, 699, 763, 639, 119, 004), \\ A_{7:12}^{\text{rot}} &= (559, 429, 226, 119, 926, 030). \end{aligned}$$

In this case the left carry is $c_1 = -1$; the right endpoint remains zero. Retaining both endpoints in (10.4) preserves the integer equality as the twelve coefficients change position.

Finally, perturbing K_1 leaves the entire recurrence from $j = 12$ to $j = 2$ unchanged. At $j = 1$, the sum decreases from seven to six, with zero quotient in both cases. This gives

$$\begin{aligned} A_{1:6}^{\text{pert}} &= (006, 297, 352, 569, 283, 800), \\ A_{7:12}^{\text{pert}} &= (997, 285, 105, 472, 380, 663). \end{aligned}$$

□

The four controls separate transport direction, boundary fibre, coefficient position, and local response. Their interpretation retains the finite domain $N = 12$. Tail transport and normalization of the complete section are developed next, with the compatibilities required by passage to arbitrary depth.

10.1.5 Prolongation and canonical normalization of the complete section

Prolongation preserves the periodic dodecaphase record and receives the complete regional sequences. Let

$$p = \sum_{j \geq 1} P_j B^{-j}, \quad e_0 = \sum_{j \geq 1} E_j B^{-j}, \quad f = \sum_{j \geq 1} \Phi_j B^{-j}.$$

Each series is a realization of the corresponding internal character. The integer parts already obtained are 3, 2, and 1, so $p = \pi - 3$, $e_0 = e - 2$, and $f = \varphi - 1$.

Proposition 10.4 (Existence and uniqueness of complete normalization). *The series*

$$y = p + e_0 - f - \kappa_{\text{per}} = \sum_{j \geq 1} (P_j + E_j - \Phi_j - K_j) B^{-j} \quad (10.5)$$

converges absolutely. On the domain of the obtained blocks, $0 < y < 10^{-2}$. There is a unique canonical expansion (A_j) of y , with no eventually constant 999 tail, and a unique bounded integer sequence (c_j) satisfying (10.3) with $c_1 = 0$. These carries belong to the invariant set $\mathcal{C} = \{-2, -1, 0, 1, 2\}$.

Proof. The coefficients of the series are bounded in absolute value by $2(B - 1)$, so a geometric series dominates it. If $Z_j = P_j + E_j - \Phi_j - K_j$ and $R_j(Z) = \sum_{r \geq 0} Z_{j+r} B^{-r-1}$, each tail is the difference of two sums of two canonical tails and belongs to $(-2, 2)$. The first coefficient is $Z_1 = 141 + 718 - 618 - 234 = 7$; hence $y = (7 + R_2(Z))/1000 \in (0.005, 0.009)$.

Let (A_j) be the canonical expansion of y . Define

$$c_j = R_j(Z) - R_j(A).$$

Multiplying equality of the two series by B^{j-1} and separating their first $j - 1$ terms gives

$$c_j = \sum_{r=1}^{j-1} (A_r - Z_r) B^{j-1-r} \in \mathbb{Z},$$

and $c_1 = 0$. Furthermore, $R_j(A) \in [0, 1)$, so $-3 < c_j < 2$, placing all carries in $\mathcal{C}$. The tail-shift identities give $Bc_j = Z_j - A_j + c_{j+1}$, which is the required recurrence.

Conversely, any bounded solution with canonical output and $c_1 = 0$ telescopes in the limit to the same series y . The canonical expansion is unique, and once it is fixed, the tail formula determines all carries. Finally, if a finite execution is applied with $c_{j+1} \in \mathcal{C}$, then $-2000 \leq s_j \leq 2000$, so its Euclidean quotient again belongs to $\mathcal{C}$. □

The proposition makes explicit the normalization of the subsequent reader: it does not replace generation of the three sequences with a conventional series introduced as data. Once those sequences have been produced by the coinductive state, their combination and the normalization boundary are determined without consulting alpha.

The finite normalization procedure propagates the five boundaries of $\mathcal{C}$ and retains only the prefix common to all of them. If two depths have coalesced before a boundary, the deterministic right-to-left recurrence gives the same blocks to its left. This is compatibility with truncation. A boundary is not chosen because it gives favorable digits, nor is all-depth compatibility inferred from observing a long run.

The initial comparison shows why this precaution matters. Stabilized prolongation gives $c_{13}^\infty = -1$, whereas the isolated window imposed $c_{13} = 0$. Thus,

$$884 + 497 - 117 - 601 - 1 = 662$$

in the last block of the first turn. The first eleven triads do not change, but the prolongation begins

$$(007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \dots).$$

An occurrence of block 999 does not violate the canonical convention: what is excluded is an eventually constant 999 tail. The zero-boundary window ends in 663; the truncation of the complete section ends in 662. Both statements describe different and exact operations. They must not be merged into a single projection notation.

10.1.6 Operative definition and complete additive equation

Definition 10.5 (Alpha coordinate of closure). The integer route determines the coordinate

$$\alpha_A = \sum_{j \geq 1} A_j B^{-j} = p + e_0 - f - \kappa_{\text{per}}, \quad (10.6)$$

where (A_j, c_j) is the compatible canonical normalization of (10.1). In the scalar realization of the HMT coordinates,

$$\boxed{\alpha_A = \pi + e - \varphi - 4 - \kappa_{\text{per}}.} \quad (10.7)$$

The integer four is not calibrated: it is $3 + 2 - 1$, the combination of the integer parts of the three coordinates. Base thousand arises from the cubic chart, and period twelve from the dodecaphase record. The signs are those of the oriented channel and closure, and the remote boundary is that imposed by the compatible section. These are the effective origins of the coefficients of the equation.

At this level, alpha can be described as the scalar coordinate of the closure defect between the three regional realizations and the rational dodecaphase return, computed with compatible carry. The term *defect* denotes a determinate algebraic difference, not an arbitrary loss or an imperfection of the system. This description does not suppress the genealogy: without the regions, signed record, orientation, and carry law, the scalar equality would not be a sufficient exposition of the construction.

Three objects that alpha notation may accompany in subsequent readers must also be kept separate: A is a word of blocks; the negative pre-carry support is obtained from the signs of the totals before normalization; and an exceptional radial vector belongs to an incidence codomain. None is interchangeable with the real value α_A .

Signed balance and sum with incoming quotient. Positional division consists of two consecutive stages. With $K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}$, designating the periodic prolongation also denoted by K_j in (10.1), the balance before quotient transport is

$$Z_j := R_j + E_j - \Phi_j - K_j^{\text{per}}.$$

The variable s_j in (10.1) designates the sum already including the incoming contribution, so that

$$s_j = Z_j + c_{j+1}, \quad A_j = Z_j + c_{j+1} - 1000c_j.$$

The incidence configuration H_{α}^- uses the signs of the balance Z_j in the initial window; determination of the digits additionally uses the linking quotients c_{j+1} and c_j . This distinction jointly preserves the signed support and the normalization of the expansion.

Exact dependence on the truncation boundary. The regional coordinates and the dodecapasic record originate in the common APP-TRIT-TPK generation. Once produced, their positional composition has an explicit dependence on reading depth. Let $B = 1000$,

$$Z_j = P_j + E_j - \Phi_j - K_j^{\text{per}}, \quad y_N = \sum_{j=1}^N Z_j B^{-j}, \quad R_{N+1}(Z) = \sum_{r \geq 0} Z_{N+1+r} B^{-r-1}.$$

The complete series in (10.5) satisfies exactly

$$\alpha_A = y_N + B^{-N} R_{N+1}(Z), \quad -2 < R_{N+1}(Z) < 2.$$

This identity combines the contribution of the window with that of its continuation, both expressed in the same coordinates.

Proposition 10.6 (Boundary transport and canonical cylinder). *Let $A_j^{(N,t)}$ be the output of finite positional division with $c_{N+1} = t \in \mathcal{C} = \{-2, -1, 0, 1, 2\}$, and let $a_N^{(t)} = \sum_{j=1}^N A_j^{(N,t)} B^{-j}$. For the records in the domain under consideration,*

$$\alpha_A - a_N^{(t)} = B^{-N}(R_{N+1}(Z) - t), \quad |\alpha_A - a_N^{(t)}| < 4B^{-N}. \quad (10.8)$$

The quotient of the complete continuation is $c_{N+1}^{\infty} = \lfloor R_{N+1}(Z) \rfloor$. With this boundary,

$$0 \leq \alpha_A - a_N^{(c_{N+1}^{\infty})} < B^{-N},$$

so that the resulting window is the canonical prefix of order N .

Proof. Telescoping (10.3) gives $a_N^{(t)} = y_N + tB^{-N} - c_1$. Since $Z_1 = 7$, each incoming state belongs to $\mathcal{C}$, and the first dividend lies between 5 and 9, we have $c_1 = 0$. Substituting this equality into the series decomposition proves (10.8). The bound follows from $R_{N+1}(Z) \in (-2, 2)$ and $t \in \mathcal{C}$.

For complete canonical normalization, $c_{N+1}^{\infty} = R_{N+1}(Z) - R_{N+1}(A)$ and $R_{N+1}(A) \in [0, 1)$. The integrality of the quotient implies $c_{N+1}^{\infty} = \lfloor R_{N+1}(Z) \rfloor$. The error formula then becomes $\alpha_A - a_N^{(c_{N+1}^{\infty})} = B^{-N} R_{N+1}(A)$, proving membership in the canonical cylinder. $\square$

The proposition identifies the meaning of the right boundary: it is the integer part of the signed tail expressed at the scale of the next position. Computation over the five boundaries preserves this information as a finite collection of possibilities until the blocks on the left stabilize. Quantitative contraction follows from depth and tail transport; its object is the normalization of the coordinates already generated.

Graded domain of the analytic resolvent. The analytic composition exhibited in §10.2.2 acts on $\text{span}\{e_7, e_8, e_9\}$ and satisfies $Ne_9 = 0$. Its evaluation yields exactly $T_{7,9}$; degree 9 is the final component of this domain. The identity $N^3 = 0$ explains why the resolvent is expressed as three summands. To describe a continuation to higher degrees, one must additionally specify the transport between successive graded domains and its action on the initial component of each domain. Local nilpotence and global compatibility thus perform different mathematical functions.

10.1.7 Constructive determination and the function of the dodecaphasic record

The preceding definition makes it possible to specify what it means to determine α from the structure. The domain of the operation contains four records of known provenance: the three regional sequences and the integral phase record. Normalization additionally incorporates a boundary condition compatible with continuation. Once these data are fixed, Euclidean division determines each remainder and its quotient. Proposition 10.4 extends this determination to the complete system. Independence from a target physical value is therefore a property of the construction order: the value compared at the end does not participate in selecting any argument of this map.

This causal independence must be distinguished from algebraic independence between numbers. The former is established by identifying the domain, operations, and provenance of the coefficients; the latter would be a statement about polynomial relations. The result used here is the former. The construction determines a coordinate through records produced by APP-TRIT-TPK and subsequently permits verification of the relations it satisfies. This order makes it possible to use an already generated coordinate in a later operation without turning it into an external parameter.

The function of the record K is more precise than that of a numerical correction added to a formula. Its twelve positions originate in the reversible transformation of the signed record. The equation $U = \mathcal{H}_{12}K$ preserves the possibility of returning to that record, while the transverse differences and total sum allow another reconstruction. In periodic evaluation, the order of the positions determines the positional weight of each component. The same set of components, arranged in a different order, generally produces a different rational coordinate. The phase origin and orientation therefore form part of the definition of the number occurring in (10.7).

Alpha normalization composes this information with the regions. Its local identity distinguishes two successive operations. First, the oriented balance of blocks is formed; then that balance is represented by a canonical remainder and a quotient linking adjacent positions. The first operation preserves the sign of the balance and allows a support to be extracted subsequently. The second allows prefixes of the real coordinate to be formed. Both readings remain related because the state preserves the data before and after normalization. The signed support and the positional expansion are thus different realizations of an explicit arithmetic composition.

The telescoping identity expresses the global content of this composition. The interior quotients cancel in pairs when a prefix is evaluated, while the endpoint terms remain in the equality. This cancellation is a property of the weighted sum; it neither removes the quotients from the state nor allows them to be discarded before the sum is performed. The right boundary of a window receives the contribution of the continuation. For this reason, evaluating an isolated window and restricting a continued section are different operations. The change of the final block from 663 to 662, proved above, provides an explicit realization of this difference without altering the already stabilized blocks to its left.

In this sense, bidirectionality has a concrete arithmetic formulation. Generation provides compatible continuations; normalization transmits the integer information at their boundary from positions of lower weight towards those of higher weight. The equation preserves the contribution of both directions through its indices and boundary conditions. Reversibility in the fixed fibre is verified by reconstructing K from the other records and the complete quotients. This inversion certifies the consistency of the original operation; the genealogical direction remains the one that first produces the record and then the alpha coordinate.

The second determination has complementary content. In it, the immediate object is a vacancy balance, and the parameter appears as its root in an isolating domain. The positional route provides a canonical normalization of records; the analytic route studies the vanishing of an operator. Presenting both requires precisely this difference in function to be retained. Their relationships with the common state must be expressed through the refinement maps and the comparison of their complete realizations, as developed in the corresponding subsection. Agreement at a finite approximation constitutes a localized check; the identification of the complete constructions has the domain and scope of its own statement.

Finally, this organization explains alpha's place in the article. Its equation appears after regional generation and the construction of K , because it uses both results. The electronic construction uses this coordinate as a factor in its composition. The exceptional extension receives the records and their supports, in which incidence information is preserved. These two continuations return to a common provenance and develop different mathematical functions: evaluating a central section and constructing a boundary configuration. Developing each makes it possible to follow the structure through the equation, rather than restricting the comparison to its scalar value.

10.2 Second route: analytic determination through vacancies

10.2.1 Transport invariants and graded resolvent

The analytic route receives invariants of the oriented state, not the output α_A . Its sixth-order truncation is

$$P_6(x) = 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \quad (10.9)$$

and the vacancy condition uses

$$\delta = \log_{10} \frac{10}{9}.$$

The ratio 10/9 compares the decimal chart with the nine APP positions of the nonadic construction. Its logarithm belongs to this subsequent compactification reading; it is not a target value of alpha.

The genealogy of the coefficients must be preserved in full, but with its exact logical strength. Recent developments bring together a marked signature of the state. The synchronization matrix and associated census are

$$D_{\text{syn}} = \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}, \quad N_9 = 43.$$

From its determinant one extracts

$$a = -\frac{\det D_{\text{syn}}}{N_9} = \frac{172}{43} = 4.$$

The clock induced on the short gaps has secondary spectrum $\{6, 7\}$; its upper endpoint gives $b = 7$. The tritic cardinality $m = 3$ determines

$$t_* = mb = 21, \quad k_* = mb - 1 = 20.$$

The vacancy reading in the thousand chart gives

$$v_- = \lfloor 1000\delta \rfloor = 45, \quad v_+ = \lceil 1000\delta \rceil = 46.$$

These integers are typed antecedents of the truncation's evaluation. They are not obtained from its root or from a metrological approximation to it.

The lower branch additionally coincides with the determinant of the APP block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \det B_c = 45.$$

The transverse rank of the preceding chapter's record gives $r_{\text{dod}} = 11$, and the phase semicrown provides $f_{\text{ph}} = 8\binom{6}{2} = 120$. In particular, $45r_{\text{dod}} = 495$.

With $\omega = 2\pi$, the signature

$$\mathcal{I}_9 = (\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$$

feeds the graded reader

$$\begin{aligned} \mathfrak{G}_9(\mathcal{I}_9)(x) = & \omega x - \frac{b}{a}x^2 + \omega^{-1}x^3 + \frac{x^4}{k_*} - \frac{2x^5}{t_*} - \frac{x^6}{v_+} \\ & - \frac{x^7}{f_{\text{ph}}} + \frac{x^8}{v_-} + \frac{2x^9}{v_-r_{\text{dod}}}. \end{aligned} \quad (10.10)$$

Substitution recovers the coefficients of P_6 and its ninth-order prolongation. The pair 45/46 crosses the separation between the two: the upper branch occupies degree six, and the lower reappears in degrees eight and nine.

The factorization of (10.10) is formulated for a state preserving origin, channel order, and orientation. Naturality with respect to morphisms preserving those data is not equivalent to uniqueness among readers that do not preserve them. The result identifies the reader associated with the typed state and the factorization of its coefficients; its domain of naturality consists of morphisms preserving origin, channel order, and orientation.

10.2.2 Nilpotent resolvent in degrees 7, 8, and 9

The first prolongation admits an especially explicit finite operator construction. On the space with basis e_7, e_8, e_9 , let

$$Ne_7 = -\frac{8}{3}e_8, \quad Ne_8 = \frac{2}{11}e_9, \quad Ne_9 = 0, \quad v_7 = -\frac{1}{120}e_7.$$

The symbol N here denotes a nilpotent operator, not a truncation depth. Its domain and grading are fixed before the polynomial is evaluated at a number.

Lemma 10.7 (Exact nilpotent prolongation). *One has $N^3 = 0$ and*

$$(I - N)^{-1}v_7 = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}.$$

Evaluation $e_n \mapsto x^n$ produces

$$T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}. \quad (10.11)$$

Proof. The operator raises degree and vanishes on the last basis vector, so its cube is zero. The inverse is the finite resolvent $I + N + N^2$. Moreover,

$$Nv_7 = \frac{1}{45}e_8, \quad N^2v_7 = \frac{2}{495}e_9.$$

Summing the three terms and evaluating gives the formula. □

Here the denominators 120, 45, and 495 are not decimal fits: they arise from the semicrown, APP determinant, and its product with the dodecaphase rank. The transfers $-8/3$ and $2/11$, together with the sign and degree of the seed, belong to the typed reader. Preserving only cycle symmetry and replacing this operator by another symmetric operator does not necessarily preserve the coefficients.

10.2.3 Multiplicity-one root of the ninth-order truncation

Define

$$P_9 = P_6 + T_{7:9}, \quad E_{21/22}^{(9)}(x) = P_9(x) - \delta, \quad I_\alpha = (0, 10^{-2}).$$

This finite operator has its own existence and uniqueness result.

Proposition 10.8 (Multiplicity-one root of the truncation). *The operator $E_{21/22}^{(9)}$ has a unique root ξ_9 in I_α , and that root is simple. Directed rational evaluation provides the isolation*

$$\frac{729735256928380099}{10^{20}} < \xi_9 < \frac{729735256928380100}{10^{20}}. \quad (10.12)$$

Proof. The internal-coordinate bounds $3.14 < \pi < 3.15$ and $0.045 < \delta < 0.046$ suffice for existence. At zero, $E_{21/22}^{(9)}(0) = -\delta < 0$. At 10^{-2} , the linear contribution minus the negative terms exceeds 0.0626, so the operator value is positive.

For $0 \leq x \leq 10^{-2}$, discarding the positive terms of the derivative gives

$$(E_{21/22}^{(9)})'(x) \geq 2\pi - \frac{7}{2} \cdot 10^{-2} - \frac{10}{21} \cdot 10^{-8} - \frac{6}{46} \cdot 10^{-10} - \frac{7}{120} \cdot 10^{-12} > 6.24.$$

Continuity gives existence; the derivative bound gives uniqueness and simplicity. The narrower interval (10.12) comes from directed rational evaluations with opposite signs. To make them explicit, for $t > 0$ and $z = (t - 1)/(t + 1)$, we use

$$\log t = 2 \sum_{j=0}^{m-1} \frac{z^{2j+1}}{2j+1} + R_m, \quad |R_m| \leq \frac{2|z|^{2m+1}}{(2m+1)(1-z^2)}.$$

Apply this with $t = 10/9$ and $t = 10$, $m = 520$, and divide the first interval by the second to bound δ . The generated interval for π is transported through the rational operations of B_9 ; evaluation at the two endpoints of (10.12) gives a negative and a positive sign, respectively. The executable supplement reproduces these exact operations. The existence and uniqueness proof does not use that interval as data to select the coefficients. $\square$

The result concerns ξ_9 . It does not authorize writing $\xi_9 = \alpha_A$ as an exact equality of complete numbers. A truncation root may share a prefix with the root of the prolonged operator and differ at a later position. Simplicity allows control of root movement under perturbations; it does not annul those perturbations.

10.3 Compatibility of the dodecaphase and analytic representations of alpha

Compatibility is established at every depth. The reversible dodecaphase representation and the analytic representation of the alpha coordinate $E_{21/22}$ are two internal representations with distinct codomains, produced from the same APP-TRIT-TPK enriched state. The jet $E_{21/22}^{(9)}$ is an exact finite witness, but does not by itself determine the tail. The completion constructed below does not follow from the bare jet: it receives the shared coordinate of the state and expresses its analytic image through an explicit recurrence. It therefore does not constitute an independent causal selection; it constitutes a typed analytic representation of the same generated point.

This distinction preserves the causal order

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \mathcal{X}_\alpha^{\text{enr}} \longrightarrow (P, E, \Phi, U_{12}^{\text{sgn}}, K) \longrightarrow \alpha_{\text{HMT}}.$$

The coordinates P, E, Φ and the record K are prior outputs of the same enriched state. Their use in the closure of alpha is therefore a subsequent internal composition, not an injection of conventional constants.

For $r \in 6\mathbb{N}$, $r \geq 36$, let $\hat{\mathcal{X}}_r^{\text{enr}, \alpha}$ be the sector of truncated states extending the internal alpha seed fixed at R_{36} . Write

$$x_r^\alpha = (\tilde{x}_r, \chi_\alpha, \varepsilon_r^\alpha, N_r^\alpha),$$

where χ_α is the internal character of the sector, not the expressed scalar value. Each state preserves residue, quotient, carry, sheet, orientation, phase, route, cylinder, boundary, signature, record, memory, survival, provenance, and depth. If

$$d_r^\alpha = \text{Dig}_{729}(\varepsilon_r^\alpha), \quad \varepsilon_{r+6}^\alpha = 729\varepsilon_r^\alpha - d_r^\alpha, \quad N_{r+6}^\alpha = 729N_r^\alpha + d_r^\alpha,$$

the effective transition is

$$u_r^\alpha = \text{Upd}_r^\alpha \circ \text{Tra}_r^\alpha \circ \text{Sel}_r^\alpha, \quad \text{Ext}_{r+6, r}^{\alpha, \rightarrow} = \text{graph}(u_r^\alpha).$$

The functional character of these extensions, with the seed R_{36} fixed, produces a unique compatible section: phase returns while memory and depth advance. The global domain of both determinations is

$$\mathcal{X}_\alpha^{\text{enr}} := \hat{\Omega}_\alpha := \varprojlim_{r \geq 36} (\hat{\mathcal{X}}_r^{\text{enr}, \alpha}, \text{Ext}_{r+6, r}^{\alpha, \rightarrow}). \quad (10.13)$$

10.3.1 Dodecaphase closure as a complete section

For $N \geq 1$, set

$$\mathcal{W}_N := \{0, \dots, 999\}^N, \quad \mathsf{T}_{\alpha, N}(\mathbf{s}) := (A_1, \dots, A_N), \quad \mathbf{s} \in \mathcal{X}_\alpha^{\text{enr}},$$

where the blocks A_j are obtained through (10.1) with the compatible boundary $c_{N+1}^\infty = [R_{N+1}(Z)]$. For $N' \geq N$, let $\rho_{N', N} : \mathcal{W}_{N'} \rightarrow \mathcal{W}_N$ be restriction to the first N blocks.

Theorem 10.9 (Complete dodecaphase output and projective compatibility). *The collection $(\mathsf{T}_{\alpha, N})_{N \geq 1}$ is compatible:*

$$\rho_{N', N} \circ \mathsf{T}_{\alpha, N'} = \mathsf{T}_{\alpha, N} \quad (N' \geq N). \quad (10.14)$$

Its canonical cylinders

$$C_N^A(\mathbf{s}) := \left[\sum_{j=1}^N A_j 1000^{-j}, \sum_{j=1}^N A_j 1000^{-j} + 1000^{-N} \right) \cap I_\alpha \quad (10.15)$$

are nested, have diameter at most 1000^{-N} , and satisfy

$$\bigcap_{N \geq 1} C_N^A(\mathbf{s}) = \{\alpha_A(\mathbf{s})\}, \quad \alpha_A = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}. \quad (10.16)$$

Consequently, the first route expresses a unique infinite section, denoted by $\alpha_{\text{HMT}} := \alpha_A$.

Proof. Proposition 10.4 provides the unique canonical expansion and the unique bounded carry sequence. The local identity (10.3) and its telescoping form (10.4) preserve the remote boundary; Proposition 10.6 proves that restricting an extended execution recovers precisely the previous prefix. This gives (10.14). Membership of the complete sum in every cylinder follows from the tail identity, and $\text{diam } C_N^A \leq 1000^{-N} \rightarrow 0$; the intersection contains a single point. Finally, (10.7) identifies this point with the expression in (10.16). All its coefficients and signs are fixed before α is expressed. $\square$

The window $\alpha_{12}^{(0)}$, closed with $c_{13} = 0$, remains an exact finite calculation; it must not be confused with the restriction of the complete section, whose compatible boundary satisfies $c_{13}^\infty = -1$. In particular,

$$\rho_{36, 12}(\alpha_{A, 36}) = \alpha_{A, 12} \quad (10.17)$$

is an equality of compatible words, not an equality between words of different lengths or an identification of the isolated window with the limit.

10.3.2 Exact analytic characterization of order nine

Let

$$\mathbb{K} := \mathbb{Q}\left(\pi_{\text{HMT}}, \log_{10} \frac{10}{9}\right), \quad E_9(x) := E_{21/22}^{(9)}(x) = R_9(x) - \delta.$$

Equations (10.10) and (10.11) construct $R_9 = R_6 + T_{7;9}$ from the state's signature and the nilpotent resolvent; Proposition 10.8 proves that E_9 has a unique simple root $\xi_9 \in I_\alpha$, isolated by the rational interval (10.12).

Finite bidirectionality can be expressed without introducing an analytic tail. Define

$$B_9(x) := \delta + \frac{7}{4}x^2 - \frac{x^4}{20} + \frac{2x^5}{21} + \frac{x^6}{46} + \frac{x^7}{120} - \frac{x^8}{45} - \frac{2x^9}{495}, \quad (10.18)$$

$$q_x(T) := xT^2 - B_9(x)T + x^3, \quad x \in I_\alpha = (0, 10^{-2}), \quad (10.19)$$

$$J_x : \mathbb{R}^\times \longrightarrow \mathbb{R}^\times, \quad J_x(T) := \frac{x^2}{T}. \quad (10.20)$$

Proposition 10.10 (Finite quadratic equivalence and involution). *For every $x \in I_\alpha$ and every $T \in \mathbb{R}^\times$,*

$$\frac{q_x(2\pi_{\text{HMT}})}{2\pi_{\text{HMT}}} = B_9(x) - \delta = E_9(x), \quad (10.21)$$

$$J_x(J_x(T)) = T, \quad T^2 q_x(J_x(T)) = x^2 q_x(T). \quad (10.22)$$

Let $Z_x = \{T \in \mathbb{R}^\times : q_x(T) = 0\}$. Consequently, $J_x|_{Z_x} : Z_x \rightarrow Z_x$ is involutive. When $E_9(x) = 0$, it exchanges the roots $2\pi_{\text{HMT}}$ and $x^2/(2\pi_{\text{HMT}})$. This is an exact involution on the scalar fibre of the order-nine jet; it does not act on the enriched state.

Proof. Dividing $q_x(T)$ by T and substituting $T = 2\pi_{\text{HMT}}$ reproduces the definition of $B_9(x) - \delta$ term by term. The first identity in (10.22) is immediate. For the second,

$$T^2 q_x(x^2/T) = x^5 - B_9(x)x^2T + x^3T^2 = x^2 q_x(T).$$

Thus J_x preserves the root set and, by Viète's formula, exchanges the two roots whose product is x^2 . $\square$

The finite decimal witness thus retains its exact force:

$$\text{Tr}_9(\xi_9^{-1}) = \text{Tr}_9((\alpha_{12}^{(0)})^{-1}) = 137.035999084. \quad (10.23)$$

Equality (10.23) certifies the finite characterization and the preceding involution. It does not turn the root of a truncated polynomial into the root of every compatible extension.

10.3.3 The jet alone does not determine an analytic extension

Let

$$j_9 : \mathbb{K}[[x]] \longrightarrow \mathbb{K}[x]/(x^{10})$$

be truncation, and for $h \in \mathbb{K}[x]$ set

$$E_h(x) := E_9(x) + x^{10}h(x). \quad (10.24)$$

All these polynomials satisfy $j_9(E_h) = j_9(E_9)$.

Theorem 10.11 (Non-uniqueness of the root from the jet). *There is no map*

$$\mathbf{r}_9 : \mathbb{K}[x]/(x^{10}) \longrightarrow I_\alpha$$

that assigns, from the jet alone, the positive root of every polynomial extension of E_9 having a unique root in I_α . In particular, E_0 and $E_{1/729}$ have the same jet and distinct positive roots.

Proof. The bound established in the proof of 10.8 gives $E'_9(x) > 6.24$ on $[0, 10^{-2}]$. For $t \geq 0$, write $E_t(x) = E_9(x) + tx^{10}$. Then

$$E'_t(x) = E'_9(x) + 10tx^9 > 6.24,$$

whereas $E_t(0) = -\delta < 0$ and $E_t(10^{-2}) > E_9(10^{-2}) > 0$. Each E_t therefore has a unique positive root $\eta(t) \in I_\alpha$. For $t = 1/729$,

$$E_{1/729}(\xi_9) = \frac{\xi_9^{10}}{729} > 0;$$

since $E_{1/729}$ is strictly increasing and changes sign only once, its root satisfies $\eta(1/729) < \xi_9 = \eta(0)$. However, $j_9(E_{1/729}) = j_9(E_0)$. A map that factored the root through j_9 would have to assign the same point to both polynomials, a contradiction. $\square$

The variation is not merely existential. The implicit function theorem applies because $\partial_x E_t(\eta(t)) > 6.24$, and gives

$$\eta'(t) = -\frac{\eta(t)^{10}}{E'_9(\eta(t)) + 10t\eta(t)^9} < 0. \quad (10.25)$$

Thus coefficients beyond degree nine are proved to change the root even while leaving the entire jet constructed through that degree invariant. This conclusion does not deny the tail produced by the enriched state: it proves only that the root does not factor through the bare truncation j_9 . Choosing a tail h from an external conventional root would be circular. The completion below consults neither that root nor a metrological table: it directly evaluates the regional and terminal coordinates of the state. Since that evaluation agrees algebraically with the integer representation, the result is a correlated representation, not an assertion of independence between two selectors.

10.3.4 Analytic representation of the alpha coordinate

The preceding analysis of the jet requires the tail to be exhibited: merely naming a complete reader is insufficient. Let $s \in \mathcal{X}_\alpha^{\text{enr}}$. Before decimal normalization, the same state already expresses the pre-coordinate

$$a(s) := p(s) + e_0(s) - f(s) - \kappa_{\text{per}}(s) \in I_\alpha, \quad (10.26)$$

with $p = \pi_{\text{HMT}} - 3$, $e_0 = e_{\text{HMT}} - 2$, and $f = \varphi_{\text{HMT}} - 1$. Equivalently, $a = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}$. Conventional α is not introduced here: p, e_0, f are the three regional characters already generated, and κ_{per} is the rational reading of the terminal record. The integer closure applies the normalization in (10.1) to (10.26); the chart below expresses a compatible analytic representation on the same pre-coordinate. Its dependency graph receives $(p, e_0, f, \kappa_{\text{per}})$ directly: it contains no edge $\alpha_A \rightarrow \lambda$, although the equality proved below allows the two representation names to be identified subsequently.

Write $q = 1/729$, let F_s be the ordered field generated by the coordinates of s , and set $E_{9,s}(X) = E_{21/22}^{(9)}(s; X)$. Define the linking coefficient

$$\lambda(s) := -\frac{E_{9,s}(a(s))(1 - qa(s)^3)}{a(s)^{10}}. \quad (10.27)$$

All its arguments belong to the state before metrological recognition. In particular, λ is not obtained by fitting a target decimal string. Equation (10.27) must be read with its exact type: once the state's first representation has produced $a(s)$, it fixes the unique extension within the geometric class defined below. It is not a second independent selection of $a(s)$, but a correlated analytic representation of the same enriched state.

Indeed, for $\mu \in F_s$, set

$$E_{s,\mu}(X) := E_{9,s}(X) + \mu \frac{X^{10}}{1 - qX^3}.$$

Since $0 < a(s) < 10^{-2}$ and $qa(s)^3 < 1$, we have

$$E_{s,\mu}(a(s)) = 0 \iff \mu = \lambda(s). \quad (10.28)$$

Thus the coefficient seed has no freedom once the state and the class of q -geometric extensions are fixed. This uniqueness is relative to the specified class; it does not assert that the order-nine jet alone selects a tail from all possible analytic series.

The local action on blocks of three coefficients is

$$\mathcal{C}_\alpha : F_s^3 \longrightarrow F_s^3, \quad \mathcal{C}_\alpha(u, v, w) = q(u, v, w), \quad c^{(0)} = (\lambda(s), 0, 0). \quad (10.29)$$

Consequently, for every $n \geq 0$,

$$b_{10+3n} = q^n \lambda(s), \quad b_{11+3n} = b_{12+3n} = 0. \quad (10.30)$$

This is the rule that computes every coefficient beyond degree nine. No free parameter remains at each extension.

For $M \geq 0$, let

$$E_{s,M}(X) := E_{9,s}(X) + \lambda(s) \sum_{n=0}^M q^n X^{10+3n}. \quad (10.31)$$

The complete function is

$$E_{s,\infty}(X) := E_{9,s}(X) + \lambda(s) \frac{X^{10}}{1 - X^3/729}. \quad (10.32)$$

To formulate the projective system without implicit types, fix $d_M = 12 + 3M$ and define the jet fibration

$$\mathfrak{F}_M := \coprod_{s \in \mathcal{X}_\alpha^{\text{enr}}} \{s\} \times F_s[X]/(X^{d_M+1}). \quad (10.33)$$

For $M' \geq M$, restriction is

$$\tau_{M',M}(s, [P]_{d_{M'}}) := (s, [P]_{d_M}). \quad (10.34)$$

Proposition 10.12 (Naturality of the analytic recurrence). *Let*

$$\Gamma_M(s) = (s, [E_{s,M}]_{d_M}).$$

For $M' \geq M$, truncation $\tau_{M',M}$ satisfies

$$\tau_{M',M} \circ \Gamma_{M'} = \Gamma_M, \quad \Gamma_{E21}(s) = \varprojlim_M \Gamma_M(s). \quad (10.35)$$

Projection onto the three new degrees is precisely $\mathcal{C}_\alpha^{M+1}(c^{(0)})$.

Proof. Equation (10.30) fixes the block of degrees $10 + 3n, 11 + 3n, 12 + 3n$. Passing from M to $M + 1$ adds only the next block; truncating it recovers the previous polynomial exactly. Compatibility for $M' \geq M$ follows by composition. Thus the inverse limit is not a section chosen from the bare jet, but the unique orbit of $c^{(0)}$ under $\mathcal{C}_\alpha$. $\square$

10.3.5 Uniform convergence, a simple root, and rational intervals

The regional cylinders and periodic record provide bounds through rational interval arithmetic. For each coordinate

$$c \in \{\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}\},$$

let P_c be its certified nonadic prefix of one thousand digits. The datum used is not the rational P_c treated as the complete value, but the half-open cylinder and its outer closure

$$\mathcal{C}_c^{(1000)} = [P_c, P_c + \varepsilon), \quad \overline{\mathcal{C}}_c^{(1000)} = [P_c, P_c + \varepsilon], \quad \varepsilon = 10^{-1000}.$$

Directed arithmetic operates conservatively on these closures. If P_π, P_e, P_φ denote the three prefixes, this gives

$$\begin{aligned} a_- &:= P_\pi + P_e - (P_\varphi + \varepsilon) - 4 - \kappa_{\text{per}}, \\ a_+ &:= (P_\pi + \varepsilon) + (P_e + \varepsilon) - P_\varphi - 4 - \kappa_{\text{per}}, \\ A_{1000} &:= [a_-, a_+], \quad a(s) \in [a_-, a_+] = A_{1000}, \quad \text{diam } A_{1000} = 3\varepsilon. \end{aligned} \quad (10.36)$$

The natural interval extension of (10.27) then transports A_{1000} , the cylinder of π_{HMT} , and the directed rational bound for δ to a closed interval $\Lambda_{1000} \ni \lambda(s)$; none of the three tails is frozen. In particular,

$$\begin{aligned} \frac{7297352569283800997285105472380662}{10^{36}} &< a(s) < \frac{7297352569283800997285105472380663}{10^{36}}, \\ -8.711 \times 10^{-6} &< \lambda(s) < -8.709 \times 10^{-6}. \end{aligned} \quad (10.37)$$

These inequalities use certified prefixes with their rational tail bound; they receive no alpha table. The package verifier reproduces them from the three nonadic outputs and the terminal chart.

Let

$$r := \frac{10^{-6}}{729} < 1.372 \times 10^{-9}.$$

For $0 \leq x \leq 10^{-2}$, the tail beyond M satisfies

$$|E_{s,\infty}(x) - E_{s,M}(x)| \leq 8.711 \times 10^{-26} \frac{r^{M+1}}{1-r}. \quad (10.38)$$

Moreover,

$$\left| \frac{d}{dx} \left(\lambda \frac{x^{10}}{1-x^{3/729}} \right) \right| \leq 8.711 \times 10^{-6} \left(\frac{10 \cdot 10^{-18}}{1-r} + \frac{3 \cdot 10^{-24}/729}{(1-r)^2} \right) < 8.712 \times 10^{-23}. \quad (10.39)$$

The derivative tail beyond block M also admits the depth-dependent estimate

$$\sup_{0 \leq x \leq 10^{-2}} |E'_{s,\infty}(x) - E'_{s,M}(x)| \leq 8.711 \times 10^{-24} r^{M+1} \left(\frac{13+3M}{1-r} + \frac{3r}{(1-r)^2} \right) \xrightarrow{M \rightarrow \infty} 0. \quad (10.40)$$

Theorem 10.13 (Limit function and unique simple root of the correlated chart). *The sequence $E_{s,M}$ converges uniformly, together with its derivatives, to $E_{s,\infty}$ on $\bar{I}_\alpha = [0, 10^{-2}]$. We have*

$$E_{s,\infty}(a(s)) = 0, \quad \inf_{x \in \bar{I}_\alpha} E'_{s,\infty}(x) > 6.239. \quad (10.41)$$

Consequently, $a(s)$ is the unique simple root of the complete function in I_α .

Proof. The ratio of two successive terms in the tail is bounded by $r < 1$, which gives (10.38); termwise differentiation and summation of the two geometric series give (10.39) and (10.40). The jet proof establishes $E'_{9,s} > 6.24$ on $\bar{I}_\alpha$; hence the complete perturbation leaves the derivative above 6.239. Finally, substituting (10.27) into (10.32) makes the two summands evaluated at $a(s)$ cancel exactly. The function is strictly increasing; its unique root is therefore simple and coincides with that pre-coordinate. $\square$

To make identification effective at every depth, directed endpoint evaluations first certify

$$E_{s,\infty}(0) < 0 < E_{s,\infty}(10^{-2}), \quad E'_{s,\infty} \geq \mu := 6239/1000 > c := 6 \quad \text{on } [0, 10^{-2}].$$

Continuity guarantees a root in the interval, and the second inequality ensures uniqueness. Only after this existence, monotonicity, and the invariant that the bracket contains the root have been established is the mean-value theorem fallback applied. Start with $D_0 = [0, 10^{-2}]$. If $D_j = [r_j, s_j]$, let $m_j = (r_j + s_j)/2$, and let $[F_j^-, F_j^+]$ be a directed rational enclosure of $E_{s,\infty}(m_j)$, obtained by propagating the cylinders of $\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}, \delta$, and λ , with

$$0 \leq F_j^+ - F_j^- \leq \frac{c}{4} (s_j - r_j). \quad (10.42)$$

When $F_j^+ < 0$, retain the right half; when $F_j^- > 0$, retain the left half. If $F_j^- \leq 0 \leq F_j^+$, no attempt is made to decide whether the midpoint is exactly the root. Let $w_j = s_j - r_j$. The uniform bound $E'_{s,\infty} \geq c = 6$ and the mean value theorem imply that every root contained in D_j belongs to

$$D_{j+1} = D_j \cap \left[m_j - \frac{w_j}{4}, m_j + \frac{w_j}{4} \right]. \quad (10.43)$$

Indeed, let ξ_j be the root, and let $y_j = E_{s,\infty}(m_j) \in [F_j^-, F_j^+]$. Since the enclosure contains zero and satisfies (10.42), we have $|y_j| \leq cw_j/4$. The mean value theorem gives $|m_j - \xi_j| \leq |y_j|/c \leq w_j/4$, proving (10.43). The argument includes the case $y_j = 0$: the root then lies at the midpoint, but the algorithm need not recognize that equality in order to retain it.

If the available enclosure does not satisfy (10.42), the source cylinders are first refined and the evaluation repeated; no sign is asserted and no insufficient contraction is accepted. In each of the three branches, the new interval retains the root and has diameter at most $w_j/2$. For every finite target depth, directed refinement makes it possible to satisfy the width bound without an uncertified comparison of a real number with zero. Monotonicity proves inductively that

$$a(s) \in D_{j+1} \subseteq D_j, \quad \text{diam } D_{j+1} \leq \frac{1}{2} \text{diam } D_j, \quad \text{diam } D_j \longrightarrow 0, \quad (10.44)$$

where the inequality is a contraction bound, not an imposed equality. The executable certificate applies this rule at twenty-four levels in base one thousand and verifies that the entire outer closure A_{1000} , not merely its lower endpoint, lies within each expressed cylinder. Its negative tests reject an excessively wide enclosure, a reversed enclosure, a point other than the midpoint, a domain without a sign change, and a degenerate bracket. Coinductive extension allows one thousand to be replaced by any required depth.

Specifically, for $S_N = 1000^N$, the verifier computes

$$j_N^- := \lfloor S_N a_- \rfloor, \quad j_N^+ := \lfloor S_N a_+ \rfloor \quad (10.45)$$

and expresses the level only if $j_N^- = j_N^+$. This equality, which includes the upper endpoint of the outer closure, proves $A_{1000} \subset [j_N^-/S_N, (j_N^- + 1)/S_N)$ and prevents an endpoint lying exactly on the right boundary from being assigned to the cell on its left.

Let $C_N^A(s)$ be the integer cylinder in (10.15). Recursively choose $m(N) > m(N - 1)$ so that $\text{diam } D_{m(N)} < 1000^{-N}/4$, and define

$$I_{B,N} := D_{m(N)} \cap C_N^A(s). \quad (10.46)$$

Define $\ell_N := \inf I_{B,N}$ and $u_N := \sup I_{B,N}$. Both are rational; the interval is non-empty because both factors contain $a(s)$. Since the cylinder is half-open, u_N need not belong to $I_{B,N}$. Evaluation at this endpoint is understood through the continuous extension of $E_{s,\infty}$ to the outer closure. Thus,

$$I_{B,N+1} \subseteq I_{B,N} \subseteq C_N^A(s), \quad \text{diam } I_{B,N} \longrightarrow 0, \quad E_{s,\infty}(\ell_N) \leq 0 \leq E_{s,\infty}(u_N). \quad (10.47)$$

Bound (10.38) allows a truncation to be chosen that preserves the sign at every endpoint where the complete function is non-zero. If an endpoint coincides with the root, the identity for the complete function and the established membership are used; the same sign is not attributed to its truncations. Indeed, the exact remainder is

$$E_{s,\infty}(x) - E_{s,M}(x) = \lambda(s)x^{10} \frac{(qx^3)^{M+1}}{1 - qx^3}.$$

On the positive domain under consideration, where $\lambda(s) < 0$, evaluation at the root gives

$$E_{s,M}(a(s)) = -\lambda(s)a(s)^{10} \frac{(qa(s)^3)^{M+1}}{1 - qa(s)^3} > 0 \quad (M < \infty).$$

Accordingly, certification of the weak sign inequalities in (10.47) concerns $E_{s,\infty}$. Evaluation uses the complete function or the truncation together with its signed remainder. This covers both the included infimum and the potentially excluded supremum of the cylinder.

Proposition 10.14 (Compatibility squares and common cylinders). *Restriction of words and of analytic representations commutes:*

$$\begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\tau_{\alpha,N'}} & \mathcal{W}_{N'} & \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_{M'}} & \mathfrak{F}_{M'} \\ \parallel & & \downarrow \rho_{N',N} & \parallel & & \downarrow \tau_{M',M} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\tau_{\alpha,N}} & \mathcal{W}_N & \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_M} & \mathfrak{F}_M \end{array}$$

The collection $I_{B,N}$ provides the effective connection between the two squares and explicitly satisfies $I_{B,N} \subseteq C_N^A$ for every N .

Proof. The first square expresses compatibility of integer normalization; the second is Proposition 10.12. The inclusion follows from Definition (10.46), and non-emptiness and the sign inequalities follow from Theorem 10.13. No step identifies the root of a truncation with the root of an arbitrary tail. $\square$

We now define, with the prefix reader made explicit,

$$\alpha_B(s) := \arg \operatorname{zero}_{x \in I_\alpha} E_{s,\infty}(x), \quad \operatorname{pref}_N(x) := \frac{\lfloor 1000^N x \rfloor}{1000^N}, \quad \alpha_{B,N} := \operatorname{pref}_N(\alpha_B). \quad (10.48)$$

Theorem 10.15 (Projective equality of the two correlated representations). *For every N , the integer route and the analytic route express the same prefix:*

$$\alpha_{A,N} = \operatorname{pref}_N(\alpha_A) = \operatorname{pref}_N(\alpha_B) = \alpha_{B,N}. \quad (10.49)$$

In the limit,

$$\alpha_A = \arg \operatorname{zero}_{x \in I_\alpha} E_{s,\infty}(x) = \alpha_B = \alpha_{\text{HMT}}. \quad (10.50)$$

Proof. Integer normalization expresses exactly $a(s)$. Theorem 10.13 proves that the analytic chart represents the same point as its unique simple root. The half-open cylinders C_N^A fix a canonical prefix, and $I_{B,N} \subseteq C_N^A$ contains $\alpha_B = \alpha_A$. Their prefixes therefore coincide for every N ; nesting and diameters tending to zero give equality of the limits. Metrological recognition occurs afterwards. $\square$

The analytic function constitutes a correlated representation of the same state, not a second derivation causally independent of the integer closure. The equality identifies the same point through the integer closure and the correlated analytic representation, both constructed from the enriched state. The continuation is determined by the pre-coordinate and the explicit recurrence, not by the ninth-order jet alone; conventional recognition follows generation.

11 Exceptional prolongation: from dodecaphasic incidence to Moonshine

The numerical expression does not exhaust the state that generated it. In the preceding construction, APP preserves the two sheets and division with remainder and quotient; the TRIT determines the local regime and its orientation; the TPK composes selection, transport, carry, memory, and return. The Archimedean reading contracts compatible cylinders. The reading now under study instead preserves the incidence of positions and their transport between frames. Both start from the same arithmetic state with trajectory memory. An exceptional geometry is not reconstructed from the final decimals of π , φ , e , or α .

This point clarifies the meaning of the prolongation $\alpha/K \rightarrow$ exceptional incidence. The arrow does not assert that an isolated real number determines a code, a lattice, or a group. It transports the dodecaphasic record and the signed data that entered the determination of α , together with the words and frames arising from APP–TRIT–TPK. In particular, the ordered record K can be recovered from its rational constant κ under the previously fixed base, length, and origin conventions. Projection onto a support, by contrast, preserves only the selected incidences. For its part, K_{ph} denotes a different, reduced-memory reading, subsequent to the holonomy Γ_9 .

The exposition separates three mathematical operations: constructing the finite objects and their maps; prolonging them to a lattice and a vertex operator algebra; and subsequently recognizing the resulting objects through the relevant classical theorems. The names Paley, Witt, Mathieu, Golay, Leech, and Monster designate those recognitions and the explicit realizations that make them verifiable. They are not retrospective selectors of the seeds or coefficients of the generation.

The argument begins with compatibility of prefixes and frames, constructs the ternary code, and determines the flag through K and the signed record of α . Marked incidence then selects the lattice origin and the Leech neighbor. The final step realizes the vertex algebra and its automorphisms. At each stage, the structure preserved will be specified: linear combinations, supports, discriminant classes, bilinear form, or graded product. Explanations of the function of K , normalization, and the origin of norm 54 accompany their respective constructions in Sections 11.3 and 11.13.

11.1 Compatibility of the Archimedean and incidence realizations

Let X_n^{cal} be the space of admissible prefixes of length n of the arithmetic state with trajectory memory, with truncations $p_{n,m} : X_n^{\text{cal}} \rightarrow X_m^{\text{cal}}$ for $m \leq n$. The superscript indicates that the declared initial frame and its transport are preserved, not that a parameter is fitted against an experimental value. We write (w_j, f_j) for the visible six-trit word and the frame at step j . The frames satisfy a causal law

$$f_{j+1} = g_j(x)f_j,$$

where $g_j(x)$ depends on the prefix already constructed. Consequently, a prolongation does not alter earlier frames.

The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is preserved in this reading. The return of the phase resets neither the frame nor the memory of the state. Likewise, the census of 468 decimal emissions, the 243 visible six-trit words, and the ambient space $\mathbb{F}_3^6$, with 729 words, remain distinct. The ambient space will be used to construct a linear code; it does not replace the admissible domain of emissions.

For a six-trit block b , the Archimedean reading takes its integer address $\nu(b) \in \{0, \dots, 728\}$. A prefix determines

$$q_n = m + \sum_{j=0}^{n-1} \frac{\nu(b_j)}{729^{j+1}}, \quad I_n = [q_n, q_n + 729^{-n}].$$

Prefix compatibility gives nested cylinders whose diameter tends to zero. Their Cauchy class belongs first to the internal completion $\mathbb{R}_{\text{HMT}}$; its subsequent identification with the usual real line does not enter the choice of blocks.

For the incidence reading we introduce, on the same prefix, the graph map

$$\gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad w \longmapsto (w, wA). \quad (11.1)$$

Words are written as row vectors. If $A^f = fAf^{-1}$, the change of frame satisfies the concrete identity

$$\gamma_{A^f}(wf^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}). \quad (11.2)$$

The calibrated codomain retains f together with the encoded word. We therefore define

$$Q_n(x) = ((f_j, \gamma_{f_j A_w f_j^{-1}}(w_j)))_{0 \leq j < n}. \quad (11.3)$$

Proposition 11.1 (Compatibility and prolongation of the incidence reading). *If $r_{n,m}$ truncates the encoded sequences, then*

$$r_{n,m} Q_n = Q_m p_{n,m}.$$

Thus the Q_n determine a unique map Q_∞ between the respective inverse limits. In each frame, the map preserves the visible word exactly, since $\text{pr}_1 \gamma_A = \text{id}$.

Proof. Two compatible prefixes have the same words and the same transport operators through index $m - 1$. By induction, they have the same frames at those indices. The first m components of (11.3) agree, proving the square. The universal property of the inverse limit gives Q_∞ . The recovery claim is the projection onto the first six coordinates of each graph. $\square$

This local injectivity has a precise domain. It does not prove that Q_∞ recovers all the deep memory of the TPK state if only the sequence of visible words and frames has been retained. Still less does it prove injectivity after passing from words to supports. Moreover, a general change of basis in $\text{GL}_6(\mathbb{F}_3)$ does not by itself preserve ordinary Hamming weight. In a general chart, the incidence and metric of the initial chart are also transported. The support calculations below are performed in the declared Paley chart; monomial relabelings directly preserve their coordinate interpretation.

11.2 The finite realization and the ternary code

The elliptic operator of the TRIT admits the following finite realization in the six-position chart. These positions coordinatize the six units $\{1, 2, 4, 5, 7, 8\}$ modulo nine, preserved by modular transport. On $\mathbb{P}^1(\mathbb{F}_5)$, with the order $\infty, 0, 1, 2, 3, 4$ declared, let C be the matrix with zero diagonal, $C_{\infty,a} = C_{a,\infty} = 1$, and $C_{a,b} = \chi(b - a)$ at the five finite positions. Here $\chi(1) = \chi(4) = 1$, $\chi(2) = \chi(3) = -1$, and $\chi(0) = 0$. This is a coordinate realization of the quadratic structure already transported; identifying the six positions with the projective line is chart data, not a new numerical input into the generator.

The quadratic identity in this chart follows from a finite correlation. The order $(\infty, 0, 1, 2, 3, 4)$ is retained, together with the declared identification of the six positions and the character χ .

Lemma 11.2 (Correlation of the quadratic character on five positions). *For $a, b \in \mathbb{F}_5$ with $a \neq b$,*

$$\sum_{t \in \mathbb{F}_5} \chi(t - a) \chi(t - b) = -1. \quad (11.4)$$

Proof. The substitution $u = (t - a)/(b - a)$ is bijective because $b - a$ is invertible in $\mathbb{F}_5$. Furthermore,

$$(t - a)(t - b) = (b - a)^2 u(u - 1), \quad \chi((b - a)^2) = 1.$$

Multiplicativity of χ , including its extension by zero, reduces the sum to $\sum_u \chi(u(u - 1))$. The five evaluations are

u	0	1	2	3	4	(11.5)
$u(u - 1) \pmod{5}$	0	0	2	1	2	
$\chi(u(u - 1))$	0	0	-1	1	-1	

Their sum is -1 , proving (11.4). $\square$

Proposition 11.3 (Orthogonality of the six-position chart). *The matrix C satisfies*

$$C = C^\top, \quad CC^\top = 5I_6, \quad C^2 = 5I_6. \quad (11.6)$$

Proof. Since $\chi(-1) = \chi(4) = 1$, one has $\chi(b-a) = \chi(a-b)$; entries involving ∞ are also symmetric. Each row contains one zero entry and five entries of absolute value one, so its squared norm is five. For two distinct finite rows, the inner product receives +1 from the ∞ coordinate and -1 from the five finite coordinates by (11.4); their sum is zero. The inner product of the infinite row with a finite row is $\sum_{t \in \mathbb{F}_5} \chi(t) = 0 + 1 - 1 - 1 + 1 = 0$. This evaluates every entry of $CC^\top$. Symmetry converts this identity into $C^2 = 5I_6$. $\square$

The argument retains the chart order to be reduced modulo three. The quadratic identity and the subsequent graph code therefore use the same coordinates, with their incidences and supports.

The quadratic-sum identities give $C^2 = 5I_6$. Reducing modulo three yields

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \quad A_W^2 = 2I_6 = -I_6. \quad (11.7)$$

The quadratic equality realizes the elliptic regime; it does not identify the field $\mathbb{F}_3$ with an Archimedean realization of the imaginary unit. They are distinct realizations of a common operator relation, with their corresponding domains.

11.2.1 Dual neutrality and orientation of the first joint boundary

The prolongation boundary joins the three regional histories through a relation preserving the axis, lateral aggregate, saturation, and column occupancy. In the established row-vector chart, the final six-position blocks of the words of length thirty are

$$u_4^\pi = 110221, \quad u_4^e = 102011, \quad u_4^\varphi = 010200.$$

The superscripts retain the three regions of (6.1); each word is interpreted as an element of $\mathbb{F}_3^6$, with integer representatives in $\{0, 1, 2\}$. The boundary relation transports the axis by

$$z := u_5^\varphi = u_4^e = 102011, \quad a := u_4^\varphi A_W L_1^3 = 111101. \quad (11.8)$$

Here a is the oriented contrast between the two lateral channels and the axis. The corresponding lateral aggregate is

$$x + y = 210112, \quad q := x + y + z = 012120, \quad a = x + y - z = 111101 \quad \text{in } \mathbb{F}_3^6, \quad (11.9)$$

where $x = u_5^\pi$ and $y = u_5^e$. The sum and the contrast have distinct dual readings: $qA_W = 012000$ and $aA_W = 120210$.

Let $B \in M_{3,6}(\mathbb{F}_3)$ be the matrix with rows x, y, z . Its canonical lift defines, column by column,

$$c_j = \left\lfloor \frac{x_j + y_j + z_j}{3} \right\rfloor, \quad o_j = \mathbf{1}_{x_j \neq 0} + \mathbf{1}_{y_j \neq 0} + \mathbf{1}_{z_j \neq 0}.$$

The law for the boundary at depth six fixes $c_j = 1$. With the axis z and the aggregate q already determined, the integer lateral sums are

$$(x_j + y_j)_{j=1}^6 = (2, 4, 3, 4, 4, 2).$$

The minimum-occupancy rule selects, among decompositions with $x_j, y_j \in \{0, 1, 2\}$, the columns with the fewest nonzero entries, including z_j . For sum two, the extreme pairs $(0, 2), (2, 0)$ occupy one lateral

position; for sum three, the pairs (1, 2), (2, 1) occupy two; for sum four, only (2, 2) is possible. Including the axis gives $o = (2, 2, 3, 2, 3, 2)$. At this boundary, with $s = [x + y]_3$, the same rule is expressed by

$$o_j = 2 + \mathbf{1}_{\{z_j \neq 0, s_j \neq 2\}}.$$

At this boundary, saturation and occupancy are

$$c = 111111, \quad o = 223232. \quad (11.10)$$

Thus the integer equality in each column is $x_j + y_j + z_j = q_j + 3$. The carry c records the integer lift of this boundary; its domain consists of the six columns under consideration.

Proposition 11.4 (Eight distributions of a saturated boundary). *Conditions (11.8)–(11.10) leave exactly eight matrices. Their column alternatives are*

j	z_j	$x_j + y_j$ in $\mathbb{Z}$	(x_j, y_j)
1	1	2	(0, 2), (2, 0)
2	0	4	(2, 2)
3	2	3	(1, 2), (2, 1)
4	0	4	(2, 2)
5	1	4	(2, 2)
6	1	2	(0, 2), (2, 0)

(11.11)

Proof. Axis transport fixes the third row. The integer equality $x_j + y_j = q_j + 3 - z_j$, together with o_j , gives the alternatives in the table. The first, third, and sixth columns each admit two independent choices; the other three admit one. Their cardinalities have product $2 \cdot 1 \cdot 2 \cdot 1 \cdot 1 \cdot 2 = 8$, and each choice preserves every stated equality. $\square$

Neutrality is evaluated in the dual chart. With $\mathbf{1} = (1, 1, 1, 1, 1, 1)^\top$, the matrix in (11.7) determines

$$\omega = A_W \mathbf{1} = (2, 1, 1, 1, 1, 1)^\top, \quad A_W \omega = -\mathbf{1}. \quad (11.12)$$

The dual closure condition at the boundary is

$$(BA_W)\mathbf{1} = B\omega = 0 \quad \text{in } \mathbb{F}_3^3. \quad (11.13)$$

This condition acts on the collection of saturated boundaries. The quadratic relation for A_W determines transport of the dual unit; neutrality specifies which boundaries satisfy the corresponding closure.

Proposition 11.5 (Dual pair and oriented boundary). *Neutrality (11.13) leaves exactly*

$$B_- = \begin{pmatrix} 0 & 2 & 1 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 & 2 & 0 \\ 1 & 0 & 2 & 0 & 1 & 1 \end{pmatrix}, \quad B_+ = \begin{pmatrix} 2 & 2 & 2 & 2 & 2 & 0 \\ 0 & 2 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \end{pmatrix}. \quad (11.14)$$

The involution interchanging closure and propagation permutes B_- and B_+ . In the half-open chart with the APP orientation fixed for the channels, the canonical boundary is $R_{36} = B_+$.

Proof. Write the three binary choices in the first row as

$$x = (2\varepsilon_1, 2, 1 + \varepsilon_3, 2, 2, 2\varepsilon_6), \quad \varepsilon_1, \varepsilon_3, \varepsilon_6 \in \{0, 1\}.$$

The condition $x\omega = 0$ is equivalent to

$$1 + \varepsilon_1 + \varepsilon_3 - \varepsilon_6 \equiv 0 \pmod{3}.$$

The possibilities are (0, 0, 1) and (1, 1, 0). The axis row satisfies $z\omega = 0$; the lateral aggregate also has zero dual evaluation. Consequently, $y\omega = 0$ in both cases. This gives exactly the two matrices in (11.14).

The final choice applies the APP orientation of the half-open chart to this mirror pair. In that chart, the ordered representations are

$$B_- \mapsto (789, 048, 894), \quad B_+ \mapsto (793, 045, 894).$$

The representative of the fixed orientation is B_+ . This selection preserves the order of the three regions and the boundary convention of their cylinders. The algebraic reduction from eight matrices to two and the choice of oriented representative thus constitute two stages of the same construction. $\square$

The visible charges distinguish these stages precisely:

$$B_- \mathbf{1} = (0, 1, 2)^\top, \quad B_+ \mathbf{1} = (1, 0, 2)^\top, \quad B_\pm \omega = 0. \quad (11.15)$$

Visible charge and dual neutrality are evaluations on distinct vectors. The selected boundary retains three rows of six trits, with ambient indices (726, 215, 301), and supplies the junction state R_{36} for joint prolongation. Its memory includes the transported axis, saturation, column incidence, and the orientation distinguishing the sheets of the pair.

Definition 11.6 (Code of the dodecaphasic chart). The ternary code associated with (11.7) is

$$C_W = \{(w, wA_W) : w \in \mathbb{F}_3^6\} \subset \mathbb{F}_3^{12}.$$

Its set of hexads is

$$\mathcal{H} = \{\text{supp}(c) : c \in C_W, \text{wt}(c) = 6\}.$$

Proposition 11.7 (Self-duality, distance, and the Witt design). *The code C_W is self-dual with parameters $[12, 6, 6]_3$. Its weight enumerator is*

$$\sum_{c \in C_W} z^{\text{wt}(c)} = 1 + 264z^6 + 440z^9 + 24z^{12}. \quad (11.16)$$

The set $\mathcal{H}$ has 132 elements and constitutes an $S(5, 6, 12)$ system: every set of five positions is contained in a unique hexad.

Proof. The generator matrix $[I_6 \mid A_W]$ has rank six and, by the symmetry of A_W , satisfies

$$[I_6 \mid A_W][I_6 \mid A_W]^\top = I_6 + A_W^2 = 0.$$

The code is self-orthogonal with dimension half its length and is therefore self-dual. Finite evaluation of the 3^6 words gives (11.16); this is an enumeration over the explicit matrix, without real values or external targets. In particular, the minimum distance is six.

If two weight-six words have the same support, one of the signs 1, -1 occurs at least three times among the six ratios of their nonzero coordinates. Subtracting the corresponding multiple would produce a nonzero word of weight at most three, which is impossible. Hence a support corresponds to exactly the pair $\{c, -c\}$, and there are $264/2 = 132$ supports. If two distinct supports shared five positions, the same argument would cancel at least three coordinates in a union of size seven, producing weight at most four. Thus a five-element set belongs to at most one hexad. Finally,

$$132 \binom{6}{5} = 792 = \binom{12}{5},$$

so it belongs to one and only one. $\square$

The subsequent recognition is the extended ternary Golay code and the small Witt design [13]. The preceding proof explains which object is recognized. It does not replace the APP-TRIT-TPK genealogy with a search among known codes.

11.3 Dodecaphasic flag associated with K and α

Articulation of regional coordinates and closure incidence. The function of this prolongation is understood through a distinction among three operations: preserving a record, evaluating a coordinate, and extracting an incidence relation. All three act on information produced by APP–TRIT–TPK. Their articulation makes explicit the article’s structural thesis: a numerical realization has an operator provenance, and certain data of that provenance additionally admit a combinatorial and lattice realization. The equations make it possible to follow both routes with their domains and the information remaining in each.

The starting point is the terminal state with its compatible prolongations. It preserves the residues and quotients of the two APP sheets, the TRIT regime and orientation, and TPK transport with its memory. The closure, propagation, and self-scaling regions are determined in that domain before their readers establish the coordinates π, e, φ . At the same time, the oriented record of the twelve windows determines the signed vector U . The subsequent appearance of K arises from the integral transformation proved in §9.8:

$$U = \mathcal{H}_{12}K, \quad K = \frac{1}{4}\mathcal{H}_{12}U, \quad \mathcal{H}_{12}^2 = 4I_{12}. \quad (11.17)$$

The congruence defining $\mathcal{L}_H$ guarantees integrality of the inverse. Thus, K constitutes a reversible coordinatization of the signed record in the pertinent sublattice. Its positions preserve phase order; its differences D_3K, D_4K preserve transversal relations; the sum $Q(K)$ determines the uniform component. That structure explains why the record enters both numerical composition and selection of a boundary configuration.

The periodic evaluation

$$K \mapsto \kappa_{\text{per}} = \frac{\sum_{m=1}^{12} K_m 1000^{12-m}}{1000^{12} - 1}$$

keeps the twelve blocks recoverable when base, length, and phase origin are fixed. Extracting a support performs a different operation: it preserves the membership of each position in a class defined by a threshold and groups records having the same support. The distinction between these two maps is decisive. The scalar precision of κ_{per} and the incidence of B_K are two specific mathematical contents, related through the complete record from which they arise.

In the frame inherited from the dodecaphasic record, the data are

$$\begin{aligned} K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601), & (11.18) \\ w_\pi &= 010211, & w_e &= 201101, \\ w_\varphi &= 121200, & w_{\text{APP}} &= 022020. \end{aligned}$$

Applying γ_{A_W} yields the following incidences:

$$\begin{aligned} P_\pi &= \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, \\ H_e &= \{1, 3, 4, 6, 9, 10\}, \\ H_\varphi &= \{1, 2, 3, 4, 7, 9\}, \\ H_{\text{APP}} &= \{2, 3, 5, 10, 11, 12\}. \end{aligned}$$

The first is the support of a weight-nine word; the other three are hexads. Through its high-block reading, the record supplies

$$B_K = \{i : K_i \geq 729\} = \{4, 6, 9, 10\} = H_e \cap P_\pi. \quad (11.19)$$

Here the integer 729 marks a boundary of the declared block format. It is not identified merely by cardinality with the ternary ambient space used in (11.6). In this particular record, all integer thresholds between 660 and 729 produce the same support: the support recovers neither the threshold nor the twelve blocks.

Before carry, the arithmetic route to α preserves the signed vector

$$Z_0 = (7, 297, 353, -430, -715, -1199, -3, 286, -894, -528, 380, 663). \quad (11.20)$$

Its negative support is

$$H_{\alpha}^{-} = \{4, 5, 6, 7, 9, 10\}. \quad (11.21)$$

These signs are not extracted from the decimal expansion of the scalar α ; they are retained in the state preceding its numerical expression.

Normalization of alpha and its signed configuration. The coordinate of α is determined along the positional route by composing regional blocks and the dodecaphase record. Before positional division, the oriented combination has components $Z_j = B_j + E_j - \Phi_j - K_j^{\text{per}}$, and the balance with incoming quotient is $s_j = Z_j + c_{j+1}$. Normalization satisfies

$$A_j = s_j - 1000c_j = Z_j + c_{j+1} - 1000c_j, \quad 0 \leq A_j < 1000. \quad (11.22)$$

This identity simultaneously preserves the normalized block and the contribution of the transported quotient. The limit of compatible prefixes determines the scalar closure coordinate. The signed configuration, for its part, preserves which positions in the combination before normalization exhibit negative defect. In the initial window, that configuration is the vector Z_0 of (11.20), and its negative support is H_{α}^{-} .

Two contents of the same operation are thus defined. Positional normalization determines a real coordinate through its compatible remainders and quotients; negative support determines a configuration of positions in the twelve-component cycle. The sign refers to the balance of records before normalization. Its incidence reading remains available because the state preserves that balance alongside the normalized result. This articulation corresponds to the proved positional route; comparison with the analytic vacancy operators preserves the precise scope established in the chapter on α .

Proposition 11.8 (Agreement of the two flag selectors). *Within $\mathcal{H}$, the support (11.21) is the unique hexad H satisfying*

$$B_K \subset H \subset P_{\pi}, \quad (|H \cap H_e|, |H \cap H_{\varphi}|, |H \cap H_{\text{APP}}|) = (4, 3, 2).$$

Thus the signed reading and the incidence reading select the same flag $B_K \subset H_{\alpha}^{-}$.

Proof. The four hexads containing B_K are obtained by adjoining, respectively, the pairs

$$\{1, 3\}, \quad \{2, 11\}, \quad \{5, 7\}, \quad \{8, 12\}.$$

Only the second and third pairs are contained in P_{π} . For the second, the intersection with H_{φ} has size three, and that with H_{APP} has size three; it does not meet the last condition. For the third, the three intersections have sizes 4, 3, 2. This last pair produces precisely the negative support of (11.20). $\square$

It is useful to retain both the flag and its marked frame:

$$\mathcal{J}_* = (B_K \subset H_{\alpha}^{-}), \quad \tilde{\mathcal{J}}_* = (\mathcal{J}_*; P_{\pi}, H_e, H_{\varphi}, H_{\text{APP}}).$$

Union, intersection, and difference are equivariant under a simultaneous relabeling of the design. The marked class, not the position numbers taken in isolation, is the transported object.

Lemma 11.9 (Origin selected by the marked incidence). *For the preceding frame,*

$$o_* = (H_e \cap H_{\varphi}) \setminus (H_{\alpha}^{-} \cup H_{\text{APP}}) = \{1\}.$$

The rule is equivariant under every automorphism applied simultaneously to the data of $\tilde{\mathcal{J}}_$.*

Proof. The first intersection is $\{1, 3, 4, 9\}$. The subtracted union contains 3, 4, 9 and does not contain 1. Equivariance follows because a bijection commutes with set operations. $\square$

This origin will select a direction of the lattice neighbor once the A_2 metric and the positive chamber are declared. Incidence alone does not contain that metric. The lattice step will explicitly preserve both data.

From ternary coding to the marked flag. The regions also provide the codes w_π, w_e, w_φ and w_{APP} in the inherited frame. The map

$$\gamma_{A_W} : w \mapsto (w, wA_W)$$

preserves the six-component word as its first coordinate and adds its transformation by A_W . The relation $A_W^2 = -I_6$ fixes the quadratic compatibility of this realization. The next step, $c \mapsto \text{supp}(c)$, preserves the nonzero positions. For weight-six words, the support identifies exactly the pair $\{c, -c\}$, as proved by the minimum distance of the code. In this way, the quotient by sign produces the hexads, while the complete code remains available for operations requiring its symbols.

In this chart, the two constructions of the tetrad coincide:

$$\{i : K_i \geq 729\} = H_e \cap P_\pi = B_K = \{4, 6, 9, 10\}. \quad (11.23)$$

The left-hand side arises from an integer comparison of the components of K ; the right-hand side, from the intersection of the coded supports of propagation and closure. The threshold belongs to the declared block chart and remains explicit in the reader's definition. The complete record preserves the values on which it acts, even when different thresholds produce the same subset. The equality therefore expresses compatibility between specified maps.

The support H_α^- completes that tetrad to a hexad. The membership condition in support P_π , together with intersection sizes $(4, 3, 2)$ relative to $H_e, H_\varphi, H_{\text{APP}}$, selects a unique hexad. Proposition 11.8 proves that determination by the sign of Z_0 agrees with determination by incidence; Lemma 11.9 fixes the origin in the resulting frame. This information is organized as

$$\tilde{J}_* = (B_K \subset H_\alpha^-, P_\pi, H_e, H_\varphi, H_{\text{APP}}). \quad (11.24)$$

The inclusion is the flag; the four additional supports constitute its marked frame. Simultaneous permutations transport the frame and preserve its inclusions and intersections. The relevant information resides in this transported configuration and its isomorphism class.

The five regions of π here recover their internal differentiation. They share w_π as visible ternary code, but retain distinct emissions, signatures, and multiplicities. In the binary realization on a marked dodecad, the transversal region corresponds to the octad whose intersection with the dodecad is the tetrad; the other four regions correspond to lifts of the four hexads of its star. The negative region is distinguished by H_α^- , and chart order individualizes the three positive regions. This correspondence preserves regional roles and the distribution $432 + 4 \cdot 144 = 1008$, in addition to realizing the incidence partition $1 + 4$.

11.4 Isometric intertwining of the incidence representations

Incidence defines a linear map whose action can be checked position by position:

$$I : \mathbb{R}^{12} \longrightarrow \mathbb{R}^{\mathcal{H}}, \quad (Iv)(H) = \sum_{p \in H} v_p. \quad (11.25)$$

Let $V_{11} = \mathbf{1}^\perp \subset \mathbb{R}^{12}$. The two Euclidean realizations are subsequent languages for the same finite design here.

Proposition 11.10 (Isometry of the dodecaphasic screen). *We have*

$$I^\top I = 36I_{12} + 30J_{12},$$

where J_{12} is the all-ones matrix. Consequently,

$$U_W = \frac{1}{6}I|_{V_{11}} : V_{11} \xrightarrow{\sim} E_{\text{inc}} := I(V_{11})$$

is an equivariant isometry for the permutation actions of the automorphisms of the design.

Proof. Each point belongs to $r = \binom{11}{4}/\binom{5}{4} = 66$ hexads, and each pair to $\lambda_2 = \binom{10}{3}/\binom{4}{3} = 30$. These are, respectively, the diagonal and off-diagonal entries of $I^\top I$. On $\mathbf{1}^\perp$, the J_{12} term vanishes, leaving $36I$. Finally, if g is an automorphism, $p \in H$ is equivalent to $g(p) \in g(H)$; hence $I\rho_{12}(g) = \rho_{\mathcal{H}}(g)I$. $\square$

The incidence information also has a concrete spectral reading. Define

$$G_{H,H'} = \binom{|H \cap H'|}{4}.$$

Lemma 11.11 (Action of the intersection kernel). *If $J_{132,12}$ is the rectangular all-ones matrix,*

$$GI = 30I + 15J_{132,12}.$$

Thus G acts by the scalar 30 on E_{inc} .

Proof. The (H, p) entry of GI counts pairs (T, H') with $T \subset H \cap H'$, $|T| = 4$, and $p \in H'$. If $p \in H$, there are ten four-element sets T containing p , each contained in four hexads, and five not containing it, each with a unique extension of the five-element set $T \cup \{p\}$. The total is $10 \cdot 4 + 5 = 45$. If $p \notin H$, the fifteen four-element subsets of H each give one extension: the total is 15. This yields the identity. The rectangular term vanishes on V_{11} . $\square$

There is indeed a material intertwiner here: its domain, codomain, entries, and normalization are determined. For example, any orthogonal projector P already constructed on V_{11} is transported to $Q = U_W P U_W^*$ on E_{inc} , with $U_W P = Q U_W$. This statement does not identify all the operators acting on these spaces solely by their trace.

11.5 The direction selected by the dodecaphase record

The record K has a geometric function complementing its rational evaluation and support reading. Once the representation of the twelve positions has been constructed, its components determine a direction in the space of zero-mean variations. The reduction $12 \rightarrow 11$ eliminates the uniform mode; the following reduction $11 \rightarrow 10$ eliminates the direction selected by the record itself. The two operations therefore have different antecedents.

To declare the chart used completely, consider the group A_5 of even permutations of five objects and the subgroup $C_5 = \langle (1\ 2\ 3\ 4\ 5) \rangle$. The twelve left cosets $r_p C_5$ are ordered through these representatives, written as image lists:

p	r_p	p	r_p
1	(4, 3, 5, 2, 1)	7	(5, 4, 3, 2, 1)
2	(4, 2, 1, 3, 5)	8	(1, 3, 4, 2, 5)
3	(1, 2, 4, 5, 3)	9	(4, 2, 3, 5, 1)
4	(2, 5, 3, 4, 1)	10	(3, 2, 5, 4, 1)
5	(4, 3, 1, 5, 2)	11	(4, 5, 2, 3, 1)
6	(5, 1, 2, 3, 4)	12	(5, 3, 2, 4, 1).

The action by left multiplication defines an orthogonal representation $\rho : A_5 \rightarrow O(\mathbb{R}^{12})$: $\rho(a)e_p = e_q$ when $ar_p C_5 = r_q C_5$. This representation chart acts on the record already obtained; group characters are not used to choose its twelve values.

Let χ_3 be the values of the three-dimensional character

class	1	2^2	3	5_a	5_b
χ_3	3	-1	0	$(1 + \sqrt{5})/2$	$(1 - \sqrt{5})/2$.

The class 5_a contains the displayed generator of C_5 , and 5_b contains its square. This convention distinguishes the two conjugate three-dimensional sectors of the representation. Define

$$B_3 = \frac{3}{60} \sum_{a \in A_5} \chi_3(a^{-1}) \rho(a), \quad B_{11} = I_{12} - \frac{1}{12} J_{12}. \quad (11.26)$$

Both matrices are determined by finite sums in $\mathbb{Q}(\sqrt{5})$. The appearance of this quadratic field belongs to the realization of the representation, subsequent to regional generation of φ .

Lemma 11.12 (Projection of the record onto the three-dimensional sector). *One has*

$$P_3 = P_3^\top = P_3^2, \quad P_3 \mathbf{1} = 0, \quad \text{tr } P_3 = 3, \quad P_3 P_{11} = P_{11} P_3 = P_3.$$

For the record (11.18), the vector

$$u_K = P_3 P_{11} K \tag{11.27}$$

is nonzero. Its exact norm is

$$\|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0. \tag{11.28}$$

Proof. The character sum in (11.26) is the isotypic projector. The real character and substitution $a \mapsto a^{-1}$ give symmetry. The multiplicity of the sector in the action on A_5/C_5 is the dimension of its vectors fixed by C_5 ,

$$\frac{1}{5} \left(3 + 2 \frac{1 + \sqrt{5}}{2} + 2 \frac{1 - \sqrt{5}}{2} \right) = 1.$$

Its rank is therefore three. Orthogonality to the trivial character gives $P_3 \mathbf{1} = 0$ and the two compositions with P_{11} . These identities may also be verified by multiplying the matrices in the finite sum.

Finally, $\sum K_p = 6263$, and substituting the twelve components into that same sum gives

$$\|u_K\|^2 = K^\top P_3 K = \frac{1}{20} \sum_{a \in A_5} \chi_3(a^{-1}) K^\top \rho(a) K = \frac{6638585 + 2275584\sqrt{5}}{20}.$$

This is an exact evaluation of sixty specified terms, reproduced by the accompanying certificate using quadratic rational arithmetic. The two positive coefficients prove nonvanishing. $\square$

Definition 11.13 (Dodecaphase direction and transversal complement). The direction selected by K in the declared chart is $\ell_K = \mathbb{R}u_K$. Its transversal complement within $V_{11} = \mathbf{1}^\perp$ is

$$V_{10}(K) = V_{11} \cap u_K^\perp.$$

Proposition 11.14 (Orthogonal reduction determined by K). *The operator*

$$P_{10}(K) = P_{11} - \frac{u_K u_K^\top}{\|u_K\|^2} \tag{11.29}$$

is the rank-ten orthogonal projector onto $V_{10}(K)$. One has

$$V_{11} = \ell_K \oplus^\perp V_{10}(K), \quad P_{10} P_{11} = P_{10}, \quad P_{10} u_K = 0.$$

Proof. Let $E_K = u_K u_K^\top / \|u_K\|^2$. The proved nonvanishing guarantees that it is defined. It is symmetric, $E_K^2 = E_K$, has rank one, and, since $u_K \in V_{11}$, $P_{11} E_K = E_K P_{11} = E_K$. Hence

$$(P_{11} - E_K)^2 = P_{11} - 2E_K + E_K = P_{11} - E_K.$$

Its image is $V_{11} \cap u_K^\perp$, and its trace is $11 - 1 = 10$. The decomposition and remaining identities are obtained by projection onto the two orthogonal summands. $\square$

This result specifies a function of K not exhausted by its designation as a record: in addition to reversibly representing oriented data and selecting the tetrad B_K , it determines a linear polarization. The direction depends on the complete record through $P_3 K$, not only on the four points of its support. The order of record, representation, direction and dimensional reduction is thus preserved: (11.27) defines the direction, and Proposition 11.14 proves the reduction and characterizes its kernel.

The transformation has a precise covariance law. If the chart is changed simultaneously by an orthogonal permutation matrix S , $K' = SK$ and $P'_3 = SP_3S^{-1}$, then

$$u'_{K'} = Su_K, \quad P'_{10}(K') = SP_{10}(K)S^{-1}.$$

Moreover, replacing u_K by au_K , with $a \neq 0$, does not change E_K . The direction retains a polarization and does not by itself allow recovery of either the amplitude of u_K or the complete record. Reversibility of $\kappa \leftrightarrow K$ remains in its own domain.

The geometric realization established here is a reduction of inner-product spaces. Its subsequent use as a compact direction requires transporting this line to the corresponding physical realization. Section 11.21 sets out the algebraic map keeping this reduction and the lattice screen coordinated.

11.6 The Witt star and the generation of M_{12}

Definition 11.15 (Star on a four-element set). For $B \in \binom{[12]}{4}$, the Witt star is the collection of the four hexads containing B . Its petals are the pairs $H \setminus B$. The permutation σ_B fixes B and exchanges the two points of each petal.

The existence of four hexads follows from

$$\lambda_4 = \frac{\binom{8}{1}}{\binom{2}{1}} = 4.$$

Two petals cannot share a point, since two hexads would then contain the same five-element set. They are therefore four disjoint pairs partitioning $[12] \setminus B$. This description is neither a five-pointed star nor a geometric metaphor: it is a four-to-one incidence of the design.

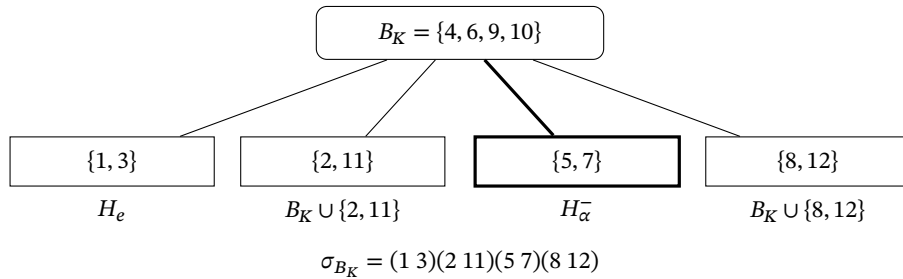


Figure 8: The concrete star of the dodecaphasic flag. Each line indicates the union of the central four-element set with the pair at its endpoint; the thick line marks the hexad also selected by the signs preceding the carry of α . The four pairs exhaust the eight outer positions.

Proposition 11.16 (The collection of stars). *The $\binom{12}{4} = 495$ permutations σ_B preserve $\mathcal{H}$. The group they generate has order 95040 and is the Mathieu group M_{12} in its action on the small Witt design.*

Proof. The preservation claim is a finite check on the 132 hexads of (11.6): for each four-element set the four petals are constructed and one verifies $\{\sigma_B(H) : H \in \mathcal{H}\} = \mathcal{H}$. Closure under composition of these permutations produces 95040 elements. In the lexicographic enumeration, it suffices to add successively six stars not yet contained in the group; the intermediate orders are 2, 6, 18, 360, 7920, 95040, and all 495 stars belong to the final group. Appendix A.2 presents the six permutations and the finite closure recurrence. Finally, one applies the recognition of the automorphism group of $S(5, 6, 12)$ as M_{12} , whose order is $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95040$. $\square$

An individual star generates a cyclic group of order two. It does not by itself generate M_{12} , still less the Monster. Nor does the existence of 495 generators of a permutation group prove that each is the transport of a specific TPK loop. That assertion requires the corresponding loop map; the result established here is the finite incidence action.

Remark 11.17 (Equality of traces does not identify operators). On V_{11} , a star has spectrum $+1$ with multiplicity seven and -1 with multiplicity four. Its normalized trace is $3/11$. A rank-three projector on the same space also has normalized trace $3/11$, but spectrum $1^3, 0^8$. They are not conjugate: their minimal polynomials and spectra differ. The intertwiner (11.25) transports defined operators; it does not turn this scalar coincidence into an equivalence of operators.

11.7 Three regional stars and the polarization of their completions

The preceding incidence construction compares the regions through one operation. The regional words arise from APP–TRIT–TPK emission; their encoding by $\gamma_{A_W}(w) = (w, wA_W)$ retains a support on the twelve positions. The register K and the pre-carry cochain add two readings of the same state: position relative to the $729 + 271$ boundary and the orientation of borrowing. The following comparison assembles these previously constructed data while retaining their causal order.

The conserved tables also use an equivalent presentation. With $D = \text{diag}(1, -1, -1, -1, -1, -1)$ over $\mathbb{F}_3$, their matrix is $A_{\text{hist}} = A_W D$, and

$$\gamma_{A_{\text{hist}}}(w) = \gamma_{A_W}(w)(I_6 \oplus D).$$

This change multiplies coordinates by units and preserves every support. The star tables therefore transfer in full to the presentation A_W used here.

Write

$$\begin{aligned} w_\pi &= 010211, & w_e &= 201101, & w_\varphi &= 121200, & w_{\text{APP}} &= 022020, \\ P_\pi &= \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, & H_e &= \{1, 3, 4, 6, 9, 10\}, \\ H_\varphi &= \{1, 2, 3, 4, 7, 9\}, & H_{\text{APP}} &= \{2, 3, 5, 10, 11, 12\}. \end{aligned}$$

The four sets in the second and third lines are the supports of the corresponding encoded words. The cochain

$$s = (7, 297, 353, -430, -715, -1199, -3, 286, -894, -528, 380, 663)$$

determines the partition

$$H_\alpha^- = \{4, 5, 6, 7, 9, 10\}, \quad H_\alpha^+ = \{1, 2, 3, 8, 11, 12\}.$$

The signs are thus fixed before decimal normalization. For a four-subset B , denote the matching of its star by

$$\mu_W(B) = \{H \setminus B : H \in \mathcal{H}, B \subset H\}.$$

A pair has type $++$, $--$, or $+-$ according as it contains zero, two, or one point of H_α^- . Its signature is $\nu(B) = (n_{++}(B), n_{--}(B), n_{+-}(B))$.

Proposition 11.18 (Comparison of the three regional intersections). *The four-subsets*

$$B_K = H_e \cap P_\pi, \quad B_\varphi = H_\varphi \cap P_\pi, \quad B_{\text{APP}} = H_{\text{APP}} \cap P_\pi$$

have the twelve completions and signatures listed in Table 11.

Proof. Applying γ_{A_W} to the four regional words yields the stated supports. For each four-subset, enumerate the hexads of $\mathcal{H}$ that contain it. The relation $\lambda_4 = 4$ ensures that the four completions in the table exhaust the star. Finally, intersection of each pair with H_α^- determines its sign. Every operation takes place within the finite design already constructed. $\square$

The three signatures distinguish concrete incidence roles. The star of B_K separates three positive pairs and one negative pair; that of B_φ contains three mixed pairs; that of B_{APP} combines all three classes. The distinction persists even when two stars share the pair $\{1, 3\}$.

Four-subset and signature	Added pair	Resulting hexad
$B_K = \{4, 6, 9, 10\}$ $\nu(B_K) = (3, 1, 0)$	$\{1, 3\}_{++}$	$\{1, 3, 4, 6, 9, 10\}$
	$\{2, 11\}_{++}$	$\{2, 4, 6, 9, 10, 11\}$
	$\{5, 7\}_{--}$	$\{4, 5, 6, 7, 9, 10\}$
	$\{8, 12\}_{++}$	$\{4, 6, 8, 9, 10, 12\}$
$B_\varphi = \{2, 4, 7, 9\}$ $\nu(B_\varphi) = (1, 0, 3)$	$\{1, 3\}_{++}$	$\{1, 2, 3, 4, 7, 9\}$
	$\{5, 11\}_{+-}$	$\{2, 4, 5, 7, 9, 11\}$
	$\{6, 12\}_{+-}$	$\{2, 4, 6, 7, 9, 12\}$
	$\{8, 10\}_{+-}$	$\{2, 4, 7, 8, 9, 10\}$
$B_{APP} = \{2, 5, 10, 11\}$ $\nu(B_{APP}) = (1, 1, 2)$	$\{1, 4\}_{+-}$	$\{1, 2, 4, 5, 10, 11\}$
	$\{3, 12\}_{++}$	$\{2, 3, 5, 10, 11, 12\}$
	$\{6, 7\}_{--}$	$\{2, 5, 6, 7, 10, 11\}$
	$\{8, 9\}_{+-}$	$\{2, 5, 8, 9, 10, 11\}$

Table 11: The three stars with the same reference polarization. Each row retains the four-subset, complementary pair, and complete hexad.

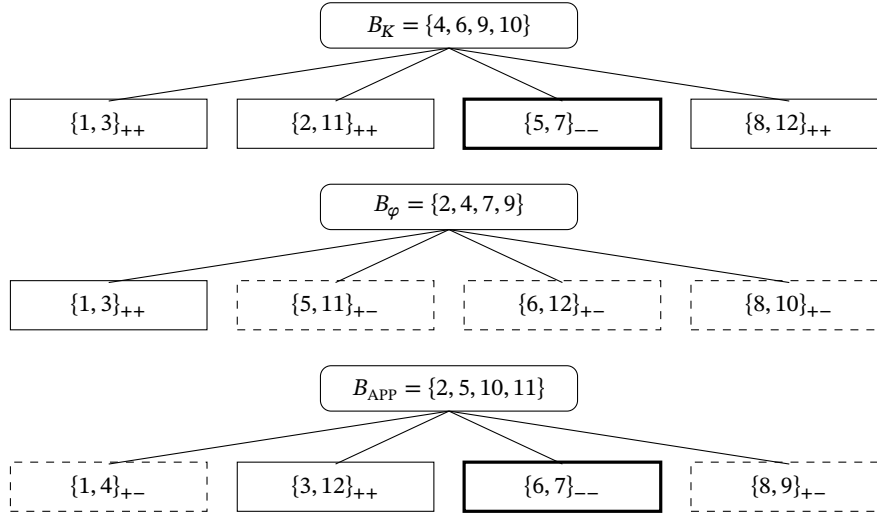


Figure 9: Comparison of the three stars. Each connection represents the union of the upper four-subset with one pair; the twelve complete hexads appear in Table 11. Thick outlines indicate negative pairs and dashed outlines mixed pairs. The connections represent incidence.

Proposition 11.19 (Coupling of corona, incidence, and sign). *For the previously constructed dodecaphasic register,*

$$B_K = \{i : K_i \geq 729\} = H_e \cap P_\pi = \{4, 6, 9, 10\}.$$

The unique hexad $H \in \mathcal{H}$ satisfying $H \cap P_\pi = B_K$ is H_e . Moreover,

$$H_\alpha^- = B_K \sqcup \{5, 7\}, \quad (B_\varphi \setminus B_K) \cap H_\alpha^- = \{7\}.$$

Proof. The register $K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601)$ reaches or exceeds 729 at precisely the four stated positions. Comparison with the supports proves the first equality. A hexad with intersection B_K belongs to the star of B_K ; among its four pairs, only $\{1, 3\}$ is disjoint from P_π . This proves the relative uniqueness of H_e . The table identifies the negative pair $\{5, 7\}$. Finally, $B_\varphi \setminus B_K = \{2, 7\}$, and the signs $s_2 = 297 > 0$, $s_7 = -3 < 0$ determine the pivot 7. $\square$

The four-subset, sign, and normalization retain distinct roles. Incidence fixes the completion H_α^- ; the Archimedean lift of the complete cochain also uses the magnitudes, positional order, and carries already established in the closure of α . This composition explains how one genealogy produces both a geometric boundary reading and a numerical reading.

11.8 Polarized classification of the 495 four-subsets

Theorem 11.20 (Signature census and regional restriction). *The $495 = \binom{12}{4}$ four-subsets are distributed as in Table 12. The last column counts hexads H with $|H \cap P_\pi| = 4$, classified by $\nu(H \cap P_\pi)$.*

$\nu = (n_{++}, n_{--}, n_{+-})$	Four-subsets	Regional hexads
(3, 1, 0)	15	9
(1, 3, 0)	15	0
(2, 2, 0)	45	9
(1, 1, 2)	180	18
(1, 0, 3)	120	18
(0, 1, 3)	120	0
Total	495	54

Table 12: Complete classification and restriction to the support P_π . The two columns count different objects: four-subsets and hexads.

Proof. The following procedure is an exact enumeration of the finite domain. Form the 3^6 words w , compute (w, wA_W) modulo 3, and retain their weight-six supports without repetition. This gives $\mathcal{H}$, with 132 elements. For each $B \in \binom{[12]}{4}$, form $\mu_W(B)$, count the intersections of its four pairs with H_α^- , and increment the entry for the resulting signature. The six counts are 15, 15, 45, 180, 120, 120. They sum to 495, and each four-subset is processed once. For the last column, traverse the 132 hexads, retain the 54 whose intersection has size four, and apply the same function ν to that intersection. The operations are set comparisons and arithmetic in $\mathbb{F}_3$; the reproducible certificate retains the complete input–output table. $\square$

The signature (3, 1, 0) occurs for fifteen four-subsets. The regional frame is characterized jointly by the corona of K , the intersection $H_e \cap P_\pi$, and the cochain sign, as established in Proposition 11.19.

11.9 Rigidity of the ordered four-support frame

Let $G_W = \langle \sigma_B : |B| = 4 \rangle$, already computed in Proposition 11.16. For an ordered collection of supports, define

$$\text{Stab}_{G_W}(S_1, \dots, S_r) = \{g \in G_W : g(S_j) = S_j \text{ for every } j\}.$$

Each support is preserved individually; internal permutations of its points remain permitted.

Theorem 11.21 (Joint rigidity of the four incidences). *The stabilizers have the following orders:*

<i>preserved data</i>	<i>order</i>
H_α^-	720
$H_\alpha^-, 6$	120
B_K	192
B_K, H_α^-	48
B_K, H_α^-, H_e	16
$H_\alpha^-, H_e, H_\varphi$	2
$H_\alpha^-, H_e, H_\varphi, H_{\text{APP}}$	1

The second row also preserves the point 6. In particular,

$$\text{Stab}_{G_W}(H_\alpha^-, H_e, H_\varphi, H_{\text{APP}}) = \{\text{id}\}.$$

Proof. Use the closure under composition of the six explicit permutations in Proposition 11.16. On its 95040 elements evaluate the predicates $g(S_j) = S_j$, together with $g(6) = 6$ where required. Summing their indicators gives 720, 120, 192, 48, 16, 2, 1, respectively. The identity satisfies every condition; the last cardinality proves that it is the unique element. The closure algorithm terminates when the set stabilizes, and the stabilizer filters are then applied to the complete group. $\square$

The table compares seven marked configurations. Two useful inclusion chains are those determined by (B_K) , (B_K, H_α^-) , (B_K, H_α^-, H_e) , and by (H_α^-) , $(H_\alpha^-, H_e, H_\varphi)$, $(H_\alpha^-, H_e, H_\varphi, H_{\text{APP}})$. Final rigidity means that the ordered regional frame fixes all residual freedom within G_W . Its domain is the action on twelve positions; subsequent transport to lattices and algebras retains the maps belonging to those realizations.

11.10 Finite incidence topology and preservation of its domain

The simplicial complex generated by the 132 hexads has face vector

$$(f_0, f_1, f_2, f_3, f_4, f_5) = (12, 66, 220, 495, 792, 132)$$

and Euler characteristic $\chi = 12 - 66 + 220 - 495 + 792 - 132 = 331$. Indeed, each five-subset has a unique hexadic extension; every smaller subset belongs to one of these five-subsets, and the maximal faces are the hexads.

The bipartite graph between four-subsets and hexads has

$$|V| = 495 + 132 = 627, \quad |E| = 495 \cdot 4 = 132 \cdot 15 = 1980.$$

It is connected. Let H, H_* be distinct hexads; their intersection has at most four points because each five-subset has a unique extension. Choose a four-subset $B \subset H$ containing $H \cap H_*$, and a point $p \in H_* \setminus H$. The unique extension H' of $B \cup \{p\}$ is joined to H by the bipartite path $H - B - H'$ and satisfies $|H' \cap H_*| > |H \cap H_*|$. Repeating the operation reaches H_* ; containing five of its points already forces equality. Every four-subset is incident to four hexads, so it too belongs to this single component. Its first Betti number is therefore

$$\beta_1 = |E| - |V| + 1 = 1354.$$

An exhaustive traversal also verifies connectedness. For unordered pairs of distinct hexads, the intersection census is

$ H \cap H' $	0	2	3	4
$\#\{H, H'\}$	66	2970	2640	2970.

The sum $8646 = \binom{132}{2}$ covers every pair.

These invariants describe the complex and its incidence graph. The junction micrograph $U_6 = A|B$ separately retains its 96 vertices, 234 edges, six components, and $\beta_1 = 234 - 96 + 6 = 144$. Likewise, compositions of the σ_B act on incidence positions, whereas Γ_0 transports the TPK state with memory. This distinction relates the readings through their effective maps and preserves, at each level, the data used by the operation.

11.11 The binary residue and the five regions of π

The next recognition uses the extended binary Golay code G_{24} , with parameters $[24, 12, 8]_2$. Its weights are 0, 8, 12, 16, 24, and the weight-eight supports form the design $S(5, 8, 24)$. Fix a dodecad D , the support of a weight-twelve word. This datum and the incidence isomorphism to be declared below belong to the realization chart; they are not used to generate or select the decimals of π .

Proposition 11.22 (Residue of a dodecad and lifting of hexads). *The set*

$$\mathcal{H}(D) = \{O \cap D : O \text{ is an octad, } |O \cap D| = 6\}$$

is an $S(5, 6, 12)$ system on D . Each of its hexads has a unique lift to an octad. For any four-element set $B \subset D$, the five octads containing it split into four residual lifts and a unique octad $O_B^\perp$ with $O_B^\perp \cap D = B$.

Proof. A five-element set $A \subset D$ belongs to a unique octad O . If $j = |O \cap D|$, the binary word given by the sum of O and D has weight $20 - 2j$. Among the permitted weights, with $0 \leq j \leq 8$, only $j = 2, 4, 6$ are possible. Since $A \subset O \cap D$, we have $j \geq 5$ and therefore $j = 6$. This gives existence and uniqueness of the residual hexad, as well as uniqueness of its lift by selecting five of its points. Moreover,

$$\lambda_4(S(5, 8, 24)) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5, \quad \lambda_4(S(5, 6, 12)) = 4.$$

The four residual hexads on B give four distinct octads. The fifth cannot meet D in six points, since it would be another residual hexad; because it contains B , its intersection has size four and equals B . $\square$

A hexad-preserving chart $\ell : [12] \rightarrow D$ identifies $\mathcal{H}$ with $\mathcal{H}(D)$. Its existence is the recognition of the design already constructed as the small Witt design; a concrete choice of ℓ fixes the relabeling. The lift

$$\iota_D : \mathcal{H} \longrightarrow \{\text{octads of } G_{24}\}, \quad H \longmapsto O \text{ such that } O \cap D = \ell(H),$$

is injective because intersection with D recovers $\ell(H)$. The domain of this injectivity is the hexads, not the record K .

The relation to the five regions of π preserves richer data than the number five. The emissions and their signatures are:

Region	Multiplicity	Emission U_6	Signature
$\perp$	432	(501, 614, 498, 169, 272, 272)	555555
$++_1$	144	(810, 923, 870, 169, 272, 272)	339555
$++_2$	144	(810, 983, 810, 169, 272, 272)	393555
$++_3$	144	(870, 923, 810, 169, 272, 272)	933555
$--$	144	(870, 923, 810, 418, 674, 521)	933717

In each three-digit decimal block, the signature uses the modular difference of the first two digits. The five emissions have the same visible tritic word $w_\pi = 010211$, but not the same regional genealogy.

The three positive regions preserve the tail 169|272|272; the negative region changes it to 418|674|521; the transverse region has a constant signature and greater multiplicity.

The marked extension of the regional chart sends the transverse region to $O_{\ell(B_K)}^\perp$, the negative region to $\iota_D(H_\alpha^-)$, and the three positive regions to the other three lifts of the star. This realizes the $4 + 1$ partition of the five octads on $\ell(B_K)$. The individual correspondence between the three positive names and the three remaining petals requires a chart order; incidence does not determine it without that datum. The precise claim is therefore a marked correspondence of regions, compatible with their roles and with the flag, not a canonical bijection inferred from the cardinality five.

The identity $432 + 4 \cdot 144 = 1008$ preserves the regional multiplicities. Also, $1008 = 36 \cdot 28$, and an octad has $\binom{8}{2} = 28$ pairs. These equalities are arithmetic catalog checks, not substitutes for the map ι_D or the regional chart.

11.11.1 Pentadic representation of the microfiber and its incidence

The arrangement of the five regions makes it possible to visualize simultaneously the unity of the ternary quotient and the differentiation preserved by the catalog. The relevant map is the restriction

$$\Psi|_{\mathcal{R}_\pi^{\text{micro}}} : \mathcal{R}_\pi^{\text{micro}} \longrightarrow \{010211\}, \quad \Psi(U) = (-U) \bmod 3.$$

Its five antecedents are the vectors U_6 in the preceding table. Figure 10 recovers their radial organization, preserving chart order, signatures, and multiplicities of the catalog in Section 11.11, computable using the recurrence and signatures in Appendix A.1.

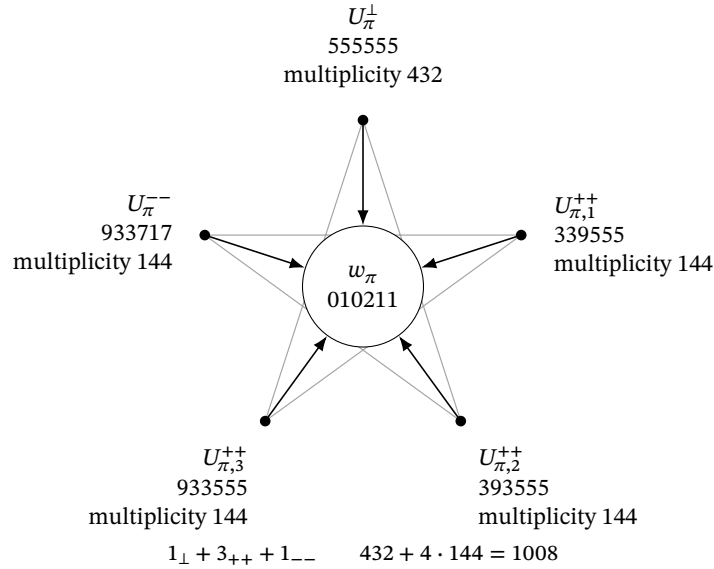


Figure 10: The five regions of π in a pentadic arrangement. Each position preserves its multiplicative signature and multiplicity; the radial arrows represent the common projection $\Psi(U) = 010211$. The star-shaped outline organizes the five positions of the chart. Their realization in the five Witt octads is performed through regional alignment and the $4 + 1$ decomposition in the text.

Projection onto the center identifies the five ternary images; the record of exterior positions retains the regional information making incidence realization possible. The distribution $1_\perp + 3_{++} + 1_{--}$ specifies one transversal region, three positive regions, and one negative region. Their normalized weights are $3/7$ and $1/7$ for each of the remaining four: transversal multiplicity is triple, even though all five images under Ψ coincide.

The link with the Witt star is formulated through the flag $B_K \subset H_\alpha^-$, the alignment ℓ , and the lift ι_D already constructed. The five regional positions are realized in the five octads over $\ell(B_K)$:

$$\begin{aligned} U_\pi^\perp &\longmapsto O_{\ell(B_K)}^\perp, \\ U_\pi^{--} &\longmapsto \iota_D(H_\alpha^-), \\ \{U_{\pi,1}^{++}, U_{\pi,2}^{++}, U_{\pi,3}^{++}\} &\longmapsto \{\iota_D(B_K \cup P) : P \in \mathcal{P}_+\}, \\ \mathcal{P}_+ &= \{\{1, 3\}, \{2, 11\}, \{8, 12\}\}. \end{aligned}$$

Assignment of the three positive positions to the three pairs is performed in an ordered chart. The transversal and negative rows are distinguished by their regional predicates and by the flag.

The two figures are thus articulated: Figure 8 shows the four hexads of the small Witt design over B_K ; Figure 10 shows the five regions that, under alignment, occupy the four residual octads and the transversal octad of the binary design. The law $4 + 1$, proved in Proposition 11.22, specifies the change in incidence. In this composition, K determines the marked tetrad, and the signature of α distinguishes one of its hexads. In addition to their common ternary value, the regions preserve the structure required for that transport toward the exceptional realization.

11.12 From the ternary code to the Niemeier lattice

The passage to dimension twenty-four need not first go through the binary code. It is performed directly from C_W by discriminant gluing. Two compatible prolongations thus remain distinct: one preserves the residue $W_{12} \subset W_{24}$; the other constructs the lattice on which the vertex operator algebra will be realized.

Let $A_2 = \mathbb{Z}a_1 \oplus \mathbb{Z}a_2$ with Gram matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \rho = a_1 + a_2, \quad \rho^2 = 2.$$

Represent the three classes of A_2^*/A_2 by

$$\varpi_0 = 0, \quad \varpi_1 = \frac{2a_1 + a_2}{3}, \quad \varpi_2 = \frac{a_1 + 2a_2}{3}.$$

The map $j \mapsto \varpi_j + A_2$ additively identifies $\mathbb{F}_3$ with this discriminant. For $c \in C_W$, write $\phi(c) = (\varpi_{c_1}, \dots, \varpi_{c_{12}})$ and define

$$N(C_W) = \bigcup_{c \in C_W} (A_2^{12} + \phi(c)). \quad (11.30)$$

Theorem 11.23 (Ternary gluing). *The object (11.30) is a positive, even, unimodular lattice of rank 24. Its root system is exactly A_2^{12} ; it is therefore recognized as the Niemeier lattice with that root system.*

Proof. Linearity of the code makes the union of classes stable under addition. Self-orthogonality annihilates the discriminant pairing: the latter is a multiple of $c \cdot d/3$ modulo $\mathbb{Z}$. Thus the lattice is integral. The norm of a glue class is $2 \text{ wt}(c)/3$ modulo $2\mathbb{Z}$. The weights in (11.16) are multiples of three, so the lattice is even.

The index of A_2^{12} in $N(C_W)$ is $|C_W| = 3^6$. Consequently,

$$\det N(C_W) = \frac{3^{12}}{(3^6)^2} = 1.$$

Positivity follows from the orthogonal sum of the twelve A_2 metrics. In a nonzero class of A_2^*/A_2 , the minimum norm is $2/3$; every nonzero word has at least six nonzero positions. Hence a vector in a nontrivial glue class has norm at least $6 \cdot 2/3 = 4$. No norm-two roots occur outside A_2^{12} . Within it, the roots are precisely those of its twelve components. $\square$

11.13 Incidence selection of the rootless unimodular neighbor

From the incidence origin to lattice modification. The marked flag determines a position through an equivariant set operation:

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

Its function is to distinguish one component among the twelve entering the lattice realization. The complete ternary code acts as a glue code in the discriminant module $(A_2^*/A_2)^{12}$. Its preimage defines $N(C_W)$, and self-duality, weights, and distance of the code translate, respectively, into properties of integrality and determinant, parity, and norm bounds for the glue classes. Here the symbolic information of the code performs a function no longer performed by the isolated support.

The A_2 metric and positive radial chamber are fixed in the lattice development. Within that chamber, the congruence required for an even index-three neighbor has twelve realizations of minimum norm 54, permuted by changing the distinguished component. The origin o_* selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 2(4^2 + 11) = 54.$$

The coefficient 4 belongs to that solution of the congruence in the declared chamber. The subscript α records the incidence provenance of the selection: its content is the frame that entered closure and now individualizes a lattice direction.

Pairing with v_α determines the index-three sublattice $N_{\alpha,0}$, and adding the class $v_\alpha/3$ produces the neighbor N_α of (11.32). This modification preserves rank and recovers unimodularity and parity; congruence selection eliminates the previous roots, and local bounds exclude roots in the new classes. Identification with Leech appears after these verifications. The binary realization of the five regions and this lattice construction are two prolongations of the common frame: the second receives the ternary code directly as glue data.

The label $o_* = 1$ of the flag is now realized as one of the twelve A_2 components. This operation declares the preceding metric and the positive radial chamber

$$v(a) = \sum_{i=1}^{12} a_i \rho_i, \quad a_i = 1 + 3b_i, \quad b_i \in \mathbb{Z}_{\geq 0}.$$

The evenness requirement for the index-three neighbor is $v(a)^2 \equiv 0 \pmod{18}$. Since

$$v(a)^2 = 2 \sum_i (1 + 3b_i)^2,$$

this congruence is equivalent to $\sum_i b_i \equiv 1 \pmod{3}$. The least norm in the chamber is obtained with a single $b_i = 1$ and all others zero; it is 54. The marked origin selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 54. \quad (11.31)$$

The word “least” refers to this positive chamber. It does not assert minimality throughout the residual class: the vector $(a_1 - 2a_2, \rho, \dots, \rho)$ represents the same class modulo three and has norm $14 + 11 \cdot 2 = 36$. The chamber restriction is part of the construction data, not an omissible conclusion.

Let $N = N(C_W)$. Define

$$N_{\alpha,0} = \{x \in N : \langle x, v_\alpha \rangle \equiv 0 \pmod{3}\}, \quad N_\alpha = N_{\alpha,0} + \mathbb{Z} \frac{v_\alpha}{3}. \quad (11.32)$$

Theorem 11.24 (Leech neighbor determined by the marked flag). *With the declared metric and chamber, N_α is a positive, even, unimodular lattice of rank 24, with no vectors of norm two. By the corresponding classification theorem, it is isometric to the Leech lattice Λ_{24} .*

Proof. The character $x \mapsto \langle x, v_\alpha \rangle \pmod{3}$ is nonzero: a simple root of a component pairs with $a_i \rho_i$ to give $a_i \equiv 1 \pmod{3}$. Hence $[N : N_{\alpha,0}] = 3$ and $\det N_{\alpha,0} = 9$. Moreover, $v_\alpha/3 \in N_{\alpha,0}^*$ and $v_\alpha \in N_{\alpha,0}$. If $v_\alpha/3$

belonged to N , the integrality of N would make all pairings with v_α divisible by three, contradicting the nonzero character. The class of $v_\alpha/3$ therefore has order three. It follows that

$$[N_\alpha : N_{\alpha,0}] = 3, \quad \det N_\alpha = 9/3^2 = 1.$$

The extension is integral, and for $x \in N_{\alpha,0}$,

$$\left(x + m \frac{v_\alpha}{3}\right)^2 = x^2 + 2m \frac{\langle x, v_\alpha \rangle}{3} + 6m^2 \in 2\mathbb{Z}.$$

Hence it is even.

It remains to exclude roots, the decisive step. The roots of N belong to A_2^{12} . Their pairings with v_α are congruent to ± 1 or ± 2 modulo three; none belongs to $N_{\alpha,0}$. For the other two classes of the neighbor, each component belongs to one of the local classes

$$A_2 + \varpi_j \pm \rho/3, \quad j = 0, 1, 2.$$

Since $4\rho/3 \equiv \rho/3 \pmod{A_2}$, this description also holds in the distinguished component. Writing a local vector as $(ua_1 + va_2)/3$, its norm is $2(u^2 - uv + v^2)/9$. The six preceding classes have nonzero residues and minimum $2/9$: representatives minimizing the quadratic form are $(\pm 1, 0)$, $(0, \pm 1)$, $(1, 1)$, $(-1, -1)$ in the appropriate residues. A vector in a new class therefore has norm at least $12 \cdot 2/9 = 8/3 > 2$. There are no roots in any of the three classes. The classification of positive even unimodular lattices of rank twenty-four identifies the unique rootless case with Λ_{24} . $\square$

The argument does not recognize Leech from a coincidence of dimension. It constructs the gluing, proves integrality, evenness, and the determinant, and eliminates roots through a local bound. Only then does it apply the classification theorem. The dependence on α is preserved in the flag selecting o_* and in the vector (11.31); the scalar value of the constant is not converted into a length of the lattice.

11.14 The vertex operator algebra and the Monster

Let $\Lambda = N_\alpha$, already recognized as Leech. The next step uses its integral lattice, not a decimal expansion. The lattice vertex operator algebra construction is written

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

Here $M(1)$ is the rank-twenty-four Heisenberg oscillator space; $\mathbb{C}_\varepsilon[\Lambda]$ is the group algebra twisted by the cocycle of the central extension of the lattice. The cocycle and the lift of the isometries are part of the construction: a bare vector-space sum does not determine the vertex products.

Negation $\lambda \mapsto -\lambda$ admits the standard lift θ . With its twisted module and the compatible parity choice, the Frenkel–Lepowsky–Meurman construction produces

$$V_{\text{HMT}}^\natural = V_\Lambda^+ \oplus (V_\Lambda^T)^+. \quad (11.33)$$

The sign $+$ denotes the even part of the lifted action. The expression includes the orbifold vertex product; it is not merely a sum of graded spaces.

Theorem 11.25 (Prolongation to the Moonshine module). *The algebra (11.33), constructed on the neighbor (11.32), is a realization of the Moonshine algebra $V^\natural$. It has central charge 24, zero weight-one component, and automorphism group isomorphic to the Monster $\mathbb{M}$. Its graded character is*

$$\text{Tr}_{V_{\text{HMT}}^\natural} q^{L_0-1} = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots, \quad q = e^{2\pi i \tau}. \quad (11.34)$$

Proof. Theorem 11.24 verifies the lattice hypotheses: positivity, evenness, unimodularity, rank twenty-four, and absence of roots. The Frenkel–Lepowsky–Meurman construction and theorem for the Leech lattice then apply. They identify the orbifold with $V^\natural$ and establish its character and its automorphism group. Composing the flag, the neighbor, and the VOA construction functor gives the realization stated here. The group is not inferred from the coefficient 196884. $\square$

The theorem used is a subsequent classical recognition with verified hypotheses, not a new proof of the FLM construction in this article [11]. Moreover, the notation $q = e^{2\pi i\tau}$ is the conventional Fourier variable used to recognize the character. It does not introduce π or e as inputs into the coinductive generator that has already determined them.

Part of the trace mechanism can be seen before the orbifold. Only the zero lattice vector is fixed by negation; each oscillator contributes an alternating sign. Hence

$$\mathrm{Tr}_{V_\Lambda} \theta q^{L_0-1} = t_2(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}.$$

This trace in the lattice algebra must not be confused with the character of $V^\natural$ without insertion. Nor does it by itself identify a Monster conjugacy class: the space, insertion, and lift must be specified before functions are compared.

Grading, unity, and automorphisms. The Frenkel–Lepowsky–Meurman construction is applied to the lattice N_α once its properties have been verified. It incorporates the oscillator space, the central extension of the lattice, the vertex product, and the sector twisted by the lift of negation. The result is the algebra $V_{\mathrm{HMT}}^\natural$ of (11.33), whose automorphism group is isomorphic to the Monster. Its action domain is thus fixed by an algebraic structure with grading and product. Borcherds’s Moonshine theorem subsequently determines the graded traces with insertion of those automorphisms [11, 12].

Unity acquires a precise realization at this level: the weight-zero component is the vacuum line and contributes the coefficient 1 of q^{-1} in the graded character. At the cylinder level, refinement maps preserve the constant function through precomposition. Every conservation statement therefore has its object and map. The development of the proofs makes visible this continuity of construction, from arithmetic operations and memory to incidence, metric, and vertex products.

This articulation places K and the closure of α in the main argument. K reversibly preserves an integral coordinate of the history; normalization of α composes regional records with that coordinate; the signed configuration determines a flag; and the marked flag selects the lattice modification on which Moonshine is realized. Proposition 11.8, Lemma 11.9 and Theorems 11.24 and 11.25 establish these stages. Reversible equivalences, information quotients and realizations adding a metric or algebraic structure thus retain their respective domains. The relation between structure, equation and value is expressed through these maps, with the attribution and scope of each result preserved.

11.15 Ternary orientation of the Leech neighbor

The order-two presentation constructed above uses negation on the Leech lattice. The same lattice, obtained from the incidence marked by K and by the dodecaphasic support of α , admits a second presentation compatible with the ternary character of the construction. To make this continuity explicit, the gluing is retained, its order-three isometry is constructed, and that isometry is shown to preserve the neighbor already selected. This step requires no alteration of K , its rational readout, or the preceding definition of α . Proposition 11.26 verifies the lattice isometry; Proposition 11.27 and Theorem 11.28 specify the order-three construction and the hypotheses of the classical identifications used.

We work in the simple-root basis of each A_2 factor, with

$$B_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \rho = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (11.35)$$

Matrix multiplication gives

$$C^3 = I_2, \quad C^2 + C + I_2 = 0, \quad C^\top B_2 C = B_2, \quad R C R = C^{-1}, \quad R^\top B_2 R = B_2.$$

These identities fix both the metric and the orientation of the action. In the discriminant $A_2^*/A_2 \cong \mathbb{F}_3$, we take the class of $\varpi = (2, 1)^\top/3$ as generator. Since $C\varpi - \varpi = (-1, 0)^\top$, C acts trivially on the discriminant; the reflection R reverses its sign.

The Paley–Witt matrix already constructed produces an element of full support through the following operation:

$$\begin{aligned} q_* &= (1, 1, 1, 1, 1, 1), & q_* A_W &= (2, 1, 1, 1, 1, 1), \\ c_* &= (q_*, q_* A_W) = (1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1) \in C_W. \end{aligned} \quad (11.36)$$

Every coordinate of c_* is nonzero. For the twelve positions, define the frames and the isometry

$$R_b(c_*) = \begin{cases} R, & (c_*)_b = 1, \\ I_2, & (c_*)_b = 2, \end{cases} \quad g_c = \bigoplus_{b=1}^{12} R_b(c_*) C R_b(c_*)^{-1}. \quad (11.37)$$

Here $R_b(c_*)$ denotes an invertible frame of an A_2 factor, not the projector P_3 onto the three-dimensional isotypic component in Section 11.5.

Proposition 11.26 (Order-three isometry of the marked neighbor). *The operator g_c is an order-three isometry with no nonzero fixed vectors. It preserves the Niemeier lattice $N = N(C_W)$ and the neighbor N_α constructed in (11.32).*

Proof. Each block of g_c is C or C^{-1} ; its minimal polynomial is $X^2 + X + 1$. This proves the order and the absence of fixed vectors. Both blocks act trivially on the discriminant, so they preserve every gluing class and hence the lattice N .

Write $v = v_\alpha = 4\rho_{o_*} + \sum_{b \neq o_*} \rho_b$, with the frames and incidence origin already fixed. Its radial coefficients are $a_{o_*} = 4$ and $a_b = 1$ for $b \neq o_*$. They all satisfy $a_b \equiv 1 \pmod{3}$. In one factor,

$$(C - I_2)\rho = (-2, -1)^T, \quad (C^{-1} - I_2)\rho = (-1, -2)^T.$$

After division by three, their discriminant classes are respectively 2 and 1. Consequently, the vectors

$$y = \frac{g_c v - v}{3}, \quad z = \frac{g_c^{-1} v - v}{3} \quad (11.38)$$

have discriminant words c_* and $-c_*$, both in C_W . Thus $y, z \in N$. The local inner product is

$$\left\langle \frac{(C^{\pm 1} - I_2)a_b \rho}{3}, a_b \rho \right\rangle = -a_b^2,$$

and summation gives $\langle y, v \rangle = \langle z, v \rangle = -(16 + 11) = -27$. Hence $y, z \in N_0 = \{x \in N : \langle x, v \rangle \equiv 0 \pmod{3}\}$. For $x \in N_0$,

$$\langle g_c x, v \rangle = \langle x, g_c^{-1} v \rangle = \langle x, v \rangle + 3\langle x, z \rangle \equiv 0 \pmod{3}.$$

Applying the argument to g_c^{-1} proves $g_c N_0 = N_0$. Finally, $g_c(v/3) = v/3 + y$ preserves the class generating $N_\alpha = N_0 + \mathbb{Z}(v/3)$. It follows that $g_c N_\alpha = N_\alpha$. $\square$

The proof preserves more than the dimension of the lattice: it uses the gluing code, its orientation, the origin of the flag, and the congruence of the radial vector. These data explain why the second presentation remains linked to the dodecaphasic record from which the construction began.

11.16 Ternary trace, twisted conformal weight, and order-three extension

Let $\Lambda = N_\alpha$, and let $\hat{g}_c$ be the standard lift of g_c to the lattice algebra $V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda]$, with the central extension and cocycle already specified. The odd order allows this lift to be chosen with order three. Each complex A_2 factor contributes the eigenvalues ζ_3 and ζ_3^2 , where $\zeta_3^2 + \zeta_3 + 1 = 0$.

Proposition 11.27 (Graded trace and type of the lift). *For $j = 1, 2$,*

$$\mathrm{Tr}_{V_\Lambda}(\hat{g}_c^j q^{L_0 - 1}) = t_3(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-12}. \quad (11.39)$$

The lowest conformal weight of the nontrivial twisted modules is 4/3, and the lift has type zero.

Proof. Only the zero vector contributes to the lattice part of the trace, because g_c and g_c^2 have no nonzero fixed vectors. At each oscillator degree, the trace on symmetric powers is the inverse determinant of $I - g_c^j q^n$. Each A_2 factor contributes $1 + q^n + q^{2n}$, proving (11.39).

The twisted-vacuum weight formula for a lattice automorphism of order three, applied to the two twelve-dimensional eigenspaces, gives

$$\rho_{\hat{g}_c} = \frac{1}{4 \cdot 3^2} (1 \cdot 2 \cdot 12 + 2 \cdot 1 \cdot 12) = \frac{4}{3}.$$

The type is the class of $3^2 \rho_{\hat{g}_c} = 12$ modulo three, which is zero. The weight formula is a result of twisted-module theory; the multiplicities and isometry entering it have been constructed here. $\square$

Theorem 11.28 (Second presentation of the Moonshine algebra). *The type-zero cyclic extension*

$$V_{(3)} = \text{Orb}_{\mathbb{Z}/3}(V_\Lambda, \hat{g}_c) \quad (11.40)$$

is isomorphic to the vertex operator algebra $V^\natural$.

Proof. Theorem 11.24 provides a positive-definite, even, unimodular, rootless lattice of rank twenty-four. Proposition 11.26 constructs an order-three fixed-point-free isometry of that same lattice; Proposition 11.27 verifies the lift and its type. The Chen–Lam–Shimakura theorem on the order-three orbifold of the Leech lattice algebra then applies [17, Theorem 1.1]. This identification is also covered by the uniform treatment of Abe–Lam–Yamada [18, Theorem 4.4 and Sections 2–4]. The application follows the construction of the lattice data; equality of characters does not replace the hypotheses of the theorem being used. $\square$

To specify the dual symmetry, write $V_{(3)} = W_0 \oplus W_1 \oplus W_2$, where W_i is the integral sector selected in the $\hat{g}_c^i$ -twisted module. The product satisfies $Y(W_i, z)W_j \subset W_{i+j \bmod 3}((z))$. Thus

$$g^\vee|_{W_i} = \zeta_3^i I \quad (11.41)$$

defines an order-three automorphism. Indeed, the factor on a product is $\zeta_3^{i+j} = \zeta_3^i \zeta_3^j$, and the vacuum and conformal vector belong to W_0 . The classical identification of this automorphism in the Monster is class $3B$, with series $T_{3B} = t_3 + 12$ [19, 12]. This normalization can be checked within the construction. The lattice character is $\text{ch } V_\Lambda = J + 24$: its initial pole is q^{-1} , its weight-one space has dimension 24, and the weight-zero modular character is determined by these data. If F is the character of the fixed subspace and H that of each integral twisted sector, projection onto invariants and conjugation of the two sectors give

$$F = \frac{J + 24 + 2t_3}{3}, \quad F + 2H = J, \quad \text{Tr } g^\vee q^{L_0-1} = F - H = t_3 + 12.$$

Its domain is the extended algebra, whereas $\hat{g}_c$ acts on the preceding lattice algebra.

11.17 Equivalence of the order-two and order-three presentations

Denote by

$$V_{(2)} = (V_\Lambda)^+ \oplus (V_\Lambda^T)^+$$

the FLM presentation in (11.33). Both presentations retain the same Leech neighbor as antecedent. Their extensions use different automorphisms and vertex products determined by the respective lift data.

Theorem 11.29 (Graded equivalence and transport of automorphisms). *With fixed vertex operator algebra isomorphisms*

$$\phi_2 : V_{(2)} \xrightarrow{\cong} V^\natural, \quad \phi_3 : V_{(3)} \xrightarrow{\cong} V^\natural,$$

the composition

$$\Phi_{23} = \phi_2^{-1} \phi_3 : V_{(3)} \xrightarrow{\cong} V_{(2)} \quad (11.42)$$

preserves the vacuum, the conformal vector, the grading, and the vertex product. Conjugation by Φ_{23} identifies the automorphism groups and preserves their graded traces.

Proof. The composition of the two isomorphisms preserves the specified structures. In particular,

$$\Phi_{23} Y_{(3)}(a, z)b = Y_{(2)}(\Phi_{23}a, z)\Phi_{23}b, \quad \Phi_{23} L_0^{(3)} = L_0^{(2)} \Phi_{23}.$$

If h is an automorphism of $V_{(3)}$, then $h' = \Phi_{23} h \Phi_{23}^{-1}$ is an automorphism of $V_{(2)}$. Equality of traces on each finite-dimensional homogeneous component gives

$$\mathrm{Tr}_{V_{(3)}} h q^{L_0-1} = \mathrm{Tr}_{V_{(2)}} h' q^{L_0-1}.$$

□

The choice of ϕ_2 and ϕ_3 is part of the statement; Φ_{23} is not presented as a canonical identification independent of those choices. Nor does it transform an order-three automorphism into the order-two involution: conjugation preserves order. Its content is that two different extensions realize the same graded structure, with explicit transport of their products, representations, and traces.

11.18 Eta functions, the Fricke involution, and vacuum normalization

For $\mathrm{Im} \tau > 0$, the traces of the two lattice automorphisms admit the expressions

$$\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n), \quad t_2 = \left(\frac{\eta(\tau)}{\eta(2\tau)} \right)^{24}, \quad t_3 = \left(\frac{\eta(\tau)}{\eta(3\tau)} \right)^{12}. \quad (11.43)$$

They follow directly from the factorizations $1 - q^{2n} = (1 - q^n)(1 + q^n)$ and $1 - q^{3n} = (1 - q^n)(1 + q^n + q^{2n})$. The order in q of the level-three quotient, before raising it to the twelfth power, is

$$\frac{1}{24} - \frac{3}{24} = -\frac{1}{12}; \quad 12 \left(-\frac{1}{12} \right) = -1.$$

Here $-1/12$ denotes the exact difference between the initial exponents of two eta functions. The operation is specified and requires no interpretation of a divergent sum as a convergent one. Equality of this scalar with a normalized defect in another realization does not identify the operators that produce it: each realization retains its own domain and operation.

Proposition 11.30 (Fricke involution on the order-two trace). *Let $w_2\tau = -1/(2\tau)$. Then*

$$t_2(w_2\tau) = \frac{2^{12}}{t_2(\tau)}, \quad T_{2A} = t_2 + 24 + \frac{2^{12}}{t_2} \quad (11.44)$$

and $T_{2A}(w_2\tau) = T_{2A}(\tau)$.

Proof. The transformation law $\eta(-1/\tau) = (-i\tau)^{1/2} \eta(\tau)$ gives

$$\frac{\eta(-1/(2\tau))}{\eta(-1/\tau)} = \sqrt{2} \frac{\eta(2\tau)}{\eta(\tau)}.$$

The twenty-fourth power yields the first identity. Substitution interchanges the two variable terms of T_{2A} and fixes the constant term. Recognition of this expression as the McKay–Thompson series of class 2A corresponds to Conway–Norton’s level-two symmetrization [19, §§4 and 6, Table 3] and the Moonshine theorem [12]. □

The modular uniformizations of levels two and three, with the normalizations above, are

$$j = \frac{(t_2 + 256)^3}{t_2^2}, \quad (11.45)$$

$$J = j - 744 = t_3 + 12 + \frac{10 \cdot 3^9}{t_3} + \frac{4 \cdot 3^{14}}{t_3^2} + \frac{3^{18}}{t_3^3}. \quad (11.46)$$

They are used as classical modular identities on the traces already constructed, not as rules for selecting the code or the neighbor. The second can also be written $j = (t_3 + 27)(t_3 + 243)^3/t_3^3$. Direct expansion of the product in (11.39) begins with

$$t_3 = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + 1188q^4 + O(q^5).$$

In particular, $[q]J = 54 + 10 \cdot 3^9 = 196884$; this is a consequence of the traces and the uniformization. The construction of $V^{\natural}$ retains its lattice and orbifold proof independently of this check of the first coefficient.

11.18.1 Ternary congruence of the trace and optimality of the exponent

The level-three identity permits comparison of every coefficient of the constructed trace with the corresponding coefficient of its modular realization. The appropriate domain for this consequence is the ring of formal power series: each coefficient depends on finitely many factors of the product, while the statement retains all degrees.

Lemma 11.31 (Integral unit of the order-three product). *The series*

$$u(q) = q t_3(q) = \prod_{n \geq 1} (1 + q^n + q^{2n})^{-12} \quad (11.47)$$

belongs to $1 + q\mathbb{Z}[[q]]$, and its inverse belongs to the same set.

Proof. Each polynomial $1 + q^n + q^{2n}$ has constant term one. For any series $v(q) = 1 + \sum_{r \geq 1} v_r q^r$ with integer coefficients, the equation $v(q)w(q) = 1$ recursively determines

$$w_0 = 1, \quad w_r = -\sum_{i=1}^r v_i w_{r-i} \quad (r \geq 1).$$

This recurrence proves existence, uniqueness and integrality of the inverse. Applied to each factor, it also proves integrality of its inverse twelfth power. Moreover, $(1 + q^n + q^{2n})^{-12} \in 1 + q^n \mathbb{Z}[[q]]$. To compute a coefficient of degree at most r , only factors with $n \leq r$ are needed; the partial products stabilize coefficientwise. Consequently, (11.47) defines an integral series with constant term one. Applying the same recurrence to u yields $u^{-1} \in 1 + q\mathbb{Z}[[q]]$. $\square$

Proposition 11.32 (Exact uniform divisibility). *For the trace t_3 and the series $J = j - 744$ related by (11.46),*

$$J - (t_3 + 12) \in 3^9 q \mathbb{Z}[[q]]. \quad (11.48)$$

The uniform exponent of divisibility by three is exactly nine:

$$[q](J - t_3 - 12) = 10 \cdot 3^9 = 196830. \quad (11.49)$$

Proof. Since $t_3 = q^{-1}u$, the lemma gives $t_3^{-r} = q^r u^{-r} \in q^r \mathbb{Z}[[q]]$ for $r = 1, 2, 3$. Identity (11.46) becomes

$$J - t_3 - 12 = 3^9 q (10u^{-1} + 4 \cdot 3^5 q u^{-2} + 3^9 q^2 u^{-3}). \quad (11.50)$$

Every term in parentheses is an integral series, proving (11.48). Its constant term is 10, since $u(0) = u^{-1}(0) = 1$, and the last two terms contain q and q^2 . This gives (11.49). As $3 \nmid 10$, the coefficient of q has 3-adic valuation nine, establishing optimality of the uniform exponent. $\square$

In particular, $[q^r]J \equiv [q^r]t_3 \pmod{3^9}$ for every $r \geq 1$. The quantifier follows from the formal identity and the integral inversion proved in the lemma. Verification through any finite degree is a specialization of this relation. The trace continues to arise from the order-three automorphism of the lattice constructed from HMT incidence; uniformization subsequently acts on that trace. The congruence preserves this order and determines an exact relation between its two graded readings.

11.19 The Fricke correspondence and differential recovery

The level-two lattice trace and its Fricke completion provide, after the exceptional construction, the coordinates

$$t = t_2, \quad s = \frac{4096}{t}, \quad T = t + s + 24, \quad t \in \mathbb{C}^\times.$$

The involution exchanges t and s . The two ordered modular readings are

$$j_1 = \frac{(t + 256)^3}{t^2}, \quad j_2 = \frac{(s + 256)^3}{s^2}.$$

These formulas use the uniformization already established in (11.45) and retain the choice of the oriented reading.

Proposition 11.33 (Correspondence and discriminant). *The following identities hold:*

$$j_1 - j_2 = (s - t)(T + 23), \tag{11.51}$$

$$j_1 + j_2 = T^2 + T - 7256, \tag{11.52}$$

$$j_1 j_2 = (T + 248)^3. \tag{11.53}$$

The two values are roots of

$$X^2 - (T^2 + T - 7256)X + (T + 248)^3 = 0,$$

whose discriminant is

$$\Delta = (T - 152)(T + 104)(T + 23)^2. \tag{11.54}$$

Proof. Expanding the first reading with $ts = 4096$ gives $j_1 = t + 768 + 48s + s^2$; the second follows by exchanging s, t . Subtraction gives $(s - t)(t + s + 47)$, and addition reduces to $(t + s)^2 + 49(t + s) - 6656$. Substitution of $t + s = T - 24$ yields the first two identities. For the product,

$$j_1 j_2 = \frac{((t + 256)(s + 256))^3}{(ts)^2} = (T + 248)^3.$$

Finally, $(s - t)^2 = (T - 24)^2 - 16384 = (T - 152)(T + 104)$. Multiplication by $(T + 23)^2$ proves (11.54). $\square$

Proposition 11.34 (Recovery of orientation from values and the first jet). *For $T \neq -23$, the ordered pair (T, j_1) recovers*

$$t = \frac{T^2 - j_1 - 3904}{T + 23}. \tag{11.55}$$

At $T = -23$, the two parameters

$$t_{\pm} = \frac{-47 \pm 45i\sqrt{7}}{2}$$

share $(T, j_1) = (-23, -3375)$. The first jet $p = dj_1/dT$, evaluated on the branch parametrized by t , separates them and recovers

$$p_{\pm} = \frac{-45 \mp 45i\sqrt{7}}{2}, \quad s = p - 1, \quad t = -46 - p. \tag{11.56}$$

Proof. The identity $s^2 = (T - 24)s - 4096$ gives

$$j_1 = (T + 23)s + T - 3352 = T^2 - 3904 - (T + 23)t.$$

For $T \neq -23$, this yields (11.55). For $T = -23$, the conditions $t + s = -47$, $ts = 4096$ give $t^2 + 47t + 4096 = 0$, with discriminant $-45^2 \cdot 7$. The same identity gives $j_1 = -3375$ at both roots.

The derivative $dT/dt = 1 - 4096/t^2$ vanishes only at $t = \pm 64$. Neither exceptional root has that value, so T is a local coordinate on both branches. Differentiating the first expression for j_1 ,

$$\frac{dj_1}{dt} = (s + 1)\frac{dT}{dt} + (T + 23)\frac{ds}{dt}.$$

At $T = -23$, this gives $p = s + 1 = -46 - t$, proving (11.56), with opposite signs for the paired indices $t_{\pm}, p_{\pm}$. $\square$

The fixed points of Fricke, $t = \pm 64$, yield respectively $(T, j) = (152, 8000)$ and $(-104, 1728)$. The coincidence at $T = -23$ instead corresponds to two distinct points of the parametrization. The field of definition follows from the equation:

$$\mathbb{Q}(t_{\pm}) = \mathbb{Q}(\sqrt{-7}), \quad \left(\frac{2t + 47}{45}\right)^2 = -7.$$

On this fibre, Fricke agrees with quadratic conjugation, with trace -47 and norm 4096 . The jet retains this field, since $(2p + 45)^2 = -45^2 \cdot 7$. Reconstruction is exact within this modular correspondence: the differential datum separates the two orientations that share their scalar values.

11.20 Compatibility of the level-two and level-five modular readers

On the modular realization subsequent to the HMT construction, fix $\text{Im } \tau > 0$, $q = e^{2\pi i \tau}$ and $q^{1/5} = e^{2\pi i \tau/5}$. The Rogers–Ramanujan reader is

$$R(q) = \frac{q^{1/5}}{1 + \frac{q}{1 + \frac{q^2}{1 + \dots}}}, \quad x = R(q)^5.$$

The modular identity of this realization, with the normalization of j used in (11.45), is

$$j = -\frac{(x^4 - 228x^3 + 494x^2 + 228x + 1)^3}{x(x^2 + 11x - 1)^5}. \quad (11.57)$$

The identity is the modular-realization premise of the following algebraic transport; the seeds, orientation and self-scaling representations remain those of APP–TRIT–TPK.

Proposition 11.35 (Pentadic quotient and branch selection). *For $x \neq 0$, let $u = x - x^{-1}$. On the domain $u \neq -11$,*

$$j = -\frac{(u^2 - 228u + 496)^3}{(u + 11)^5}. \quad (11.58)$$

The involution $x \mapsto -1/x$ preserves u . Its preimages are

$$x = \frac{u \pm \sqrt{u^2 + 4}}{2}.$$

The quadratic covering ramifies at $u = \pm 2i$; the subsequent rational map $u \mapsto j$ has degree six.

Proof. The exact factorizations are

$$\begin{aligned} x^4 - 228x^3 + 494x^2 + 228x + 1 &= x^2(u^2 - 228u + 496), \\ x^2 + 11x - 1 &= x(u + 11). \end{aligned}$$

Substitution into (11.57) proves (11.58). The fibre equation is $x^2 - ux - 1 = 0$; its discriminant gives the ramification points. The numerator of (11.58) has degree six and the denominator degree five. At $u = -11$, the numerator polynomial has value 3125, so the two have no common factor. This proves the asserted degree. $\square$

For real u , the two roots have opposite signs. On the complex domain, the branch selected in the parametrization by τ is retained. The reading j alone retains a fibre with generically six values of u .

Proposition 11.36 (Self-scaling limit and compatibility of levels). *In the real limit $q \rightarrow 1^-$,*

$$R(q) \longrightarrow \varphi_{\text{HMT}}^{-1}, \quad x \longrightarrow \varphi_{\text{HMT}}^{-5} = \frac{5\sqrt{5} - 11}{2}, \quad u \longrightarrow -11.$$

When $t = t_2(\tau)$ and u are read on the same modular parameter, they satisfy

$$(t + 256)^3(u + 11)^5 + t^2(u^2 - 228u + 496)^3 = 0. \quad (11.59)$$

Writing $\epsilon = u + 11$, the branches satisfying the stated limits obey

$$\epsilon \rightarrow 0, t \rightarrow 0 : \quad \frac{t^2}{(-\epsilon)^5} \longrightarrow \frac{2^{24}}{5^{15}}, \quad (11.60)$$

$$\epsilon \rightarrow 0, t \rightarrow \infty : \quad t\epsilon^5 \longrightarrow -5^{15}. \quad (11.61)$$

Proof. For $0 < q < 1$, the alternating positive convergents of the continued fraction bracket $R(q)$. As $q \rightarrow 1^-$, each fixed convergent tends to the corresponding convergent of the continued fraction with all numerators equal to one. The two subsequences of these convergents have limit $\varphi_{\text{HMT}}^{-1}$, in its already established quadratic realization. The bracketing proves the first limit. Taking the fifth power and using $\varphi^2 = \varphi + 1$ gives the other two.

Equating (11.45) and (11.58) proves (11.59). Substitution of $u = \epsilon - 11$ gives

$$\frac{(t + 256)^3}{t^2} = -\frac{(3125 - 250\epsilon + \epsilon^2)^3}{\epsilon^5}.$$

On the branch $t \rightarrow 0$, the identity rearranges as

$$\frac{t^2}{(-\epsilon)^5} = \frac{(t + 256)^3}{(3125 - 250\epsilon + \epsilon^2)^3}.$$

Its limit is $256^3/3125^3 = 2^{24}/5^{15}$. On the branch $t \rightarrow \infty$, multiplication by ϵ^5 and expression of the left-hand side as $t\epsilon^5(1 + 256/t)^3$ give the second limit. $\square$

The relation retains the branches of a correspondence over the same j . The choice of τ , or of the corresponding lift, fixes the pair being transported. The pentadic pole and Fricke inversion retain their respective domains and operations.

The three operations appearing in this prolongation have different domains. The dual symmetry $g^\vee$ acts on sectors of the vertex algebra; Fricke acts on the modular parameter and its function; radius duality interchanges momentum and winding in the compact lattice sector. Their reciprocity formulas can be coordinated through the corresponding maps, but the name “duality” does not replace those maps. This distinction allows the construction to continue toward dimensional screens while preserving the incidence, grading, and origin of each operation.

11.21 Exceptional incidence and coordination of the screens

The direction selected by K and the Leech lattice arise from different representations of the constructed incidence. The former determines the orthogonal reduction $12 \rightarrow 11 \rightarrow 10$; the latter enters the isotropic reduction $26 \rightarrow 24$. Their coordination preserves the preceding space, its relations and the information conveyed by each projection.

Sections 18 and 19 develop this coordination jointly. The former brings together the involution of integer coordinates, unitary conjugation with its domains and the change of residual section retaining the quotient memory. The latter constructs the morphism $\mathcal{R}_M^{\text{alg}}$ of (19.5) and proves the isomorphism of the quotient presentations in Theorem 19.6. The graph preserves the coordinate in V_{11} together with its projection in V_{10} , while the isotropic quotient recovers the Leech lattice. The respective kernels and codomains are thereby preserved.

The action ellipse is constructed before duality is applied and fixes its dimensionless radius together with the reciprocal radius (Section 17). The compact sector then connects exceptional incidence with a determined quadratic form. The fields, action and operators of the subsequent physical realization are presented in their own domain, preserving this algebraic antecedent and its internal proofs.

11.22 Graded Moonshine traces and representation of the Monster

For $g \in \text{Aut}(V_{\text{HMT}}^{\mathfrak{h}})$, the trace with insertion is

$$T_g(\tau) = \text{Tr}_{V_{\text{HMT}}^{\mathfrak{h}}} g q^{L_0 - 1}.$$

The monstrous Moonshine theorem identifies these series as Hauptmoduln of the corresponding genus-zero groups [12, Theorem 1.1]. The character J , the group $\mathbb{M}$, and the collection $g \mapsto T_g$ are realizations of the algebra already constructed. They do not form a causal sequence in which one first guesses J and then infers the Monster.

The unit has a precise algebraic location here. The weight-zero subspace is the vacuum line, $V_0^{\mathfrak{h}} = \mathbb{C}\mathbf{1}$, and the coefficient of q^{-1} in J is therefore 1. In weight one the dimension is zero, accounting for the absence of the constant term in $J = j - 744$. A morphism of vertex algebras preserves the vacuum vector. This property provides the technical meaning of preservation of the unit at this level; the unit of cylindrical observables is preserved by the precomposition proved earlier. The relation between the two levels must follow the construction maps with their units.

In weight two, the equality

$$196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

appears. The first decomposition corresponds to the trivial conformal line and the irreducible component of dimension 196883 in the Monster representation. The other expressions have lattice-theoretic or arithmetic readings. Their equality does not automatically identify their summands with TPK regions, words, or routes. Doing so would require presenting the maps between those spaces and proving that they respect the relevant actions.

In particular, an isometry $N_\alpha \simeq \Lambda_{24}$ induces a realization of (11.33), and a choice of isomorphism with $V^{\mathfrak{h}}$ transports its automorphism group. This does not construct an action of the Monster on the digits of π , on the blocks of K , or on all TPK histories. Such an additional assertion would require a representation ρ_{TPK} and a map F with an explicit square

$$F\rho_{\text{TPK}}(g) = \rho_{V^{\mathfrak{h}}}(g)F,$$

including domain, codomain, and preserved information. The absence of that step from the present claim does not remove the incidence intertwiner U_W , the flag, or the neighbor already constructed; it only delimits a different promotion, namely an action on the source state.

The chain established in this section can be summarized, keeping its transport data visible, as

$$\begin{aligned} & \text{APP-TRIT-TPK and arithmetic state with trajectory memory} \longrightarrow (w_j, f_j; K, Z_0) \\ & \longrightarrow (C_W, \mathcal{H}; B_K \subset H_{\alpha}^-, P_{\pi}, H_e, H_{\varphi}, H_{\text{APP}}) \\ & \longrightarrow (N(C_W), o_*, \text{metric and chamber}) \longrightarrow N_{\alpha} \longrightarrow V_{\text{HMT}}^{\natural}. \end{aligned}$$

At the incidence level, the stars generate M_{12} , and the residual chart realizes the regional $4 + 1$ structure in W_{24} . At the lattice and VOA level, the construction gives Leech and the Moonshine module. These are prolongations of the same marked source, with different objects and maps. Causal continuity does not require identifying their codomains or erasing the explicit choices of frame, chamber, and lift.

The finite and lattice proofs establish the passage from dodecaphase incidence to the hypotheses of the recognition theorems. The resulting composition realizes the Moonshine algebra and its automorphism group on the specified domains. The assertion retains the stated choices of frame, metric, chamber, and lift; it neither identifies these automorphisms with an action on digits nor adds experimental validation of the constants.

12 Hilbert-space realization of the nonadic phase and preservation of the unit

The construction of Triadic Modular Holography first provides the states, their transitions, and the restriction maps. The Hilbert-space realization adds a linear structure and an inner product to their discrete coordinates. This order determines what each operator represents: the basis labels arise from the nonadic construction; linear operations subsequently act on those labels. Genealogical memory remains in the TPK state and accompanies the projections used to represent it.

12.1 Phase fiber, labeled basis, and inner product

For an integer depth $d \geq 1$, let

$$\mathcal{C}_{9,d} = (\mathbb{Z}/9\mathbb{Z})^d, \quad \mathcal{H}_d = \ell^2(\mathcal{C}_{9,d}), \quad \mathcal{A}_d = B(\mathcal{H}_d). \quad (12.1)$$

The inner product is $\langle f, g \rangle = \sum_x \overline{f(x)}g(x)$. The functions δ_x form an orthonormal basis labeled by the phase coordinates. Thus $\dim \mathcal{H}_d = 9^d$ and $\mathcal{A}_d \cong M_{9^d}(\mathbb{C})$. The matrix algebra represents this finite fiber; routes, quotients, and memory omitted by a phase projection still belong to the starting state.

Coordinate factorization induces the unitary map

$$\mathcal{H}_{d+1} \longrightarrow \mathcal{H}_d \otimes \mathbb{C}^9, \quad \delta_{(x,r)} \longmapsto \delta_x \otimes \delta_r. \quad (12.2)$$

Define the normalized trace $\tau_d(a) = 9^{-d} \text{Tr}(a)$. Finite counting and normalization are distinguished: $\text{Tr}(I) = 9^d$, whereas $\tau_d(I) = 1$. Likewise, the group $(\mathbb{Z}/9\mathbb{Z})^3$ and the vector space $\mathbb{F}_3^6$ both have 729 elements, but retain different algebraic laws. Realization of a phase fiber does not replace the ternary encoding of APP emissions.

12.1.1 Encoding trit pairs and addition with carry

The ternary reading of APP-TRIT-TPK emissions has a six-coordinate ambient space over $\mathbb{F}_3$. Grouping its coordinates into three pairs organizes this space through three nonadic digits. The grouping preserves the position and order of the trits; the addition it represents must also transport their carry.

Let $T = \{0, 1, 2\}$ be the set of ternary representatives. For a class $x \in \mathbb{Z}/9\mathbb{Z}$, write $\tilde{x} \in \{0, \dots, 8\}$ for its canonical representative. Define

$$\beta : T^2 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad \beta(a, b) = [a + 3b]_9. \quad (12.3)$$

Here $[s]_m$, when used as a digit, denotes the representative in $\{0, \dots, m-1\}$ of the residue class of s .

Proposition 12.1 (Inverse and transport of addition). *The map β is a bijection, with inverse*

$$\beta^{-1}(x) = \left([\tilde{x}]_3, \left\lfloor \frac{\tilde{x}}{3} \right\rfloor \right). \quad (12.4)$$

The operation on T^2 transported from nonadic addition is

$$(a, b) \oplus (c, d) = \left([a + c]_3, \left[b + d + \left\lfloor \frac{a + c}{3} \right\rfloor \right]_3 \right). \quad (12.5)$$

With this operation, β is an isomorphism of additive groups. The identity is $(0, 0)$, and the inverse of (a, b) is

$$\left([-a]_3, \left[-b + \left\lfloor \frac{-a}{3} \right\rfloor \right]_3 \right). \quad (12.6)$$

Proof. Euclidean division of each $0 \leq \tilde{x} \leq 8$ by three gives a unique remainder $a \in T$ and a unique quotient $b \in T$, with $\tilde{x} = a + 3b$. This proves bijectivity and (12.4).

For addition, write $a + c = 3k + r$, where $r = [a + c]_3$ and $k = \lfloor (a + c)/3 \rfloor$. Then

$$(a + 3b) + (c + 3d) = r + 3(b + d + k).$$

Reducing $b + d + k$ modulo three changes the right-hand side only by a multiple of nine. Hence $\beta((a, b) \oplus (c, d)) = \beta(a, b) + \beta(c, d)$. Bijectivity transports associativity, commutativity and the identity. Finally, the division $-a = 3\lfloor -a/3 \rfloor + [-a]_3$, applied to $-a - 3b$, gives (12.6). $\square$

The carry is already visible in

$$(2, 0) \oplus (1, 0) = (0, 1), \quad \beta(2, 0) + \beta(1, 0) = [3]_9.$$

Vector addition in $\mathbb{F}_3^2$ would produce $(0, 0)$ in this same example. Moreover, $(1, 0)$ has order nine under $\oplus$, whereas its order under vector addition is three. These operations specify the two algebraic structures carried by the same set of nine representatives.

Corollary 12.2 (Cubic chart of the ternary ambient space and projection fibres). *The coordinatewise map*

$$\begin{aligned} \beta^{\times 3} : T^6 &\longrightarrow (\mathbb{Z}/9\mathbb{Z})^3, \\ (t_1, t_2, t_3, t_4, t_5, t_6) &\longmapsto (\beta(t_1, t_2), \beta(t_3, t_4), \beta(t_5, t_6)) \end{aligned} \quad (12.7)$$

is a bijection between sets of $729 = 3^6 = 9^3$ elements. Projection $\text{pr}_{12}(X, Y, Z) = (X, Y)$ onto the 81 nonadic positions has exactly nine elements in each fibre.

Proof. The product of the three inverses in (12.4) inverts (12.7). For fixed X, Y , the third coordinate ranges over the nine classes of $\mathbb{Z}/9\mathbb{Z}$; hence $\text{pr}_{12}^{-1}(X, Y) = \{(X, Y, Z) : Z \in \mathbb{Z}/9\mathbb{Z}\}$, with the asserted cardinality. Retaining Z together with X, Y allows the full inverse to recover the six ordered trits. $\square$

The domain of this count is the complete ambient space of 729 words. The catalogue of 468 decimal emissions and its admissible image of 243 ternary words retain their own emission maps and restrictions. The restriction of $\beta^{\times 3}$ to that image remains injective, whereas its projection-fibre sizes are determined by intersection with the ambient fibre. The uniform statement of nine elements belongs to the complete cube.

The resulting two-dimensional chart recovers a labelled representation of the previously constructed APP support. The additive and multiplicative sheets are additional functions on this support, with the representatives and quotients declared in the kernel. The third coordinate preserves the datum omitted by facial projection; routes and temporal memory remain in the TPK state from which the reading proceeds. The finite encoding thereby specifies the information it transports and distinguishes it from the complete history to which it belongs.

12.2 Translation and phase: finite Weyl representation

In the complex realization subsequent to the constructions of π and the structure $i^2 = -1$, fix $\omega = \exp(2\pi i/9)$. On $\mathcal{H}_1$, let

$$(Xf)(r) = f(r - 1), \quad (Zf)(r) = \omega^r f(r). \quad (12.8)$$

The first operation shifts the phase coordinate; the second evaluates it through a character. Both are unitary and satisfy

$$X^9 = Z^9 = I, \quad ZX = \omega XZ. \quad (12.9)$$

Indeed, $(ZXf)(r) = \omega^r f(r - 1)$, whereas $(XZf)(r) = \omega^{r-1} f(r - 1)$. The character appears here as a realization of the phase already constructed and does not select the TPK numerical registers.

12.2.1 Oriented area and central extension of translations

The translation support is $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. The mixed APP curvature constructed in Section 4.4 assigns +1 to the oriented plaquette of the sum-product pair and -1 to the pair in the opposite order. Its bilinear extension to two displacements $u = (a, b)$ and $v = (c, d)$ is the alternating form

$$\sigma_{\text{ar}}(u, v) = ad - bc \quad \text{in } \mathbb{Z}/9\mathbb{Z}. \quad (12.10)$$

Indeed, alternation makes the evaluation vanish on two displacements along the same axis. Values on the oriented basis are $\sigma_{\text{ar}}((1, 0), (0, 1)) = 1$ and $\sigma_{\text{ar}}((0, 1), (1, 0)) = -1$; distributing both arguments gives (12.10). The same formula is the sum of plaquettes of any oriented cellular chain realizing that parallelogram: interior edges cancel. Changing an integer representative by a multiple of nine preserves its representation in $\mathbb{Z}/9\mathbb{Z}$.

The order $X^a Z^b$ of the already defined operators fixes a section of the central extension. Denote its composition factor by

$$c_W(u, v) = bc \quad \text{in } \mathbb{Z}/9\mathbb{Z}. \quad (12.11)$$

This cocycle preserves the order of the two operations, while σ_{ar} preserves the orientation of the parallelogram.

Proposition 12.3 (Central extension and commutator sign). *The set $\mathfrak{H}_9 = (\mathbb{Z}/9\mathbb{Z})^3$, with product*

$$(a, b, t)(c, d, s) = (a + c, b + d, t + s + bc), \quad (12.12)$$

is a group of order 729. The map

$$\varrho(a, b, t) = \omega^t X^a Z^b$$

is a faithful unitary representation on $\mathcal{H}_1$. With the convention $[g, h] = ghg^{-1}h^{-1}$, one has

$$[\varrho(a, b, 0), \varrho(c, d, 0)] = \omega^{bc-ad} I = \omega^{-\sigma_{\text{ar}}((a,b),(c,d))} I. \quad (12.13)$$

The center and the translation group form the exact sequence

$$0 \longrightarrow \mathbb{Z}/9\mathbb{Z} \xrightarrow{t \mapsto (0,0,t)} \mathfrak{H}_9 \xrightarrow{(a,b,t) \mapsto (a,b)} X_9 \longrightarrow 0.$$

Proof. For $w = (e, f)$, the cocycle identity reads

$$c_W(u, v) + c_W(u + v, w) = bc + (b + d)e = de + b(c + e) = c_W(v, w) + c_W(u, v + w).$$

It proves associativity of (12.12); the identity is $(0, 0, 0)$, and the inverse of (a, b, t) is $(-a, -b, -t + ab)$. The three residues are free, giving order $9^3 = 729$. From $ZX = \omega XZ$ it follows that

$$(X^a Z^b)(X^c Z^d) = \omega^{bc} X^{a+c} Z^{b+d},$$

which proves the representation property. If $\omega^t X^a Z^b = I$, its action on δ_r first forces $a = 0$. Evaluation at $r = 0$ gives $t = 0$, and evaluation at $r = 1$ gives $b = 0$, since ω has order nine. The representation is therefore faithful. Comparing the products in both orders gives the factor $bc - ad$. A central element must satisfy $bc - ad = 0$ for every (c, d) . Taking successively $(c, d) = (1, 0), (0, 1)$ yields $b = a = 0$. This identifies the center and proves exactness. $\square$

The Wilson phase for orientation (12.10) is $\omega^{\sigma_{\text{ar}}(u,v)}$; the commutator of the ordered section $X^a Z^b$ expresses its inverse. In particular, $c_W(u, v) - c_W(v, u) = -\sigma_{\text{ar}}(u, v)$. This identity relates two determined conventions without identifying the polarized cocycle with the alternating form. The extension belongs to the translation group of the APP support. Its representation acts on the phase fiber; the trajectory register and carries retain their residence in the TPK state that produced that fiber. The group order 729 and the cardinality 729 of the ternary ambient space preserve the different types established in (12.1). The center $\mathbb{Z}/9\mathbb{Z}$ of this extension records the commutator phase. The memory C_{12} of the sequence $0 \rightarrow C_{12} \rightarrow C_{108} \rightarrow C_9 \rightarrow 0$, constructed in Proposition 9.1, belongs to the dodecaphasic calendar. Each extension preserves its own composition law and projection.

Proposition 12.4 (Algebra generated by phase and translation). *The 81 operators $X^a Z^b$, with $0 \leq a, b < 9$, form a basis of $\mathcal{A}_1 = M_9(\mathbb{C})$. In particular, the representation (12.8) is irreducible.*

Proof. If $a \not\equiv 0 \pmod{9}$, the matrix $X^a Z^b$ has no diagonal entries. If $a = 0$, its trace is $\sum_{r=0}^8 \omega^{br}$, equal to 9 for $b = 0$ and 0 otherwise. By (12.9), the product $(X^a Z^b)^* X^{a'} Z^{b'}$ is a factor of modulus one times $X^{a'-a} Z^{b'-b}$. Therefore,

$$\tau_1((X^a Z^b)^* X^{a'} Z^{b'}) = \delta_{a,a'} \delta_{b,b'}.$$

These are 81 linearly independent matrices in a space of dimension 81. They generate the whole matrix algebra, whose commutant consists of scalars; irreducibility follows. $\square$

Theorem 12.5 (Uniqueness with fixed primitive central character). *Every irreducible unitary representation of the group presented by $X^9 = Z^9 = c^9 = I$, c central, and $ZX = cXZ$, in which c acts as ωI , is unitarily equivalent to (12.8).*

Proof. Let $v \neq 0$ be an eigenvector of Z , with eigenvalue λ . Since $Z^9 = I$, λ is a power of ω . The vectors $X^k v$, $0 \leq k < 9$, have eigenvalues $\omega^k \lambda$, distinct because the character is primitive. They are orthogonal, and their span is invariant under X , Z , and the center. Irreducibility forces it to be the entire space. Reordering the basis to begin with eigenvalue 1, Z is diagonal with entries $1, \omega, \dots, \omega^8$, and X is cyclic translation. Normalizing the basis produces the asserted unitary equivalence. $\square$

Primitivity of the character and irreducibility are part of the statement. At depth d , operators acting on different factors commute, and tensor products of the preceding bases generate $\mathcal{A}_d$. The phase structure is thus represented at each level without identifying phase return with return of the complete genealogical state.

12.3 Two lifts and two different normalizations

Factorization (12.2) permits preserving the entire new fiber or fixing one of its labels. For $p_j = |\delta_j\rangle\langle\delta_j|$, define

$$\iota_d(a) = a \otimes I_9, \quad J_{d,j}(a) = a \otimes p_j. \quad (12.14)$$

Proposition 12.6 (Preservation of unit and trace). *The inclusion ι_d is unital and preserves normalized trace. The inclusion $J_{d,j}$ is a corner embedding and satisfies*

$$\tau_{d+1}(\iota_d(a)) = \tau_d(a), \quad \iota_d(I) = I, \quad \tau_{d+1}(J_{d,j}(a)) = \frac{\tau_d(a)}{9}, \quad J_{d,j}(I) = I \otimes p_j.$$

Proof. The trace of a tensor product is the product of the traces. Since $\text{Tr}(I_9) = 9$ and $\text{Tr}(p_j) = 1$, dividing by 9^{d+1} produces both formulas. The images of the identity are read directly from (12.14). $\square$

The unital inclusion preserves the set of 9 phases and its global normalization. The corner preserves a marked continuation and its support projector. Both operations are legitimate, but transmit different data to the next depth.

12.4 Inductive closure and tracial realization

Theorem 12.7 (Nonadic uniformly hyperfinite algebra). *The norm closure of the union through the inclusions ι_d is*

$$\mathcal{A}_{9^\infty} = \varinjlim (\mathcal{A}_d, \iota_d) \cong \bigotimes_{n=1}^{\infty} M_9(\mathbb{C}).$$

It is a UHF algebra of type $9^\infty = 3^\infty$, with a unique normalized trace τ , whose restriction to $\mathcal{A}_d$ is τ_d .

Proof. The inclusions are injective and unital. The closure of the increasing union of matrix algebras is precisely the UHF construction. The compatibility in Proposition 12.6 defines τ on the union; positivity and norm one allow its extension to the closure. Every normalized trace on the limit restricts to the unique normalized trace on each matrix algebra. Since the union is dense, those restrictions determine the trace uniquely. $\square$

Let $\mathcal{H}_\tau$ be the completion of $\mathcal{A}_{9^\infty}$ for $\langle a, b \rangle_\tau = \tau(a^*b)$, and let π_τ be its representation by left multiplication. The vector defined by the identity is denoted Ω_τ .

Theorem 12.8 (Tracial factor of complete refinement). *The weak closure $\mathcal{R}_9 = \pi_\tau(\mathcal{A}_{9^\infty})''$ is a hyperfinite factor of type II_1 . The nonadic inclusions preserve its unit and normalized trace.*

Proof. The product trace extends to a faithful finite normal trace in the tracial representation. To see that the center is trivial, let E_d be the tracial expectation onto $\pi_\tau(\mathcal{A}_d)$. On a tensor of length $n > d$, it takes the trace of the last $n - d$ factors. These maps extend continuously to the corresponding orthogonal projections in L^2 . If z is central, $E_d(z)$ commutes with $\pi_\tau(\mathcal{A}_d)$, by bimodularity of E_d . It is therefore scalar and equal to $\tau(z)I$. Density of the matrix union implies $E_d(z) \rightarrow z$ in L^2 , hence $z = \tau(z)I$. The algebra is thus a finite factor. It contains unital matrix algebras of unbounded sizes 9^d , so it is not a finite matrix factor. It is of type II_1 . It is hyperfinite because the same union of matrices is weakly dense. Preservation of unit and trace follows from each finite inclusion and persists upon taking closures. $\square$

12.5 Integer ranks and continuous tracial dimension

In $\mathcal{A}_d$, a projection P has normalized dimension $\tau_d(P) = \text{rank}(P)/9^d$. Ranks remain integers at every finite depth. Their compatible normalizations allow the following passage to the limit.

Proposition 12.9 (Realization of each tracial dimension). *For each $t \in [0, 1]$, there exists a projection $P_t \in \mathcal{R}_9$ with $\tau(P_t) = t$.*

Proof. Let $m_d = \lfloor 9^d t \rfloor$. Then $0 \leq m_{d+1} - 9m_d \leq 8$. After choosing a diagonal projection of rank m_d , its inclusion has rank $9m_d$; add $m_{d+1} - 9m_d$ diagonal positions in its complement. This yields an increasing sequence of projections within $\mathcal{R}_9$. Its strong limit is a projection P_t , and normality of the trace gives

$$\tau(P_t) = \lim_{d \rightarrow \infty} \frac{\lfloor 9^d t \rfloor}{9^d} = t.$$

$\square$

Here t denotes the dimension coordinate represented in the closure, not an input to the APP generator. The assertion describes the scope of the resulting tracial realization. To compare it with a particular observable, it remains necessary to identify the projection and the genealogical information it preserves.

By contrast, the isometries $V_{d,j}\xi = \xi \otimes \delta_j$ realize $J_{d,j}(a) = V_{d,j}aV_{d,j}^*$. In the limit of a marked route, the finite spaces are dense. The corners approximate all finite-rank operators; their norm closure is $\mathcal{K}(\mathcal{H}_\infty)$, and their bicommutant is $B(\mathcal{H}_\infty)$, of type I_∞ . This difference preserves the operational content of the two choices in (12.14): selection of one continuation and unital prolongation of the entire fiber have different limits.

Proposition 12.4 realizes the matrix algebra through the phase operators. Theorems 12.7 and 12.8 establish the unital inductive closure and its tracial realization, with explicit choices of inclusion. This is the operator-theoretic framework used subsequently for coordinate interchange and the compact Hamiltonian.

12.6 Polar transport of incidence and preservation of readers

The finite realization of the oriented incidence complex receives the boundary and orientation constructed in the APP-TRIT-TPK state. Once the inner products of this realization are fixed, let $\mathcal{H}_0, \mathcal{H}_1$ be its finite-dimensional spaces and

$$d : \mathcal{H}_0 \longrightarrow \mathcal{H}_1, \quad \Delta_0 = d^*d, \quad \Delta_1 = dd^*.$$

The incidence and return readers are

$$\rho_{\text{int}}(v) = (dv, d^*dv), \quad \rho_{\text{ref}}(a) = (d^*a, dd^*a).$$

The symbols Δ_0, Δ_1 here denote the Laplacians of this Hilbert-space realization. Orientation, metrics and the domain of d are part of its data.

Proposition 12.10 (Unitary equivalence of the positive sectors). *Let*

$$\mathcal{H}_0^+ = (\ker d)^\perp, \quad \mathcal{H}_1^+ = (\ker d^*)^\perp = \text{Ran } d.$$

The map

$$\mathcal{U} = d(\Delta_0|_{\mathcal{H}_0^+})^{-1/2} : \mathcal{H}_0^+ \longrightarrow \mathcal{H}_1^+ \quad (12.15)$$

is unitary and satisfies, on those sectors,

$$\mathcal{U}\Delta_0 = \Delta_1\mathcal{U}, \quad \mathcal{U}f(\Delta_0) = f(\Delta_1)\mathcal{U} \quad (12.16)$$

for every function f defined on the common finite positive spectrum.

Proof. The reduced singular value decomposition is $d = V\Sigma W^*$, where the columns of W and V are orthonormal bases of $\mathcal{H}_0^+$ and $\mathcal{H}_1^+$, and Σ is diagonal with strictly positive entries. Hence

$$\Delta_0|_{\mathcal{H}_0^+} = W\Sigma^2W^*, \quad \Delta_1|_{\mathcal{H}_1^+} = V\Sigma^2V^*, \quad \mathcal{U} = VW^*.$$

It follows that $\mathcal{U}^*\mathcal{U} = I_{\mathcal{H}_0^+}$ and $\mathcal{U}\mathcal{U}^* = I_{\mathcal{H}_1^+}$. Evaluating f on the same diagonal Σ^2 proves (12.16). If $d = 0$, both positive sectors are zero spaces and the identities have their unique interpretation between zero-dimensional spaces. $\square$

For $v \in \mathcal{H}_0^+$ and $h, t \geq 0$, quadratic energy, the resolvent and the semigroup are preserved:

$$\langle \mathcal{U}v, \Delta_1\mathcal{U}v \rangle = \langle v, \Delta_0v \rangle, \quad (12.17)$$

$$\mathcal{U}(I + h\Delta_0)^{-1} = (I + h\Delta_1)^{-1}\mathcal{U}, \quad (12.18)$$

$$\mathcal{U}e^{-t\Delta_0} = e^{-t\Delta_1}\mathcal{U}. \quad (12.19)$$

The first identity follows from unitarity and intertwining. The other two are instances of (12.16), with $f(x) = (1 + hx)^{-1}$ and $f(x) = e^{-tx}$.

12.6.1 Realization on the tritic cycle

In the orthonormal bases of vertices and oriented edges of the three-element cycle,

$$d = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix}.$$

Matrix multiplication gives

$$\Delta_0 = \Delta_1 = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} = 3I - J_3 = 2I - C - C^{-1},$$

where every entry of J_3 equals one and C is the cyclic permutation. The constant vector spans the kernel; on its orthogonal complement, $J_3 = 0$. Thus the spectrum is $\{0, 3, 3\}$ and $\mathcal{U} = d/\sqrt{3}$ on the positive sector. The two readers share this spectrum because they arise from the same incidence, while retaining their vertex and edge spaces.

12.6.2 Orientation and compatibility of refinements

The replacement $d \mapsto -d$ preserves Δ_0, Δ_1 and changes $\mathcal{U}$ to $-\mathcal{U}$. More generally, for a reversal of edge orientations represented by a diagonal unitary matrix E ,

$$d' = Ed, \quad \Delta'_0 = \Delta_0, \quad \Delta'_1 = E\Delta_1E^*, \quad \mathcal{U}' = E\mathcal{U}.$$

These equalities follow by using $E^*E = I$ in the definitions. Orientation is thus retained by the transporter even when the spectral readings coincide.

In a collection of refinements, the maps $J_m^{(0)} : \mathcal{H}_{0,m}^+ \rightarrow \mathcal{H}_{0,m+1}^+$ and $J_m^{(1)} : \mathcal{H}_{1,m}^+ \rightarrow \mathcal{H}_{1,m+1}^+$ must retain domains and norms and satisfy

$$\mathcal{U}_{m+1}J_m^{(0)} = J_m^{(1)}\mathcal{U}_m$$

in order to transport this identification across levels. The proposition proves finite equivalence of the positive sectors; the preceding square expresses the additional refinement condition. Kernels, nonlinear constraints and passages to the limit retain their own proofs.

13 Arithmetic–geometric coupling and bilateral recovery

The dodecaphase record of Section 9 admits arithmetic, incidence-theoretic, and operator-theoretic readings. We call the generated record K the *holographic arithmetic–geometric coupling constant*, retaining the order of its twelve windows, the phase origin, and the orientation. Its rational reading $\kappa = K_{\text{int}}/(1000^{12} - 1)$ encodes that record; it does not replace its types or the enriched history that produces it. Hadamard inversion recovers its signed coordinate; the incidence-theoretic reading determines the Witt tetrad. The following constructions make two further functions explicit: identifying operators on its cyclic carrier and preserving information between two isometric realizations. These operations follow the realization of the continuum; they are not rules that introduce conventional constants into APP–TRIT–TPK.

13.1 Exact encoding and cyclic reference

Proposition 13.1 (Recovery of the record from its rational reading). *With base $b = 1000$, length twelve, origin, and orientation fixed, the map*

$$(k_1, \dots, k_{12}) \mapsto \kappa = \frac{\sum_{j=1}^{12} k_j b^{12-j}}{b^{12} - 1}, \quad 0 \leq k_j < b,$$

is injective. From κ , one recovers the integer $n = (b^{12} - 1)\kappa$ and, by successive Euclidean divisions, the twelve blocks, including leading zeros. An absolute error strictly smaller than $1/[2(b^{12} - 1)]$ permits the same recovery by rounding.

Proof. The fixed-length integer expansion in base b is unique. Two distinct integers are separated by at least one; their rational readings are separated by at least $1/(b^{12} - 1)$. The strict bound places the approximate datum within the rounding cell of a unique integer. $\square$

Now consider $K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601)$, the cyclic shift S of its positions, and $O_K = (K, SK, \dots, S^{11}K)$ on $\mathbb{R}^{12}$. The shift of positions is identified neither with the entire enriched evolution nor, solely by virtue of its periodicity, with an exceptional symmetry.

Theorem 13.2 (Invertibility of the orbital reference). *The matrix O_K is invertible, and*

$$589609\|c\|^2 \leq \|O_K c\|^2 \leq 6263^2\|c\|^2.$$

Its centred orbit has rank eleven.

Proof. The Gram matrix $O_K^* O_K$ is circulant. Its first row, obtained from the twelve inner products $\langle K, S^j K \rangle$, is

$$(4251257, 2999323, 3163385, 3287920, 3216553, 3344743, \\ 2950064, 3344743, 3216553, 3287920, 3163385, 2999323).$$

Evaluation at the twelve characters of the cycle yields the eigenvalues

$$6263^2, \quad 697225, \quad 589609 \ (2), \quad 1407529 \ (2), \quad 1053157 \ (2), \\ (1248025 - 345420\sqrt{3}) \ (2), \quad (1248025 + 345420\sqrt{3}) \ (2),$$

where parentheses indicate multiplicities. The smallest value is 589609: for the only comparison that is not immediate, $658416 > 345420\sqrt{3}$, as is seen by squaring both sides. The triangle inequality for the Fourier transform and $K_j \geq 0$ give the maximum $(\sum K_j)^2 = 6263^2$. Positivity of every mode proves invertibility and the bounds. Centring the record removes only the uniform mode. $\square$

Corollary 13.3 (Identification of an operator from its responses). *A linear operator $T : \mathbb{R}^{12} \rightarrow \mathbb{R}^{12}$ is determined by $Y = (TK, TSK, \dots, TS^{11}K)$, through $T = YO_K^{-1}$. If $TS = ST$, the complete vector response $y = TK$ suffices, since $T = O_y O_K^{-1}$. For a perturbation E of Y , $\|\hat{T} - T\| \leq \|E\|/\sqrt{589609}$.*

Proof. We have $Y = TO_K$. Under commutation, $TS^jK = S^jy$, hence $Y = O_y$. The bound follows from $\|O_K^{-1}\| = 1/\sqrt{589609}$. The norms are the Euclidean norm and its induced operator norm. $\square$

The response y contains twelve components. The statement does not attribute the recovery of an arbitrary operator to a single scalar: it specifies the observation and commutation condition that make identification possible.

13.2 Conservation and reconstruction of two channels

Let H, H' be Hilbert spaces, and let $\mathcal{B}, \mathcal{C} : H \rightarrow H'$ be bounded linear isometries. For $a > 0$, define

$$p = a^2 + a + 1, \quad N = (2a + 1)p = (a + 1)^3 + a^3, \quad Q_- = a\mathcal{B} - \mathcal{C}, \quad Q_+ = (a + 1)\mathcal{B} + \mathcal{C}.$$

Theorem 13.4 (Bilateral balance and recovery). *The following identities hold:*

$$(a + 1)Q_-^*Q_- + aQ_+^*Q_+ = NI_H, \quad W_a v = \frac{1}{\sqrt{N}} \begin{pmatrix} \sqrt{a+1} Q_- v \\ \sqrt{a} Q_+ v \end{pmatrix}, \quad W_a^* W_a = I_H.$$

The data $x = Q_- v$, $y = Q_+ v$ determine

$$\mathcal{B}v = \frac{x + y}{2a + 1}, \quad \mathcal{C}v = \frac{ay - (a + 1)x}{2a + 1}, \quad v = \mathcal{B}^* \frac{x + y}{2a + 1}.$$

Proof. Setting $X = \mathcal{B}^* \mathcal{C} + \mathcal{C}^* \mathcal{B}$, we obtain

$$Q_-^* Q_- = (a^2 + 1)I - aX, \quad Q_+^* Q_+ = ((a + 1)^2 + 1)I + (a + 1)X.$$

The weights cancel X , and the remaining coefficient is $2a^3 + 3a^2 + 3a + 1 = N$. Normalization gives the isometry. Adding the two channels recovers $(2a + 1)\mathcal{B}v$; the combination $ay - (a + 1)x$ recovers $(2a + 1)\mathcal{C}v$. Finally, $\mathcal{B}^* \mathcal{B} = I_H$. $\square$

The balance also holds in infinite dimension: only bounded operators are composed. The image of W_a is closed, and its orthogonal projector is $W_a W_a^*$. Thus a normalized pair has an input precisely when it belongs to that image; its orthogonal component measures the compatibility defect.

The factorization

$$W_a = R_a E_a, \quad E_a v = \frac{1}{\sqrt{p}} \begin{pmatrix} \sqrt{a(a+1)} \mathcal{B}v \\ \mathcal{C}v \end{pmatrix}, \quad R_a = \frac{1}{\sqrt{2a+1}} \begin{pmatrix} \sqrt{a} & -\sqrt{a+1} \\ \sqrt{a+1} & \sqrt{a} \end{pmatrix}$$

is verified by block multiplication; R_a acts on $H' \oplus H'$, with each scalar entry multiplying $I_{H'}$. In the nonadic partition $a = 8$, the weights are $72/73$ and $1/73$; the factor $73 = 8 \cdot 9 + 1$ follows from the general law $a(a + 1) + 1$, rather than being an additional primitive datum. The corresponding balance is $9\|Q_- v\|^2 + 8\|Q_+ v\|^2 = 17 \cdot 73 \|v\|^2$.

13.3 Transport of observables and incoming memory

Theorem 13.5 (Observables common to both channels). *For a bounded self-adjoint operator F on H' ,*

$$W_a^* \text{diag}(F, F) W_a = \frac{a(a + 1)\mathcal{B}^* F \mathcal{B} + \mathcal{C}^* F \mathcal{C}}{p}.$$

If $\mathcal{B}$ is unitary and $\mathcal{C} = \mathcal{B}R$, with R unitary, and $G = \mathcal{B}^ F \mathcal{B}$, the observable is preserved if and only if $GR = RG$.*

Proof. Expanding $(a + 1)Q_-^* F Q_- + aQ_+^* F Q_+$ cancels the cross terms. The other two coefficients are $a(a + 1)(2a + 1)$ and $2a + 1$. Dividing by N gives the formula. In the stated case, equality with G is equivalent to $R^* GR = G$; unitarity makes this equivalent to $GR = RG$. $\square$

The norm is obtained by taking $F = I$. A general observable requires the commutation condition: preserving the norm does not mean preserving every observable without transformation.

Theorem 13.6 (Unitary extension with memory). *If $H' = H$, $\mathcal{B} = I$, and $\mathcal{C} = U$ is unitary, set*

$$A = \frac{\sqrt{a+1}}{\sqrt{N}} Q_-, \quad D = \frac{\sqrt{a}}{\sqrt{N}} Q_+, \quad \mathcal{R}_a = \begin{pmatrix} A & -D^* \\ D & A^* \end{pmatrix}.$$

Then $\mathcal{R}_a$ is unitary on $H \oplus H$, its first column is W_a , and it preserves $\|v\|^2 + \|m\|^2$ for every input (v, m) . The inverse is

$$v = A^* z_1 + D^* z_2, \quad m = -D z_1 + A z_2.$$

Proof. The operators A, D are normal and commute with each other and with their adjoints, since they are polynomials in U, U^* . The balance gives $A^* A + D^* D = I$, and normality gives $AA^* + DD^* = I$. The cross terms in both products $\mathcal{R}_a^* \mathcal{R}_a$ and $\mathcal{R}_a \mathcal{R}_a^*$ vanish; the diagonal blocks are the identity. Applying the adjoint to (z_1, z_2) yields the inverse. $\square$

Record recovery, orbital identification, and unitary extension specify three distinct levels of information. The first preserves the blocks together with their chart; the second identifies operators under the stated hypotheses; the third transports state and memory jointly. This hierarchy allows the same generated structure to be used without confusing a reduced reading with the state in its entirety.

14 Determinantal structure and oriented action sections

The representations of π, φ, e, α come from the APP-TRIT-TPK state of Sections 2–10. The dodecaphasic register of 9 preserves its orientation, its residues and quotients, and the memory of return (4.6). Two successive operations are defined on that register: a local norm determines the valuation of the action section, and a character with regional continuation determines its coordinates before and after return. Both operations transmit their rules and normalizations to the angular construction.

14.1 Local quotient and determinantal norm

The local step-three block of the dodecaphasic register, identified in 9.8, is

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad G_H = \mathbb{Z}^4 / H_4 \mathbb{Z}^4. \quad (14.1)$$

The quotient corresponds to this local block. The three blocks of the complete register preserve their positions; the norm operation that follows acts on the sheets of G_H .

Lemma 14.1 (Structure of the local quotient). *The Smith normal form of H_4 is $\text{diag}(1, 2, 2, 4)$. Consequently,*

$$G_H \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \quad |G_H| = 16.$$

Proof. The greatest common divisor of the entries is one. The minors of order two are even, and one has value -2 , so their greatest common divisor is two. The identities $H_4 H_4^T = 4I$ and $\det H_4 = -16$ show that the cofactors are ± 4 ; the third determinantal divisor is four. The fourth is 16. The successive quotients of 1, 2, 4, 16 are 1, 2, 2, 4; their product determines the order of the quotient. $\square$

Definition 14.2 (Determinantal action readout). The scalar section α , constant on the sheets of G_H , and one application of the self-scale φ determine

$$\Sigma_H = \varphi N_{G_H}(\alpha), \quad N_{G_H}(\alpha) = \prod_{g \in G_H} \alpha = \alpha^{16}. \quad (14.2)$$

Its decimal valuation is

$$\text{ord}_{10}(\Sigma_H) = \lfloor \log_{10} \Sigma_H \rfloor.$$

The sixteenth power comes from the number of sheets once the multiplicative norm has been fixed. The norm and the single application of φ are part of the declared readout: the group order, considered in isolation, admits other functionals.

Theorem 14.3 (Decimal valuation of the action section). *The preceding readout, applied to the HMT coordinates already generated, satisfies*

$$10^{-34} < \Sigma_H < 10^{-33}, \quad \text{ord}_{10}(\Sigma_H) = -34. \quad (14.3)$$

Proof. We use the rational enclosures of the preceding representations,

$$\frac{1618}{1000} < \varphi < \frac{1619}{1000}, \quad \frac{7297}{10^6} < \alpha < \frac{7298}{10^6}. \quad (14.4)$$

The function $(x, a) \mapsto xa^{16}$ is increasing in both positive variables. The integer inequalities

$$1618 \cdot 7297^{16} > 10^{65}, \quad 1619 \cdot 7298^{16} < 10^{66}$$

give

$$10^{-34} < \frac{1618}{1000} \left(\frac{7297}{10^6} \right)^{16} < \Sigma_H < \frac{1619}{1000} \left(\frac{7298}{10^6} \right)^{16} < 10^{-33}.$$

The definition of the valuation completes the proof. The enclosures are subsequent checks on coordinates already produced; the determinantal operation precedes any chart of units. $\square$

14.2 Action character and regional continuation

The central chart preserves the matrix

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \lambda_+ = 9, \quad \lambda_- = 5.$$

The calendar has length 12, the step-three orbit has length four, and the census quotient used is $104976/1944 = 54$. These invariants determine the coefficients assigned to degrees 2, 3, 4, 5 of the character:

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \quad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \quad \frac{7}{48} = \frac{\text{tr}(B_c)/2}{4 \cdot 12}, \quad \frac{1}{54} = \frac{1944}{104976}. \quad (14.5)$$

The assignment to those degrees and the alternating signs belong to the definition of the analytic character. The operational content brings together the support, orientation, and that assignment rule.

Definition 14.4 (Local character and regional return readout). On the representations α, φ, π , the fifth-order character, regional transfer, and its continuation are

$$H_5 = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \quad (14.6)$$

$$D_A = \exp\left(-\frac{100\pi\alpha}{9}\right), \quad (14.7)$$

$$\mathcal{C}_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \quad (14.8)$$

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi} \mathcal{C}_\pi(\alpha). \quad (14.9)$$

The integer 169 is the regional selector of the representations constructed in 6; the first strict extension of its continuation has weight D_A , and each concatenation multiplies the weight by D_A . The initial normalization is unity.

The continuation is formulated before evaluation at $z = \alpha$. With D_A already produced by its readout, write $\mathcal{C}(z) = 169z^6 + \mathcal{R}(z)$, where $\mathcal{R}(z) \in z^7\mathbb{R}[[z]]$. Concatenation increases the degree by one; the initial weight and the transfer rule give

$$\mathcal{R}(z) = D_A z^7 + D_A z \mathcal{R}(z). \quad (14.10)$$

Proposition 14.5 (Uniqueness of the normalized continuation). Equation (14.10) determines a unique series:

$$\mathcal{C}(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z} = 169z^6 + \sum_{n=7}^{\infty} D_A^{n-6} z^n. \quad (14.11)$$

Its realization is analytic for $|D_A z| < 1$.

Proof. The constant term of $1 - D_A z$ is one, so that element is invertible in the ring of formal power series. Solving (14.10) gives the rational expression. Equivalently, the coefficient of degree seven is D_A , and each subsequent coefficient is D_A times the preceding one. The geometric series converges absolutely in the indicated disk. $\square$

Normalization of the first weight enters the uniqueness result. The degree-six impulse and the concatenation ratio, without that initial weight, are also compatible with

$$169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}.$$

The unit rule fixes $\lambda = 1$ before evaluation. For $0 < \alpha < 1$, we have $0 < D_A < 1$ and $D_A \alpha < 1$; therefore,

$$\frac{D_A \alpha^7}{1 - D_A \alpha} = \sum_{n=0}^{\infty} D_A^{n+1} \alpha^{n+7}. \quad (14.12)$$

After the term of index N , the exact remainder is $D_A^{N+2} \alpha^{N+8} / (1 - D_A \alpha)$. The rational expression thus preserves the complete continuation at any precision.

14.3 Positive sections and action per turn

Let $\mathcal{L}_S$ be the oriented action line and $\mathcal{U}_S$ a positive basis. Its sections before and after return are

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S. \quad (14.13)$$

Return preserves the regional compensation $90\mathcal{C}_\pi(\alpha)/\pi$ and the ratio

$$\delta_{\text{ret}} = H_5 - \eta_{\text{ret}}, \quad R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5}{\eta_{\text{ret}}}. \quad (14.14)$$

Proposition 14.6 (Positivity and covariance of the action ratio). *On the intervals (14.4), $0 < \eta_{\text{ret}} < H_5$, and $R_{\text{act}} > 1$. The ratio remains fixed under a change of the common positive basis $\mathcal{U}_S$.*

Proof. The terms of $\mathcal{C}_\pi$ are positive. To bound H_5 , it suffices that $\alpha > 0.007$, $\alpha < 0.008$, and $\varphi < 1.619$. The square-root term in (14.6) is greater than one, and the sum of the magnitudes of its negative terms is less than 0.008001; hence $H_5 > 0.99$. With $D_A < 1$ and $\pi > 3$,

$$0 < \frac{90}{\pi} \mathcal{C}_\pi < 30 \left(169(0.008)^6 + \frac{(0.008)^7}{1 - 0.008} \right) < 10^{-6}.$$

Thus, $\eta_{\text{ret}} > 0.989999 > 0$ and $\eta_{\text{ret}} < H_5$. A change of basis divides both action coordinates by the same positive factor, which cancels in their ratio. $\square$

The structural determination of angular action is the pair $(\hbar_{\text{pre}}, \hbar_{\text{ret}})$ with its identification by return. It preserves the valuation of the local norm, the action character, and the regional continuation. In each section $s \in \{\text{pre}, \text{ret}\}$, the action of a complete turn is

$$h_s = 2\pi \hbar_s. \quad (14.15)$$

The subsequent chart $\mathcal{U}_S \mapsto J$ expresses those sections in action units. The normalized radius to be constructed preserves its independence from this common choice of units.

15 Angular representation and channel orientation

The section $\hbar_{\text{ret}}$ belongs to the action line $\mathcal{L}_S$; η_{ret} is its dimensionless coefficient in the basis specified in (14.13). The angular representation then acts on $(\alpha, \eta_{\text{ret}})$. Writing $[t]_{360}$ for the class of t in $\mathbb{R}/(360\mathbb{Z})$, define

$$\mathcal{P}_{\text{ang}}(\alpha, \eta_{\text{ret}}) = ([1000\alpha]_{360}, [2(\eta_{\text{ret}} + \alpha)]_{360}). \quad (15.1)$$

Nonadic completion determines the factor 1000; bidirectional closure determines the factor two. The circular chart of 360 units preserves the partitions $12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120$. On the positive branch, the lifted representatives are

$$A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha). \quad (15.2)$$

The subscript distinguishes these angular coordinates from the blocks of the dodecaphase record.

Proposition 15.1 (Shift of the base-thousand representation). *Let $\alpha = \sum_{n \geq 1} b_n 1000^{-n}$, with $b_n \in \{0, \dots, 999\}$, be the compatible expansion already generated. Its lifted coordinate satisfies*

$$A_{\text{deg}} = b_1 + \sum_{n \geq 1} b_{n+1} 1000^{-n}.$$

The representation preserves the blocks from the second onward and places the initial block in the integer part.

Proof. The series converges absolutely. Multiplying each term by 1000 and separating the initial term gives the identity. The operation acts on the constructed expansion, with its blocks and memory; the subsequent circular representation takes its class modulo 360. $\square$

15.1 Circular charts and covariance of the action

Definition 15.2 (Charts of the angular coordinate). The degree chart expresses a complete turn by 360 units; the radian chart, by 2π . The coordinate change and its inverse are

$$\theta_A = \frac{\pi}{180} A_{\text{deg}}, \quad \theta_C = \frac{\pi}{180} C_{\text{deg}}^*, \quad A_{\text{deg}} = \frac{180}{\pi} \theta_A, \quad C_{\text{deg}}^* = \frac{180}{\pi} \theta_C. \quad (15.3)$$

A traversal with its turn count retained uses the lifted coordinates. The circular quotient expresses its class modulo the corresponding turn.

The record retains orientation, turns, and transport memory. The final phase and the lift are different representations: the latter distinguishes traversals sharing the former. Each action expression specifies which of these coordinates it uses.

Proposition 15.3 (Covariance of the action readout). *For a section $s \in \{\text{pre}, \text{ret}\}$ and a lifted angular coordinate, one has*

$$\mathcal{S}_s(\theta) = \hbar_s \theta_{\text{rad}} = h_s \frac{\theta_{\text{deg}}}{360}. \quad (15.4)$$

One positive turn expresses h_s .

Proof. Substituting $\theta_{\text{rad}} = \pi \theta_{\text{deg}}/180$ and $h_s = 2\pi \hbar_s$ gives

$$\hbar_s \frac{\pi \theta_{\text{deg}}}{180} = \frac{h_s \theta_{\text{deg}}}{360}.$$

For one turn, $\theta_{\text{deg}} = 360$, yielding h_s . $\square$

The periodicity of a representation retains its own condition: if $U(2\pi) = -I$, operator restoration occurs at 4π . The normalization of action per turn in (15.4) and the return of the represented state thus retain their respective domains.

15.2 Positive channels and reversal of the odd coordinate

In either chart, the channels are expressed through

$$q_{\pm} = \exp[-(\theta_A \pm \theta_C)] = \exp\left[-\frac{\pi}{180}(A_{\text{deg}} \pm C_{\text{deg}}^*)\right]. \quad (15.5)$$

Proposition 15.4 (Recovery of the angular coordinates). *In the lifted real chart, the positive channels determine uniquely*

$$A_{\text{deg}} = -\frac{90}{\pi} \log(q_+ q_-), \quad C_{\text{deg}}^* = \frac{90}{\pi} \log \frac{q_-}{q_+}. \quad (15.6)$$

The interchange $(q_+, q_-) \mapsto (q_-, q_+)$ preserves A_{deg} and reverses C_{deg}^ .*

Proof. The sum and difference of the logarithms in (15.5) are $-\pi A_{\text{deg}}/90$ and $-\pi C_{\text{deg}}^*/90$, respectively. The real logarithm is single-valued on the positive channels. Their product is symmetric, and their quotient is inverted upon interchange. $\square$

The antipodal involution $r \mapsto r + 6 \pmod{12}$ of the record forms six pairs. Uniform distribution of the odd coordinate over those pairs assigns $C_{\text{deg}}^*/6$ to each; their sum recovers C_{deg}^* . Unit conversion is applied exactly once to each argument. In particular, $C_{\text{deg}}^*/A_{\text{deg}} = \theta_C/\theta_A$ when $A_{\text{deg}} \neq 0$, while the preceding state retains the lifted angles and their memory.

16 Dodecaphasic germ and recovery of unimodal branches

The oriented wheel U_{12}^{sign} , received from the TPK return, determines, in the uniform sector $\eta = 0$, the weights $w_m = 1/12$. Fix the phase functional and its ordered real representatives

$$\phi_m = \pi_{\text{HMT}} \frac{3m-1}{18}, \quad m = 1, \dots, 12. \quad (16.1)$$

The real lift of the phases is part of the reader: the following moments are evaluated on these representatives. The construction proceeds from the wheel, its orientation, weights and functional; a modification of any of these data modifies the resulting collection.

16.1 Normalization and analytic profile

Let $a_m = (3m-1)/18$ and $D_0 = \frac{1}{12} \sum_{m=1}^{12} a_m^2$. The integer sums

$$\sum_{m=1}^{12} (3m-1)^2 = 5394 = 6 \cdot 899, \quad \sum_{m=1}^{12} (3m-1)^4 = 4294614$$

give $D_0 = 899/648$. The quadratic normalization condition $s_0^2 \pi_{\text{HMT}}^2 D_0 = 2$ fixes

$$s_0 = \frac{36}{\pi_{\text{HMT}} \sqrt{899}}, \quad k_m = s_0 \phi_m = \frac{2(3m-1)}{\sqrt{899}}.$$

The angular scale cancels in this normalization before iteration of the profile

$$u_0(x) = 1 - \frac{1}{12} \sum_{m=1}^{12} \cos(k_m x), \quad f_\lambda(x) = 1 - \lambda u_0(x). \quad (16.2)$$

Proposition 16.1 (Exact representation and criticality). *The profile u_0 is an even entire function and satisfies*

$$u_0(x) = x^2 - \frac{715769}{2424603} x^4 + O(x^6).$$

With $y = x/\sqrt{899}$, its closed expression is

$$u_0(x) = 1 - \frac{\cos(37y) \sin(36y)}{12 \sin(3y)} = 1 - \frac{\sin(73y) - \sin y}{24 \sin(3y)}, \quad (16.3)$$

where the quotients extend through their removable singularities. For $0 < \lambda \leq 2/u_0(1)$, f_λ is an even, real-analytic self-map of $[-1, 1]$, with a unique quadratic critical point at zero and negative Schwarzian away from that point.

Proof. The finite cosine sum is entire and even. Its Taylor expansion uses

$$\frac{1}{12} \sum_m k_m^2 = 2, \quad \frac{1}{24 \cdot 12} \sum_m k_m^4 = \frac{715769}{2424603},$$

obtained from the preceding integer sums. For the closed form, the progression of arguments is $(6m-2)y$. Summing the geometric series of exponentials and taking the real part gives

$$\sum_{m=1}^{12} \cos((6m-2)y) = \frac{\sin(36y)}{\sin(3y)} \cos(37y).$$

The identity $2 \cos(37y) \sin(36y) = \sin(73y) - \sin y$ proves (16.3). The finite sum defines its extension at every zero of the denominator.

For $0 < x \leq 1$, all arguments satisfy $0 < k_m x \leq 70/\sqrt{899} < \pi_{\text{HMT}}$. Hence

$$u'_0(x) = \frac{1}{12} \sum_m k_m \sin(k_m x) > 0, \quad u'''_0(x) = -\frac{1}{12} \sum_m k_m^3 \sin(k_m x) < 0.$$

Parity gives the opposite signs on the negative interval and proves uniqueness of the critical point. Moreover, $f_\lambda''(0) = -2\lambda \neq 0$. Since $0 \leq u_0(x) \leq u_0(1)$ on $[-1, 1]$, the stated interval for λ gives $-1 \leq f_\lambda(x) \leq 1$. Finally,

$$S(f_\lambda) = \frac{u_0'''}{u_0'} - \frac{3}{2} \left(\frac{u_0''}{u_0'} \right)^2 < 0 \quad (0 < |x| \leq 1).$$

The first quotient is negative on both sides of the origin, and the remaining term is nonpositive. $\square$

16.2 Branchwise inversion and itinerary memory

Proposition 16.2 (Recovery of the folding). *Let $\lambda > 0$ and $v = (u_0|_{[0,1]})^{-1}$. For $y \in [1 - \lambda u_0(1), 1]$, the preimage on a branch is*

$$x = \sigma v\left(\frac{1-y}{\lambda}\right), \quad \sigma = \operatorname{sgn} x. \quad (16.4)$$

At $y = 1$, the preimage is zero and $\sigma = 0$ is fixed.

Proof. Continuity and strict monotonicity of u_0 on $[0, 1]$ provide the inverse v . The equation $y = 1 - \lambda u_0(x)$, together with parity, determines $|x| = v((1-y)/\lambda)$; the sign determines orientation. At $y = 1$, uniqueness of the minimum of u_0 gives $x = 0$. Away from the critical point, the inverse is differentiable. Its derivative becomes singular at the origin, where $u_0(x) = x^2 + O(x^4)$. $\square$

For a finite distribution on pairs $\pm x_i$, with distinct $x_i > 0$, let $p_i^\pm$ be their probabilities, $p_i = p_i^+ + p_i^-$, $Y = f_\lambda(X)$ and $\Sigma = \operatorname{sgn} X$. A mass at zero is also allowed; its branch-uncertainty contribution is zero. With natural logarithms,

$$H(X | Y, \Sigma) = 0, \quad (16.5)$$

$$H(X | Y) = H(\Sigma | Y) = \sum_{i: p_i > 0} p_i h(p_i^+ / p_i), \quad (16.6)$$

$$h(p) = -p \log p - (1-p) \log(1-p). \quad (16.7)$$

The proof groups the distribution by the fibres of f_λ . Monotonicity distinguishes the moduli x_i , and (16.4) distinguishes the two signs within each fibre. With equal probabilities in each pair and no mass at zero, the uncertainty is $\log 2$.

If $0 < \lambda \leq 2/u_0(1)$, the second iteration is defined on the same interval and satisfies

$$H(X | f_\lambda^2(X)) = H(X | f_\lambda(X)) + H(f_\lambda(X) | f_\lambda^2(X)).$$

Indeed, both intermediate maps are deterministic, and the chain rule for the entropy of $(X, f_\lambda(X))$, conditioned on $f_\lambda^2(X)$, gives the identity. Successive recovery uses the signs at each step. For $\lambda = 0$, the map is constant and additionally requires retention of the input modulus.

16.3 Scope of the rescaled return

For $a = -f(1) \neq 0$, let $h_a(x) = -ax$. The operator

$$\mathcal{R}(f) = h_a^{-1} \circ f^2 \circ h_a$$

is a rescaled return on $[-1, 1]$ when the interval $J = h_a([-1, 1])$ belongs to the domain of f^2 and satisfies $f^2(J) \subseteq J$. The scaling h_a is bijective; the two-step return retains its interpretation with the itinerary and the map f . Reconstruction of an iterative root from $\mathcal{R}(f)$ and a is an additional condition. Proposition 16.1 establishes the germ and its critical class. Existence and uniqueness of a fixed point of the renormalizer, its stable and unstable spectral decomposition, and transversality of this collection are the additional hypotheses required for passage to a universal renormalization limit.

17 Action ellipse and determination of the normalized radius

The angular readout and the action section retain the same APP–TRIT–TPK antecedent. The form constructed below uses the representations (15.2) and a positive section $\hbar = \hbar_s$, with $s \in \{\text{pre}, \text{ret}\}$, from (14.13). The angular ratio will determine the form; the action section will preserve its common scale.

17.1 Domain and labelled semiaxes

For brevity, let $A = A_{\text{deg}}$ and $C^* = C_{\text{deg}}^*$. In the chart $A \neq 0$, $\hbar > 0$, and $|C^*/A| < 1$, define

$$\xi = \frac{C^*}{A}, \quad \eta_{\text{el}} = \text{artanh } \xi, \quad a_S = \sqrt{2\hbar} e^{\eta_{\text{el}}/2}, \quad b_S = \sqrt{2\hbar} e^{-\eta_{\text{el}}/2}. \quad (17.1)$$

The semiaxes and the balanced coordinates Q, P have units of the square root of action. The axes remain labelled: the sign of η_{el} determines which is larger. The curve and the product of its semiaxes are

$$\gamma(\theta) = (a_S \cos \theta, b_S \sin \theta), \quad \frac{Q^2}{a_S^2} + \frac{P^2}{b_S^2} = 1, \quad a_S b_S = 2\hbar. \quad (17.2)$$

Proposition 17.1 (Membership of the representations in the elliptic chart). *The positive branch of (15.2), with the intervals (14.4), satisfies $A > 0$, $0 < C^* < A$, and $\hbar_s > 0$. Reversal of the odd coordinate preserves $|C^*/A| < 1$.*

Proof. Proposition 14.6 gives $0 < \eta_{\text{ret}} < H_5$ and positivity of both action sections. For an upper bound on H_5 , (14.4) implies $\alpha < 1/125$ and $\varphi > 8/5$. Retaining only the positive terms of (14.6) yields

$$H_5 < \frac{\sqrt{5}}{2} + \frac{9}{16 \cdot 125^2} + \frac{7}{48 \cdot 125^4}.$$

Since $5 < 81/16$, one has $\sqrt{5}/2 < 9/8$. Moreover,

$$\frac{9}{16 \cdot 125^2} + \frac{7}{48 \cdot 125^4} = \frac{421882}{11718750000} < \frac{1}{1000}.$$

Thus $H_5 < 9/8 + 1/1000 < 2$. Using $\alpha > 7/1000$ now gives

$$0 < C^* = 2(\eta_{\text{ret}} + \alpha) < 2\left(2 + \frac{1}{125}\right) = \frac{502}{125} < 7 < 1000\alpha = A.$$

The channel interchange in 15.4 replaces C^* by $-C^*$ and preserves A ; it therefore preserves the absolute inequality. $\square$

17.2 Parametric phase and oriented area

Proposition 17.2 (Area law and transport of angular coordinates). *The parametric phase θ of γ and its polar angle ϕ , continuously lifted from zero, satisfy*

$$\phi(\theta) = \arg(a_S \cos \theta + ib_S \sin \theta), \quad \frac{d\phi}{d\theta} = \frac{a_S b_S}{a_S^2 \cos^2 \theta + b_S^2 \sin^2 \theta} > 0. \quad (17.3)$$

In a chart where both tangents are defined, $\tan \phi = (b_S/a_S) \tan \theta$. The oriented area swept from phase zero is

$$\mathcal{A}(\theta) = \frac{1}{2} \int_0^\theta (QP' - PQ') dt = \hbar \theta, \quad \mathcal{A}(1) = \hbar, \quad \mathcal{A}(2\pi) = h_s. \quad (17.4)$$

Proof. The derivative of the argument of a nonzero vector is $(QP' - PQ')/(Q^2 + P^2)$. For the curve (17.2), the numerator equals $a_S b_S = 2\hbar$; this proves the derivative formula and its positivity. The quotient P/Q gives the tangent relation; the continuous lift preserves quadrants and turns. The integral equals $a_S b_S \theta/2 = \hbar \theta$. Finally, $h_s = 2\pi \hbar_s$ gives the identity for the complete turn. $\square$

One radian of parametric phase sweeps an area $\hbar$. The polar angle retains its own derivative in (17.3). In polar coordinates, the equivalent law is

$$d\mathcal{A} = \frac{1}{2}\rho_\partial(\phi)^2 d\phi, \quad \rho_\partial(\phi) = \left(\frac{\cos^2 \phi}{a_S^2} + \frac{\sin^2 \phi}{b_S^2} \right)^{-1/2}.$$

The radial formula follows by substituting $(Q, P) = \rho(\cos \phi, \sin \phi)$ into (17.2). With the orientation of γ , the integral of the canonical one-form retains the sign

$$\oint_\gamma P dQ = -\pi a_S b_S = -h_s. \quad (17.5)$$

Indeed, $P dQ = -a_S b_S \sin^2 \theta d\theta$, and the integral of $\sin^2 \theta$ over one turn is π .

17.3 Shape reciprocity and symplectic orientation

Define the matrices

$$D(\eta) = \begin{pmatrix} e^\eta & 0 \\ 0 & e^{-\eta} \end{pmatrix}, \quad S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (17.6)$$

Proposition 17.3 (Quadratic form and action under reciprocity). *Both transports satisfy $T^\top D(-\eta)T = D(\eta)$. For $\omega = dQ \wedge dP$, the interchange S_2 is antisymplectic and the rotation J_2 is symplectic:*

$$S_2^* \omega = -\omega, \quad J_2^* \omega = \omega.$$

If $\mathcal{I}(\gamma) = \oint_\gamma P dQ$ on a closed curve, then

$$\mathcal{I}(S_2 \gamma) = -\mathcal{I}(\gamma), \quad \mathcal{I}(J_2 \gamma) = \mathcal{I}(\gamma). \quad (17.7)$$

Proof. Matrix multiplication interchanges the two diagonal entries of D , proving both congruences. The determinants of S_2, J_2 are $-1, 1$; this gives the transformation of the area form. For $\lambda = P dQ$, direct calculation gives

$$S_2^* \lambda = d(PQ) - \lambda, \quad J_2^* \lambda = \lambda - d(PQ).$$

The integral of the exact differential over the closed curve is zero, proving both action identities. $\square$

Orientation distinguishes the two transports even though they express the same quadratic form, including in the circular case. The labelling of the axes and the sign of the action therefore belong to the data preserved in passing to the lattice representation.

17.4 Normalized radius and lattice-readout matrix

Proposition 17.4 (Determination and recoverability of the shape radius). *The labelled ellipse determines a unique positive radius*

$$R_\star = \sqrt{\frac{b_S}{a_S}} = e^{-\eta_{\text{el}}/2} = \left(\frac{A - C^*}{A + C^*} \right)^{1/4}, \quad \tau_{\text{el}} = iR_\star^2. \quad (17.8)$$

The angular ratio is recovered through

$$\frac{C^*}{A} = \frac{1 - R_\star^4}{1 + R_\star^4}, \quad R_\star^4 = \frac{499\alpha - \eta_{\text{ret}}}{501\alpha + \eta_{\text{ret}}}. \quad (17.9)$$

Interchanging the semiaxes transforms $(\eta_{\text{el}}, R_\star, \tau_{\text{el}})$ into $(-\eta_{\text{el}}, R_\star^{-1}, -1/\tau_{\text{el}})$ and preserves $a_S b_S = 2\hbar$.

Proof. Dividing the semiaxes gives $b_S/a_S = e^{-\eta_{\text{el}}}$. The map $R \mapsto R^2$ is bijective on the positive reals, so the positive root is unique. For $\xi = C^*/A \in (-1, 1)$,

$$2 \operatorname{artanh} \xi = \log \frac{1 + \xi}{1 - \xi};$$

hence $R_\star^4 = (1 - \xi)/(1 + \xi)$. Solving for ξ proves the first identity in (17.9). Substituting $A = 1000\alpha$ and $C^* = 2(\eta_{\text{ret}} + \alpha)$ gives

$$\frac{A - C^*}{A + C^*} = \frac{998\alpha - 2\eta_{\text{ret}}}{1002\alpha + 2\eta_{\text{ret}}} = \frac{499\alpha - \eta_{\text{ret}}}{501\alpha + \eta_{\text{ret}}}.$$

The interchange inverts the semiaxis ratio and changes the sign of η_{el} . The identity $-1/(iR_\star^2) = iR_\star^{-2}$ gives the modular transport. The product of the semiaxes remains fixed. $\square$

On the positive branch, $0 < R_\star < 1$; the reversed shape chart expresses $R_\star^{-1} > 1$. The angular ratio is recovered from the radius, while the absolute angles and their history remain in the enriched state. The circular shape corresponds to $C^*/A = 0$, with $R_\star = 1$. Changing the positive action section, while retaining the angular pair, modifies the common scale of the semiaxes and preserves their ratio.

Proposition 17.5 (Lattice form determined by the ellipse). *The matrix of normalized squared semiaxes is*

$$D(\eta_{\text{el}}) = \frac{1}{2\hbar} \begin{pmatrix} a_S^2 & 0 \\ 0 & b_S^2 \end{pmatrix} = \begin{pmatrix} R_\star^{-2} & 0 \\ 0 & R_\star^2 \end{pmatrix}. \quad (17.10)$$

Its evaluation on an integer pair produces

$$E_\star(m, w) = \begin{pmatrix} m & w \end{pmatrix} D(\eta_{\text{el}}) \begin{pmatrix} m \\ w \end{pmatrix} = \frac{m^2}{R_\star^2} + w^2 R_\star^2. \quad (17.11)$$

The equation of the ellipse multiplied by $2\hbar$ uses, instead, the inverse matrix $D(-\eta_{\text{el}})$.

Proof. The squares of the semiaxes in (17.1), divided by $2\hbar$, are $e^{\eta_{\text{el}}}$ and $e^{-\eta_{\text{el}}}$. By (17.8), these are $R_\star^{-2}$ and $R_\star^2$. Multiplication by the integer vector proves (17.11). Finally, $2\hbar/a_S^2 = e^{-\eta_{\text{el}}}$ and $2\hbar/b_S^2 = e^{\eta_{\text{el}}}$, identifying the matrix of the elliptic equation. $\square$

The radius $R_\star$ is a dimensionless representation of the labelled shape. The polar radius ρ_∂ belongs to the square-root-of-action line. Representation by a spatial radius requires its subsequent dimensional chart; this preserves the shape already determined and the independence of its construction from a prescribed length.

17.5 Oriented depth moment

The realization of the depth register with a quarter-turn defines the parametric moment

$$\mathcal{C}_2(r) = \sum_{k \geq 0} \frac{(-1)^k r^{2k+1}}{(2k+1)^2}, \quad 0 < r < 1, \quad \mathcal{D} = r \frac{d}{dr}.$$

Local uniform convergence permits termwise differentiation:

$$\mathcal{D}\mathcal{C}_2(r) = \arctan r, \quad \mathcal{D}^2\mathcal{C}_2(r) = \frac{r}{1+r^2}.$$

Indeed, the first differentiated series is the integral from zero of the geometric series $1/(1+r^2)$. A second logarithmic differentiation gives the last equality. The conditions $\mathcal{C}_2(0) = \mathcal{D}\mathcal{C}_2(0) = 0$ fix the integration constants and give

$$\mathcal{C}_2(r) = \int_0^r \frac{\arctan t}{t} dt = \int_0^r \frac{\log(r/t)}{1+t^2} dt.$$

Domination by $\sum_{n \geq 1} n^{-2}$ retains the limit $\mathcal{C}_2(1) = G_{\text{Cat}}$. The second derivative at $r = 1$ is understood through its Abel limit.

17.6 Pell dilation, Catalan moment and oriented return

The preceding parametric moment $\mathcal{C}_2$ retains the depth variable; write $G_{\text{Cat}} = \mathcal{C}_2(1)$. Its following specialization receives the scale of the Pell spectral screw

$$V^{-1}SV = \lambda_P i S, \quad \lambda_P = 2 + \sqrt{3}, \quad \rho = \lambda_P^{-1} = 2 - \sqrt{3}.$$

The covariance law is the operator hypothesis of this specialization; it is understood on the common invariant domain of S and the invertible transport V . The unit λ_P belongs to $\mathbb{Q}(\sqrt{3})$; the unit $3 + 2\sqrt{2}$ of the compression realization belongs to $\mathbb{Q}(\sqrt{2})$ and retains its own reader. On its scalar transverse orbit,

$$z_n = z_0(\lambda_P i)^n, \quad r_n = r_0 \lambda_P^n, \quad \theta_n = \theta_0 + \frac{\pi_{\text{HMT}} n}{2}.$$

For $z_0 \neq 0$, eliminating n gives the suspension

$$r(\theta) = r_0 \exp\left(\frac{2 \log \lambda_P}{\pi_{\text{HMT}}}(\theta - \theta_0)\right).$$

The inverse branch has multiplier $-i\rho$. This scale realization receives the screw as its operator antecedent and retains its orientation.

Proposition 17.6 (Spectral reading and evaluation at the contractive scale). *On $\mathcal{H} = \ell^2(\mathbb{N}_{>0})$, let*

$$N\delta_n = n\delta_n, \quad U_4\delta_n = i^n\delta_n, \quad Q_4 = \frac{U_4 - U_4^*}{2i}, \quad C_4 = \frac{U_4 + U_4^*}{2}.$$

For $0 < r < 1$ and $\nu \in \mathbb{C}$, the traces

$$F_\nu(r) = \text{Tr}(N^{-\nu} r^N Q_4), \quad H_\nu(r) = \text{Tr}(N^{-\nu} r^N C_4)$$

converge absolutely and locally uniformly in (ν, r) . One has $F_2(r) = \mathcal{C}_2(r)$ and

$$\mathcal{C}_2(\rho) = \frac{2}{3}G_{\text{Cat}} - \frac{\pi_{\text{HMT}}}{12} \log \lambda_P, \tag{17.12}$$

$$\mathcal{D}\mathcal{C}_2(\rho) = \frac{\pi_{\text{HMT}}}{12}, \quad \mathcal{D}^2\mathcal{C}_2(\rho) = \frac{1}{4}. \tag{17.13}$$

Proof. The diagonal basis gives the series with coefficients $\sin(n\pi/2)$ and $\cos(n\pi/2)$. On a compact set of ν and with $r \leq r_0 < 1$, their absolute values are bounded by $n^M r_0^n$ for some M , a summable series. Selection of the odd indices in the first trace produces exactly $\mathcal{C}_2(r)$.

The tangent subtraction formula gives $\rho = \tan(\pi_{\text{HMT}}/12)$, as a subsequent angular recognition of the Pell unit. For $0 < \theta < \pi/2$, integration by parts of the moment gives

$$\mathcal{C}_2(\tan \theta) = \theta \log(\tan \theta) - \int_0^\theta \log(\tan x) dx.$$

The boundary term at zero vanishes. Subtracting the Fourier series of $\log(2 \sin x)$ and $\log(2 \cos x)$, and integrating with Abel regularization, gives

$$-\int_0^\theta \log(\tan x) dx = \sum_{\substack{n \geq 1 \\ 2 \nmid n}} \frac{\sin(2n\theta)}{n^2}.$$

The regularization converges in the integral because the logarithmic singularity at zero is integrable; the integrated series is dominated by $\sum n^{-2}$.

For odd n , let $\chi_{-4}(n) = (-1)^{(n-1)/2}$. The identity, checked in the six odd residue classes modulo twelve, is

$$2 \sin(n\pi_{\text{HMT}}/6) = \chi_{-4}(n) + 3 \mathbf{1}_{3|n} \chi_{-4}(n/3).$$

Its sum with weight n^{-2} gives $\frac{1}{2}(1 + 3/9)G_{\text{Cat}} = 2G_{\text{Cat}}/3$. Taking $\theta = \pi_{\text{HMT}}/12$ and $\log \rho = -\log \lambda_P$ yields the first evaluation. The others follow from $\mathcal{D}\mathcal{C}_2(r) = \arctan r$, $\mathcal{D}^2\mathcal{C}_2(r) = r/(1 + r^2)$ and $\rho + \rho^{-1} = 4$. $\square$

The real component of the same trace retains

$$H_2(r) = \frac{1}{4} \sum_{k \geq 1} \frac{(-r^2)^k}{k^2}, \quad \mathcal{D}H_2(r) = -\frac{1}{2} \log(1 + r^2), \quad \mathcal{D}^2 H_2(r) = -\frac{r^2}{1 + r^2}.$$

These formulas follow by selecting the even indices and differentiating the absolutely convergent series. Hence

$$\mathrm{Tr}(N^{-2} r^N U_4) = H_2(r) + i\mathcal{C}_2(r).$$

The designation $\mathrm{Li}_2(z) = \sum_{n \geq 1} z^n/n^2$ subsequently recognizes this trace as $\mathrm{Li}_2(ir)$. In particular,

$$\mathrm{Tr}(N^{-2} \rho^N U_4) = \frac{1}{4} \mathrm{Li}_2(-\rho^2) + i \left(\frac{2}{3} G_{\mathrm{Cat}} - \frac{\pi_{\mathrm{HMT}}}{12} \log \lambda_P \right).$$

Replacing U_4 by U_4^* preserves the real part and reverses the imaginary part.

Proposition 17.7 (Composition of depth and orientation). *For $0 < r, s < 1$, $a, b \in \mathbb{Z}/4\mathbb{Z}$ and $m \geq 0$, let $B(r, a) = r^N U_4^a$. Then*

$$B(r, a)B(s, b) = B(rs, a + b), \quad B(r, 1)^m = r^{mN} U_4^m.$$

In particular, $B(r, 1)^4 = r^{4N} \neq I$ for $0 < r < 1$.

Proof. Both factors are diagonal in the same basis. On δ_n , their product multiplies by $r^n i^{an} s^n i^{bn} = (rs)^n i^{(a+b)n}$. The power formula follows by induction, and $U_4^4 = I$ gives the fourth return. The factor r^{4N} remains distinct from the identity. $\square$

The power reader of the inverse multiplier $-i\rho$ is $B(\rho, -1)$. Its imaginary moment has the opposite orientation to that of $B(\rho, 1)$, whereas the damped depth is the same. The transport V , observable $\mathcal{S}$ and reader U_4 retain their respective spaces: the preceding composition concerns the orbit multipliers and their diagonal reading. Phase returns after four steps, while accumulated contraction remains recorded.

18 T-duality: integer coordinates, change of section, and operator equivalence

The arithmetic decomposition of APP preserves the residual representative and its quotient. TPK evolution also transports its orientation and the memory of boundary crossings. Interchanging the two coordinates requires preserving this distinction and specifying the domain on which the operation is again defined. An arithmetic involution is first constructed on that domain; its quadratic form subsequently admits a representation by self-adjoint operators.

18.1 Stable domain of the interchange

In the positive residual section,

$$r_9^+(n) = 1 + ((n - 1) \bmod 9), \quad Q_9^+(n) = \frac{n - r_9^+(n)}{9}, \quad n = r_9^+(n) + 9Q_9^+(n). \quad (18.1)$$

An interchange can place an arbitrary quotient in the first coordinate. Its stable domain is therefore $\mathcal{D}_0 = \mathbb{Z}^2 \times \mathbb{R}_{>0}$, not merely the section $1 \leq r \leq 9$. Extending the domain makes it possible to define a total operation; the admissibility of a pair as the readout of a TPK route separately retains its selection rules.

Definition 18.1 (Normalized compact form and interchange). For $d = (m, w, R_0) \in \mathcal{D}_0$, let

$$E_{R_0}(m, w) = \frac{m^2}{R_0^2} + w^2 R_0^2, \quad T_0(m, w, R_0) = (w, m, R_0^{-1}). \quad (18.2)$$

The symbols m, w name the two transported integer coordinates; R_0 is the positive dimensionless scale of the form. Their interpretation through momentum, winding, and physical radius is established by a subsequent realization map.

Theorem 18.2 (Involution and preservation of the quadratic form). *The map T_0 is an involution of $\mathcal{D}_0$, and*

$$E_{R_0^{-1}}(w, m) = E_{R_0}(m, w).$$

Proof. Two interchanges restore (m, w) , and two inversions restore R_0 . Moreover,

$$E_{R_0^{-1}}(w, m) = w^2 R_0^2 + \frac{m^2}{R_0^2} = E_{R_0}(m, w).$$

□

The equivalence holds for each radius and its reciprocal. At $R_0 = 1$, the radius remains fixed and the operation acts on the same form. A specific pair is fixed by the full involution precisely when, in addition, $m = w$.

18.2 Compact Hamiltonian and self-adjoint domain

Let $\mathcal{H}_{\text{lat}} = \ell^2(\mathbb{Z}^2)$, with orthonormal basis $\{|m, w\rangle\}$. Define

$$H(R_0)|m, w\rangle = E_{R_0}(m, w)|m, w\rangle, \quad (18.3)$$

$$D(H(R_0)) = \left\{ \psi : \sum_{m, w \in \mathbb{Z}} E_{R_0}(m, w)^2 |\psi_{m, w}|^2 < \infty \right\}. \quad (18.4)$$

Finitely supported vectors belong to the domain and form a core for the diagonal operator. The associated quadratic form has domain

$$D(H(R_0)^{1/2}) = \left\{ \psi : \sum_{m, w} E_{R_0}(m, w) |\psi_{m, w}|^2 < \infty \right\}.$$

Proposition 18.3 (Self-adjointness and compact resolvent). *The operator $H(R_0)$ is self-adjoint and nonnegative. Its eigenvalues are $E_{R_0}(m, w)$, counted with multiplicity, and $(H(R_0) + I)^{-1}$ is compact.*

Proof. A multiplication operator defined by a real sequence is symmetric on (18.4). If η belongs to the domain of the adjoint, testing the adjoint identity against each vector $|m, w\rangle$ forces its image to have coordinates $E_{R_0}(m, w)\eta_{m,w}$. Membership in ℓ^2 is exactly condition (18.4); hence the adjoint has the same domain and the same action. Nonnegativity follows by summing the nonnegative weights.

With $c_{R_0} = \min(R_0^{-2}, R_0^2) > 0$, we have $E_{R_0}(m, w) \geq c_{R_0}(m^2 + w^2)$. Outside increasing finite sets, the coefficients $(1 + E_{R_0}(m, w))^{-1}$ tend uniformly to zero. Truncating the resolvent to those sets produces finite-rank operators converging in norm, which proves its compactness. The diagonal basis determines the stated spectrum. $\square$

Theorem 18.4 (Unitary conjugation, including domains). *The interchange $U_T|m, w\rangle = |w, m\rangle$ is unitary, $U_T^2 = I$, and satisfies*

$$U_TD(H(R_0)) = D(H(R_0^{-1})), \quad U_TH(R_0)U_T^{-1} = H(R_0^{-1}). \quad (18.5)$$

It also transports the quadratic-form domains.

Proof. The interchange is an involutive permutation of the orthonormal basis. For a sequence ψ , the identity in Theorem 18.2 gives

$$\sum_{m,w} E_{R_0^{-1}}(m, w)^2 |(U_T\psi)_{m,w}|^2 = \sum_{m,w} E_{R_0}(m, w)^2 |\psi_{m,w}|^2.$$

This yields equality of domains. On each coordinate, $U_TH(R_0)U_T^{-1}$ multiplies by $E_{R_0}(w, m) = E_{R_0^{-1}}(m, w)$. The same sum, with the first power of the weight, proves the result for the quadratic forms. $\square$

The scalar equality and the operator equality are two realizations of the same involution. The second preserves the domains that allow the first to be interpreted as spectral equivalence, not merely as an equality on finitely supported vectors.

18.3 Specialization to the radius of the action ellipse

The radius $R_\star$ determined in Section 17 fixes the realization of this collection that corresponds to the angular representations of the state. The symbol R_0 allows the transport results to be expressed for the entire positive collection; composition with the ellipse evaluates that collection at $R_0 = R_\star$ and also preserves its reciprocal image.

Corollary 18.5 (Elliptic collection stable under duality). *The set*

$$\mathcal{D}_{\text{el}} = \mathbb{Z}^2 \times \{R_\star, R_\star^{-1}\}$$

is stable under τ_0 . Its Hamiltonians are self-adjoint, nonnegative, and have compact resolvent. The interchange U_T transports the domain of $H(R_\star)$ onto that of $H(R_\star^{-1})$, including their form domains.

Proof. The positivity of $R_\star$ and its unique determination are established in Section 17. Inversion permutes $R_\star$ and $R_\star^{-1}$, while interchange preserves $\mathbb{Z}^2$. Proposition 18.3 and Theorem 18.4 apply on these two fibres. If $R_\star = 1$, the two fibres coincide; otherwise the operator intertwines two distinct fibres of the same collection. $\square$

The common action scale determines the area of the ellipse; its semiaxis ratio determines $R_\star$. This separation makes it possible to preserve the area and the common action scale while interchanging the semiaxes and obtaining the reciprocal radius. The index interchange on $\ell^2(\mathbb{Z}^2)$ and the transports of the action plane act on different spaces; their respective properties are proved on their own domains.

18.4 Change of representative and quotient memory

The zero section uses $\bar{r} \in \{0, \dots, 8\}$. The change that preserves the integer in (18.1) is

$$C(r, Q) = (\bar{r}, Q + \mathbf{1}_{\{r=9\}}). \quad (18.6)$$

The modification of the quotient is part of the operation. For $n = 9$, the two pairs are $(9, 0)$ and $(0, 1)$. Their uncorrected energies are $81/R_0^2$ and R_0^2 . Representing the same integer therefore does not imply equality of those two insertions into a form that distinguishes the coordinates.

The equivalence relation is preserved by defining

$$L_9 = \mathbb{Z}(9, -1), \quad S(x_1, x_2) = (x_2, x_1), \quad q_{R_0}(x) = \frac{x_1^2}{R_0^2} + x_2^2 R_0^2.$$

For $x \in \mathbb{Z}^2$, let

$$\mathcal{E}_{R_0}^{L_9}([x]) = \min_{k \in \mathbb{Z}} q_{R_0}(x + k(9, -1)). \quad (18.7)$$

This construction operates on classes. The memory distinguishing individual representatives remains in the TPK state preceding the quotient; the function in (18.7) determines the invariant evaluation of the chosen class.

Theorem 18.6 (Covariant duality of sections). *The energy in (18.7) is well defined. The transformation*

$$\mathsf{T}_{\text{cov}}([x]_{L_9}, R_0) = ([Sx]_{SL_9}, R_0^{-1})$$

is a bijection between $(\mathbb{Z}^2/L_9) \times \mathbb{R}_{>0}$ and $(\mathbb{Z}^2/SL_9) \times \mathbb{R}_{>0}$, with inverse given by the second interchange, and

$$\mathcal{E}_{R_0^{-1}}^{SL_9}([Sx]) = \mathcal{E}_{R_0}^{L_9}([x]).$$

Proof. Replacing x by $x + k_0(9, -1)$ reindexes the candidates for the minimum. The minimum exists: the function of k is quadratic and has leading coefficient $81/R_0^2 + R_0^2 > 0$. The change $(9, Q) \mapsto (0, Q+1)$ has difference in L_9 , so it represents the same class. Interchange transports the equivalence relation to SL_9 , and

$$\min_{\ell \in L_9} q_{R_0^{-1}}(Sx + S\ell) = \min_{\ell \in L_9} q_{R_0}(x + \ell).$$

Finally, $S^2 = I$ recovers the initial quotient and radius. $\square$

The minimum can be evaluated exactly using two candidates. If $x = (m, w)$, the real minimum of the quadratic occurs at

$$k_* = \frac{wR_0^4 - 9m}{81 + R_0^4}.$$

The integer minimum is attained at $\lfloor k_* \rfloor$ or $\lceil k_* \rceil$. This formula does not alter the equivalence; it makes its finite evaluation explicit and allows the effect of changing section to be checked without discarding the quotient.

18.5 Radius normalization in the string realization

Let $\alpha' > 0$ be a scale of the string realization. This symbol denotes the string scale and is kept distinct from the fine-structure constant α . Define

$$\mathsf{T}_{\alpha'}(m, w, R) = \left(w, m, \frac{\alpha'}{R}\right), \quad \mathcal{N}_{\alpha'}(m, w, R) = \left(m, w, \frac{R}{\sqrt{\alpha'}}\right). \quad (18.8)$$

Theorem 18.7 (Exact conjugation by the scale). *Normalization satisfies*

$$\mathcal{N}_{\alpha'} \mathsf{T}_{\alpha'} = \mathsf{T}_0 \mathcal{N}_{\alpha'}.$$

If

$$E_{\text{str}}(m, w, R) = \frac{m^2}{R^2} + \frac{w^2 R^2}{\alpha'^2},$$

then $E_{\text{str}} = \alpha'^{-1} E \circ \mathcal{N}_{\alpha'}$. The combinations

$$p_L = \frac{m}{R} + \frac{wR}{\alpha'}, \quad p_R = \frac{m}{R} - \frac{wR}{\alpha'}$$

transform as $p_L \mapsto p_L$ and $p_R \mapsto -p_R$.

Proof. Both compositions in the first equality give $(w, m, \sqrt{\alpha'}/R)$. On the other hand,

$$E_{R/\sqrt{\alpha'}}(m, w) = \frac{\alpha' m^2}{R^2} + \frac{w^2 R^2}{\alpha'} = \alpha' E_{\text{str}}(m, w, R).$$

Substituting $(w, m, \alpha'/R)$ into the expressions for p_L, p_R gives the two asserted transformations, respectively. $\square$

The conjugation fixes what it means to express the same result in a dimensional chart. In the elliptic collection, the radii are

$$R = \sqrt{\alpha'} R_{\star}, \quad R^{\vee} = \sqrt{\alpha'} R_{\star}^{-1}.$$

Their product is α' . The internal determination of the dimensionless semiaxis ratio and the length-squared scale of the realization serve different functions: the former fixes the shape; the latter expresses its radii in a dimensional unit.

The preceding normalization allows the same result to be expressed with normalized or physical radius. Assigning a tension and a unit to α' belongs to the realization map and has its own origin; it is not a datum entering the production of APP-TRIT-TPK registers.

18.6 Relationship with the modular transformation

Radius inversion has a precise representation on the imaginary axis of the upper half-plane. Let $\iota(R_0) = iR_0^2$ and $S(\tau) = -1/\tau$. Then

$$S \circ \iota(R_0) = \iota(R_0^{-1}). \quad (18.9)$$

The equality follows from $-1/(iR_0^2) = iR_0^{-2}$. For the modular function $J = j - 744$, whose invariance is preserved in the Moonshine development, it follows that $J(iR_0^2) = J(iR_0^{-2})$.

The map ι expresses a semiconjugacy of parameters. The interchange U_T acts on the coordinates of $\ell^2(\mathbb{Z}^2)$; S acts on the upper half-plane; the dual automorphism of an orbifold acts on its graded sectors. Writing out their domains makes it possible to compose specific relationships between them while preserving the information of each representation. Theorems 18.4, 18.6 and 18.7 establish, respectively, unitary conjugation, the change of residual section and scale normalization, with the stated operator domains.

19 Dimensional screens and coordinated state reduction

The ranks appearing in the exceptional prolongation and in compactification belong to different objects. Relating them requires retaining their presentation: generators, relations, inner product, and selected direction. Dimension is then a consequence of the channel module and its relations. Its transport is formulated through maps acting on that module, as well as on the state determining it.

19.1 Channels, relations, and effective dimension

Let $\mathcal{S}_n$ be the set of surviving states of depth n , with compatible restrictions $r_{m,n} : \mathcal{S}_m \rightarrow \mathcal{S}_n, m \geq n$. The boundary datum $\beta_n(x)$ preserves the information required to decide compatibility of a prolongation: sheet, orientation, quotient, memory, and incidence class according to the readout considered. The restrictions satisfy

$$r_{\ell,n} = r_{m,n} r_{\ell,m}, \quad \beta_n r_{m,n} = b_{m,n} \beta_m.$$

These equalities are the naturality conditions of the presentation and retain the same data under restriction in stages or directly.

Definition 19.1 (Channel screen). Fix a field $\mathbb{K}_n$, and let $M_n(x)$ be the space generated by the prolongation channels and $\mathcal{R}_n(x)$ the subspace of relations. The screen is

$$\Sigma_n(x) = (\mathbb{K}_n, M_n(x), \beta_n(x), \mathcal{R}_n(x)),$$

and its effective dimension is

$$d_{\Sigma_n}(x) = \dim_{\mathbb{K}_n}(M_n(x)/\mathcal{R}_n(x)). \quad (19.1)$$

If $M_n(x) \cong \mathbb{K}_n^{g_n}$ and the columns of $R_n(x)$ generate the relations, then $d_{\Sigma_n}(x) = g_n - \text{rank } R_n(x)$.

Proposition 19.2 (Invariance of presentation and transport). *Dimension (19.1) is preserved under invertible changes of basis. For two screens over the same field $\mathbb{K}_n = \mathbb{K}_m = \mathbb{K}$, if the linear map $F_\gamma : M_n(x) \rightarrow M_m(T_\gamma x)$ sends $\mathcal{R}_n(x)$ into $\mathcal{R}_m(T_\gamma x)$, it induces a unique linear map between the screen quotients.*

Proof. A change of generators and relations transforms R into URV , with U, V invertible, preserving its rank. For descent, define $\bar{F}_\gamma([v]) = [F_\gamma(v)]$. If two representatives differ by a relation, linearity implies that their images differ by a relation of the terminal screen; the map is well defined. The quotient projection is surjective and guarantees uniqueness. $\square$

The dimensional change during transport is determined by

$$\Delta d(\gamma; x) = d_{\Sigma_m}(T_\gamma x) - d_{\Sigma_n}(x).$$

This expression retains the state and its channel presentation. An isolated rank permits reconstruction of neither the relations nor the map that produced it.

19.2 Ternary puncturing and syndrome screen

The code $C_W = [12, 6, 6]_3$, constructed from the exceptional incidence, produces another screen by puncturing one coordinate. Denote that map by $p : C_W \rightarrow \mathbb{F}_3^{11}$ and its image by $G_{11} = p(C_W)$.

Proposition 19.3 (Punctured code and perfect partition). *The map p is injective. The code G_{11} has parameters $[11, 6, 5]_3$, and its Hamming balls of radius 2 partition $\mathbb{F}_3^{11}$.*

Proof. A vector in the kernel of p would have support contained in the sole removed coordinate. The minimum distance 6 of C_W excludes every nonzero vector of that form. Thus the dimension remains 6. Puncturing decreases weight by at most one, so the minimum distance of the image is at least 5. The balls of radius 2 are therefore disjoint. Each contains

$$1 + 2 \binom{11}{1} + 2^2 \binom{11}{2} = 1 + 22 + 220 = 243$$

vectors. Since $|G_{11}| = 3^6$ and $3^6 \cdot 243 = 3^{11}$, the balls cover the ambient space. To verify that the distance is exactly 5, take a vector of weight 3. Its decoding ball cannot have centre zero and has a centre of weight at most $3 + 2 = 5$; that centre must have weight at least 5. A word of weight 5 is thereby obtained. $\square$

The integer $243 = 3^{11-6}$ is here the number of classes of the syndrome quotient $\mathbb{F}_3^{11}/G_{11}$. Its function is fixed by the puncturing and distance of the code; equality in cardinality with another fibre does not replace a map between them. This combinatorial screen and the following real screen retain their own operations.

19.3 The dodecaphase direction and rank reduction

The construction in 11.5 starts from the record K , already obtained by APP-TRIT-TPK, and the representation chart specified on $\mathbb{R}^{12}$. In that chart,

$$V_{11} = \mathbf{1}^\perp, \quad P_{11} = I_{12} - \frac{1}{12}J_{12}, \quad u_K = P_3 P_{11} K. \quad (19.2)$$

Here J_{12} is the all-ones matrix and P_3 is the isotypic projector specified by the action of A_5 . The finite sum defining P_3 , its representatives, and the generation of K are retained in the cited section of this same article. In particular,

$$\|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0.$$

The positive norm defines $\ell_K = \mathbb{R}u_K$, and the projector

$$P_{10} = P_{11} - \frac{u_K u_K^\top}{\|u_K\|^2} \quad (19.3)$$

has image $V_{10} = V_{11} \cap u_K^\perp$.

Proposition 19.4 (Two orthogonal reductions). *One has*

$$\mathbb{R}^{12} = \mathbb{R}\mathbf{1} \oplus^\perp V_{11}, \quad V_{11} = \ell_K \oplus^\perp V_{10}, \quad \dim V_{10} = 10.$$

Moreover, $P_{10}P_{11} = P_{10}$, $P_{10}u_K = 0$, and P_{10} is an orthogonal projector.

Proof. Proposition 11.14, proved before the screens are constructed, establishes idempotence, symmetry, rank, and the two operator identities for this same u_K and this same P_{10} . The first decomposition follows from $P_{11} = I - J_{12}/12$; the second is the orthogonal decomposition of that proposition. The single proof of the reduction is thus retained, and its two stages are made explicit here. $\square$

The first reduction removes only the uniform mode. The second removes the direction determined by K in the fixed representation. If the chart is simultaneously changed by an orthogonal matrix S , the vector transforms as $u_K \mapsto Su_K$ and the projector as $P_{10} \mapsto SP_{10}S^{-1}$. This covariance preserves the geometric operation although its coordinates change.

19.4 Lorentzian lattice screen

Let $\Lambda = N_\alpha$ be the Leech lattice constructed in 11.13. Consider the integral hyperbolic plane $U = \mathbb{Z}e_+ \oplus \mathbb{Z}e_-$, with

$$\langle e_+, e_+ \rangle = \langle e_-, e_- \rangle = 0, \quad \langle e_+, e_- \rangle = 1.$$

The lattice $L_{26} = \Lambda \oplus U$ is even and unimodular, of signature $(25, 1)$. Each vector is written $y = \lambda + ae_+ + be_-$. Since $\langle y, e_+ \rangle = b$,

$$e_+^\perp = \Lambda \oplus \mathbb{Z}e_+, \quad e_+^\perp / \mathbb{Z}e_+ \cong \Lambda. \quad (19.4)$$

The quotient map is $q_{24}(\lambda + ae_+) = \lambda$; it is surjective and has kernel $\mathbb{Z}e_+$. The rank decreases by two through an orthogonality restriction followed by an isotropic quotient. This is therefore an operation of a different type from (19.3).

19.5 Joint screen morphism

With the domain $\mathcal{D}_0$ of 18.1, let

$$\mathcal{G}_K = \{(v_{11}, P_{10}v_{11}) : v_{11} \in V_{11}\},$$

$$\mathcal{S}_{\text{HMT}} = \mathbb{R}^{12} \times \mathcal{D}_0 \times e_+^\perp, \quad \mathcal{S}_M^{\text{alg}} = \mathcal{G}_K \times \mathcal{D}_0 \times \Lambda.$$

Definition 19.5 (Algebraic map of coordinated screens). Define

$$\mathcal{R}_M^{\text{alg}}(v, d, y) = ((P_{11}v, P_{10}v), d, q_{24}(y)). \quad (19.5)$$

Theorem 19.6 (Quotient isomorphism and intertwining of duality). *The map (19.5) induces an isomorphism*

$$\overline{\mathcal{R}}_M^{\text{alg}} : (\mathbb{R}^{12}/\mathbb{R}\mathbf{1}) \times \mathcal{D}_0 \times (e_+^\perp/\mathbb{Z}e_+) \longrightarrow \mathcal{S}_M^{\text{alg}}.$$

It intertwines the transformations acting by $\mathbb{T}_0$ on the factor $\mathcal{D}_0$ and by the identity on the other two factors. The form $E_{R_0}(m, w)$ is preserved under that duality.

Proof. The map $[v] \mapsto P_{11}v$ is an isomorphism from $\mathbb{R}^{12}/\mathbb{R}\mathbf{1}$ onto V_{11} . Since $P_{10}P_{11} = P_{10}$, the map $[v] \mapsto (P_{11}v, P_{10}v)$ identifies that quotient with $\mathcal{G}_K$; its inverse takes the first component and its class. By (19.4), q_{24} induces the lattice isomorphism of the third factor. The second is transported by the identity. The product of the three maps proves the isomorphism.

The projectors and q_{24} do not modify d , while duality acts only on d . The two compositions are therefore identical. Preservation of the form is Theorem 18.2. $\square$

The graph $\mathcal{G}_K$ retains the preceding coordinate v_{11} together with its projection. The theorem proves equivalence of the quotient presentations; the isolated reduction $V_{11} \rightarrow V_{10}$ retains its kernel ℓ_K . This precision establishes what information the isomorphism retains and what information is removed by each projection composing it.

Corollary 19.7 (Screens over the elliptic collection). *The same morphism restricts to an isomorphism*

$$(\mathbb{R}^{12}/\mathbb{R}\mathbf{1}) \times \mathcal{D}_{\text{el}} \times (e_+^\perp/\mathbb{Z}e_+) \cong \mathcal{G}_K \times \mathcal{D}_{\text{el}} \times \Lambda,$$

and preserves the intertwining of duality.

Proof. The morphism and its inverse preserve the second factor exactly. Corollary 18.5 proves that duality preserves $\mathcal{D}_{\text{el}}$. Thus both the bijection and the already proved intertwining square restrict to it. $\square$

19.6 Coordination with the exceptional realization

On the domain $\mathcal{X}_{\text{HMT}}$, with phase, orientation, quotient, boundary, incidence, grading, and memory, the readouts ε_{exc} and ε_{scr} extract the exceptional and screen data. The corridor constructs the graded algebra $\mathfrak{B}(x) \cong V^{\natural}$, with its vertex product and automorphisms; $\mathfrak{S}_M(x)$ assembles the preceding screens, quotients, and operators. The joint assignment $x \mapsto (\mathfrak{B}(x), \mathfrak{S}_M(x))$ therefore has an algebra as its first component and a collection of spaces and maps as its second. The Monster action belongs to the first; the reduction directed by K and compact duality belong to the second. The two readouts establish their common provenance, and each construction retains its type. Theorem 19.6 gives the isomorphism factor by factor; its proof identifies the inverses, kernels and operations preserved by each component.

20 Action on trajectories and eleven-dimensional dynamical realization

Extending the structure to dimensional dynamics requires joint transport of the state, channels and boundary data. This section begins with a functional defined on TPK trajectories and its exact composition law. The worldsheet and eleven-dimensional codomain are introduced afterwards, with the maps that must preserve that composition and the operators already constructed.

20.1 Descent of an evolution and sufficiency of the observed datum

Let $q : \mathcal{X}_{\text{HMT}} \rightarrow Y$ be a surjective readout of states with memory, and let $U : \mathcal{X}_{\text{HMT}} \rightarrow \mathcal{X}_{\text{HMT}}$ be a complete evolution. An evolution on observed values exists precisely when the datum preserved by q suffices to determine the next observed value.

Theorem 20.1 (Exact criterion for descent along fibres). *There exists a unique map $\bar{U} : Y \rightarrow Y$ satisfying $qU = \bar{U}q$ if and only if*

$$q(x) = q(y) \implies q(Ux) = q(Uy). \quad (20.1)$$

Proof. If $\bar{U}$ exists, applying the same function to $q(x) = q(y)$ proves the condition. Conversely, define $\bar{U}(q(x)) = q(Ux)$. Condition (20.1) guarantees independence of the representative. Surjectivity of q defines the map on all of Y and proves uniqueness. $\square$

The theorem determines what memory a readout must preserve. When two representatives have different observed images, the retained datum may be enlarged, the domain restricted, or a multivalued correspondence used. These are distinct mathematical choices. In particular, return of a phase coordinate does not authorize identification of the evolutions of two histories retaining different memories.

20.2 Cocycle of directional and arithmetic changes

An admissible TPK trajectory carries, on each edge e_j , a direction d_j and an arithmetic type t_j . To count changes in a concatenation, the preceding datum is also preserved. The augmented edge is $\tilde{e}_j = (e_j, d_{j-1}, t_{j-1})$; the pair (d_0, t_0) belongs to the initial boundary. Define

$$w_{\text{giro}}(\tilde{e}_j) = \mathbf{1}_{\{d_j \neq d_{j-1}\}}, \quad w_{\text{tipo}}(\tilde{e}_j) = \mathbf{1}_{\{t_j \neq t_{j-1}\}}.$$

For $\tilde{\gamma} = (\tilde{e}_1, \dots, \tilde{e}_R)$, let

$$\mathbf{I}(\tilde{\gamma}) = \left(\sum_{j=1}^R w_{\text{giro}}(\tilde{e}_j), \sum_{j=1}^R w_{\text{tipo}}(\tilde{e}_j) \right) = (N_{\text{giro}}, N_{\text{tipo}}). \quad (20.2)$$

Two trajectories compose when their endpoints agree and the preceding datum of the second agrees with the terminal datum of the first.

Proposition 20.2 (Additivity with preserved boundary). *For every composable pair,*

$$\mathbf{I}(\tilde{\gamma}\tilde{\eta}) = \mathbf{I}(\tilde{\gamma}) + \mathbf{I}(\tilde{\eta}). \quad (20.3)$$

Thus, $\mathbf{I}$ is an additive cocycle with values in $\mathbb{N}^2$ on the category of augmented trajectories. For every $\lambda > 0$, the functional

$$I_\lambda(\tilde{\gamma}) = N_{\text{giro}}(\tilde{\gamma}) + \lambda N_{\text{tipo}}(\tilde{\gamma})$$

is also additive.

Proof. Concatenation does not change the preceding datum of any edge: for the first edge of $\tilde{\eta}$, the composition condition makes it agree with the terminal datum of $\tilde{\gamma}$. The sum over all edges splits exactly into the two sums in (20.3). Applying the linear functional $(a, b) \mapsto a + \lambda b$ proves the last statement. $\square$

Boundary information is an effective part of the proof. If two one-edge segments are evaluated separately while ignoring the connection between them, both may have zero turn count even though their directions differ. The concatenated path then contains a turn. Preserving (d_{j-1}, t_{j-1}) records that term on the first edge of the second segment and restores additivity.

For fixed endpoints and boundary data, the minimum-interaction principle selects minimizers of I_λ among admissible routes. The principle is a selection rule, whereas the following result establishes a finite-existence property.

Proposition 20.3 (Finite existence of minimizers). *If the set $\Gamma(a, b)$ of admissible trajectories with the fixed boundary is finite and nonempty, there exists $\gamma_* \in \Gamma(a, b)$ such that $I_\lambda(\gamma_*) \leq I_\lambda(\gamma)$ for all $\gamma \in \Gamma(a, b)$.*

Proof. The image of $\Gamma(a, b)$ under I_λ is a nonempty finite set of real numbers, which has a minimum element and at least one preimage. $\square$

The proposition does not select the value of λ , nor does it confuse existence with uniqueness. The value and selection rule used in a particular realization must be part of its declared data. The additive functional recovered here quantifies changes of the path itself; its transport to a dimensional action is a later operation.

20.3 Extended persistence and worldsheet

Realization of a route as an extended object distributes its genealogy along an internal coordinate. For a string, the history has two local coordinates σ^a : one internal spatial coordinate and one temporal coordinate. A realization is described by a map $X : \Sigma \rightarrow \mathcal{M}$, a metric h_{ab} on the worldsheet and background fields in the target space. Conditions on $\partial\Sigma$ belong to the variational domain. For a brane of spatial dimension p , the history is a worldvolume of dimension $p + 1$.

The variational realization of the string is expressed by the Polyakov action

$$S_P[X, h] = -\frac{T_s}{2} \int_\Sigma d^2\sigma \sqrt{-h} h^{ab} g_{MN}(X) \partial_a X^M \partial_b X^N, \quad (20.4)$$

where T_s is tension. The notation T_s distinguishes it from duality T_0 , cardinal transport T_d and complete evolution U .

Composition of augmented trajectories already has the property proved in 20.2. A continuous realization of that composition must specify the map from paths to configurations, its transport of boundary conditions and the normalization comparing I_λ with the continuous action. Integral (20.4) specifies the codomain and its fields; its parameters do not retrospectively select TPK emissions or registers.

20.4 Eleven-dimensional realization and graded operators

The reduction directed by K provides $V_{11} = \ell_K \oplus V_{10}$, and Theorem 19.6 coordinates this reduction with the lattice screen and duality. The physical realization is typed through

$$\Xi_M : \mathcal{S}_M^{\text{alg}} \longrightarrow \mathcal{X}_{11}^{\text{phys}}.$$

The codomain carries a metric g_{MN} , a three-form C_{MNP} , a gravitino ψ_M , an action, gauge constraints and supersymmetry transformations. An operator realization includes

$$\mathfrak{S}_M = (\mathcal{H}_{\text{phys}}, H, Q), \quad \mathcal{H}_{\text{phys}} = \mathcal{H}_+ \oplus \mathcal{H}_-.$$

The grading is expressed through an involution Γ , equal to $+I$ on $\mathcal{H}_+$ and $-I$ on $\mathcal{H}_-$. The supercharge is odd, $\Gamma Q = -Q\Gamma$, and the Hamiltonian is nonnegative. The relation $Q^2 = H$ is understood as an equality of operators, including their domains.

Let $\mathcal{H}_{\text{HMT}}$ be a Hilbert-space realization of the relevant genealogical states, with operators $\hat{H}, \hat{Q}$. An isometry $W : \mathcal{H}_{\text{HMT}} \rightarrow \mathcal{H}_{\text{phys}}$ must transport domains, grading and observables. The operator conditions are written

$$W\hat{H} = HW, \quad W\hat{Q} = QW, \quad WU_T^{\text{HMT}} = U_T^{\text{phys}}W. \quad (20.5)$$

Theorem 20.4 (Compatible transport of the supercharge). *Suppose $\hat{Q}^2 = \hat{H}$, $WD(\hat{Q}) \subset D(Q)$, and the intertwining relations of (20.5) hold on their domains. Then, for $\psi \in D(\hat{H}) = D(\hat{Q}^2)$,*

$$W\psi \in D(Q^2), \quad Q^2W\psi = HW\psi.$$

If physical transport of the radii also respects (18.5), the transported compact sector preserves its spectrum under reciprocal radii.

Proof. Since $\psi \in D(\hat{Q}^2)$, both ψ and $\hat{Q}\psi$ belong to $D(\hat{Q})$. Thus, $W\psi \in D(Q)$ and $QW\psi = W\hat{Q}\psi \in D(Q)$. One may apply Q again, and

$$Q^2W\psi = W\hat{Q}^2\psi = W\hat{H}\psi = HW\psi.$$

Spectral equality of the compact sector is transported by the isometry and the unitary T-intertwiner; on the closed image of W , this is a unitary equivalence of the corresponding realizations. $\square$

The theorem explicitly preserves the operators and the domain on which the identity is proved. The grading organizes two sectors; construction of fermionic fields and their occupation relations has its own structure. Thermal distributions and the species catalogue are not used in the proof of the screens or in the compact duality of this article.

20.5 Compatible compactification and conclusion of the development

A compactification of the state is described through a prolongable section x_∞ , a map κ_{comp} to the chosen moduli space and a transport $\mathcal{U}$ of the metric, orientation, boundary and memory data. The subscript distinguishes this map from the rational readout κ of the dodecaphasic register. Its compatibility requires four concrete properties:

1. the finite restrictions of x_∞ belong to admissible TPK prolongations;
2. the compactification map intertwines the dualities, $\kappa_{\text{comp}}\mathbb{T} = \mathbb{T}_M\kappa_{\text{comp}}$;
3. observables and the spectrum descend to the chosen quotients according to criterion 20.1;
4. the realization transports the direction ℓ_K , boundary conditions and operator domains, and preserves the declared action and supersymmetry equations.

The preceding proofs provide the phase Hilbert-space realization, tracial closure, screens with their quotient maps, radius involution, its unitary equivalence and the additive trajectory cocycle. The eleven-dimensional realization specifies the fields and compatibilities it uses when receiving that structure.

Conditions for physical identification. Complete physical identification must additionally specify the tension and action normalization, construct the oscillator sector and its amplitudes, control anomalies and establish a consistent low-energy gravitational limit for each stated compactification. The string action and change of scale used here are expressed in (20.4) and (18.8), respectively. Theorem 20.4 proves the supercharge identity on the image of W , under the domains and intertwining relations in its statement. The conditions for complete physical identification do not alter this identity or the other algebraic proofs: they delimit the physical realization to which those results are transported.

The exceptional and dimensional prolongations remain linked by their common origin. The Monster acts on the graded construction $V^{\natural}$; the compact Hamiltonian and screens receive data from other maps of the same state. This formulation preserves the positive content of the dual realization, with its operations, proofs and the information transmitted at each stage: Theorem 11.25 establishes the exceptional identification, and Theorems 18.4 and 19.6 establish compact equivalence and screen coordination.

21 Conclusions

The discrete structure of the continuum developed in this article supports an exceptional realisation reaching $V^{\natural}$ and an operator-theoretic realisation preserving compact duality. The connection is constructed: the APP–TRIT–TPK state transmits orientation, sheets, residues, quotients, boundary data and memory to its numerical coordinates, incidence register, and phase and action representations. The central contribution relates these realisations through codes, gluings, vertex products, projectors and intertwining operators with specified domains.

The joint construction of the continuum supplies the antecedent of these maps. The same extensions produce the measure on histories, the sheetwise Gram identity with its terminal remainder, the cancellation defect, rectangular incidence and the natural transport of the memory complex. The assembly retains all continuations and satisfies $D^2 = 0$ on its generators; its determinant line retains degrees and bases. Compatible weights, isometric transport, the survival remainder and the conditions for a scalar determinant reading retain the specific roles assigned to them by the proofs. These correlated operations constitute the common structure preceding numerical, exceptional and dimensional representations.

The arithmetic and incidence connection is specified by results with explicit domains. The defect laws distinguish modulus refinement, periodic repetition, and increasing dimension. The aggregate operator preserves an integral sublattice of index two. In its quadratic block $3H$, the normalized unit H has norm one; the Pell recurrence determines H^n , while the original block evolves as $3^n H^n$. The bijection between pairs of trits and nonadic digits preserves carry, while projection of the ambient cube has nine elements per fibre. The alphabet of 42 blocks, regional junctions, and correlations of ordered records make explicit the information preceding linear representations.

The five regional closures attain the minimum of eight in the finite problem of covering the four prescribed edges. The graph retains emission labels, and the quotient retains their incidences between phase classes. Oriented reversal makes the reader v_{120} odd and preserves the even parity of v_{270} ; these identities make explicit the response of memory to a change of orientation.

The regional comparison determines three polarised configurations: the stars of B_K , B_φ and B_{APP} have signatures $(3, 1, 0)$, $(1, 0, 3)$ and $(1, 1, 2)$, respectively. The corona $\{i : K_i \geq 729\} = B_K = H_e \cap P_\pi$ links the numerical register to an incidence four-subset, and the signed cochain fixes $H_{\bar{\alpha}} = B_K \sqcup \{5, 7\}$. The complete census places the signature $(3, 1, 0)$ at fifteen four-subsets. Rigidity belongs to the ordered frame $(H_{\bar{\alpha}}, H_e, H_\varphi, H_{APP})$: its stabiliser in G_W is the identity. This result fixes the permutation freedom on the twelve positions; the values of K retain their determination by the integer register and its reconstruction operations.

The integer lift of the regional signature supplies a complementary result. The identities $p + e + f = q + 3c$ and $p + e - f = a + 3h$ retain the quotients accompanying the residues, and the carry cocycle makes their composition compatible with association of the sum. In the complete table of 27 columns, the triples $(0, 0, 0)$ and $(1, 2, 0)$ have $q = a = 0$, whereas their pairs (c, h) are $(0, 0)$ and $(1, 1)$. Each reduction $\bar{c}, \bar{h}$, adjoined separately to the evaluation of $1, p, e, f$, increases the rank from four to five. The complete reading (q, a, c, h) itself retains fibres: $(1, 2, 0)$ and $(2, 1, 0)$ produce the same datum $(0, 0, 1, 1)$. The transport identities and cylinder bounds therefore specify both the additional information retained by integer memory and the multiplicity of antecedents remaining in each reading. This articulation jointly preserves the numerical content and incidence of the regional history.

The identity $C^2 = 5I_6$ follows from quadratic correlation evaluated at the five finite positions. Saturation and dual neutrality leave a pair of boundaries whose representative depends on the declared orientation. Closure controls distinguish that selection from variations of the boundary and of register position.

The exceptional result follows the incidence of K through to the vertex algebra. The dodecaphasic register is constructed before its rational reading and support selection. Its signed configuration enters the Witt flag; the ternary code determines the gluing of A_2^{12} , and the flag with the specified metric, origin and chamber determines the Leech neighbour. The order-two and order-three extensions of its lattice algebra realise $V^{\natural}$ by application of the corresponding construction theorems, whose hypotheses are verified in the lattice corridor. The chosen isomorphisms transport the vacuum, conformal vector, grading, product and traces; they do not identify an order-two automorphism with an order-three automorphism. The Monster acts on $V^{\natural}$, and Moonshine determines its graded traces. The rational reading of the register and the digits of the constants have other domains: their connection with the algebra is the constructed chain of objects and maps, not an action on their digits.

In the exceptional realization, the rational identity proves $J - (t_3 + 12) \in 3^9 q\mathbb{Z}[[q]]$; its first coefficient determines optimality of the uniform exponent. The formal proof retains its scope at every degree, while truncated checks retain their role as finite comparisons.

The same K determines a geometric polarisation and a recovery reference. In the fixed real representation, the nonzero vector u_K determines a line in V_{11} ; its exact squared norm normalizes the rank-one term of the rank-ten orthogonal projector, whose covariance law is specified. The full dodecaphasic orbit is invertible and identifies operators from their responses with exact stability bounds. Its rational encoding preserves the twelve blocks in the fixed chart; bilateral balance recovers the state from two channels defined by isometries, and its unitary extension jointly recovers state and incoming memory. Invariance of other observables retains the proved commutation condition. The arithmetic–geometric role of K therefore comprises incidence, direction and identification in addition to its scalar evaluation.

The unital extension of the Weyl representation establishes a continuous realisation of tracial dimension. The phase and shift operators generate each finite matrix algebra, and full-fibre inclusions preserve the unit and normalised trace. The nonadic uniformly hyperfinite closure has a tracial representation whose weak closure is a hyperfinite factor of type II_1 ; normalised ranks attain every tracial dimension in the unit interval within it. A marked continuation uses another inclusion and produces a different closure. The result identifies the refinement operation responsible for preserving global normalisation.

The action and angular pair determine the compact form rather than receiving a prescribed radius. Normalised regional renewal and the positive action sections fix the common scale of the ellipse; its angular coordinates fix the semiaxis ratio. The unique positive radius satisfies

$$R_{\star}^4 = \frac{A - C^*}{A + C^*} = \frac{499\alpha - \eta_{\text{ret}}}{501\alpha + \eta_{\text{ret}}}.$$

Membership in the elliptic chart and the choice of positive root are proved. The semiaxis product preserves action; their exchange transforms the radius into $R_{\star}^{-1}$. The angular charts, axis orientation and action scale thus retain their respective roles in the quadratic form.

The established T-duality is a scalar and operator equivalence. A single involution preserves the compact form and unitarily conjugates the self-adjoint Hamiltonians of reciprocal radii, including their domains and quadratic-form domains. Covariance of the residual section requires transport of L_9 ; on the corresponding quotient, the minimum energy is well defined and preserves duality. String normalisation transports these identities to dimensional radii without changing the generated shape radius. The elliptic collection contains both reciprocal fibres and preserves the restriction of domains and of the screen isomorphism. Modular semiconjugacy on the imaginary axis connects their parameters and functions while retaining the proper domain of the operators in each realisation.

The screens express the dimensional hierarchy through specified maps. The reductions $12 \rightarrow 11 \rightarrow 10$ and $26 \rightarrow 24$ arise, respectively, from removal of the uniform mode and the direction u_K , and from orthogonal restriction followed by the isotropic quotient of the Lorentzian lattice. The joint morphism identifies the quotient presentations and intertwines the compact sector. The graph retains the coordinate in V_{11} together with its projection; each individual projection retains its kernel. Ternary puncturing adds the length-eleven combinatorial screen with distance five and a perfect partition. These operations determine the ranks, rather than an identification between spaces of coincident dimension.

The extension towards M-theory has a determined algebraic support and a precise dynamical condition. Changes of direction and arithmetic type along trajectories, composed with their preceding boundary datum, form an exactly additive cocycle. The worldsheet, Polyakov action and codomain of eleven-dimensional fields subsequently receive this structure through the realisation maps. The transport theorem preserves $Q^2 = H$ on the image of the isometry under the stated intertwining and domain conditions. Complete physical identification includes the fields, tensions and action normalisation, the oscillator sector and its amplitudes, anomaly control and the low-energy gravitational limit of each compactification (Section 20.5). These requirements fix the dynamics into which the proved structures are transported.

A common chain is thus established linking the discrete construction of the continuum with the Moonshine algebra, dodecaphasic polarisation, tracial realisation, compact duality and action composition. Its maps determine what information remains, which quotient is taken and which equations are preserved. This is the mathematical content received by the dimensional realisations and on which the developments of statistics, radiation and species structure continue.

A Finite enumerations of emissions and Witt stars

The two enumerations in this appendix concern distinct objects. The first classifies the cursor signatures of APP emission; the second computes the group generated by the stars of the Witt design. Their domains and rules are retained so that each table can be reproduced without being used as an input to the calculation.

A.1 Initial-condition indices and signature recurrence

Let $i, j \in \{0, \dots, 8\}$ and $a \in \{0, 1, 2, 3\}$, with direction order (N, E, S, O) and vectors

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The index

$$\text{id}(i, j, a) = 4(9i + j) + a$$

ranges over exactly $\{0, \dots, 323\}$. Its inverse is obtained from $a = \text{id} \bmod 4$, $j = \lfloor \text{id} / 4 \rfloor \bmod 9$ and $i = \lfloor \text{id} / 36 \rfloor$. Two indices, one per sheet, enumerate the 324^2 joint conditions in (4.1); this domain is not identified with the 81 positions.

For each index, the cursor starts at (i, j) , with vector v_a . The instants $t = 1, \dots, 54$ are traversed in six windows of nine instants. Write $r_t = 1 + ((t - 1) \bmod 9)$. The rule is:

1. On the additive sheet, an operation occurs when $r_t \in \{1, 4, 7\}$: $\rho_9(i + j + 2)$ is added to the window accumulator, and the cursor moves by v_a , modulo nine.
2. On the multiplicative sheet, an operation occurs when $r_t \in \{2, 5, 8\}$: $\ell_9(\rho_9((i + 1)(j + 1)))$ is added, and the cursor moves by v_a , modulo nine. The extension of ℓ_9 by zero outside the units, stated in Section 4, is retained.
3. At all other instants, that cursor does not move. After instants 27 and 54, v_a is replaced by $-v_a$.
4. At the end of each window, its accumulator is recorded modulo ten, and only the accumulator is reset; neither position nor orientation is reset.

This defines two functions on the 324 indices,

$$s : \{0, \dots, 323\} \longrightarrow \{0, \dots, 9\}^6, \quad e : \{0, \dots, 323\} \longrightarrow \{0, \dots, 9\}^6.$$

The multiplicity of a signature is the number of indices with that image. The witness index in each table is the least index in its fibre. Complete preimages are recovered by applying the recurrence to all 324 indices and retaining those that produce the stated signature.

Table 13: Additive signatures: image of s , multiplicity and witness index.

Class	Signature	Multiplicity	Witness index
1	122564	18	17
2	122898	18	24
3	212645	18	5
4	212988	18	0
5	221456	18	29
6	221889	18	12
7	456221	18	28
8	465898	18	21
9	546988	18	9
10	564122	18	16
11	645212	18	4
12	654889	18	33

Class	Signature	Multiplicity	Witness index
13	889221	18	13
14	889654	18	32
15	898122	18	25
16	898465	18	20
17	988212	18	1
18	988546	18	8

Table 14: Multiplicative signatures: image of e , multiplicity and witness index.

Class	Signature	Multiplicity	Witness index
1	000000	108	8
2	177393	8	1
3	177555	8	54
4	177771	8	11
5	339555	8	6
6	339771	8	15
7	339933	8	59
8	393177	8	0
9	393393	8	45
10	393555	8	40
11	555177	8	52
12	555339	8	4
13	555393	8	41
14	555555	24	84
15	555717	8	14
16	555771	8	18
17	555933	8	24
18	717555	8	12
19	717717	8	21
20	717933	8	19
21	771177	8	9
22	771339	8	13
23	771555	8	16
24	933339	8	57
25	933555	8	26
26	933717	8	17

The sums of multiplicities are $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$. Joint emission is obtained by applying (4.3) to every pair of signatures. The inverse through hundreds and tens proved in Section 4 identifies the 468 emissions and gives their multiplicities as the product of the two fibres. The subsequent map (4.4) produces the 243 visible words. The initial domain, emission, its fibres and the ternary quotient are thus specified separately.

A.2 Explicit generators and closure of the stars

The matrix A_W of (11.7) is retained, and the following words are formed in lexicographic order:

$$c(w) = (w, wA_W), \quad w \in \mathbb{F}_3^6.$$

They are grouped by weight, and the weight-six supports are retained. This produces the enumerator (11.16) and the 132 hexads of $\mathcal{H}$. For each tetrad $B \subset \{1, \dots, 12\}$, the four pairs $H \setminus B$, with $B \subset H \in \mathcal{H}$, define σ_B : it fixes B and interchanges the two points in each pair. This yields the collection of 495 stars in Proposition 11.16.

The six generators used in its closure are expressed by their lists of images of $1, \dots, 12$:

	1	2	3	4	5	6	7	8	9	10	11	12
g_1	1	2	3	4	12	11	9	10	7	8	6	5
g_2	1	2	3	12	5	10	11	9	8	6	7	4
g_3	1	2	3	11	10	6	8	7	12	5	4	9
g_4	1	2	12	4	5	9	8	7	6	11	10	3
g_5	1	12	3	4	5	8	10	6	11	7	9	2
g_6	12	2	3	4	5	7	6	11	10	9	8	1

Each row is verified to be a star by comparison with the σ_B already constructed. For $j = 1, \dots, 6$, the closure is computed by

$$G_j^{(0)} = \{\text{id}\}, \quad G_j^{(n+1)} = G_j^{(n)} \cup \{g_i \circ p : 1 \leq i \leq j, p \in G_j^{(n)}\}.$$

The increasing sequence stabilizes in the finite set $\text{Sym}(12)$. Since the generators are involutions, its stable value is exactly the group G_j they generate. With the convention $(g_i \circ p)(a) = g_i(p(a))$, their orders are

j	1	2	3	4	5	6
$ G_j $	2	6	18	360	7920	95040.

Finally, each of the 495 stars is checked to belong to G_6 , and all are checked to preserve $\mathcal{H}$. The first inclusion, together with membership of the six generators in the collection, equates the two generated groups. Preservation is checked through $\{\sigma_B(H) : H \in \mathcal{H}\} = \mathcal{H}$. This is the finite enumeration used in the proof of 11.16; the subsequent identification as M_{12} retains the external reference stated there.

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